



AppZone m2mb Sample Apps

80000NT11840A Rev. 1 - 2021-01-29

TELIT
TECHNICAL
DOCUMENTATION

1 AppZone m2mb Sample Apps

Package Version: **1.1.26-CxX**

Minimum Firmware Version: **25.30.008.0**

1.1 Features

This package goal is to provide sample source code for common activities kick-start.

2 Quick start

2.1 Deployment Instructions

To manually deploy the Sample application on the devices perform the following steps:

1. Have **25.30.008.0** FW version flashed (AT#SWPKGV will give you the FW version)
2. Copy m2mapz.bin to /mod/

`AT#M2MWRITE="/mod/m2mapz.bin",<size>,1`

where <size> is in bytes
3. Configure the module to run the downloaded binary as default app:
`AT#M2MRUN=2,m2mapz.bin`
4. Restart the module and if no AT commands are sent within **10** seconds, start the app: `AT+M2M=4,10`

2.2 References

More info on

- [Getting started with ME910C1](#) (doc ID 80529NT11661A)
- [How to run applications with AppZone](#)

2.3 Known Issues

None

2.4 Contact Information, Support

For general contact, technical support services, technical questions and report documentation errors contact Telit Technical Support at: TS-EMEA@telit.com.

For detailed information about where you can buy the Telit modules or for recommendations on accessories and components visit:

<http://www.telit.com>

Our aim is to make this guide as helpful as possible. Keep us informed of your comments and suggestions for improvements.

Telit appreciates feedback from the users of our information.

2.5 Troubleshooting

- Application does not work/start:
 - Delete application binary and retry
`AT#M2MDEL="/mod/m2mapz.bin"`
 - Delete everything, reflash and retry
`AT#M2MDEL="/mod/m2mapz.bin"`
`AT#M2MDEL="/mod/appcfg.ini"`
- Application project does not compile
 - Right click on project name
 - Select Properties
 - Select AppZone tab
 - Select the right plugin (firmware) version
 - Press "Restore Defaults", then "Apply", then "OK"
 - Build project again
- Application project shows missing symbols on IDE
 - Right click on project name
 - Select Index
 - Select Rebuild. This will regenerate the symbols index.

2.6 Making source code changes

2.6.1 Folder structure

The applications code follow the structure below:

- `hdr`: header files used by the application
 - `app_cfg.h`: the main configuration file for the application
- `src`: source code specific to the application
- `azx`: helpful utilities used by the application (for GPIOs, LOGGING etc)
 - `hdr`: generic utilities' header files
 - `src`: generic utilities' source files
- `Makefile.in`: customization of the Make process

2.7 Import a Sample App into an IDE project

Consider that the app HelloWorld that prints on Main UART is a good starting point. To import it in a project, please follow the steps below:

On IDE, create a new project: "File"-> "New" -> "Telit Project"

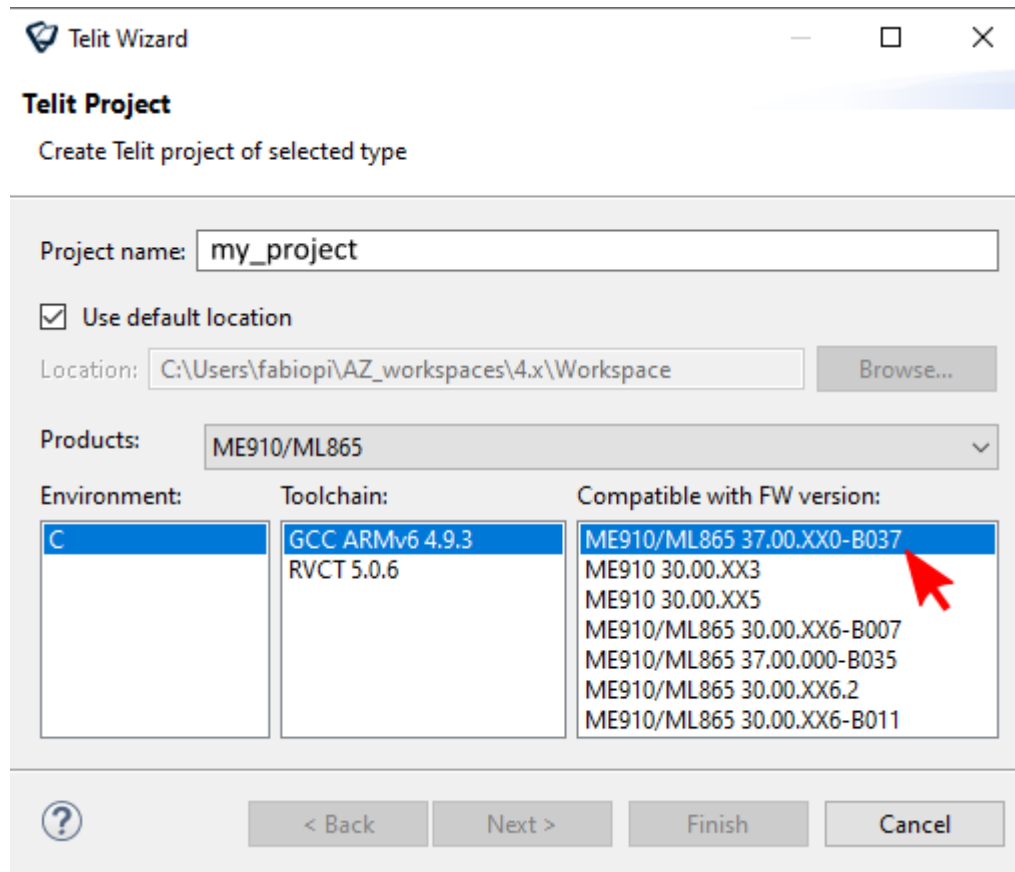


Figure 1

Select the preferred firmware version (e.g. 30.00.xx7) and create an empty project.

in the samples package, go in the HelloWorld folder (e.g. AppZoneSampleApps-MAIN_UART\HelloWorld), copy all the files and folders in it (as src, hdr, azx) and paste them in the root of the newly created IDE project. You are now ready to build and try the sample app on your device.

2.8 Heap and starting address

By default, every application defines a memory HEAP size and its start address in memory.

They are usually provided by the linking phase of the build process:

```
"[...]arm-none-eabi-ld" --defsym __ROM=0x40000000 --defsym __HEAP_PUB_SIZE=0x40000 --defsym
```

`__ROM` is the default starting address, `__HEAP_PUB_SIZE` is the default HEAP size in bytes. Both are expressed in hexadecimal format.

These values can be customized through makefile variables:

HEAP=<new size in bytes>

ROM_START=<new address>

IMPORTANT allowed address range is 0x40000000 - 0x4FFFF000.

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3 Applications

3.1 MISC

Applications that provide usage examples for various functionalities, without prints

3.1.1 GPIO toggle example

Sample application showcasing GPIO usage with M2MB API

Features

- How to open a gpio in output mode and change its status

3.2 BASIC

Basic applications showing simple operations with minimum code overhead

3.2.1 Basic Hello World (Aux UART)

The application prints “Hello World!” on Auxiliary UART every 2 seconds using

Features

- How to open Auxiliary UART as an output channel
- How to print messages out of the channel

3.2.1.1 Application workflow

M2MB_main.c

- Open Auxiliary UART with **m2mb_uart_open** function
- write a welcome message using **m2mb_uart_write**
- write “Hello World!” every 2 seconds in a while loop, using **m2mb_uart_write**

```
Start Hello world Application [ version: 2.000000 ]
Hello world 2.0 [ 000001 ]
Hello world 2.0 [ 000002 ]
Hello world 2.0 [ 000003 ]
Hello world 2.0 [ 000004 ]
Hello world 2.0 [ 000005 ]
Hello world 2.0 [ 000006 ]
Hello world 2.0 [ 000007 ]
Hello world 2.0 [ 000008 ]
Hello world 2.0 [ 000009 ]
```

Figure 2

3.2.2 Basic Hello World (Main UART)

The application prints “Hello World!” on Main UART every 2 seconds using

Features

- How to open Main UART as an output channel
- How to print messages out of the channel

3.2.2.1 Application workflow

M2MB_main.c

- Open Main UART with `m2mb_uart_open` function
- write a welcome message using `m2mb_uart_write`
- write “Hello World!” every 2 seconds in a while loop, using `m2mb_uart_write`

```
Start Hello world Application [ version: 2.000000 ]
Hello world 2.0 [ 000001 ]
Hello world 2.0 [ 000002 ]
Hello world 2.0 [ 000003 ]
Hello world 2.0 [ 000004 ]
Hello world 2.0 [ 000005 ]
Hello world 2.0 [ 000006 ]
Hello world 2.0 [ 000007 ]
Hello world 2.0 [ 000008 ]
Hello world 2.0 [ 000009 ]
```

Figure 3

3.2.3 Basic Hello World (USB0)

The application prints “Hello World!” on USB 0 every 2 seconds using

Features

- How to open USB 0 as an output channel
- How to print messages out of the channel

3.2.3.1 Application workflow

M2MB_main.c

- Open USB 0 with **m2mb_usb_open** function
- write a welcome message using **m2mb_usb_write**
- write “Hello World!” every 2 seconds in a while loop, using **m2mb_usb_write**

```
Start Hello world Application [ version: 2.000000 ]
Hello world 2.0 [ 000001 ]
Hello world 2.0 [ 000002 ]
Hello world 2.0 [ 000003 ]
Hello world 2.0 [ 000004 ]
Hello world 2.0 [ 000005 ]
Hello world 2.0 [ 000006 ]
Hello world 2.0 [ 000007 ]
Hello world 2.0 [ 000008 ]
Hello world 2.0 [ 000009 ]
```

Figure 4

3.2.4 Basic Task

The application shows how to create and manage tasks with m2mb APIs. Debug prints on MAIN UART (can be changed in M2MB_Main function)

Features

- How to create a new task using m2mb APIs
- How to start the task and send messages to it
- how to destroy the task

3.2.4.1 Application workflow

M2MB_main.c

- Open UART
- Print welcome message
- Configure and create message queue for task
- Configure and create task
- Send 2 messages to the task queue

task_entry_function

- Receive messages from the task queue in a loop
- Print the message data when one arrives

```
Starting Basic Task demo app. This is v1.0.8 built on Apr 16 2020 06:40:40.
Successfully created a queue area buffer of 720 bytes.
Queue successfully created.
Creating the task...
Task created and ready to receive messages!
[DEBUG] 16.88 M2MB_main:411 - M2MB_main{M2M_DamsStart}$ Sending a message to the task...
[DEBUG] 16.88 M2MB_main:125 - task_entry_function(mytask)$ Received a message with a 5 bytes payload: <hello>
[DEBUG] 18.90 M2MB_main:420 - M2MB_main{M2M_DamsStart}$ Sending a second message to the task...
[DEBUG] 18.90 M2MB_main:430 - M2MB_main{M2M_DamsStart}$ Result code at the end: 0
[DEBUG] 18.91 M2MB_main:125 - task_entry_function(mytask)$ Received a message with a 5 bytes payload: <world>
Clearing resources...
Done. App complete
```

Figure 5

3.2.5 UART USB tunnel example

Sample application that opens a tunnel between main UART and USB0 port.

Features

- Opens Main UART port with a callback function
- Opens USB0 port with a callback function
- Creates a simple task to manage data exchange between ports

3.2.5.1 Application workflow

M2MB_main function

- Create Main UART handle and configure its parameters
- Create USB0 handle and configure its parameters
- Create the data management task
- Write **READY** on both ports when the tunneling is ready

USB_Cb

- When data are received on the USB0 port, retrieve the available amount and send the value to the data management task with the proper command

UART_Cb

- When data are received on the Main UART port, retrieve the available amount and send the value to the data management task with the proper command

dataTask_Cb

- if command is TASK_UART_READ_AND_USB_WRITE, read the requested amount from the Main UART port and write it on USB0
- if command is TASK_USB_READ_AND_UART_WRITE, read the requested amount from the USB0 port and write it on Main UART

UART output received from USB0 (in RED, the user input data from UART)

```
READYHello from UART
Hello from USB0
```

USB0 output received from UART (in RED, the user input data from USB0)

```
READYHello from UART
Hello from USB0
```

3.3 C++

Applications that provide usage examples with C++

3.3.1 Logging C++

Sample application showcasing how to create a C++ OO code, providing a logging class (equivalent to the one in Logging demo)

Features

- how to define a class object
- how to instantiate and call the class from a C++ main
- how to configure makefile flags to build the application

3.3.1.1 Application workflow

M2MB_main.c

- Call C++ main function

main.cpp

- Create a Logger class instance and set it to USB/UART/UART_AUX
- Print one message for every log level

```
[TRACE] 15.20 main.cpp:68 - cpp_main{M2M_DamsStart}$ C++ Trace print example
[DEBUG] 15.22 main.cpp:69 - cpp_main{M2M_DamsStart}$ C++ Debug print example
C++ Info print example
[WARN ] 15.22 main.cpp:71 - cpp_main{M2M_DamsStart}$ C++ Warning print example
[ERROR] 15.23 main.cpp:72 - cpp_main{M2M_DamsStart}$ C++ Error print example
[CRITICAL] 15.24 main.cpp:73 - cpp_main{M2M_DamsStart}$ C++ Critical print example
```

Figure 6

3.3.2 C++ method to function pointer

Sample application showing how to manage class methods as function pointers.
Debug prints on MAIN_UART

Features

- how to define a class object with a generic method with the same prototype as a m2mb callback function (in this case, a hw timer callback)
- how to use a single static function in the class workspace to call multiple class instances method by using "this" as argument in the timer creation
- how to configure the static function to convert the input parameter with a static cast and call the input class instance method

3.3.2.1 Application workflow

M2MB_main.c

- Call C++ main function

main.cpp

- Create two HwTimer class instance with different timeouts
- Start both timers.
- Each will expire at a different time, and both m2mb timers will call the static function, which will run the appropriate class instance method as callback.

```
Starting C++ method as function pointer example. This is v1.0.11-C1 built on Jul  9 2020 14:58:25.
Timer "first" created with 1000 ms timeout
Timer "second" created with 1500 ms timeout
[DEBUG] 18.70 hwtimer:167 - start{M2M_DamsStart}$ Starting "first" timer
[DEBUG] 18.71 hwtimer:167 - start{M2M_DamsStart}$ Starting "second" timer
In the static timer callback. Calling class method...
[DEBUG] 19.73 hwtimer:179 - timer_cb{pubTspt_0}$
Timer "first" class callback called. Class instance: 0x400212e0; handle: 0x4002b288
In the static timer callback. Calling class method...
[DEBUG] 20.25 hwtimer:179 - timer_cb{pubTspt_0}$
Timer "second" class callback called. Class instance: 0x400212c8; handle: 0x4002b30c
```

Figure 7

3.4 AUX UART

Applications that provide usage examples for various functionalities, log output on Auxiliary UART

3.4.1 Alarm example

Sample application that shows how to set an alarm to wake-up module. Debug prints on **AUX UART**

Features

- How to set an alarm
- How to use it to turn on module

Application workflow

M2MB_main.c

- Init RTC
- Wait for registration
- Get current date and time
- Call function set_alarm
- Init Power and turn off module

```
12:34:48.532- Start Alarm demo application. This is v1.1.19 built on Sep 9 2024 12:34:02.
12:34:48.532-
12:34:48.532- Waiting for registration...
12:34:48.532- Module is registered!
12:34:48.532- NETWORK OPERATOR (mcc mnc): 222 01, Rat: 0
12:34:48.532- Module system time is: 2024-09-09, 12:33:35
12:34:48.532- Setting alarm in 2 minutes
12:34:48.532- Alarm will be set at: 2024-09-09, 12:35:35
12:34:48.741-
12:34:48.741- Wait 5 seconds and then Turn off module...
12:34:53.752- Power off moduley
```

Figure 8

3.4.2 ATI (AT Instance)

Sample application showing how to use AT Instance functionality (sending AT commands from code). The example supports both sync and async (using a callback) modes. Debug prints on **AUX UART**

Features

- How to open an AT interface from the application
- How to send AT commands and receive responses on the AT interface

3.4.2.1 Application workflow, sync mode

M2MB_main.c

- Open USB/UART/UART_AUX
- Init AT0 (first AT instance)
- Send AT+CGMR command
- Print response.
- Release AT0

at_sync.c

- Init ati functionality and take AT0
- Send AT+CGMR command, then read response after 2 seconds, then return it
- Deinit ati, releasing AT0

```
Starting AT demo app. This is v1.0.7 built on Apr 1 2020 15:12:58.
[DEBUG] 17.15 at_sync.c:53 - at_cmd_sync_init{M2M_DamsStart}$ m2mb_ati_init() on instance 0
Sending command AT+CGMR in sync mode
[DEBUG] 17.16 at_sync.c:79 - send_sync_at_command{M2M_DamsStart}$ Sending AT Command: AT+CGMR
Command response: <AT+CGMR
MOB.950004-B008

OK
>

[DEBUG] 19.21 at_sync.c:61 - at_cmd_sync_deinit{M2M_DamsStart}$ m2mb_ati_deinit() on instance 0
Application end
```

Figure 9

3.4.2.2 Application workflow, async mode

M2MB_main.c

- Open USB/UART/UART_AUX
- Init AT0 (first AT instance)
- Send AT+CGMR command
- Print response.

- Release AT0

at_async.c

- Init ati functionality and take AT0, register AT events callback
- Send AT+CGMR command, wait for response semaphore (released in callback), then read it and return it
- Deinit ati, releasing AT0

```
Starting AT demo app. This is v1.0.7 built on Apr 1 2020 15:07:45.
[DEBUG] 17.13 at_async.c:116 - at_cmd_async_init{M2M_DamsStart}$ m2mb_ati_init() on instance 0
Sending command AT+CGMR in async mode
[DEBUG] 17.15 at_async.c:153 - send_async_at_command{M2M_DamsStart}$ Sending AT Command: AT+CGMR
[DEBUG] 17.15 at_async.c:169 - send_async_at_command{M2M_DamsStart}$ waiting command response...
[DEBUG] 17.17 at_async.c:88 - at_cmd_async_callback{pubTspt_0}$ Callback - available bytes: 25
[DEBUG] 17.18 at_async.c:181 - send_async_at_command{M2M_DamsStart}$ Receive response...
Command response: <AT+CGMR
MOB.950004-B008

OK
>

[DEBUG] 17.19 at_async.c:136 - at_cmd_async_deinit{M2M_DamsStart}$ m2mb_ati_deinit() on instance 0
Application end
```

Figure 10

3.4.3 AWS demo

Sample application showcasing AWS IoT Core MQTT communication. Debug prints on **AUX UART**

Features

- How to check module registration and enable PDP context
- How to load certificates into device SSL session storage
- How to configure MQTT client parameters
- How to connect to AWS server with SSL and exchange data over a topic

3.4.3.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage MQTT client and start it

aws_demo.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Init MQTT client
- Configure it with all parameters (Client ID, PDP context ID, keepalive timeout...)
- Initialize the TLS parameters (TLS1.2) and auth mode (server+client auth in the example)
- Create SSL context
- Read certificates files and store them
- Connect MQTT client to broker
- Subscribe to topic
- Publish 10 messages with increasing counter
- Print received message in mqtt_topc_cb function
- Disconnect MQTT client and deinit it
- Disable PDP context

3.4.3.2 How to get started with AWS IoT

- Go to [AWS console](#) and create an account if one is not available yet.
- Go to **IoT Core** section
- Go to **Secure > Policies** section
- Create a new policy, which describes what the device will be allowed to do (e.g. subscribe, publish)
- Give it a name, then configure it using the configuration below (it is possible to copy/paste by clicking on **Add statements** section, then **Advanced mode**) :

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Action": [
        "iot:Publish",
        "iot:Subscribe",
        "iot:Connect",
        "iot:Receive"
      ],
      "Effect": "Allow",
      "Resource": [
        "*"
      ]
    }
  ]
}
```

- Click on create to complete the policy creation.
- Go to **Manage** section
- Press **Create**, then **Create a single thing**
- Give the new thing a name, then click on Next
- Select **One-click certificate creation (recommended)** by clicking on **Create certificate**
- Once presented with the **Certificate created** page, download all certificates and keys
- Click on the **Activate** button to enable the certificate authentication of the newly created device
- Click on **Attach a policy** and select the policy created in a previous step

For further information, please refer to the full [AWS IoT documentation](#)

3.4.3.3 Application setup

- Set **CLIENTCERTFILE** and **CLIENTKEYFILE** defines in **aws_demo.c** file in order to match the certificate and key created in the previous section.
- Set **AWS_BROKER_ADDRESS** to the correct AWS URL. It can be retrieved from AWS IoT Manage > Things > Interact in the HTTPS Rest API Endpoint URL.
- Set **CLIENT_ID** to the desired Client ID for your AWS device
- (Optional) if required, change **CACERTFILE** to match the one to be used.

3.4.3.4 Device setup

The application requires the certificates (provided in sample app **certs** subfolder) to be stored in **/mod/ssl_certs/** folder. It can be created with

```
AT#M2MMKDIR=/mod/ssl_certs
```

Certificates can then be loaded with

```
AT#M2MWRITE="/mod/ssl_certs/preload_CACert_01.crt",1468 AT#M2MWRITE="/mod/ssl_certs/IoT.crt",1646
```

providing the file content in RAW mode (for example using the “Transfer Data” button in Telit AT Controller)

For client certificates, the commands will be

```
AT#M2MWRITE="/mod/ssl_certs/xxxxx.crt",yyyy
AT#M2MWRITE="/mod/ssl_certs/xxxxx.key",zzzz
```

PLEASE NOTE: always verify the file sizes to be used in the commands above as they might change

```

Starting AWS IoT Core MQTT demo app. This is v1.1.5 built on Apr 30 2021 09:05:17.
[DEBUG] 15.51 aws_demo:607 - AWS_Task{MQTT_TASK}$ Init MQTT client for AWS
[DEBUG] 15.52 aws_demo:265 - PrepareSSLEnvironment{MQTT_TASK}$ m2mb_ssl_config SNI succeeded
[DEBUG] 15.52 aws_demo:271 - PrepareSSLEnvironment{MQTT_TASK}$ Root CA cert file /mod/ssl_certs/preload_CACert_01.crt
[DEBUG] 15.52 aws_demo:297 - PrepareSSLEnvironment{MQTT_TASK}$ Buffer successfully received from file. 1468 bytes were loaded.
[DEBUG] 15.52 aws_demo:308 - PrepareSSLEnvironment{MQTT_TASK}$ Cross Signed CA cert file /mod/ssl_certs/Amazon-IoT.crt
[DEBUG] 15.52 aws_demo:334 - PrepareSSLEnvironment{MQTT_TASK}$ Buffer successfully received from file. 1646 bytes were loaded.
[DEBUG] 15.52 aws_demo:360 - PrepareSSLEnvironment{MQTT_TASK}$ Client certificate file /mod/ssl_certs/ab71 -certificate.pem.crt
[DEBUG] 15.52 aws_demo:384 - PrepareSSLEnvironment{MQTT_TASK}$ Buffer successfully received from file. 1224 bytes were loaded.
[DEBUG] 15.52 aws_demo:396 - PrepareSSLEnvironment{MQTT_TASK}$ Client Key file /mod/ssl_certs/ab71 -private.pem.key
[DEBUG] 15.52 aws_demo:422 - PrepareSSLEnvironment{MQTT_TASK}$ Buffer successfully received from file. 1679 bytes were loaded.

SSL environment preparation completed
[DEBUG] 15.52 aws_demo:726 - AWS_Task{MQTT_TASK}$ Waiting for registration...
[DEBUG] 15.52 aws_demo:514 - NetCallback{pubTspt_0}$ Module is registered
[DEBUG] 15.52 aws_demo:738 - AWS_Task{MQTT_TASK}$ PDP context initialization
[DEBUG] 17.55 aws_demo:753 - AWS_Task{MQTT_TASK}$ Activate PDP with APN web.omnitel.it on CID 1....
[DEBUG] 18.37 aws_demo:557 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 18.37 aws_demo:561 - PdpCallback{pubTspt_0}$ IP address: 109.114.102.21

Connecting to Server <angy83rl5oizs-ats.iot.eu-west-2.amazonaws.com>:8883...
Done.
[DEBUG] 27.87 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{ "ID": 2, "data": { "RSSI": -72, "tm": "2021-04-30,09:05:50" } }> to topic
device/35308109f /updates
[DEBUG] 27.87 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 30.94 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{ "ID": 3, "data": { "RSSI": -72, "tm": "2021-04-30,09:05:53" } }> to topic
device/35308109f /updates
[DEBUG] 30.94 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 33.99 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{ "ID": 4, "data": { "RSSI": -72, "tm": "2021-04-30,09:05:56" } }> to topic
device/35308109f /updates
[DEBUG] 34.00 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 37.00 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{ "ID": 5, "data": { "RSSI": -72, "tm": "2021-04-30,09:05:59" } }> to topic
device/35308109f /updates
[DEBUG] 37.00 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 40.03 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{ "ID": 6, "data": { "RSSI": -72, "tm": "2021-04-30,09:06:02" } }> to topic
device/35308109f /updates
[DEBUG] 40.03 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 43.05 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{ "ID": 7, "data": { "RSSI": -72, "tm": "2021-04-30,09:06:05" } }> to topic
device/353081f /updates
[DEBUG] 43.05 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 46.13 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{ "ID": 8, "data": { "RSSI": -72, "tm": "2021-04-30,09:06:08" } }> to topic
device/3530810f /updates
[DEBUG] 46.13 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 49.15 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{ "ID": 9, "data": { "RSSI": -72, "tm": "2021-04-30,09:06:11" } }> to topic
device/3530810f /updates
[DEBUG] 49.15 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 52.19 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{ "ID": 10, "data": { "RSSI": -72, "tm": "2021-04-30,09:06:14" } }> to topic
device/3530810f /updates
[DEBUG] 52.19 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 55.22 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{ "ID": 11, "data": { "RSSI": -72, "tm": "2021-04-30,09:06:17" } }> to topic
device/353081f /updates
[DEBUG] 55.22 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.

Disconnecting from MQTT broker..
[DEBUG] 58.27 aws_demo:878 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 58.27 aws_demo:908 - AWS_Task{MQTT_TASK}$ application exit
[DEBUG] 58.27 aws_demo:918 - AWS_Task{MQTT_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 58.27 aws_demo:924 - AWS_Task{MQTT_TASK}$ Application complete.

```

Figure 11

Data received from a subscriber:

```

device/353081f /updates 1
QoS 0

device/3530810 /updates 1
30-04-2021 09:06:00.32760467 QoS 0
{ "ID": 2, "data": { "RSSI": -72, "tm": "2021-04-30,09:05:50" } }

```

Figure 12

3.4.4 App Manager

Sample application showing how to manage AppZone apps from m2mb code. Debug prints on **AUX UART**

Features

- How to get how many configured apps are available
- How to get the handle to manage the running app (change start delay, enable/disable)
- How to create the handle for a new binary app, enable it and set its parameters
- How to start the new app without rebooting the device, then stop it after a while.

3.4.4.1 Prerequisites

This app will try to manage another app called “second.bin”, which already exists in the module filesystem and can be anything (e.g. another sample app as GPIO toggle). the app must be built using the flag ROM_START=

in the Makefile to set a different starting address than the main app (by default, 0x40000000). For example, 0x41000000.

3.4.4.2 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- get a non existing app handle and verify it is NULL
- get the current app handle, then get the start delay **set in the INI file (so persistent)**
- change the current app delay value **in the INI file**
- verify that the change has been stored
- get current app state
- create an handle for a second application binary.
- add it to the INI file
- set its execution flag to 0
- get the delay time and the state from INI file for the new app
- get the current set address for the new app
- set the app delay **in RAM, INI will not be affected.**
- start the new app without reboot, using the right set delay
- wait some time, then get the app state and the used RAM amount
- wait 10 seconds, then stop the second app.
- set its execution flag to 1 so it will run at next boot.

```
Starting App Manager demo app. This is v1.0.14-C1 built on Sep 24 2020 12:33:25.  
There are 2 configured apps.  
Not existing app handle test (should be 0): 0x0  
Manager app handle: 0x809e20e0  
Manager app delay from nv memory: 5 seconds  
  
Changing Manager app delay time (on non volatile configuration) to 5 seconds..  
Manager app delay from nv memory is now 5 seconds  
Manager app state is M2MB_APPMNG_STATE_RUN  
  
Trying to get Second app handle...  
Second app handle is valid  
2nd app delay from nv memory is 1  
2nd app current state is M2MB_APPMNG_STATE_READY  
Second app current address is 0x41000000  
Setting volatile Second app delay (not stored in nvm) to 0 seconds...  
Starting Second app on the fly (without reboot)...  
Waiting 2 seconds...  
2nd app current state is M2MB_APPMNG_STATE_RUN  
Second app is running!  
Second App is using 475136 bytes of RAM  
Stopping Second app now...  
wait 10 seconds...  
2nd app current state is M2MB_APPMNG_STATE_STOP  
Set permanent run permission for Second app.  
Done. Second App will also run from next boot-up
```

Figure 13

3.4.5 App update OTA via FTP

Sample application showcasing Application OTA over FTP with AZX FTP. Debug prints on **AUX UART**

Features

- How to check module registration and activate PDP context
- How to connect to a FTP server
- How to download an application binary and update the local version

The app uses a predefined set of parameters. To load custom parameters, upload the `ota_config.txt` file (provided in project's `/src` folder) in module's `/mod` folder, for example with

```
AT#M2MWRITE="/mod/ota_config.txt",<filesize>
```

3.4.5.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage app OTA and start it

ftp_utils.c

- Set parameters to default
- Try to load parameters from `ota_config.txt` file
- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Initialize FTP client
- Connect to FTP server and log in
- Get new App binary file size on remote server
- Download the file in `/mod` folder, with the provided name
- Close FTP connection
- Disable PDP context
- Update applications configuration in **app_utils.c**

app_utils.c

- Set new application as default
- Delete old app binary
- Restart module

```
Starting FTP APP OTA demo app. This is v1.0.7 built on Apr 7 2020 17:04:05.
[DEBUG] 21.23 ftp_utils.c:447 - msgFTPTask{FTPOTA_TASK}$ INIT
[DEBUG] 21.25 ftp_utils.c:152 - readConfigFromFile{FTPOTA_TASK}$ Reading parameters from file
[DEBUG] 21.26 ftp_utils.c:154 - readConfigFromFile{FTPOTA_TASK}$ Opening /mod/ota_config.txt in read mode..
Set APN to: <<web.omnitel.it>>
Set FTP URL to: <<ftp.telit.com>>
Set FTP PORT to: 21
Set FTP USER to: <<_ _ _ _>>
Set FTP PASS to: <<_ _ _ _>>
Set FTP FILE URI to: <</samples/APP_OTA/helloworld.bin>>
Set LOCAL FINAL APP NAME to: <<helloworld.bin>>
Set LOCAL ORIGINAL APP NAME to: <<m2mapz.bin>>
[DEBUG] 23.53 ftp_utils.c:464 - msgFTPTask{FTPOTA_TASK}$ m2mb_os_ev_init success
[DEBUG] 23.54 ftp_utils.c:470 - msgFTPTask{FTPOTA_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.55 ftp_utils.c:478 - msgFTPTask{FTPOTA_TASK}$ Waiting for registration...
[DEBUG] 23.56 ftp_utils.c:371 - NetCallback{pubTspt_0}$ Module is registered to network
[DEBUG] 23.56 ftp_utils.c:491 - msgFTPTask{FTPOTA_TASK}$ Pdp context activation
[DEBUG] 23.57 ftp_utils.c:495 - msgFTPTask{FTPOTA_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 25.61 ftp_utils.c:504 - msgFTPTask{FTPOTA_TASK}$ Activate PDP with APN web.omnitel.it on cid 3....
[DEBUG] 26.30 ftp_utils.c:398 - PdpCallback{pubTspt_0}$ Context active
[DEBUG] 26.30 ftp_utils.c:401 - PdpCallback{pubTspt_0}$ IP address: 176.246.110.148
Start ftp client...
[DEBUG] 27.36 ftp_utils.c:533 - msgFTPTask{FTPOTA_TASK}$ Connected.
[DEBUG] 28.87 ftp_utils.c:546 - msgFTPTask{FTPOTA_TASK}$ FTP login successful.
Get remote file /samples/APP_OTA/helloworld.bin size
[DEBUG] 29.31 ftp_utils.c:568 - msgFTPTask{FTPOTA_TASK}$ Done. File size: 116224.
Starting download of remote file /samples/APP_OTA/helloworld.bin into local /mod/helloworld.bin
/samples/APP_OTA/helloworld.bin 4.68% 5440
/samples/APP_OTA/helloworld.bin 9.36% 10880
/samples/APP_OTA/helloworld.bin 14.04% 16320
/samples/APP_OTA/helloworld.bin 18.72% 21760
/samples/APP_OTA/helloworld.bin 23.40% 27200
/samples/APP_OTA/helloworld.bin 28.08% 32640
/samples/APP_OTA/helloworld.bin 32.76% 38080
/samples/APP_OTA/helloworld.bin 37.44% 43520
/samples/APP_OTA/helloworld.bin 42.13% 48960
/samples/APP_OTA/helloworld.bin 46.81% 54400
/samples/APP_OTA/helloworld.bin 51.49% 59840
/samples/APP_OTA/helloworld.bin 56.17% 65280
/samples/APP_OTA/helloworld.bin 60.85% 70720
/samples/APP_OTA/helloworld.bin 65.53% 76160
/samples/APP_OTA/helloworld.bin 70.21% 81600
/samples/APP_OTA/helloworld.bin 74.89% 87040
/samples/APP_OTA/helloworld.bin 79.57% 92480
/samples/APP_OTA/helloworld.bin 84.25% 97920
/samples/APP_OTA/helloworld.bin 88.93% 103360
/samples/APP_OTA/helloworld.bin 93.61% 108800
/samples/APP_OTA/helloworld.bin 97.42% 113220
[DEBUG] 43.54 ftp_utils.c:608 - msgFTPTask{FTPOTA_TASK}$ download successful.
FTP quit...
[DEBUG] 43.77 ftp_utils.c:632 - msgFTPTask{FTPOTA_TASK}$ Deactivating PDP
[DEBUG] 43.77 ftp_utils.c:642 - msgFTPTask{FTPOTA_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 44.20 ftp_utils.c:407 - PdpCallback{pubTspt_0}$ Context deactive
[DEBUG] 45.44 app_utils.c:76 - update_app{FTPOTA_TASK}$ Application successfully configured.
[DEBUG] 45.45 app_utils.c:82 - update_app{FTPOTA_TASK}$ Deleting old application /mod/m2mapz.bin
Starting. This is v1.0.7 built on Apr 7 2020 17:02:52. LEVEL: 2

Start Hello world Application [ version: 2.000000 ]

Hello world 2.0 [ 000001 ]
Hello world 2.0 [ 000002 ]
Hello world 2.0 [ 000003 ]
```

Figure 14

3.4.6 cJSON example:

Sample application showcasing how to manage JSON objects. Debug prints on **AUX UART**

Features

- How to read a JSON using cJSON library
- How to write a JSON
- How to manipulate JSON objects

3.4.6.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Parse an example string into a JSON object and print the result in a formatted string
- Print some test outcomes (e.g. non existing item correctly not found)
- Retrieve single elements from the parsed JSON object and use them to format a descriptive string
- Delete the JSON object
- Create a new JSON object appending elements to it
- Print the result JSON string from the object

```

Starting Logging demo app. This is v1.0.7 built on Apr  7 2020 08:33:03.
And here is what we got:
{
  "name":      "Atlantic Ocean",
  "format":    {
    "type":     "salt",
    "volume":   310410900,
    "depth":    -8486,
    "volume_percent": 23.300000,
    "tide":     -3.500000,
    "calm":     false,
    "life":     ["plankton", "corals", "fish", "mammals"]
  }
}
inexistent key not found
name found: Atlantic Ocean
format found (null)
Our JSON string contains info about an ocean named Atlantic Ocean,
has a volume of 310410900 km^3 of salt water with -8486 meters max depth,
represents 23.3% of total oceans volume,
has an average low tide of -3.5 meters,
hosts a huge number of living creatures such as plankton, corals, fish, mammals,
and is not always calm.

Let's build a TR50 command with a property.publish and an alarm.publish for MQTT (no auth).
And here is what we got:
{
  "1": {
    "command": "property.publish",
    "params": {
      "thingKey": "mything",
      "key": "mykey",
      "value": 123.144000
    }
  },
  "2": {
    "command": "alarm.publish",
    "params": {
      "thingKey": "mything",
      "key": "mykey",
      "state": 3,
      "msg": "Message."
    }
  }
}
END.

```

Figure 15

3.4.7 Crypto Elliptic Curve Cryptography (ECC) example

Sample application showcasing how to manage Elliptic Curve Cryptography functionalities. Debug prints on **AUX UART**

Features

- How to initialize ECC contexts A (Alice) and B (Bob). Alice is emulating a remote host, from which a public key is known.
- How to generate keypairs for contexts and export public keys
- how to export keyblobs from a context (a keyblob is encrypted with hw specific keys, and can only be used on the module where it was created)
- How to save a keyblob in secured TrustZone.
- How to reload a keyblob from the TrustZone into an initialized context
- How to sign a message with ECDSA from context B (Bob) and verify it from another context A (Alice) with the signature and public key of Bob.
- How to make Bob and Alice derive a shared session keys using each other's public key.
- How to make Bob and Alice create an AES context with the newly created shared keys, encode data and decode it on the other side

3.4.7.1 Application workflow

M2MB_main.c

- Create Bob ECC context, create a keypair and export it in a keyblob
- Open a file in secured Trust Zone, then store the keyblob in it.
- Destroy Bob ECC context
- Recreate Bob ECC context, open the file from Trust Zone and read the keyblob.
- Import the keyblob in Bob context.
- Export Bob public key
- Create Alice ECC context, to simulate an external host. Generate a keypair and export the public key.
- Sign a message with Bob context, generating a signature.
- Use Alice to verify the signed message using Bob's signature and public key
- Derive a shared key for Bob, using Alice's public key
- Create an AES context for Bob
- Import the shared key into the AES context
- Encrypt a message using Bob's AES context.

- Derive a shared key for Alice, using Bob's public key
- Create an AES context for Alice
- Import the shared key into the AES context
- Decrypt the message using Alice's AES context.
- Check the decrypted message and the original one match
- Clear all resources

```
Starting Crypto ECC demo app. This is v1.0.9-C1 built on May 11 2020 16:30:23.
Bob (local) and Alice (remote) scenario
Bob's keypair generated
Bob's keyblob length is 224
Bob exported the keyblob to be securely stored.

Bob already had an item in Secure Data Area, it was removed to create a new one
Bob securely saved the keyblob in Secure Data Area
Releasing resources

Close Bob's context...
Done. Now Bob context does not exist anymore.

Re-initialize Bob Context and load the keyblob from the secure zone
Bob securely loaded the keyblob from the SDA
Import keyblob in Bob's context..
Done. Now export Bob's public key...
Bob's public key successfully exported

Alice's keypair generated
Alice's public key successfully exported

Bob's message signed with ECDSA!
Alice verified bob's message with his pubkey and signature!

-----
Bob and Alice will now exchange a message with AES encrypt
-----

Bob retrieved the generated shared key size
Bob's shared keyblob length is: 32. Allocate the required memory to store it.
Bob created a shared key using Alice's public key!

Bob created an AEX context to exchange encrypted data with Alice
Bob's AES context imported the shared keyblob
Bob Encrypted the message using AES and the shared key!
Encrypted data:
 94EE531E3B84B2A4EF05502186BFF5DA

Alice retrieved the generated shared key size
Alice's shared keyblob length is: 32. Allocate the required memory to store it.
Alice created a shared key using Bob's public key!

Alice created an AEX context to exchange encrypted data with Bob
Alice's AES context imported the shared keyblob
Alice decrypted the message using AES and the shared key!
Decrypted:
 414094941E8942A4445548035BFAE943

Original, plain message:
 414094941E8942A4445548035BFAE943

Plain and decrypted messages match!
```

Figure 16

3.4.8 EEPROM 24AA256

Sample application showing how to communicate with a MicroChip 24AA256T I2C EEPROM chip using azx eeprom utility APIs. Debug prints on **AUX UART**

Setup

This demo application requires that: - A0, A1, and A2 pins (1,2,3 chip pins) are connected to ground (pin 4) for device address 0xA0 - Pin 7 (WP) is connected to ground - Pin 6 (SCL) is connected to module GPIO 3 - Pin 5 (SDA) is connected to module GPIO 2 - Pin 4 is connected to one of the ground pins of the module - Pin 8 is connected to 1v8 supply (e.g. VPWRMON pin on the module)

Features

- Initialize the logs on the output channel
- configure the EEPROM utility, setting the slave address and the memory parameters (page size, memory size)
- Write single bytes on a random address
- Read written bytes as a page
- Write data using pages
- Read the new data using pages
- Read again using sequential reading
- Read a single byte from a specific address
- Read next byte using read from current address
- Erase the EEPROM
- Deinit EEPROM utility

3.4.8.1 Application workflow

M2MB_main.c

- call `azx_eeprom_init()` to set the utility parameters (SDA and SCL pins, page and memory sizes)
- call `azx_eeprom_writeByte()` to store a single byte with value "5" at the address 0x0213
- call `azx_eeprom_writeByte()` to store a single byte with value "6" at the address 0x0214
- call `azx_eeprom_readPages()` from address 0x0213 to retrieve the 2 bytes from the EEPROM
- call `azx_eeprom_writePages` to write 1024 bytes from a buffer, starting from address 0x00
- call `azx_eeprom_readPages()` again, to read 256 bytes from address 0x00
- call `azx_eeprom_readSequentially()` to read 256 bytes from 0x00 by without pages (less overhead on I2C protocol)

- call `azx_eeprom_readByte()` to get a single byte from address 0x00
- call `azx_eeprom_readByteFromCurrentAddress()` to get a byte from next address (0x01)
- call `azx_eeprom_eraseAll()` to completely erase the EEPROM memory (this writes 0xFF in each byte)
- call `azx_eeprom_readPages` from address 0x0213 to get 2 bytes and verify the values have been written to 0xFF
- call `azx_eeprom_deinit` to close the eeprom handler and the I2C channel

```
Starting I2C EEPROM 24AA256T demo app. This is v1.0.13-C1 built on Nov  3 2020 16:28:23.
Configuring the I2C device...
Opening I2C channel /dev/I2C-160 ( device address is 0xA0 )
Writing 1 byte at address 0x0213...
Done.
Writing 1 byte at address 0x0214...
Done.
Reading the 2 bytes from address 0x0213...
Done. Data: [0xFF 0xFF]

Writing 1024 bytes at address 0x0000..
Done.

Reading 256 bytes from address 0x0000...
Done. Data:
<<ABCDEFGHIJKLMNOPQRSTUVWXYZ.....abcdefghijklmnopqrstuvwxyz.....

Reading 256 bytes sequentially from address 0x0000...
Done. Data:
<<ABCDEFGHIJKLMNOPQRSTUVWXYZ.....abcdefghijklmnopqrstuvwxyz.....

Reading 1 byte from address 0x0000...
Done. Data: 'A'

Reading 1 byte from current address (should be 0x0001)...
Done. Data: 'B'

[DEBUG] 17.47  M2MB_main:177 - run_I2C_EEPROM_Demo{M2M_DamsStart}$ Erasing all the eeprom...
[DEBUG] 28.05  M2MB_main:185 - run_I2C_EEPROM_Demo{M2M_DamsStart}$ Done

Reading the 2 bytes from address 0x0213...
Done. Data: [0xFF 0xFF]

Deinit EEPROM...
Done
```

Figure 17

3.4.9 Easy AT example

Sample application showcasing Easy AT functionalities. Debug prints on **AUX UART**

Features

- Shows how to register custom commands

The application adds two custom commands to the list of available ones:

- AT#MYCMD
- AT#MYINPUT

3.4.9.1 AT#MYCMD

This is a simple parameter-waiting command. It expects one string parameter, and will print it on the logging interface once received. The command simply returns OK

3.4.9.2 AT#MYINPUT

This command expects a numeric parameter, which indicates how many bytes will be received over the interface at most (the command will provide a prompt indicating it is waiting data). Then the data management callback will print when data is received, and if CTRL+Z (0x1A in hex) is received, it will complete the process, printing in the log interface what was received. sending ESC will terminate the process discarding any pending data.

3.4.10 Protect and AT command with a pwd using EasyAT functionality example

Sample application showing how to protect an AT command with a pwd using EasyAT functionality. Debug prints on **AUX UART**

Features

- Shows how to register a new custom command and overwrite an existing one
- AT#M2MADMIN (new command)
- AT#M2MWRITE (existing one)

Application workflow

M2MB_main.c

- Entry point that calls EasyAT initialization function

3.4.10.1 AT#M2MADMIN

This command allows to enter a pwd to lock/unlock #M2MWRITE command. In the example app the new pwd is saved in the trustzone

AT#M2MADMIN="pwd",["newpwd"]

where mode: 0 - unlock command #m2mwrite 1 - lock command #m2mwrite 2 - change pwd

```
13:40:22.898- Starting Easy AT admin demo app. This is v1.1.19 built on Jul 24 2024 13:39:26.
13:40:22.898- Easy_at_init start
13:40:22.898- [DEBUG] 20.19 at_common:375 - easy_at_init{M2M_DamsStart}$ m2mb_atp_init succeeded
13:40:22.898- Register AT#M2MWRITE
13:40:22.898- Register AT#M2MADMIN
13:40:32.444- [DEBUG] 29.90 at_common:187 - _easy_at_task_entry{M2M_AT}$ task entry, message event type: CMD
13:40:32.444- [ERROR] 29.91 at_hash_M2M:74 - M2MWRITE_AT_Callback{M2M_AT}$ Operation not allowed!
13:40:37.464- [DEBUG] 34.93 at_common:187 - _easy_at_task_entry{M2M_AT}$ task entry, message event type: CMD
13:40:37.464- [DEBUG] 34.94 at_hash_M2M:223 - M2MADMIN_AT_Callback{M2M_AT}$ admin callback
13:40:37.464- Read form secure item res: 0
13:40:37.464- parameter is <mypassword>
13:40:37.464- Operation is <0>
13:40:37.464- hash: <89E01536AC207279409D4DE1E5253E01F4A1769E696DB0D6062CA9B8F56767C8>
13:40:37.464- right password!
13:40:37.909- [DEBUG] 35.31 at_hash_M2M:309 - M2MADMIN_AT_Callback{M2M_AT}$ Release SUCCESS was OK
13:41:04.285- [DEBUG] 62.27 at_common:187 - _easy_at_task_entry{M2M_AT}$ task entry, message event type: CMD
13:41:04.285- [DEBUG] 62.28 at_hash_M2M:223 - M2MADMIN_AT_Callback{M2M_AT}$ admin callback
13:41:04.285- parameter is <mypassword>
13:41:04.285- Operation is <1>
13:41:04.285- hash: <89E01536AC207279409D4DE1E5253E01F4A1769E696DB0D6062CA9B8F56767C8>
13:41:04.285- right password!
13:41:04.285- [DEBUG] 62.30 at_hash_M2M:362 - M2MADMIN_AT_Callback{M2M_AT}$ Release SUCCESS was OK
13:41:11.611- [DEBUG] 69.69 at_common:187 - _easy_at_task_entry{M2M_AT}$ task entry, message event type: CMD
13:41:11.611- [ERROR] 69.71 at_hash_M2M:74 - M2MWRITE_AT_Callback{M2M_AT}$ Operation not allowed!
```

Figure 18

3.4.11 Events

Sample application showcasing events setup and usage. Debug prints on **AUX UART**

Features

- How to setup OS events with a custom bitmask
- How to wait for events and generate them in callback functions to synchronize blocks of code

3.4.11.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create an event handler
- Create a timer to generate an event, with a 2 seconds expiration time
- Wait for a specific event bit on the event handler
- At timer expiration, set the same event bit and verify that the code flow went through after the event.

```
Starting Events demo app. This is v1.0.7 built on Apr  7 2020 08:44:29.  
[DEBUG] 20.55 M2MB_main.c:171 - M2MB_main{M2M_DamsStart}$ m2mb_os_ev_init success  
Set the timer attributes structure success.  
Timer successfully created  
[DEBUG] 20.57 M2MB_main.c:125 - setup_timer{M2M_DamsStart}$ Start the timer, success.  
[DEBUG] 22.60 M2MB_main.c:60 - hwTimerCb{pubTspt_0}$ Timer Callback, generate event!  
[DEBUG] 22.61 M2MB_main.c:183 - M2MB_main{M2M_DamsStart}$ event occurred!
```

Figure 19

3.4.12 Events - Barrier (multi events)

Sample application showcasing how to setup and use multiple events to create a barrier. Debug prints on **AUX UART**

Features

- How to setup OS events to be used as a barrier
- How to wait for multiple events in the same point, and generate them in call-back functions to synchronize blocks of code

3.4.12.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create an event handler
- Create a timer to generate an event, with a 3 seconds expiration time
- Create another timer to generate an event, with a 6 seconds expiration time
- Start both timers
- Wait for both event bits on the event handler (each one will be set by one of the timers)
- At first timer expiration, set the first event bit and verify that the code flow does not procede.
- At second timer expiration, set the second event bit and verify that the code flow went through after the event (implementing a barrier).

```
Starting Barrier demo app. This is v1.0.7 built on Apr 7 2020 08:48:30.
[DEBUG] 20.01 M2MB_main.c:179 - M2MB_main{M2M_DamsStart}$ m2mb_os_ev_init success
Set the timer attributes structure success.
Timer successfully created with 3000 timeout (ms)
Set the timer attributes structure success.
Timer successfully created with 6000 timeout (ms)
[DEBUG] 23.08 M2MB_main.c:66 - hwTimerCb1{pubTspt_0}$ Timer Callback, generate event 1!
[DEBUG] 26.12 M2MB_main.c:75 - hwTimerCb2{pubTspt_0}$ Timer Callback, generate event 2!
[DEBUG] 26.13 M2MB_main.c:214 - M2MB_main{M2M_DamsStart}$ BOTH events occurred!
```

Figure 20

3.4.13 FOTA example

Sample application showcasing FOTA usage with M2MB API. Debug prints on **AUX UART**

Features

- How download a delta file from a remote server
- How to apply the delta and update the module firmware

3.4.13.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a main task to manage connectivity.
- create a fota task to manage FOTA and start it with INIT option

fota.c

fotaTask()

- Initialize FOTA system then reset parameters.
- Check current FOTA state, if not in IDLE, return error.
- Send a message to mainTask so networking is initialized.
- after PdPCallback() notifies the correct context activation, configure the fota client parameters such as FTP server URL, username and password
- get delta file from server. when it is completed, FOTADownloadCallback is called.
- If delta download went fine, check it.
- If delta file is correct, apply it. Once complete, restart the module.

mainTask()

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context. Event will be received on **PdP-Callback** function
- Disable PDP context when required to stop the app

PdpCallback()

- When PDP context is enabled, send a message to fotaTask to start the download


```

Starting FOTA demo app. This is v1.1.7 built on Jun 11 2021 12:20:43.
[DEBUG] 23.60 fota:187 - fotaTask{FOTA_TASK}$ Init FOTA...

Session file not present, procede with FOTA...
[DEBUG] 23.61 fota:236 - fotaTask{FOTA_TASK}$ m2mb_fota_reset PASS
[DEBUG] 23.61 fota:260 - fotaTask{FOTA_TASK}$ m2mb_fota_state_get M2MB_FOTA_STATE_IDLE
[DEBUG] 23.62 fota:379 - mainTask{MAIN_TASK}$ INIT
[DEBUG] 23.62 fota:392 - mainTask{MAIN_TASK}$ m2mb_os_ev_init success
[DEBUG] 23.63 fota:398 - mainTask{MAIN_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.63 fota:405 - mainTask{MAIN_TASK}$ Waiting for registration...
[DEBUG] 23.64 fota:131 - NetCallback{pubTspt_0}$ Module is registered to network
[DEBUG] 23.65 fota:418 - mainTask{MAIN_TASK}$ Pdp context initialization
[DEBUG] 25.70 fota:431 - mainTask{MAIN_TASK}$ Activate PDP with APN web.omnitel.it on cid 1....
[DEBUG] 35.42 fota:152 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 35.43 fota:155 - PdpCallback{pubTspt_0}$ IP address: 2.41.116.139

[DEBUG] 35.43 fota:285 - fotaTask{FOTA_TASK}$
Trying to download "samples/FOTA/37.00.003.3_to_37.00.003.1_ME310G1_NANVWWAU.bin" delta file...
[DEBUG] 35.45 fota:295 - fotaTask{FOTA_TASK}$ m2mb_fota_get_delta OK - Waiting for the completion callback
[DEBUG] 119.43 fota:96 - FOTADownloadCallBack{pubTspt_0}$ FOTA download Success - performing packet validation...
[DEBUG] 119.44 fota:301 - fotaTask{FOTA_TASK}$ Validating delta file...
[DEBUG] 156.36 fota:317 - fotaTask{FOTA_TASK}$ Packet is valid, start update...
[DEBUG] 156.40 fota:329 - fotaTask{FOTA_TASK}$ m2mb_fota_start PASS
[DEBUG] 158.36 fota:342 - fotaTask{FOTA_TASK}$
Rebooting...After reboot there will be the new FW running on module!

#OTAEV: Module Upgraded To New Fw
Starting FOTA demo app. This is v1.1.7 built on Jun 11 2021 12:20:43.
[DEBUG] 29.24 fota:187 - fotaTask{FOTA_TASK}$ Init FOTA...

Session file is already present, stop.

```

Figure 21

3.4.14 FOTA_FTP_client example

Sample application that shows how to download a delta file from an FTP server, stores it in the FOTA partition and deploys it. Debug prints on **AUX UART**

Features

- How to download a delta file from FTP server using FTP client
- How to store directly delta file in the FOTA partition
- How to deploy delta file to upgrade module fw.

Application workflow

M2MB_main.c

- Print welcome message
- Create a main task to manage connectivity, delta download and deployment

ftp_test.c

msgFTPTask()

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context. Event will be received on
- Initialize FOTA system then reset parameters.
- After PDP context activation notified by PdPCallback() configure fota client parameters as FTP server url, username and password and SSL
- Get delta file from server and store it directly in the FOTA partition
- If delta download went fine, check it (m2mb_fota_update_package_check_setup) and if it's correct apply it (m2mb_fota_start).
- Once completed restart module.

PdpCallback()

- When PDP context is enabled, send a message to fotaTask to start the download

buf_data_cb_OTA()

- Handles data reception and writing in the FOTA partition (one block size at a time)

```

yStarting FOTA delta file FTP download and deploy demo app. This is v1.1.19 built on May 28 2024 14:41:14.
14:45:06.588- [DEBUG] 19.69 ftp_test:915 - msgFTPTask{FTP_TASK}$ INIT
14:45:06.588- Set APN to: <<ibox.tim.it>>
14:45:06.588- Set APN_USER to: <<>>
14:45:06.588- Set APN_PASS to: <<>>
14:45:06.588- Set PDP_CDX to: 1
14:45:06.588- Set FTP_ADDR to: <<ftp.telit.com>>
14:45:06.588- Set FTP_PORT to: 21
14:45:06.588- Set FTP_USER to: <<[REDACTED]>>
14:45:06.588- Set FTP_PASS to: <<[REDACTED]>>
14:45:06.588- Set ENABLE_TLS to: <<0>>
14:45:06.588- Set AUTH_TYPE to: <<2>>
14:45:06.588- Set CA_CERT_PATH to: <</mod>>
14:45:06.807- Set CLIENT_CERT_PATH to: <</mod>>
14:45:06.807- Set CLIENT_KEY_PATH to: <</mod>>
14:45:06.807- Set REMOTE_FOLDER to: <<robertaga>>
14:45:06.807- Set DLTOBUF_FILE to: <<delta_dummy_6_B710.bin>>
14:45:06.807- Set FOTA_STATUS_FILE to: <</mod/fota_stat.txt>>
14:45:06.807- [DEBUG] 19.78 ftp_test:932 - msgFTPTask{FTP_TASK}$ m2mb_os_ev_init success
14:45:06.807- [DEBUG] 19.79 ftp_test:938 - msgFTPTask{FTP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
14:45:06.807- [DEBUG] 19.80 ftp_test:946 - msgFTPTask{FTP_TASK}$ Waiting for registration...
14:45:06.807- [DEBUG] 19.82 ftp_test:840 - NetCallback{pubTspt_0}$ Module is registered to network
14:45:06.807-
14:45:06.807- Check FOTA upgrade status in file system
14:45:06.807-
14:45:06.807- Module current fw version is:
14:45:06.807- 37.00.216-B010-POC.210000
14:45:06.807- MOC.200006-B010
14:45:06.807- POC.210000
14:45:06.807- AOC.210000
14:45:07.248- [WARN ] 20.37 ftp_test:334 - FOTAStatusInit{FTP_TASK}$ File doesn't exist create it, first app execution
14:45:07.248- File created, store current fw version and fota upgrade flag=0
14:45:08.315-
14:45:08.315- Read stored data
14:45:08.315-
14:45:08.315- FOTA Flag: 0 Module previous fw version:
14:45:08.315- 37.00.216-B010-POC.210000
14:45:08.315- MOC.200006-B010
14:45:08.315- POC.210000
14:45:08.315- AOC.210000
14:45:08.315- Fw to be upgraded...
14:45:08.315- [DEBUG] 21.44 ftp_test:1009 - msgFTPTask{FTP_TASK}$ Pdp context activation
14:45:08.315- [DEBUG] 21.45 ftp_test:1013 - msgFTPTask{FTP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
14:45:10.426- [DEBUG] 23.50 ftp_test:1022 - msgFTPTask{FTP_TASK}$ Activate PDP with APN ibox.tim.it on cid 1....
14:45:10.855- [DEBUG] 24.05 ftp_test:867 - PdpCallback{pubTspt_0}$ Context active
14:45:10.855- [DEBUG] 24.07 ftp_test:870 - PdpCallback{pubTspt_0}$ IP address: 2.193.245.246
14:45:10.855- Start ftp client...
14:45:10.855- Connecting to host: <ftp.telit.com>
14:45:11.294- Retrieved address: 185.86.42.214
14:45:11.294- socket 0x4003d2a4 ok

```

Figure 22

3.4.15 FOTA from Local File example

Sample application that shows how perform FOTA upgrade using a delta file stored into file system. Debug prints on **AUX UART**

Features

- How to store and get FOTA upgrade information to/from a file
- How to get delta file from module file system
- How to apply the delta and update module firmware

Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Check if module has been already upgraded or needs to be upgraded reading FOTA upgrade status from a file
- Create a fota task to manage FOTA and start it with INIT option

smartFotaTask() - Initialize FOTA system then reset parameters. - Get FOTA partition size and block size - Copy delta file from file system to FOTA partition. when it is completed, FOTADownloadCallback is called. - If delta file is correct, apply it. Once complete, write FOTA status flag and current fw version to a file, restart the module.

```
Starting FOTA application from local delta file demo app. This is v1.0.14-C1 built on May 16 2022 15:50:38.

Check FOTA upgrade status in file system

Module current fw version is:
37.00.214-POC.210001
MOC.200003
POC.210001
AOC.210000
[WARN ] 13.83 M2MB_main:190 - FOTAUpgradeStatusInit{M2M_DamsStart}$ File doesn't exist create it, first app execution
File created, store current fw version and fota upgrade flag=0

Read stored data

FOTA Flag: 0 Module previous fw version:
37.00.214-POC.210001
MOC.200003
POC.210001
AOC.210000

Start FOTA process
[DEBUG] 15.42 M2MB_main:450 - smartFotaTask{FOTA_TASK}$ INIT
[DEBUG] 15.43 M2MB_main:461 - smartFotaTask{FOTA_TASK}$ m2mb_fota_init success
[DEBUG] 15.44 M2MB_main:464 - smartFotaTask{FOTA_TASK}$
Get block and FOTA partition size
OTA blockSize: 131072
OTA partitionSize: 11272192
[DEBUG] 15.47 M2MB_main:472 - smartFotaTask{FOTA_TASK}$
Copy delta file from File system to FOTA partition
[DEBUG] 15.47 M2MB_main:277 - copyFromFileSystemToDelta{FOTA_TASK}$ File size: 746552
[DEBUG] 15.87 M2MB_main:480 - smartFotaTask{FOTA_TASK}$ Delta file writing completed
[DEBUG] 15.88 M2MB_main:484 - smartFotaTask{FOTA_TASK}$
Delta file check...
[DEBUG] 15.89 M2MB_main:347 - check_fota_delta{FOTA_TASK}$ -- check_fota_delta...
[DEBUG] 54.73 M2MB_main:366 - check_fota_delta{FOTA_TASK}$ Fota check integrity PASS
[DEBUG] 54.74 M2MB_main:492 - smartFotaTask{FOTA_TASK}$ ...delta file OK
[DEBUG] 54.75 M2MB_main:496 - smartFotaTask{FOTA_TASK}$
--> Start update...
[DEBUG] 56.83 M2MB_main:403 - startUpdate{FOTA_TASK}$
Reboot module to start delta deployment
```

Figure 23

3.4.16 FTP

Sample application showcasing FTP client demo with AZX FTP. Debug prints on **AUX UART**

Features

- How to check module registration and activate PDP context
- How to connect to a FTP server
- How to exchange data with the server

3.4.16.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage FTP client and start it

ftp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Init FTP client and set the debug function for it
- Connect to the server
- Perform log in
- Check remote file size and last modification time
- Download file from server to local filesystem. A data callback is set to report periodic info about the download status
- Upload the same file to the server with a different name. A data callback is set to report periodic info about the upload status
- Download another file content in a buffer instead of a file. A data callback is set to report periodic info about the download status
- Close the connection with FTP server
- Disable PDP context

```

Starting FTP demo app. This is v1.0.7 built on Apr 7 2020 11:17:36.
[DEBUG] 21.23 ftp_test.c:290 - msgFTPTask{FTP_TASK}$ INIT
[DEBUG] 21.23 ftp_test.c:304 - msgFTPTask{FTP_TASK}$ m2mb_os_ev_init success
[DEBUG] 21.23 ftp_test.c:310 - msgFTPTask{FTP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 21.23 ftp_test.c:318 - msgFTPTask{FTP_TASK}$ Waiting for registration...
[DEBUG] 21.25 ftp_test.c:214 - NetCallback{pubTspt_0}$ Module is registered to network
[DEBUG] 21.26 ftp_test.c:331 - msgFTPTask{FTP_TASK}$ Pdp context activation
[DEBUG] 21.27 ftp_test.c:335 - msgFTPTask{FTP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.31 ftp_test.c:344 - msgFTPTask{FTP_TASK}$ Activate PDP with APN web.omnitel.it on cid 3...
[DEBUG] 24.09 ftp_test.c:241 - PdpCallback{pubTspt_0}$ Context active
[DEBUG] 24.10 ftp_test.c:244 - PdpCallback{pubTspt_0}$ IP address: 176.244.166.181
Start ftp client...
[DEBUG] 24.82 ftp_test.c:373 - msgFTPTask{FTP_TASK}$ Connected.
[DEBUG] 26.32 ftp_test.c:386 - msgFTPTask{FTP_TASK}$ FTP login successful.
Get remote file /samples/pattern_big.txt size
[DEBUG] 26.69 ftp_test.c:428 - msgFTPTask{FTP_TASK}$ Done. File size: 20026.
Get remote file /samples/pattern_big.txt last modification date
[DEBUG] 26.89 ftp_test.c:450 - msgFTPTask{FTP_TASK}$ Done. File last mod date: 20200407090654
.

Starting download of remote file /samples/pattern_big.txt into local /mod/_pattern_big.txt
/samples/pattern_big.txt 47.54% 9520
/samples/pattern_big.txt 100.00% 20026
[DEBUG] 29.75 ftp_test.c:488 - msgFTPTask{FTP_TASK}$ download successful.
[DEBUG] 29.76 ftp_test.c:522 - msgFTPTask{FTP_TASK}$
Local file /mod/_pattern_big.txt size: 20026

Starting upload of local file /mod/_pattern_big.txt
/mod/_pattern_big.txt 81.81% 16384
Upload successful.

Starting download of remote file /samples/pattern.txt into local buffer
Getting remote file /samples/pattern.txt size..
[DEBUG] 32.97 ftp_test.c:583 - msgFTPTask{FTP_TASK}$ Done. File size: 988.
Starting download of remote file /samples/pattern.txt to buffer
[DEBUG] 34.08 ftp_test.c:145 - buf_data_cb{FTP_TASK}$ Received START event
[DEBUG] 34.09 ftp_test.c:149 - buf_data_cb{FTP_TASK}$ Received DATA: 988 bytes on buffer 0x400399e0
[DEBUG] 34.26 ftp_test.c:153 - buf_data_cb{FTP_TASK}$ Received END event
[DEBUG] 34.26 ftp_test.c:623 - msgFTPTask{FTP_TASK}$ Download successful. Received 988 bytes<<<
0 |-----| |-----| |-----| |-----| |-----| *
1 | A | | A | | A | | A | | A | | *
2 | AAA | | AAA | | AAA | | AAA | | AAA | | *
3 | AAAAA | | AAAAA | | AAAAA | | AAAAA | | AAAAA | | *
4 | AAAAAAA | | AAAAAAA | | AAAAAAA | | AAAAAAA | | AAAAAAA | | *
5 | AAAAAAAA | | AAAAAAAA | | AAAAAAAA | | AAAAAAAA | | AAAAAAAA | | *
6 | AAAAAAA | | AAAAAAA | | AAAAAAA | | AAAAAAA | | AAAAAAA | | *
7 | AAAAA | | AAAAA | | AAAAA | | AAAAA | | AAAAA | | *
8 | AAA | | AAA | | AAA | | AAA | | AAA | | *
9 | A | | A | | A | | A | | A | | *
10 |-----| |-----| |-----| |-----| |-----| *
11 | | | | | | | *
12 |-----| |-----| |-----| |-----| |-----|

```

Figure 24

3.4.17 File System example

Sample application showcasing M2MB File system API usage. Debug prints on **AUX UART**

Features

- How to open a file in write mode and write data in it
- How to reopen the file in read mode and read data from it

3.4.17.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Open file in write mode
- Write data in file
- Close file
- Reopen file in read mode
- Read data from file and print it
- Close file and delete it

```
Starting FileSystem demo app. This is v1.0.7 build on Mar 26 2020 09:50:19. LEVEL: 2
Opening /my_text_file.txt in write mode..
Buffer written successfully into file. 15 bytes were written.
Closing file.
Opening /my_text_file.txt in read only mode..
Received 15 bytes from file:
<Hello from file>
Closing file.
Deleting File
File deleted
App Completed
```

Figure 25

3.4.18 GNSS example

Sample application showing how to use GNSS functionality. Debug prints on **AUX UART**

Features

- How to enable GNSS receiver on module
- How to collect location information from receiver

Note: on MEx10G1 product family both M2MB_GNSS_SERVICE_NMEA_REPORT and M2MB_GNSS_SERVICE_POSITION_REPORT services are available, while on ME910C1 product family only M2MB_GNSS_SERVICE_POSITION_REPORT is available

3.4.18.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print a welcome message
- Create GNSS task and send a message to it

gps_task.c - Init Info feature and get module type - Init gnss, enable position/NMEA report and start it. - When a fix or a NMEA sentence is available, a message will be printed by the GNSS callback function

```
Starting GNSS demo app. This is v1.1.4 built on Oct  1 2021 15:27:44.
Model: ME910C1-E2
m2mb_gnss_enable, POSITION OK
m2mb_gnss_start OK, waiting for position/nmea sentences...
latitude_valid: 1 - latitude: 45.713643
longitude_valid: 1 - longitude: 13.738041
altitude_valid: 1 - altitude: 195.000000
uncertainty_valid: 1 - uncertainty: 95.000000
velocity_valid: 1 - codingType: 0
speed_horizontal: 0.650000
bearing: 0.000000
timestamp_valid: 1 -timestamp: 1633095357439
speed_valid: 1 - speed: 1.471360

***** Wait 120 seconds and then stop GPS *****
```

Figure 26

3.4.19 GPIO interrupt example

Sample application showing how to use GPIOs and interrupts. Debug prints on **AUX UART**

Features

- How to open a GPIO in input mode with interrupt
- How to open a second GPIO in output mode to trigger the first one

3.4.19.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Open GPIO 4 as output
- Open GPIO 3 as input and set interrupt for any edge (rising and falling). **A jumper must be used to short GPIO 3 and 4 pins.**
- Toggle GPIO 4 status high and low every second
- An interrupt is generated on GPIO 3

```
Starting GPIO interrupt demo app. This is v1.0.7 built on Mar 26 2020 16:33:01.  
Setting gpio 3 interrupt...  
Setting GPIO 4 HIGH  
CALLBACK->Interrupt on GPIO 3! Value: 1  
Setting GPIO 4 LOW  
CALLBACK->Interrupt on GPIO 3! Value: 0  
Setting GPIO 4 HIGH  
CALLBACK->Interrupt on GPIO 3! Value: 1  
Setting GPIO 4 LOW  
CALLBACK->Interrupt on GPIO 3! Value: 0  
Setting GPIO 4 HIGH  
CALLBACK->Interrupt on GPIO 3! Value: 1  
Setting GPIO 4 LOW  
CALLBACK->Interrupt on GPIO 3! Value: 0
```

Figure 27

3.4.20 GTP example

Sample application that shows how to get the position using GTP feature. Debug prints on **AUX UART**

Features

- How to init and enable GTP feature
- How to get the position using GTP

Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Init NET functionality and wait for module to be registered
- Init PDP functionality and set APN on CID1
- Init GTP functionality
- Check if GTP is already enabled. If not enable it and reboot module
- If GTP is enabled, get position

```
Starting GTP demo app. This is v1.1.8 built on Dec 20 2022 15:29:19.
[DEBUG] 15.08 M2MB_main:237 - M2MB_main{M2M_DamsStart}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 15.09 M2MB_main:245 - M2MB_main{M2M_DamsStart}$ Waiting for registration...
[DEBUG] 15.10 M2MB_main:253 - M2MB_main{M2M_DamsStart}$ Pdp context setting
[DEBUG] 15.11 M2MB_main:82 - NetCallback{pubTspt_0}$ Module is registered to cell 0x201
[DEBUG] 15.11 M2MB_main:257 - M2MB_main{M2M_DamsStart}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 15.13 M2MB_main:268 - M2MB_main{M2M_DamsStart}$ m2mb_pdp_deinit returned M2MB_RESULT_SUCCESS

Check if GTP has been already enabled
[DEBUG] 16.19 M2MB_main:312 - M2MB_main{M2M_DamsStart}$ GTP status: 1 => Enabled
[DEBUG] 16.20 M2MB_main:317 - M2MB_main{M2M_DamsStart}$ Get the position...

m2mb_gnss_GTP OK, waiting for position...
[DEBUG] 107.96 M2MB_main:160 - gnssCallbackFN{pubTspt_0}$ gnssCallback event: 1
GTP position got
latitude: 41.900002
longitude: 12.500000
altitude: 0.000000
altitudeMeanSeaLevel: -0.000000
accuracy: 0.000000
speed: -0.000000
bearing: -0.000000
timestamp: 0
verticalAccuracy 0.000000
speedAccuracy -0.000000
bearingAccuracy: -0.000000
```

Figure 28

3.4.21 General_INFO example

Sample application prints some Module/SIM information as IMEI, fw version, IMSI and so on; it prints also some information about registration. Debug prints on **AUX UART**

Features

- How to print some Module information as IMEI, FW version etc
- How to print some SIM information as IMSI, ICCID
- How to get and print some informatio about Module registration as Netowrk Operator, AcT, RSSI, etc

3.4.21.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Init NET functionality
- Init INFO functionality
- Get and print Module and SIM info
- Wait form module to register to network
- Get and print registration INFO

```

Starting. This is v1.1.4 built on Mar 31 2021 09:56:03. LEVEL: 2

Start General INFO application [ version: 1.000000 ]

=====
MODULE ME910C1-E2 INFO
=====
MANUFACTURER: Telit
IMEI: 353080091125422
MODEM FIRMWARE VERSION: MOB.700005
PACKAGE VERSION:
30.00.709-B005-P0B.700100
MOB.700005
P0B.700100
A0B.700000

=====
SIM INFO
=====
IMSI: 222015602268648
ICCID: 89390100001138084906

=====
Waiting for registration...

=====
Module is registered to HOME network cellID 0x5221
NETWORK OPERATOR (mcc mnc): 222 01
Network Technology 2G (AcT: 0) RSSI: -81

```

Figure 29

3.4.22 HTTP Client

Sample application showing how to use HTTPs client functionalities. Debug prints on **AUX UART**

Features

- How to check module registration and activate PDP context
- How to initialize the http client, set the debug hook function and the data callback to manage incoming data
- How to perform GET, HEAD or POST operations (GET also with single range support)

NOTE: the sample app has an optional dependency on `azx_base64.h` if basic authentication is required (refer to `HTTP_BASIC_AUTH_GET` define in `M2MB_main.c` for further details)

3.4.22.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage HTTP client and start it

httpTaskCB

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create HTTP client options and initialize its functionality
- Create HTTP SSL config and initialize the SSL options
- Configure data management options for HTTP client
- Apply all configurations to HTTP client
- Perform a GET request to a server
- Disable PDP context

DATA_CB

- Print incoming data
- Set the abort flag to 0 to keep going.

```

Starting HTTP(s) client demo app. This is v1.1.15 built on Nov 28 2022 12:12:27.
[DEBUG] 17.12 read_param:101 - readConfigFromFile(HttpClient)$ Reading parameters from file
[DEBUG] 17.13 read_param:103 - readConfigFromFile(HttpClient)$ Opening /mod/HTTP_Client_config.txt in read mode..
Set APN to: <web.omnitel.it>
Set APN USER to: <>
Set APN PASS to: <>
Set CID to: 3
Set CACERTFILE to: <>
Set CLIENTCERTFILE to: <>
Set CLIENTKEYFILE to: <>
Set REQUEST_TYPE to: 2
Set SERVER to: <http://linux-ip.net>
[DEBUG] 17.16 m2mb_HTTP_t:267 - activatePdp(HttpClient)$ m2mb_os_ev_init success
[DEBUG] 17.17 m2mb_HTTP_t:273 - activatePdp(HttpClient)$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 17.18 m2mb_HTTP_t:281 - activatePdp(HttpClient)$ Waiting for registration..
[DEBUG] 17.19 m2mb_HTTP_t:101 - NetCallback{pubTspt_0}$ Module is registered to cell 0x468E!
[DEBUG] 17.20 m2mb_HTTP_t:295 - activatePdp(HttpClient)$ Pdp context initialization
[DEBUG] 19.26 m2mb_HTTP_t:305 - activatePdp(HttpClient)$ Activate PDP with APN web.omnitel.it....
[DEBUG] 20.16 m2mb_HTTP_t:128 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 20.18 m2mb_HTTP_t:131 - PdpCallback{pubTspt_0}$ IP address: 176.243.212.114
Performing a GET request...
Connecting to linux-ip.net:80//
Socket connected!
<!DOCTYPE html>
<html>
<head>
  <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
  <meta name="author" content="Martin A. Brown" />
  <meta name="robots" content="index, follow"/>

  <meta property="og:title" content="http://linux-ip.net/" />
  <meta property="og:url" content="http://linux-ip.net/" />
  <meta property="og:site_name" content="http://linux-ip.net/" />
  <meta property="og:type" content="website"/>

  <link rel="canonical" href="http://linux-ip.net" />

  <title>http://linux-ip.net/</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <link rel="stylesheet" type="text/css" href="//netdna.bootstrapcdn.com/font-awesome/4.0.3/css/font-awesome.css" />
  <link rel="stylesheet" type="text/css" href="//netdna.bootstrapcdn.com/twitter-bootstrap/2.3.2/css/bootstrap-combined.min.css" />
  <link rel="stylesheet" type="text/css" href="http://linux-ip.net/theme/css/main.css" />
</head>
<body>
  <div id="site-footer">
    <div class="row-fluid">
      <div class="span10 offset1">
        <address>
          <p>
            Powered by <a href="http://getpelican.com/">Pelican</a>
            and <a href="http://python.org">Python</a>.
            Theme based on <a href="http://github.com/jsliang/pelican-fresh">Fresh</a>
            by <a href="http://jsliang.com/">jsliang</a>
          </p>
        </address>
      </div>
    </div>
  </div>
</body>
</html>
Result: 200
[DEBUG] 26.80 m2mb_HTTP_t:137 - PdpCallback{pubTspt_0}$ Context deactivated!

```

Figure 30

3.4.23 HW Timer (Hardware Timer)

The sample application shows how to use HW Timers M2MB API. Debug prints on **AUX UART**

Features

- How to open configure a HW timer
- How to use the timer to manage recurring events

3.4.23.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create hw timer structure
- Configure it with 100 ms timeout, periodic timer (auto fires when expires) and autostart
- Init the timer with the parameters
- Wait 10 seconds
- Stop the timer

TimerCb

- Print a message with an increasing counter

```
Starting HW Timers demo app. This is v1.0.7 built on Mar 26 2020 13:04:14.
[DEBUG] 14.06 M2MB_main.c:114 - M2MB_main{M2M_DamsStart}$ Set the timer attributes structure: success.
Timer successfully created
Start the timer, success.
[DEBUG] 14.18 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [0]
[DEBUG] 14.28 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [1]
[DEBUG] 14.38 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [2]
[DEBUG] 14.48 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [3]
[DEBUG] 14.58 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [4]
[DEBUG] 14.69 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [5]
[DEBUG] 14.79 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [6]
[DEBUG] 14.88 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [7]
[DEBUG] 14.98 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [8]
[DEBUG] 15.08 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [9]

[DEBUG] 23.90 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [96]
[DEBUG] 24.01 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [97]
[DEBUG] 24.11 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [98]
Stop a running timer: success
Application end
```

Figure 31

3.4.24 Hello World

The application prints “Hello World!” over selected output every two seconds. Debug prints on **AUX UART**, using AZX log example functions

Features

- How to open an output channel using AZX LOG sample functions
- How to print logging information on the channel using AZX LOG sample functions

3.4.24.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print “Hello World!” every 2 seconds in a while loop

```
Starting. This is v1.0.7 built on Mar 26 2020 09:34:16. LEVEL: 2
Start Hello world Application [ version: 2.000000 ]
Hello world 2.0 [ 000001 ]
Hello world 2.0 [ 000002 ]
Hello world 2.0 [ 000003 ]
Hello world 2.0 [ 000004 ]
Hello world 2.0 [ 000005 ]
Hello world 2.0 [ 000006 ]
Hello world 2.0 [ 000007 ]
Hello world 2.0 [ 000008 ]
Hello world 2.0 [ 000009 ]
```

Figure 32

3.4.25 I2C example

Sample application showing how to communicate with an I2C slave device. Debug prints on **AUX UART**

Features

- How to open a communication channel with an I2C slave device
- How to send and receive data to/from the slave device

Setup

- Connect sensor VDD to 1v8 supply (e.g. Vaux/PwrMon pin of the module)
- Connect sensor GND to a GND pin of the module
- Connect sensor SDA to module GPIO2
- Connect sensor SCL to module GPIO3

3.4.25.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Open I2C bus, setting SDA and SCL pins as 2 and 3 respectively
- Set registers to configure accelerometer -Read in a loop the 6 registers carrying the 3 axes values and show the g value for each of them

```
Starting I2C demo app. This is v1.0.7 built on Mar 26 2020 16:50:40.
Configuring the Kionix device...
opening channel /dev/I2C-30
[DEBUG] 20.18 M2MB_main.c:218 - test_I2C{M2M_DamsStart}$|-
WHOAMI content: 0x01
Configuring I2C Registers - Writing 0x4D into 0x1D register (CTRL_REG3)...
Write: success

I2C reading data from 0x1D register (CTRL_REG3)...
Read: success.
Accelerometer Enabled. ODR tilt: 12.5Hz, ODR directional tap: 400Hz, ORD Motion Wakeup: 50Hz
Configuring I2C Registers - Writing 0xC0 into 0x1B register (CTRL_REG1)...
Write: success

I2C reading data from 0x1B register (CTRL_REG1)...
Read: success.
Accelerometer Enabled. Operative mode, 12bit resolution
I2C read axes registers
-----
Reading Success.

X: -0.050 g
Y: -0.046 g
Z: 1.006 g
Reading Success.

X: -0.049 g
Y: -0.044 g
Z: 1.004 g
Reading Success.

X: -0.052 g
Y: -0.044 g
Z: 1.007 g
Reading Success.

X: -0.048 g
Y: -0.045 g
Z: 1.005 g
```

Figure 33

3.4.26 I2C Combined

Sample application showing how to communicate with an I2C slave device with I2C raw mode. Debug prints on MAIN UART

Features

- How to open a communication channel with an I2C slave device
- How to send and receive data to/from the slave device using raw mode API

Setup

- Connect sensor VDD to 1v8 supply (e.g. Vaux/PwrMon pin of the module)
- Connect sensor GND to a GND pin of the module
- Connect sensor SDA to module GPIO2
- Connect sensor SCL to module GPIO3

3.4.26.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Open I2C bus, setting SDA and SCL pins as 2 and 3 respectively
- Set registers to configure accelerometer - Read in a loop the 6 registers carrying the 3 axes values and show the g value for each of them

```
Starting I2C raw demo app. This is v1.0.13-C1 built on Jul 30 2020 11:28:18.
Configuring the I2C device...
Opening I2C channel /dev/I2C-30 ( device address is 0x0F << 1 )
Accelerometer Enabled. ODR tilt: 12.5Hz, ODR directional tap: 400Hz, ORD Motion Wakeup: 50Hz
Accelerometer Enabled. Operative mode, 12bit resolution
I2C read axes registers
-----
X: 0.000 g
Y: 0.000 g
Z: 0.000 g

X: -0.270 g
Y: 0.016 g
Z: 0.917 g

X: -0.268 g
Y: 0.013 g
Z: 0.925 g

X: -0.271 g
Y: 0.015 g
Z: 0.922 g

X: -0.267 g
Y: 0.016 g
Z: 0.918 g

X: -0.274 g
Y: 0.019 g
Z: 0.915 g
```

Figure 34

3.4.27 LWM2M

Sample application showcasing LWM2M client usage with M2MB API. Debug prints on **AUX UART**

Features

- Configure LWM2M client and enable it
- Create an instance of a custom object
- Set an integer value on a read only resource
- Set two integer values on a multi-instance read only resource
- write a string on a read/write resource
- Manage exec requests from the portal
- Manage write, read and monitoring resources changed from the portal

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6", "<user apn>"
```

Depending on the Mobiler Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.4.27.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask - Check registration status

- Initialize LWM2M client,
- Check for XML file fo custom object
- Enable unsolicited messages from client
- Create a task (lwm2m_taskCB is its callback function)to manage events from Portal
- Enable LwM2M client

- Create a new instance for the custom object
- Wait for client to register to Portal
- Performs all operations (set, read, get, write) on the related resources
- Performs a set with notify ack enabled
- Wait for events from server

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)
- Run lwm2m_taskCB when a monitored resource changes, to manage the action to be done

3.4.27.2 Device Profile upload

LWM2M resources demo device profile must be imported to have a real-time update of resources values on the LWM2M browser.

To do so, import the file `json/lwm2m_resources_demo.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

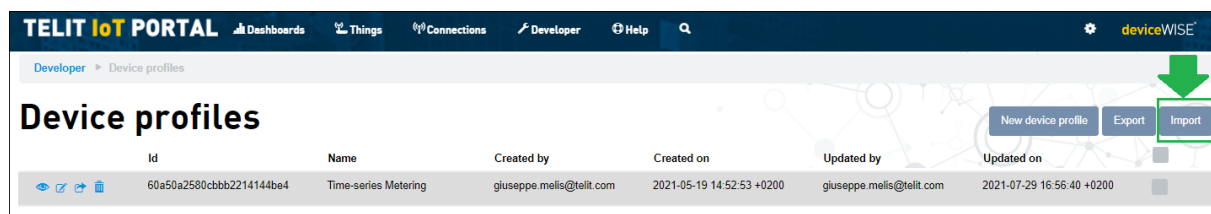


Figure 35

3.4.27.3 Custom Object configuration

The XML file content must be loaded on the Telit IoT Portal for the demo application to be fully executed.

First, enter Developer section from the top menu



Figure 36

Choose Object Registry

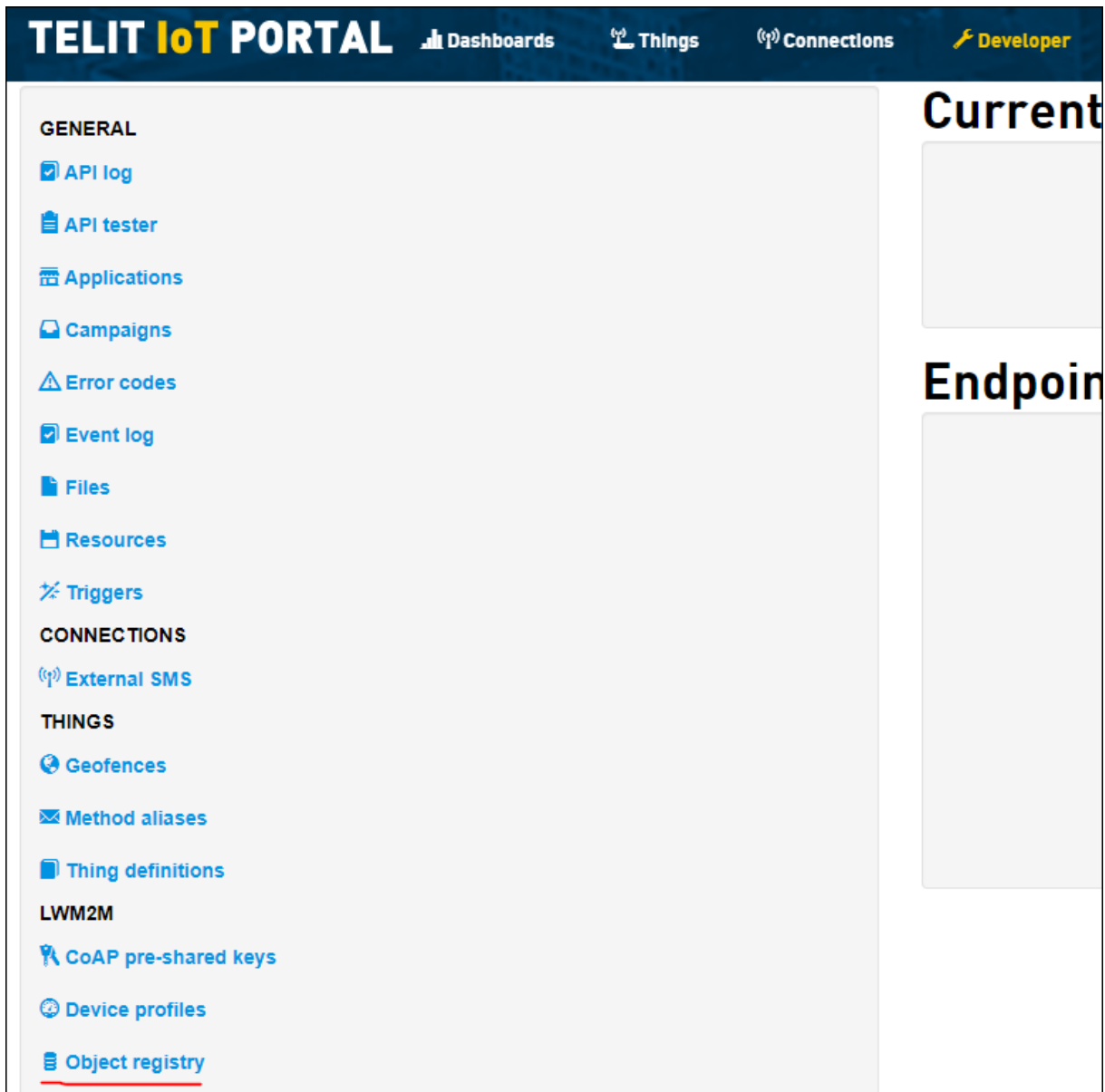


Figure 37

Create a New Object

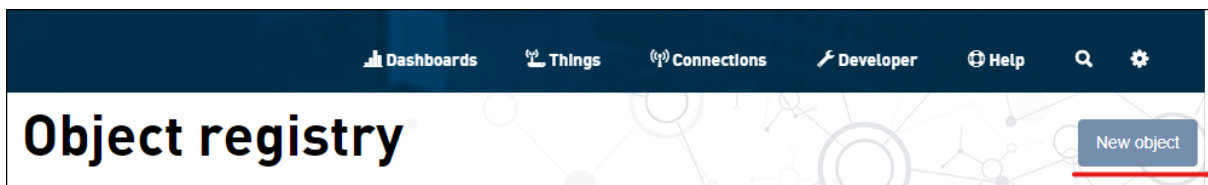
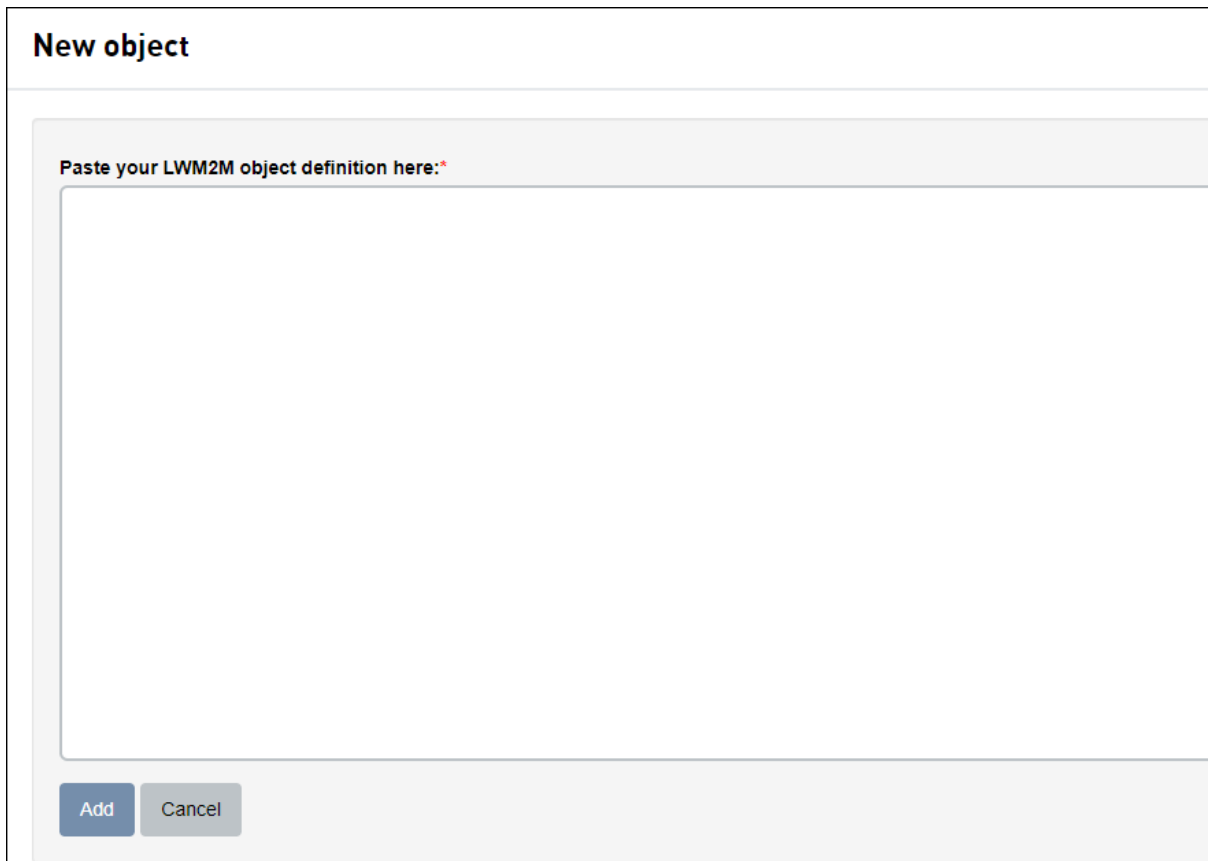


Figure 38

Copy the xml file content and paste it in the new Object form



New object

Paste your LWM2M object definition here:*

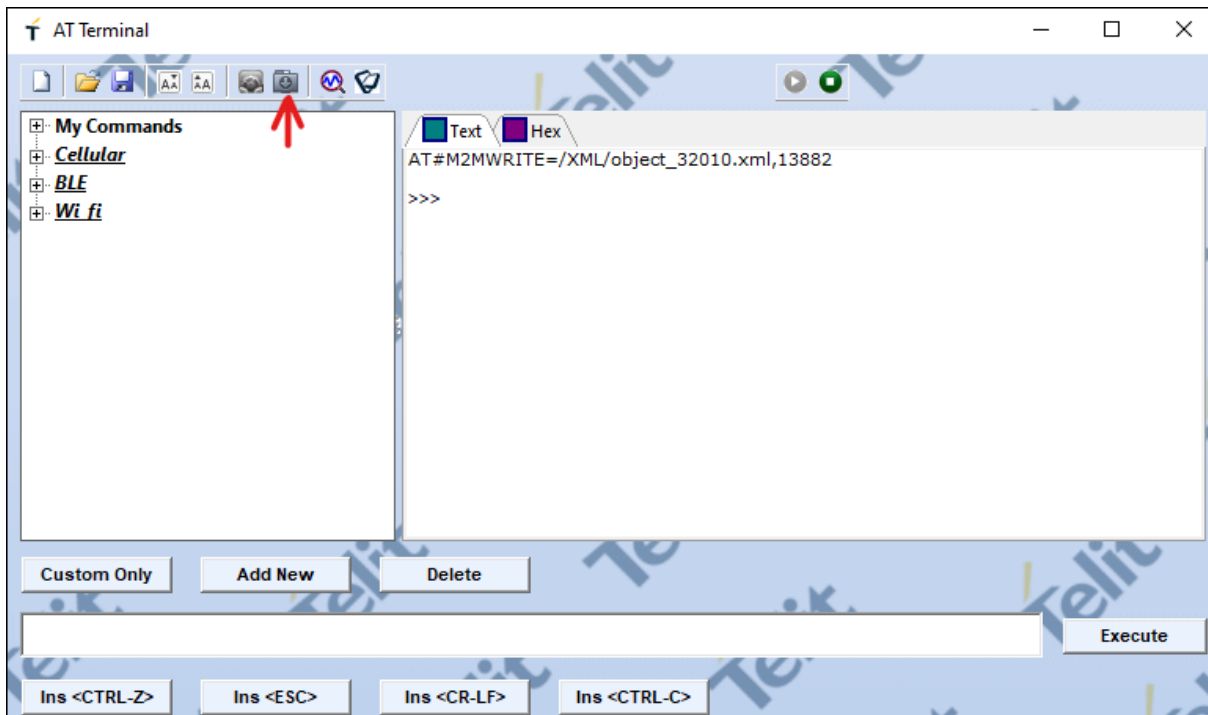
Add Cancel

Figure 39

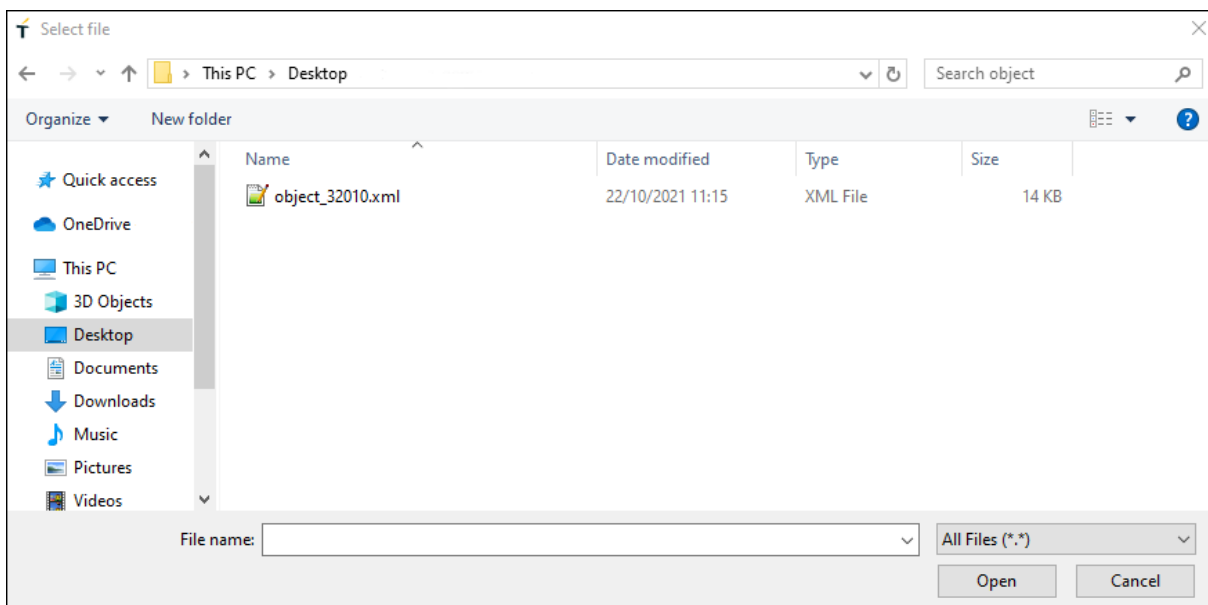
Also, the application requires the XML file `xml/object_32010.xml` (provided with the sample files) to be stored in module's `/XML/` folder. It can be done with

```
AT#M2MWRITE=/XML/object_32010.xml,<size_in_bytes>
```

To load the XML file in the module, Telit AT Controller (TATC) can be used. Once the command above is issued, press the load content button:

**Figure 40**

Select the file from your computer

**Figure 41**

The file is successfully loaded on the module

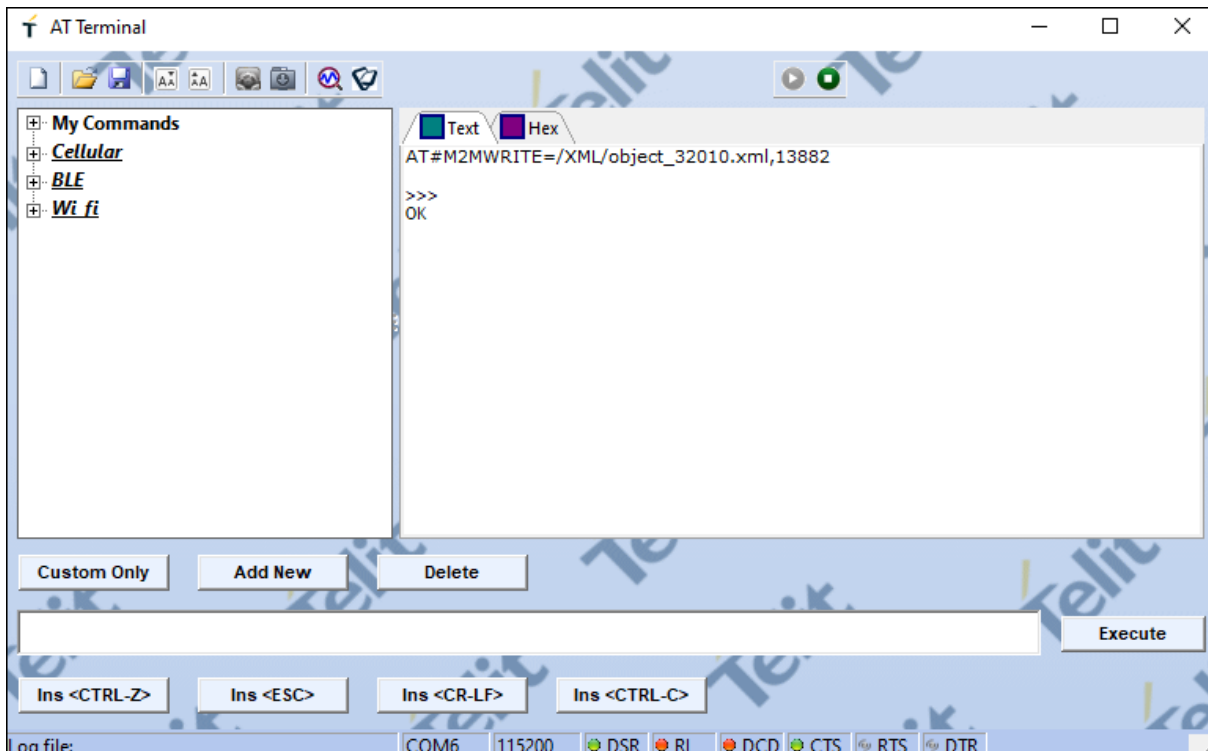


Figure 42

3.4.27.4 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on “**Things**” section, click “**New Thing**” button in the top right corner.

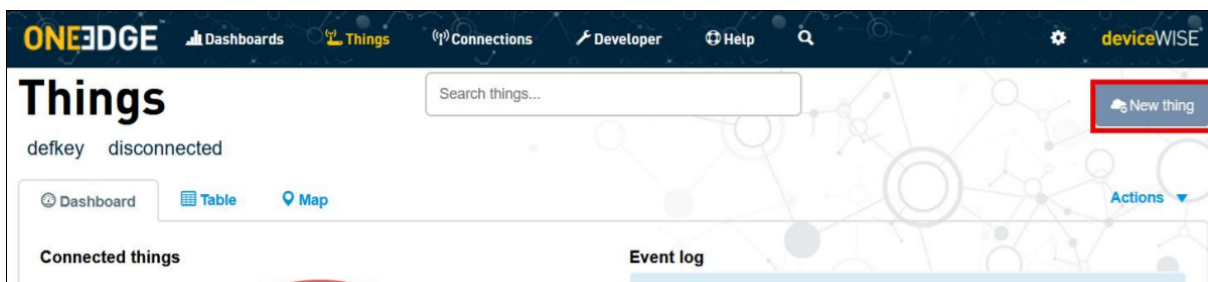


Figure 43

In the Create a new thing dialog, select “Telit Module”

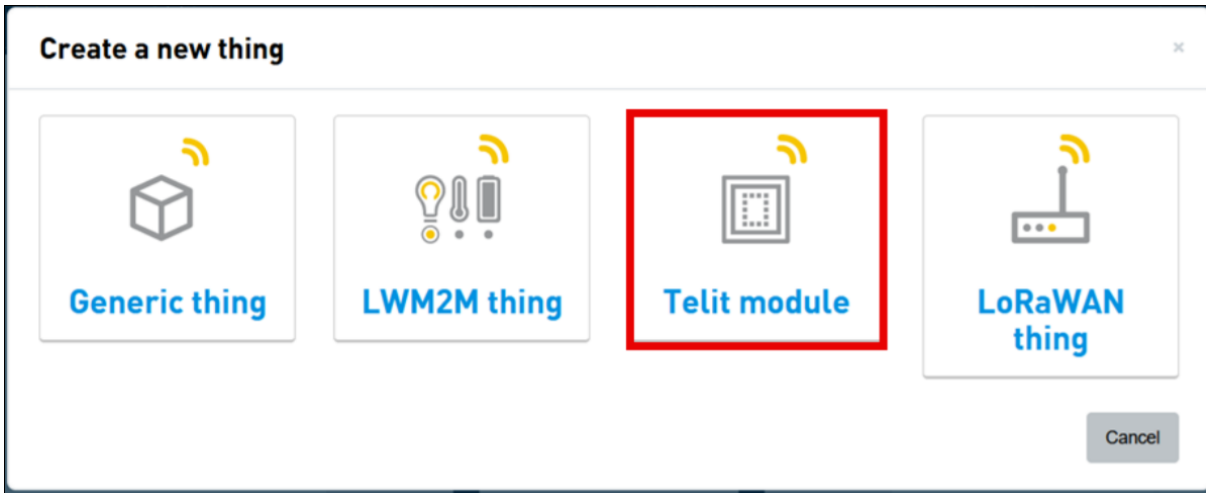


Figure 44

A dialog appears: select “Default” thing definition

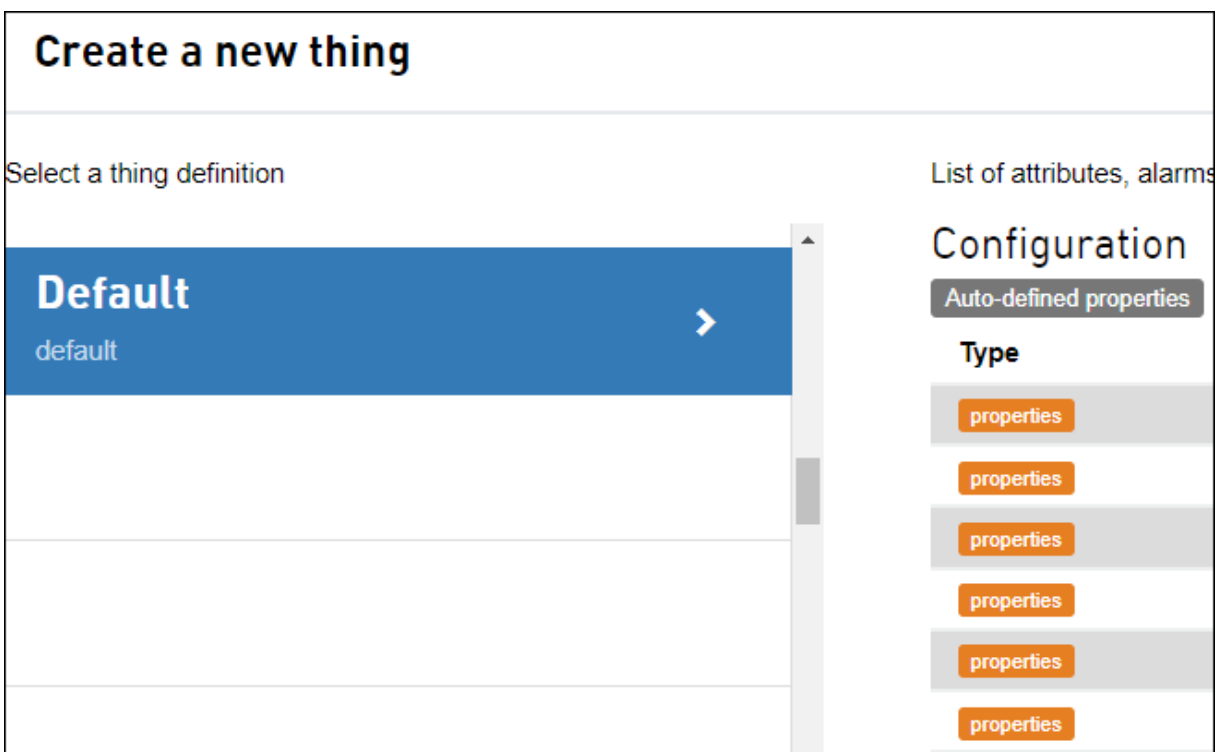


Figure 45

In the following screen, provide the Telit ID as “Identifier” Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the

screenshot below

Create a new thing

Setup
LWM2M
Identification
VAR
Other
Back

LWM2M
Device profile*
LwM2M resources demo

Add Cancel

Figure 46

Click **“Add”** to complete the new thing creation procedure.

If the Thing already exists, its device profile can be changed by following the steps shown in the picture below

TELIT IoT PORTAL Dashboards Things Connections Developer Help

Things > ep_353080091125356

ep_353080091125356

Overview Details LWM2M LWM2M Device Management Firmware Attributes Remote AT Events Files Tunnels API usage Actions

Identity NDWfxhWyZGatw114
Device profile Device Management Profile - Troubleshooting
Remote address 141.226.117.241:3702
Protocol LWM2M-CoAP (DTLSv12)
Binding UQ
Lifetime 630

Object browser **Set profile** Queue history Wakeup

Set LWM2M profile

Change profile*
LwM2M resources demo

Set Cancel

Figure 47

3.4.27.5 Application execution example

```

Starting lwm2m demo. This is v1.1.10 built on Oct 22 2021 10:49:52.
On OneEdge portal, be sure that the demo Device Profile has been uploaded on the IoT Portal
Initializing resources...

LWM2M enable result OK
[DEBUG] 38.20 lwm2m_demo:998 - lwm2mIndicationCB{pubTspt_0}$ Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED
GET STATUS.
IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

```

Figure 48

```
=====
READ-ONLY RESOURCES
=====

Setting integer resource {32010/0/2} value to 50 on LWM2M client.
Reading integer resource {32010/0/2} value on LWM2M client.
Resource /32010/0/2/0 changed!
Reading integer resource {32010/0/2/0} value on LWM2M client.
---Integer value is now 50
Integer data in {32010/0/2/0} resource was updated to new value: 50
-----

Setting integer resource {32010/0/22/0} value to 10 on LWM2M client.
Resource /32010/0/22/0 changed!
Reading integer resource {32010/0/22/0} value on LWM2M client.
Integer data in {32010/0/22/0} resource was updated to new value: 10
Setting integer resource {32010/0/22/1} value to 11 on LWM2M client.
Resource /32010/0/22/1 changed!
Reading integer resource {32010/0/22/1} value on LWM2M client.
Integer data in {32010/0/22/1} resource was updated to new value: 11
-----

Setting double resource {32010/0/3} value to 20.500000 on LWM2M client.
Reading double resource {32010/0/3} value on LWM2M client.
Resource /32010/0/3/0 changed!
Reading double resource {32010/0/3/0} value on LWM2M client.
---Double value is now 20.500000
Float data in {32010/0/3/0} resource was updated to new value: 20.500000
-----

Setting boolean resource {32010/0/4} value to 1 on LWM2M client.
Reading boolean resource {32010/0/4} value on LWM2M client.
Resource /32010/0/4/0 changed!
Reading boolean resource {32010/0/4/0} value on LWM2M client.
---Boolean value is now true
Boolean data in {32010/0/4/0} resource was updated to new value: true
-----
```

```

Current time in seconds from the epoch: 1634900084
Setting timestamp resource {32010/0/6} value to 1634900084 on LWM2M client.
Reading timestamp resource {32010/0/6} value on LWM2M client.
Resource /32010/0/6/0 changed!
Reading timestamp resource {32010/0/6/0} value on LWM2M client.
---Timestamp value is now 1634900084
Time data in {32010/0/6/0} resource was updated to new value: 1634900084 (2021/10/22T10:54:44+00:00)
-----

Setting opaque resource {32010/0/5} on LWM2M client.
Resource /32010/0/5/0 changed!
Reading opaque resource {32010/0/5/0} on LWM2M client.
Opaque data in {32010/0/5/0} resource was updated to new content (10 bytes):
00 01 02 03 04 05 06 07 08 09
Reading opaque resource {32010/0/5} on LWM2M client.
---Opaque content is now (10 bytes): 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09
-----

Setting string resource {32010/0/1} value to Hello World! on LWM2M client.
Reading string resource {32010/0/1} value on LWM2M client.
Resource /32010/0/1/0 changed!
Reading integer resource {32010/0/1/0} value on LWM2M client.
---String content is now: <Hello World!>
String data in {32010/0/1/0} resource was updated to new content: <Hello World!>
-----

```



```
=====
READ-WRITE RESOURCES
=====

Writing integer resource {32010/0/12} value to 50 on LWM2M client.
Reading integer resource {32010/0/12} value on LWM2M client.
Resource /32010/0/12/0 changed!
Reading integer resource {32010/0/12/0} value on LWM2M client.
---Integer value is now 50
Integer data in {32010/0/12/0} resource was updated to new value: 50
-----

Writing double resource {32010/0/13} value to 20.500000 on LWM2M client.
Reading double resource {32010/0/13} value on LWM2M client.
Resource /32010/0/13/0 changed!
Reading double resource {32010/0/13/0} value on LWM2M client.
---Double value is now 20.500000
Float data in {32010/0/13/0} resource was updated to new value: 20.500000
-----

Writing boolean resource {32010/0/14} value to 1 on LWM2M client.
Reading boolean resource {32010/0/14} value on LWM2M client.
Resource /32010/0/14/0 changed!
Reading boolean resource {32010/0/14/0} value on LWM2M client.
---Boolean value is now true
Boolean data in {32010/0/14/0} resource was updated to new value: true
-----
```

```

Current time in seconds from the epoch: 1634900125
Writing timestamp resource {32010/0/16} value to 1634900125 on LWM2M client.
Reading timestamp resource {32010/0/16} value on LWM2M client.
Resource /32010/0/16/0 changed!
Reading timestamp resource {32010/0/16/0} value on LWM2M client.
---Timestamp value is now 1634900125
Time data in {32010/0/16/0} resource was updated to new value: 1634900125 (2021/10/22T10:55:25+00:00)
-----
Writing opaque resource {32010/0/15} on LWM2M client.
Resource /32010/0/15/0 changed!
Reading opaque resource {32010/0/15/0} on LWM2M client.
Opaque data in {32010/0/15/0} resource was updated to new content (10 bytes):
00 01 02 03 04 05 06 07 08 09
Reading opaque resource {32010/0/15} on LWM2M client.
---Opaque content is now (10 bytes): 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09
-----
Writing string resource {32010/0/11} value to <Hello World!> on LWM2M client.
Reading string resource {32010/0/11} value on LWM2M client.
Resource /32010/0/11/0 changed!
Reading integer resource {32010/0/11/0} value on LWM2M client.
---String content is now: <Hello World!>

```

```
=====
WRITE-ONLY RESOURCES
=====

Writing integer resource {32010/0/42} value to 50 on LWM2M client.
String data in {32010/0/11/0} resource was updated to new content: <Hello World!>
Getting integer resource {32010/0/42} value on LWM2M client.
Resource /32010/0/42/0 changed!
Getting integer resource {32010/0/42/0} value on LWM2M client.
---Integer value is now 50
Integer data in {32010/0/42/0} resource was updated to new value: 50
-----

Writing double resource {32010/0/43} value to 20.500000 on LWM2M client.
Getting double resource {32010/0/43} value on LWM2M client.
Resource /32010/0/43/0 changed!
Getting double resource {32010/0/43/0} value on LWM2M client.
---Double value is now 20.500000
Float data in {32010/0/43/0} resource was updated to new value: 20.500000
-----

Writing boolean resource {32010/0/44} value to 1 on LWM2M client.
Getting boolean resource {32010/0/44} value on LWM2M client.
Resource /32010/0/44/0 changed!
Getting boolean resource {32010/0/44/0} value on LWM2M client.
---Boolean value is now true
Boolean data in {32010/0/44/0} resource was updated to new value: true
-----
```

```

Current time in seconds from the epoch: 1634900163
Writing timestamp resource {32010/0/46} value to 1634900163 on LWM2M client.
Getting timestamp resource {32010/0/46} value on LWM2M client.
Resource /32010/0/46/0 changed!
Getting timestamp resource {32010/0/46/0} value on LWM2M client.
---Timestamp value is now 1634900163
Time data in {32010/0/46/0} resource was updated to new value: 1634900163 (2021/10/22T10:56:03+00:00)
-----

Writing opaque resource {32010/0/45} on LWM2M client.
Resource /32010/0/45/0 changed!
Getting opaque resource {32010/0/45/0} on LWM2M client.
Opaque data in {32010/0/45/0} resource was updated to new content (10 bytes):
00 01 02 03 04 05 06 07 08 09
Getting opaque resource {32010/0/45} on LWM2M client.
---Opaque content is now (10 bytes): 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09
-----

Writing string resource {32010/0/41} value to <Hello World!> on LWM2M client.
Getting string resource {32010/0/41} value on LWM2M client.
Resource /32010/0/41/0 changed!
Getting integer resource {32010/0/41/0} value on LWM2M client.
---String content is now: <Hello World!>
String data in {32010/0/41/0} resource was updated to new content: <Hello World!>
-----
Resources operations examples done.
-----

-----
Will perform a SET with notify ACK enabled.
-----
Enable notify ack
Setting integer resource {32010/0/2} value to 60 on LWM2M client.
Resource /32010/0/2/0 changed!
Reading integer resource {32010/0/2/0} value on LWM2M client.
Integer data in {32010/0/2/0} resource was updated to new value: 60
ACK received from server!
Reading integer resource {32010/0/2} value on LWM2M client.
---Integer value is now 60
Disable notify ack
Done.

Waiting for events from the OneEdge portal. Please write on monitored resources or call an 'exec' one.

```

After the Demo completes the initialization, it is possible to access the object resources from the section Things: select your device, then “LwM2M” tab of OneEdge IoT portal:

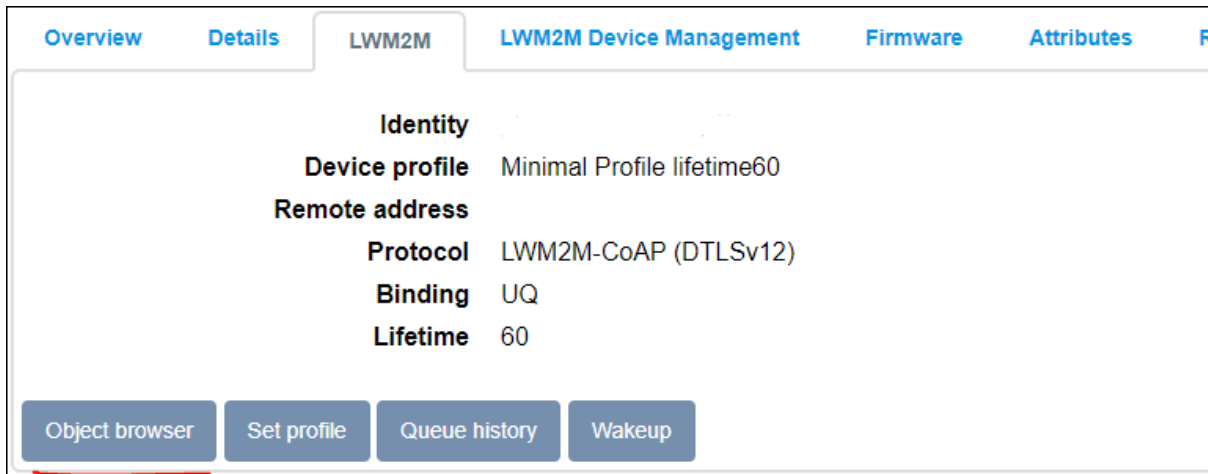


Figure 49

An instance of the object will be present and the resources can be modified.

m2mb lwm2m demo		/35000					
Instance 0	/35000/0		Observe	Read	Write	Attributes	Delete
● Read Only String	/35000/0/1		Observe	Read	Attributes		
● Read Only Integer	/35000/0/2		Observe	Read	Attributes		
● Read Only Float	/35000/0/3		Observe	Read	Attributes		
● Read Only Boolean	/35000/0/4		Observe	Read	Attributes		
● Read Only Opaque	/35000/0/5		Observe	Read	Attributes		
● Read Only Time	/35000/0/6		Observe	Read	Attributes		
● Read Only Objlnk	/35000/0/7		Observe	Read	Attributes		
● Read Write String	/35000/0/11		Observe	Read	Write	Attributes	
● Read Write Integer	/35000/0/12		Observe	Read	Write	Attributes	
● Read Write Float	/35000/0/13		Observe	Read	Write	Attributes	
● Read Write Boolean	/35000/0/14		Observe	Read	Write	Attributes	
● Read Write Opaque	/35000/0/15		Observe	Read	Write	Attributes	
● Read Write Time	/35000/0/16		Observe	Read	Write	Attributes	
● Read Write Objlnk	/35000/0/17		Observe	Read	Write	Attributes	
● Read only MultiInstance String	/35000/0/21		Observe	Read	Attributes		
● Read only MultiInstance Integer	/35000/0/22		Observe	Read	Attributes		
● Read only MultiInstance Float	/35000/0/23		Observe	Read	Attributes		
● Read only MultiInstance Boolean	/35000/0/24		Observe	Read	Attributes		
● Read only MultiInstance Opaque	/35000/0/25		Observe	Read	Attributes		
● Read only MultiInstance Time	/35000/0/26		Observe	Read	Attributes		
● Read only MultiInstance Objlnk	/35000/0/27		Observe	Read	Attributes		
● Read Write MultiInstance String	/35000/0/31		Observe	Read	Write	Attributes	
● Read Write MultiInstance Integer	/35000/0/32		Observe	Read	Write	Attributes	
● Read Write MultiInstance Float	/35000/0/33		Observe	Read	Write	Attributes	
● Read Write MultiInstance Boolean	/35000/0/34		Observe	Read	Write	Attributes	
● Read Write MultiInstance Opaque	/35000/0/35		Observe	Read	Write	Attributes	
● Read Write MultiInstance Time	/35000/0/36		Observe	Read	Write	Attributes	
● Read Write MultiInstance Objlnk	/35000/0/37		Observe	Read	Write	Attributes	
● Executable 1	/35000/0/101		Exec				
● Executable 2	/35000/0/102		Exec				

Figure 50

For example, executing the two Exec Resources at the bottom of the list, the application will react accordingly:

```
Info Exec Ind: 32010/0/101/0  
Asked to execute resource 101  
  
Resource /32010/0/11/0 changed!
```

Figure 51

Writing a string resource (id /32010/0/11), the application will notify the change

```
Reading integer resource {32010/0/11/0} value on LWM2M client.  
  
String data in {32010/0/11/0} resource was updated to new content: <Hello from the IoT Portal!>
```

Figure 52

3.4.28 LWM2M FOTA ACK management

Sample application showcasing LWM2M client FOTA events and ACKs management via APIs. Debug prints on **AUX UART**

Features

- Configure LWM2M client and enable it (if not already running)
- Enable monitoring on object 5 (Firmware Update)
- Configure lwm2m FOTA parameters to enable ACK requirement for each operation (Download, Update)
- Wait for an externally generated FOTA request to handle the steps by sending the acknowledgements when needed

Please refer to “80654NT11889A OneEdge Firmware Management Application Note” for further details about FOTA workflow. Get in touch with TS-OneEdge support to receive the document.

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6","<user apn>"
```

Depending on the Mobile Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.4.28.1 Device Profile upload

Minimal FOTA profile (short lifetime) device profile must be imported and selected to improve the responsiveness of the FOTA operations

To do so, import the file `json/lwm2m_fota_profile_short.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

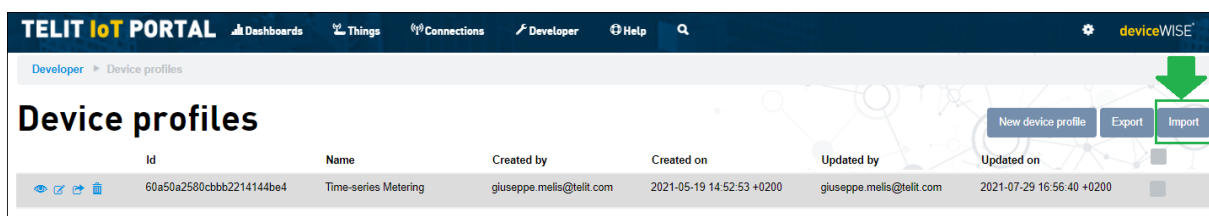


Figure 53

3.4.28.2 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on **“Things”** section, click **“New Thing”** button in the top right corner.

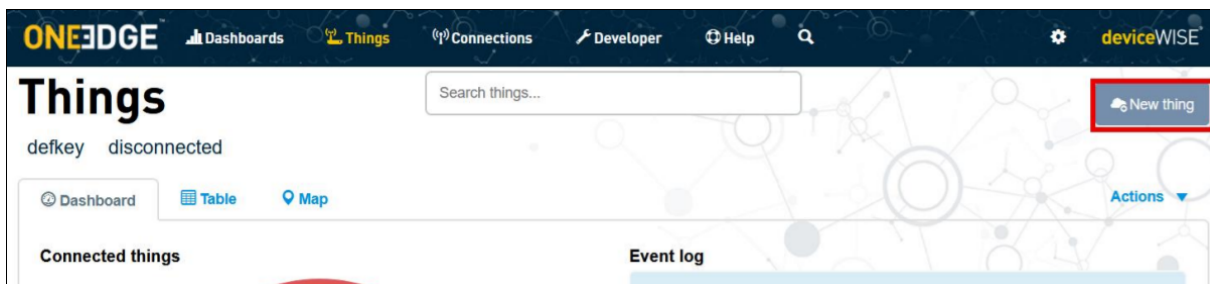


Figure 54

In the Create a new thing dialog, select “Telit Module”

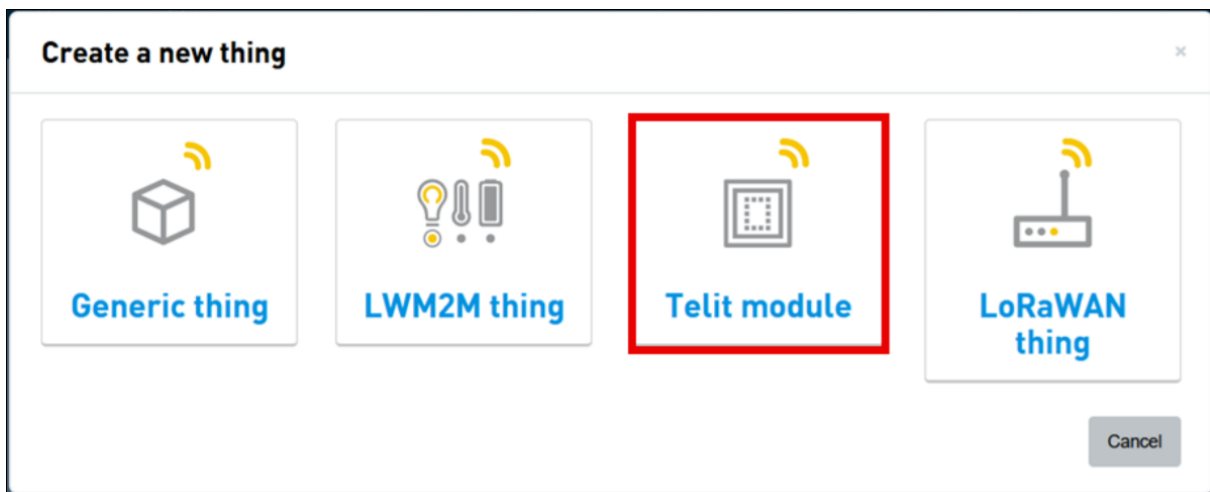


Figure 55

A dialog appears: select “Default” thing definition

Create a new thing

Select a thing definition

Default
default

List of attributes, alarms

Configuration

Auto-defined properties

Type

properties

properties

properties

properties

properties

properties

Figure 56

In the following screen, provide the Telit ID as “Identifier” Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the screenshot below

Create a new thing

Setup

LWM2M

Identification

VAR

Other

Back

LWM2M

Device profile*

Minimal FOTA profile (short lifetime)

Add Cancel

Figure 57

Click **“Add”** to complete the new thing creation procedure.

3.4.28.3 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask

- Initialize LWM2M client,
- Enable unsolicited messages from client
- Enable FOTA ACK configuration
- Create a task (lwm2m_taskCB is its callback function)to manage events from Portal
- Check if Lwm2M client is running, if not, enable it
- Wait for client to register to Portal
- Wait for FOTA events to arrive, and when the Update request arrives, notify the user that everything must stop before sending the last ACK (as the module will start updating and then reboot)

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)
- Run lwm2m_taskCB when a monitored resource changes, to manage the action to be done

3.4.28.4 Application execution example

```
Starting lwm2m FOTA management demo. This is v1.1.14 built on Jan 21 2022 14:39:47.
Initializing resources...
[DEBUG] 22.14 lwm2m_demo:283 - at_cmd_async_init{LWM2M_TASK}$ m2mb_ati_init() on instance 1

IF Status: M2MB_LWM2M_IF_STATE_DISABLED
Client Status: M2MB_LWM2M_CL_STATE_DISABLED
Enable lwm2m client
LWM2M enable result OK
Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

Waiting for events from the OneEdge portal. Please start a FOTA process.
```

Figure 58

```
Resource /5/0/1/0 changed!
[DEBUG] 108.42 lwm2m_demo:544 - lwm2m_taskCB{LWM2M_TASK}$ Firmware update event!
FOTA process is waiting for Download ACK, sending it!
Waiting for Update ACK!
Before sending the ACK to apply FOTA, stop any ongoing operation (filesystem, sockets)

Asked to perform UPDATE step, stopping all operations now.
Application complete.

----- Info, DEVICE REBOOT FOR FOTA -----
?????????
#OTAEV: Module Upgraded To New Fw
```

FOTA request arrives from server, ACK is sent for Download and Update. Device restarts with the new firmware version

```
Starting lwm2m FOTA management demo. This is v1.1.14 built on Jan 21 2022 14:39:47.
Initializing resources...
[DEBUG] 26.61 lwm2m_demo:283 - at_cmd_async_init{LWM2M_TASK}$ m2mb_ati_init() on instance 1
[WARN ] 34.98 lwm2m_demo:738 - lwm2mIndicationCB{pubTspt_0}$ write res 1

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
Enable already done, skip. State: M2MB_LWM2M_IF_STATE_ACTIVE
Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

Waiting for events from the OneEdge portal. Please start a FOTA process.
```

Application restarts and is ready for a new FOTA operation

3.4.29 LWM2M FOTA ACK management (AT URCs)

Sample application showcasing LWM2M client FOTA events and ACKs management via AT URCs. Debug prints on **AUX UART**

Features

- Configure LWM2M client and enable it (if not already running)
- Enable monitoring on object 5 (Firmware Update)
- Configure lwm2m FOTA parameters to enable ACK requirement for each operation (Download, Update)
- Wait for an externally generated FOTA request to handle the steps by sending the acknowledgements when needed

Please refer to “80654NT11889A OneEdge Firmware Management Application Note” for further details about FOTA workflow. Get in touch with TS-OneEdge support to receive the document.

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6","<user apn>"
```

Depending on the Mobile Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.4.29.1 Device Profile upload

Minimal FOTA profile (short lifetime) device profile must be imported and selected to improve the responsiveness of the FOTA operations

To do so, import the file `json/lwm2m_fota_profile_short.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

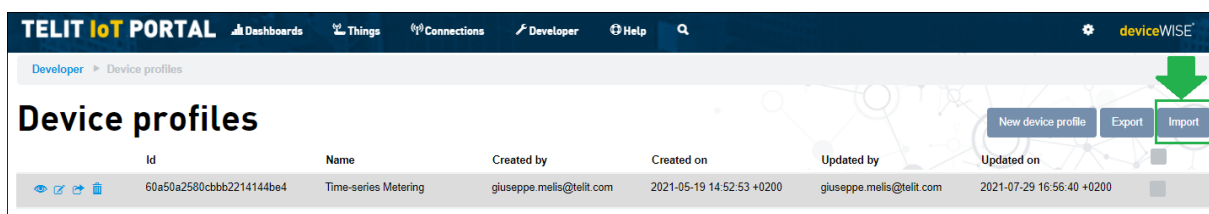


Figure 59

3.4.29.2 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on **“Things”** section, click **“New Thing”** button in the top right corner.

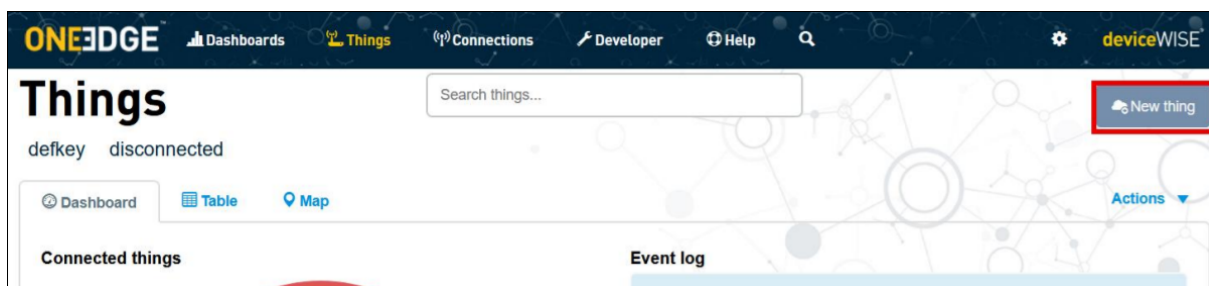


Figure 60

In the Create a new thing dialog, select “Telit Module”

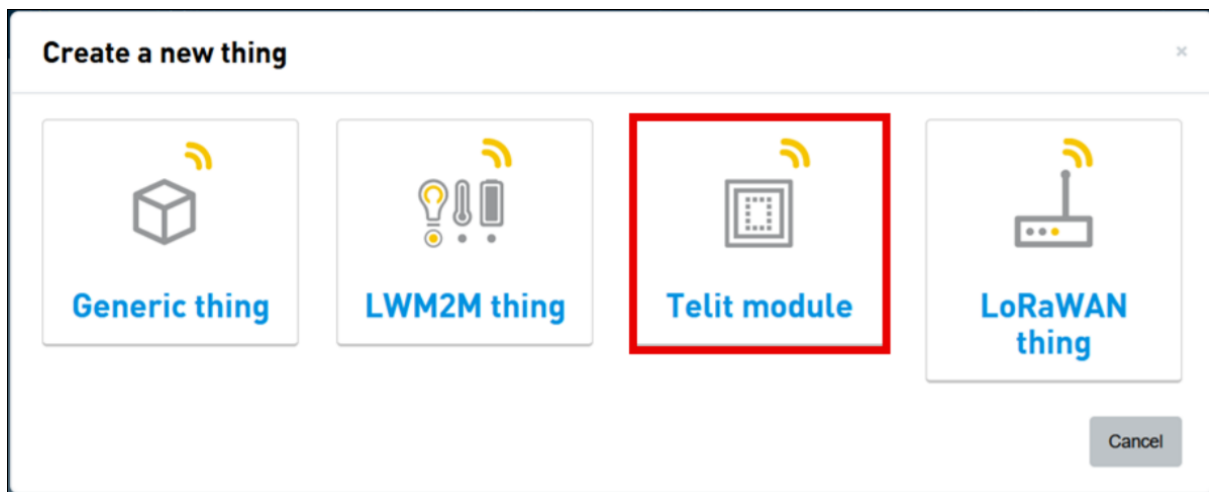


Figure 61

A dialog appears: select “Default” thing definition

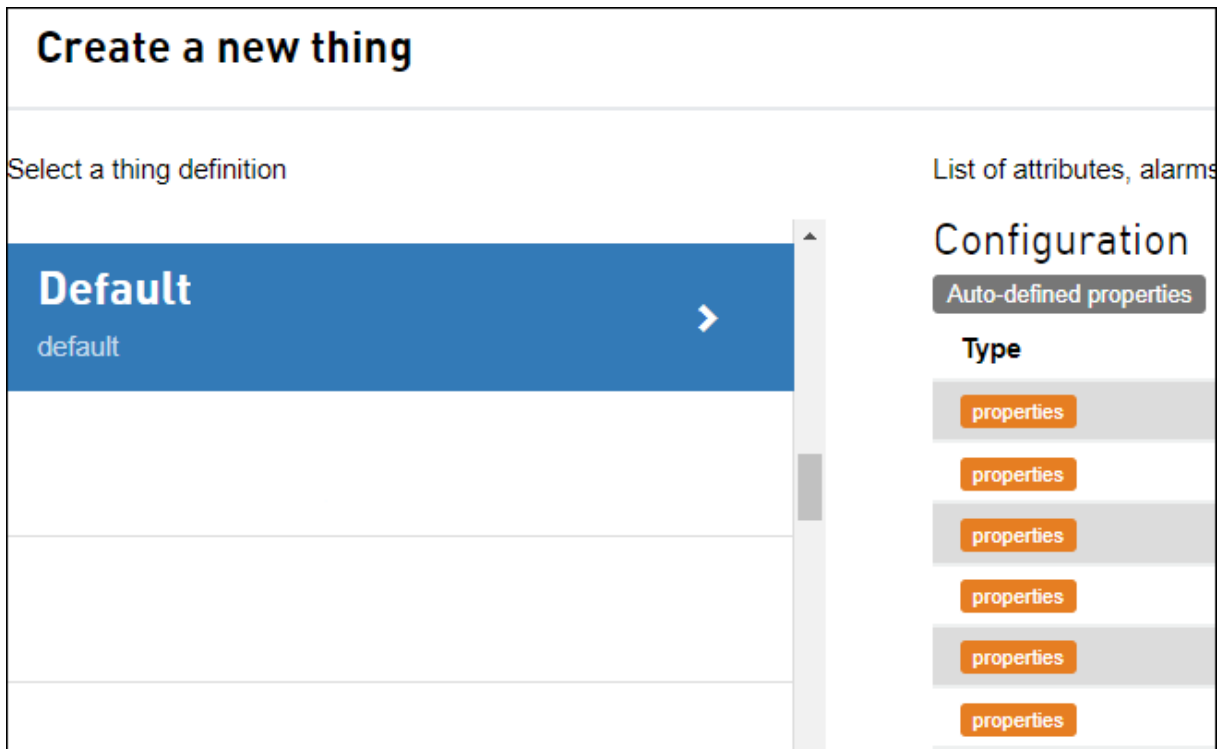


Figure 62

In the following screen, provide the Telit ID as “Identifier” Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the screenshot below

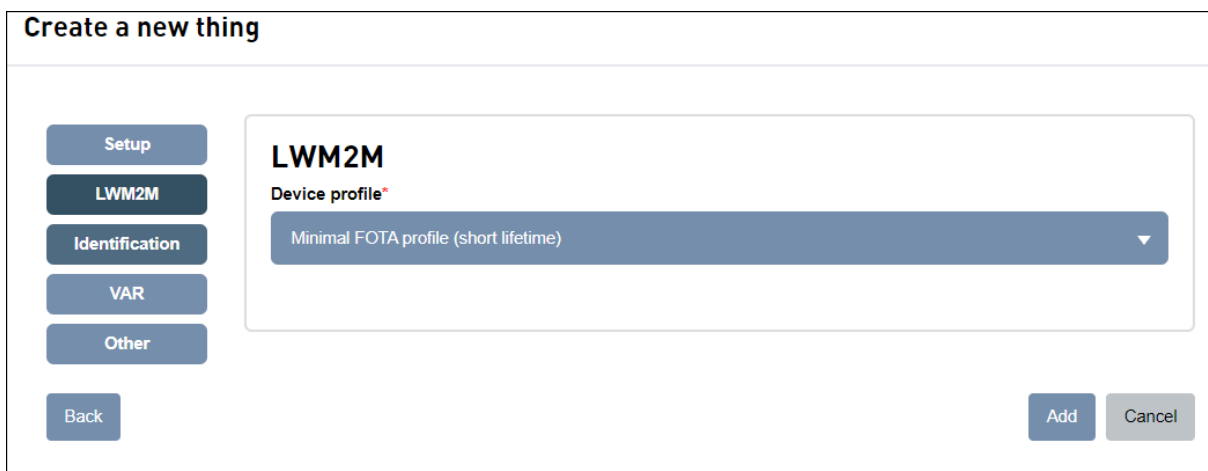


Figure 63

Click **“Add”** to complete the new thing creation procedure.

3.4.29.3 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask

- Initialize LWM2M client,
- Enable unsolicited messages from client
- Enable FOTA ACK configuration
- Create a task (lwm2m_taskCB is its callback function)to manage events from Portal
- Check if Lwm2M client is running, if not, enable it
- Wait for client to register to Portal
- Wait for FOTA events to arrive, and when the Update request arrives, notify the user that everything must stop before sending the last ACK (as the module will start updating and then reboot)

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)
- Run lwm2m_taskCB when a monitored resource changes, to manage the action to be done

3.4.29.4 Application execution example

```
Starting lwm2m FOTA management demo. This is v1.1.14 built on Jan 21 2022 14:39:47.
Initializing resources...
[DEBUG] 22.14 lwm2m_demo:283 - at_cmd_async_init{LWM2M_TASK}$ m2mb_ati_init() on instance 1

IF Status: M2MB_LWM2M_IF_STATE_DISABLED
Client Status: M2MB_LWM2M_CL_STATE_DISABLED
Enable lwm2m client
LWM2M enable result OK
Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

Waiting for events from the OneEdge portal. Please start a FOTA process.
```

Figure 64

```
Resource /5/0/1/0 changed!
[DEBUG] 108.42 lwm2m_demo:544 - lwm2m_taskCB{LWM2M_TASK}$ Firmware update event!
FOTA process is waiting for Download ACK, sending it!
Waiting for Update ACK!
Before sending the ACK to apply FOTA, stop any ongoing operation (filesystem, sockets)

Asked to perform UPDATE step, stopping all operations now.
Application complete.

----- Info, DEVICE REBOOT FOR FOTA -----
?????????
#OTAEV: Module Upgraded To New Fw
```

FOTA request arrives from server, ACK is sent for Download and Update. Device restarts with the new firmware version

```
Starting lwm2m FOTA management demo. This is v1.1.14 built on Jan 21 2022 14:39:47.
Initializing resources...
[DEBUG] 26.61 lwm2m_demo:283 - at_cmd_async_init{LWM2M_TASK}$ m2mb_ati_init() on instance 1
[WARN ] 34.98 lwm2m_demo:738 - lwm2mIndicationCB{pubTspt_0}$ write res 1

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
Enable already done, skip. State: M2MB_LWM2M_IF_STATE_ACTIVE
Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

Waiting for events from the OneEdge portal. Please start a FOTA process.
```

Application restarts and is ready for a new FOTA operation

3.4.30 LWM2M OBJ_GET AND OBJ_SET

Sample application showcasing LWM2M client m2mb_lwm2m_objget and m2mb_lwm2m_objset M2MB APIs usage. Debug prints on **AUX UART**

Features

- Configure LWM2M client and enable it
- Create an instance of a custom object
- Create a Json string
- Set string, integer, float, boolean, timestamp and opaque values with m2mb_lwm2m_objset
- Get all resources values with m2mb_lwm2m_objget
- Manage write, read and monitoring resources changed from the portal

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6", "<user apn>"
```

Depending on the Mobiler Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.4.30.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask - Check registration status

- Initialize LWM2M client,
- Check for XML file fo custom object
- Enable unsolicited messages from client
- Create a task (lwm2m_taskCB is its callback function)to manage events from Portal
- Enable Lwm2M client

- Create a new instance for the custom object
- Wait for client to register to Portal
- Performs obj_set and obj_get operations on the related resources
- Wait for events from server

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)
- Run lwm2m_taskCB when a monitored resource changes, to manage the action to be done

3.4.30.2 Device Profile upload

LWM2M resources demo device profile must be imported to have a real-time update of resources values on the LWM2M browser.

To do so, import the file `json/lwm2m_resources_demo.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

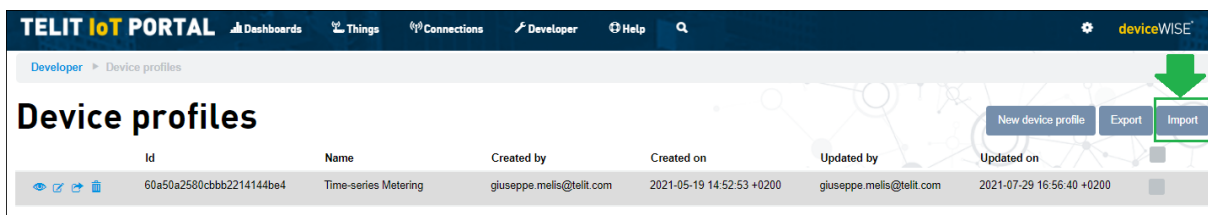


Figure 65

3.4.30.3 Custom Object configuration

The XML file content must be loaded on the Telit IoT Portal for the demo application to be fully executed.

First, enter Developer section from the top menu



Figure 66

Choose Object Registry

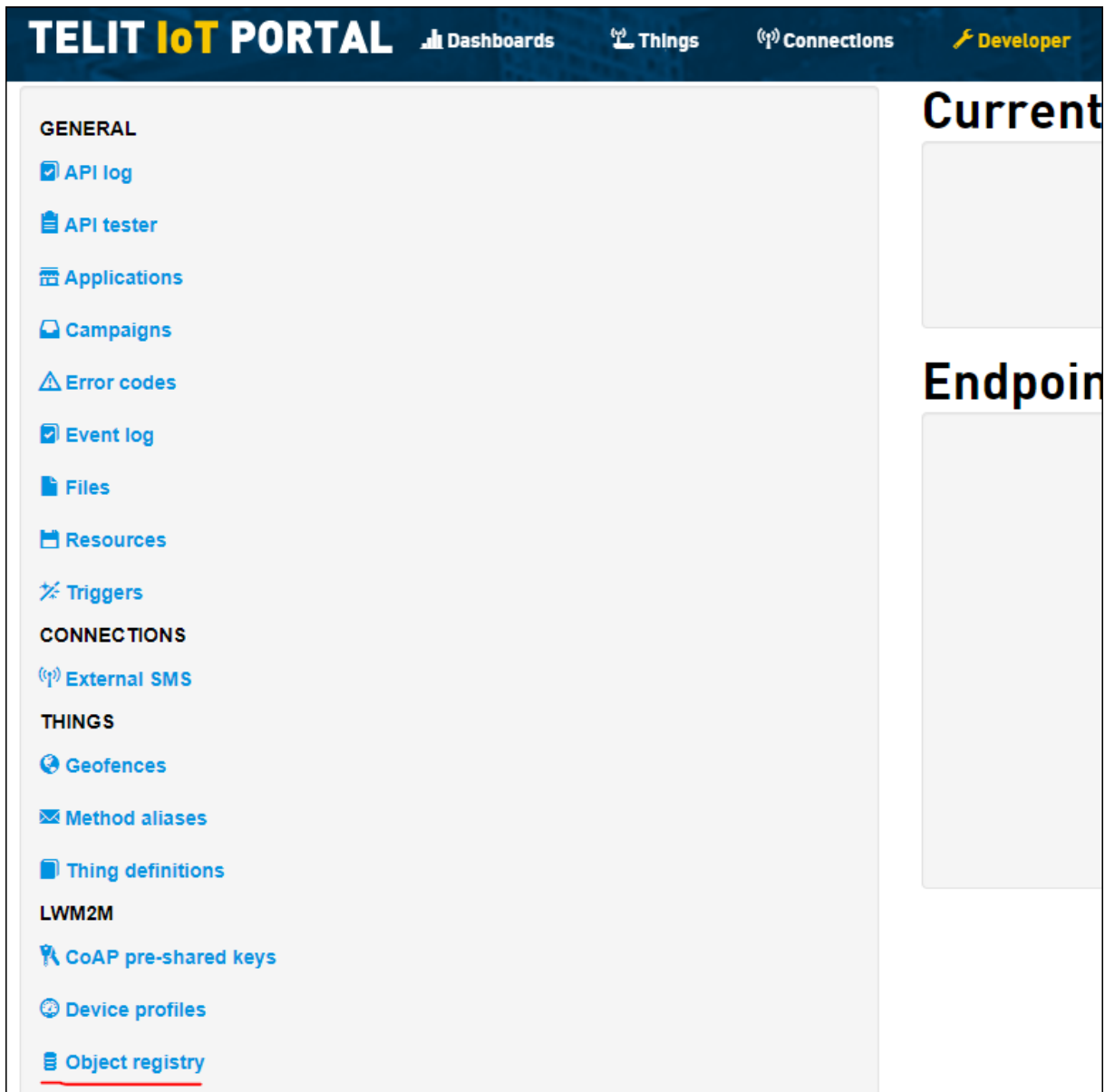


Figure 67

Create a New Object

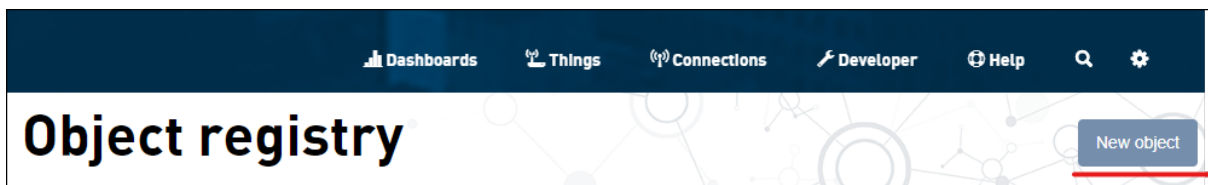
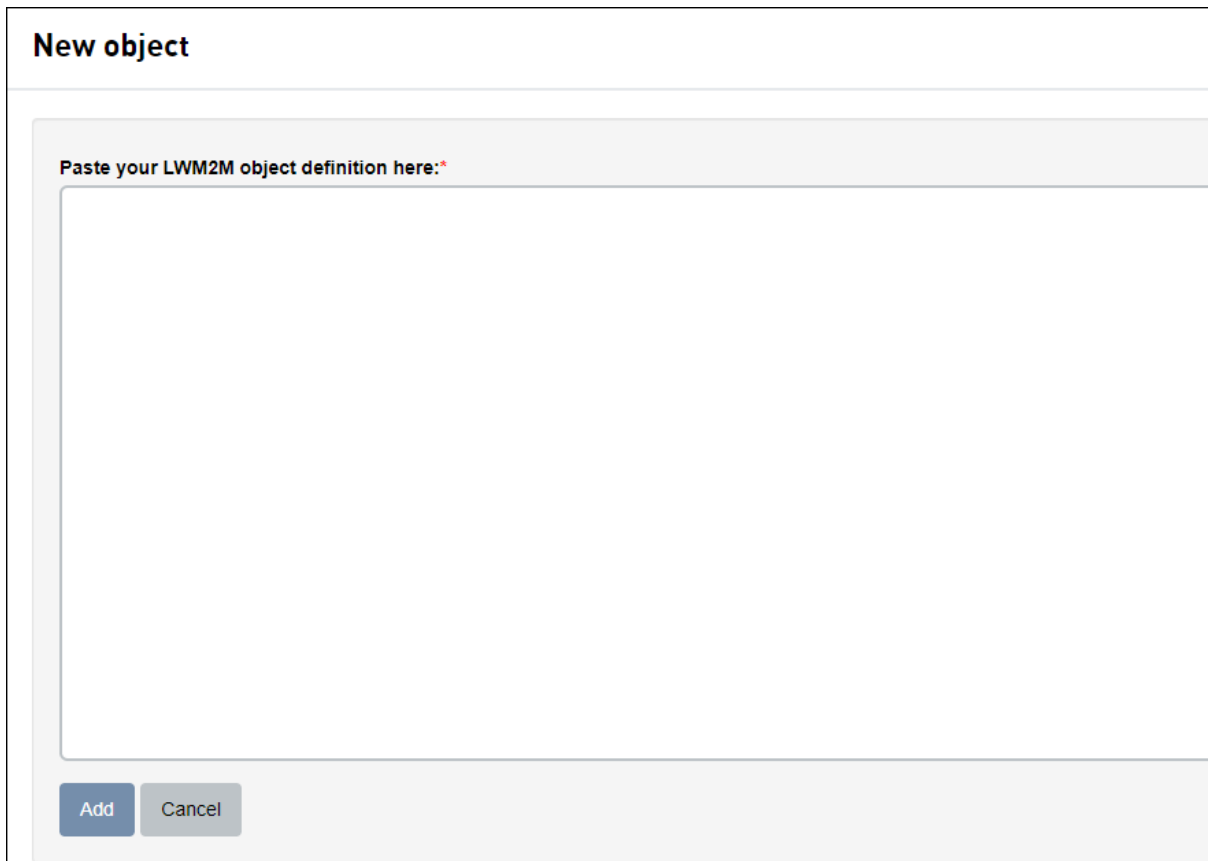


Figure 68

Copy the xml file content and paste it in the new Object form



New object

Paste your LWM2M object definition here:*

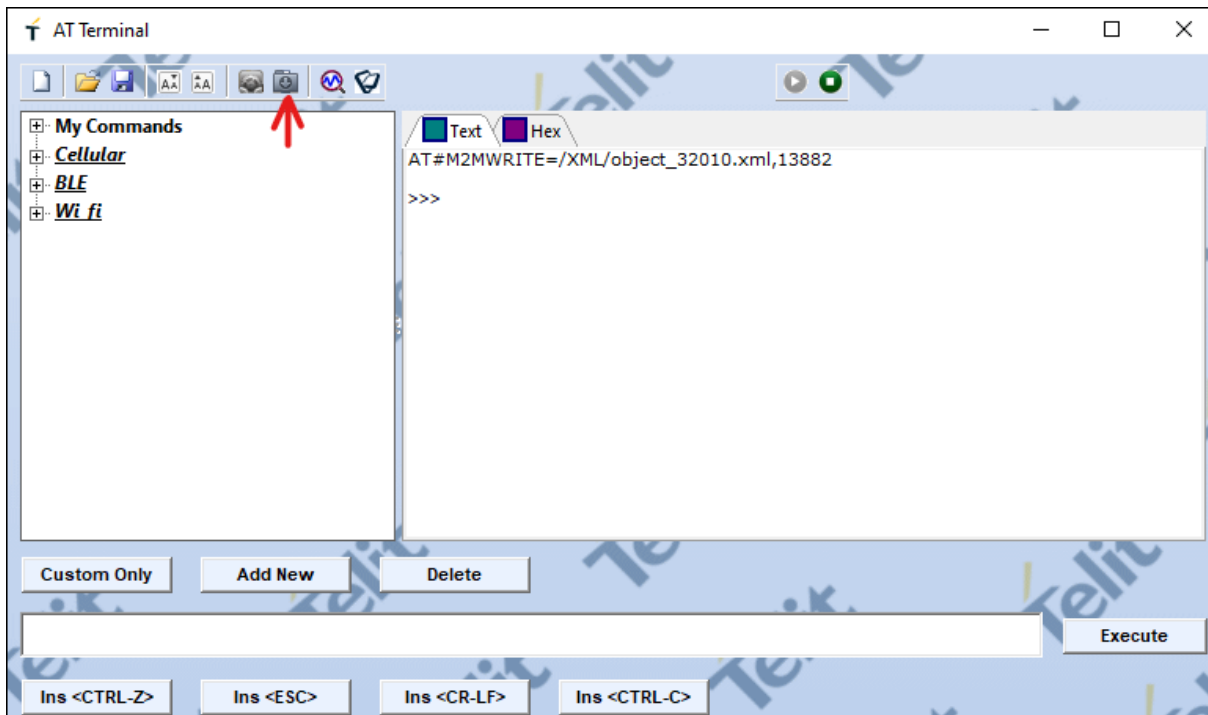
Add Cancel

Figure 69

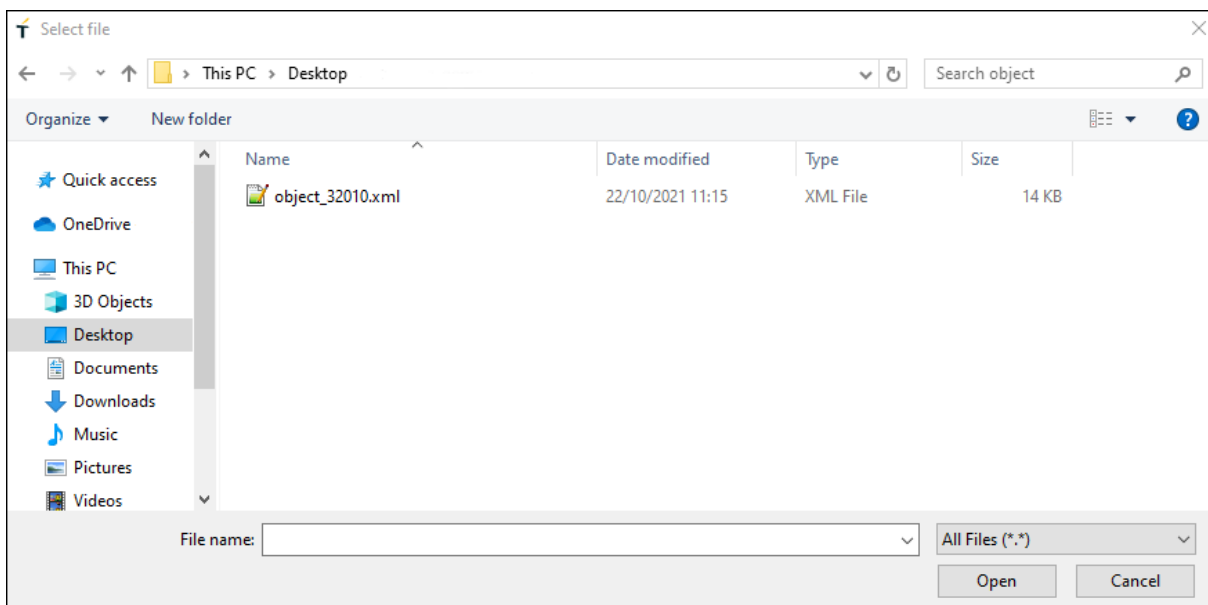
Also, the application requires the XML file `xml/object_32011.xml` (provided with the sample files) to be stored in module's `/XML/` folder. It can be done with

```
AT#M2MWRITE=/XML/object_32011.xml,<size_in_bytes>
```

To load the XML file in the module, Telit AT Controller (TATC) can be used. Once the command above is issued, press the load content button:

**Figure 70**

Select the file from your computer

**Figure 71**

The file is successfully loaded on the module

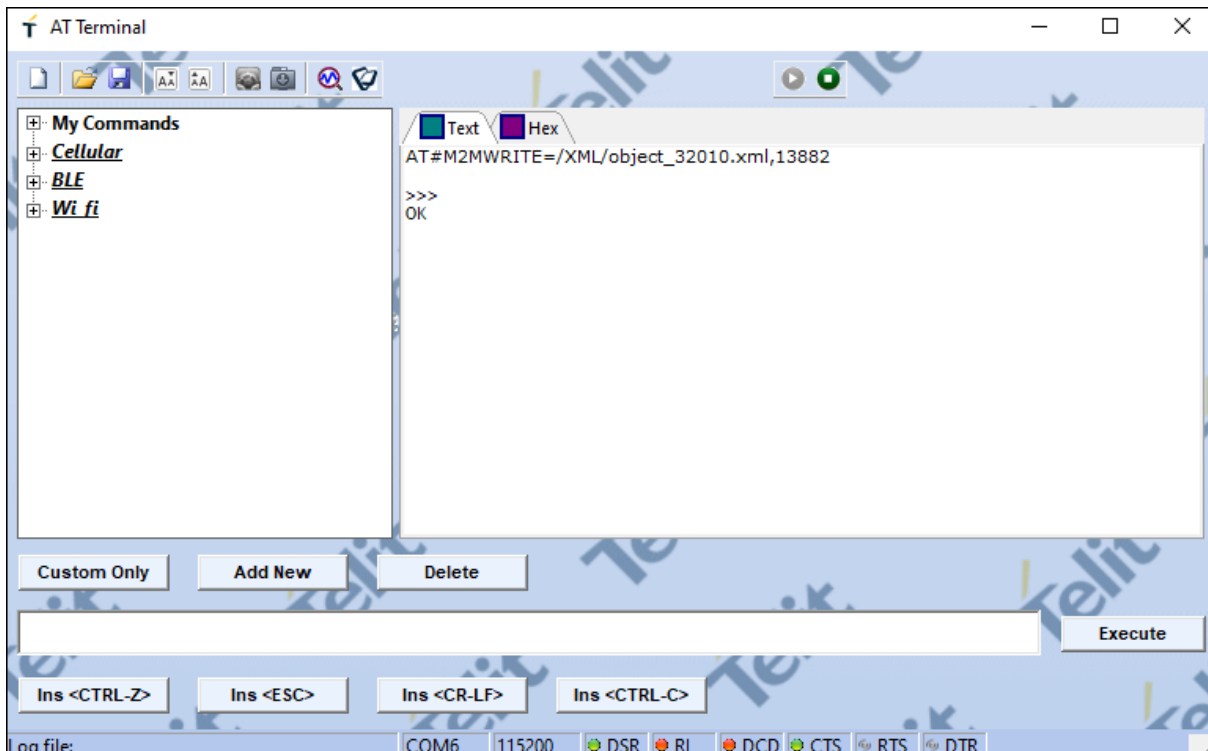


Figure 72

3.4.30.4 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on “**Things**” section, click “**New Thing**” button in the top right corner.

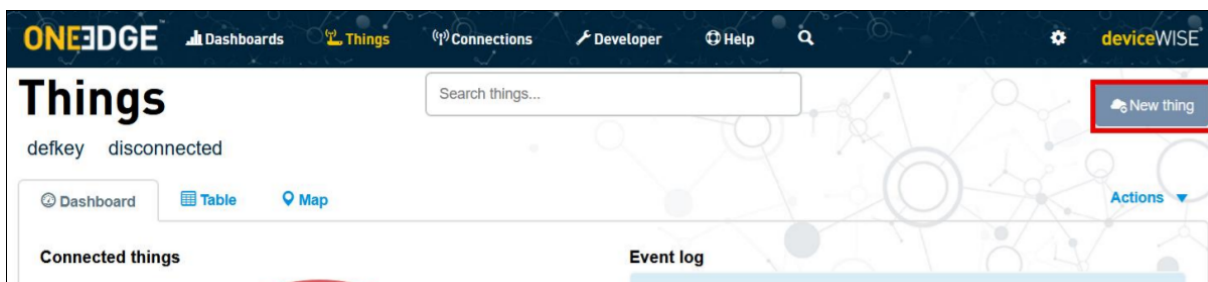


Figure 73

In the Create a new thing dialog, select “Telit Module”

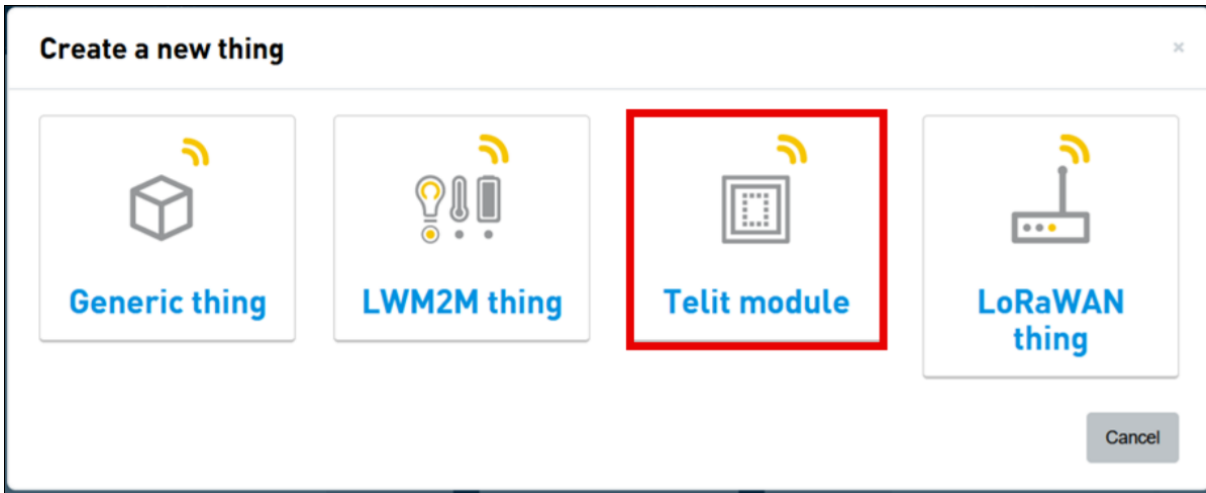


Figure 74

A dialog appears: select “Default” thing definition

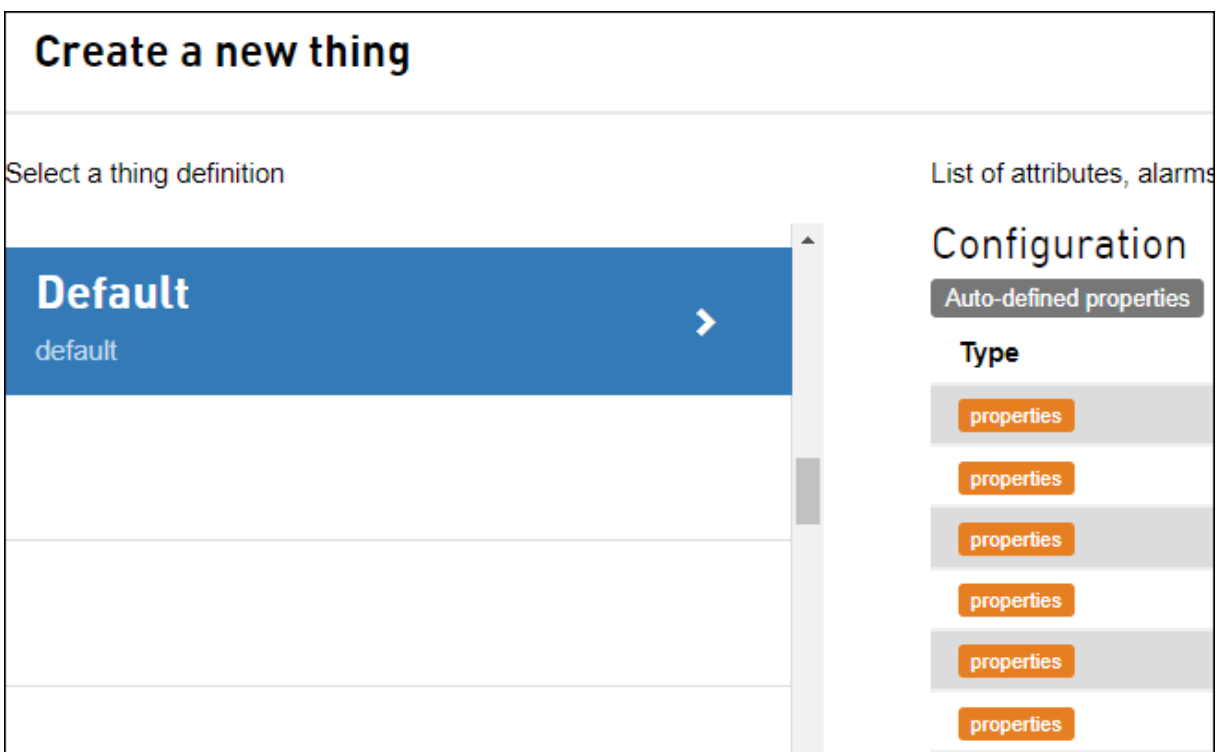


Figure 75

In the following screen, provide the Telit ID as “Identifier” Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the

screenshot below

Figure 76

Click **“Add”** to complete the new thing creation procedure.

If the Thing already exists, its device profile can be changed by following the steps shown in the picture below

Figure 77

3.4.30.5 Application execution example


```
Starting lwm2m demo. This is v1.1.10 built on Oct 22 2021 10:49:52.
On OneEdge portal, be sure that the demo Device Profile has been uploaded on the IoT Portal
Initializing resources...

LWM2M enable result OK
[DEBUG] 38.20 lwm2m_demo:998 - lwm2mIndicationCB{pubTspt_0}$ Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED
GET STATUS.
IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED
```

Figure 78

```
Starting lwm2m demo. This is v1.1.11 built on Nov 24 2021 13:32:52.
On OneEdge portal, be sure that the 'lwm2m resources objset - objget demo' Device Profile has been uploaded on the IoT Portal

Initializing resources...
LWM2M enable result OK
[DEBUG] 29.42 lwm2m_demo:593 - lwm2mIndicationCB{pubTspt_0}$ Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED
GET STATUS.
IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

m2mb_lwm2m_objset SUCCESS

Resource /32011/0/11/0 changed!

Reading string resource {32011/0/11/0} value on LWM2M client.

Resource /32011/0/12/0 changed!
Resource /32011/0/13/0 changed!
Resource /32011/0/15/0 changed!
Resource /32011/0/16/0 changed!
Resource /32011/0/31/0 changed!
Resource /32011/0/31/1 changed!
Resource /32011/0/31/2 changed!
Resource /32011/0/34/0 changed!
Resource /32011/0/34/1 changed!
Resource /32011/0/34/2 changed!
Resource /32011/0/34/3 changed!
```

```
String data in {32011/0/11/0} resource was updated to new content: <hello world>
Reading integer resource {32011/0/12/0} value on LWM2M client.
Integer data in {32011/0/12/0} resource was updated to new value: 70
Reading double resource {32011/0/13/0} value on LWM2M client.
Float data in {32011/0/13/0} resource was updated to new value: -12.500000
Reading opaque resource {32011/0/15/0} on LWM2M client.
Opaque data in {32011/0/15/0} resource was updated to new content (11 bytes):
68 65 6C 6C 6F 20 77 6F 72 6C 64
Reading timestamp resource {32011/0/16/0} value on LWM2M client.
Time data in {32011/0/16/0} resource was updated to new value: 1620828702 (2021/05/12T14:11:42+00:00)
Reading string resource {32011/0/31/0} value on LWM2M client.
String data in {32011/0/31/0} resource was updated to new content: <Greetings>
Reading string resource {32011/0/31/1} value on LWM2M client.
String data in {32011/0/31/1} resource was updated to new content: <from>
Reading string resource {32011/0/31/2} value on LWM2M client.
String data in {32011/0/31/2} resource was updated to new content: <AppZone!>
Reading boolean resource {32011/0/34/0} value on LWM2M client.
Boolean data in {32011/0/34/0} resource was updated to new value: true
Reading boolean resource {32011/0/34/1} value on LWM2M client.
Boolean data in {32011/0/34/1} resource was updated to new value: false
Reading boolean resource {32011/0/34/2} value on LWM2M client.
Boolean data in {32011/0/34/2} resource was updated to new value: true
Reading boolean resource {32011/0/34/3} value on LWM2M client.
Boolean data in {32011/0/34/3} resource was updated to new value: false
Waiting 5 seconds before the m2mb_lwm2m_objget
Waiting: 5
Waiting: 4
Waiting: 3
Waiting: 2
Waiting: 1
m2mb_lwm2m_objget SUCCESS
Content:
<{"bn":"/32011/0/",
"e":[
{"n":"11","sv":"hello world"},
{"n":"12","v":70},
{"n":"13","v":-12.500000},
{"n":"14","bv":false},
{"n":"15","sv":"aGVsbG8gd29ybGQ="},
{"n":"16","v":1620828702},
{"n":"17","ov":"0:0"},
{"n":"31/0","sv":"Greetings"},
{"n":"31/1","sv":"from"},
{"n":"31/2","sv":"AppZone!"},
{"n":"31/3","sv":""},
{"n":"31/4","sv":""},
{"n":"31/5","sv":""},
{"n":"31/6","sv":""},
{"n":"31/7","sv":""},
{"n":"31/8","sv":""},
{"n":"31/9","sv":""},
{"n":"34/0","bv":true},
{"n":"34/1","bv":false},
{"n":"34/2","bv":true},
{"n":"34/3","bv":false},
{"n":"34/4","bv":false},
{"n":"34/5","bv":false},
{"n":"34/6","bv":false},
{"n":"34/7","bv":false},
{"n":"34/8","bv":false},
{"n":"34/9","bv":false}]
}>
```

After the Demo completes the initialization, it is possible to access the object re-

sources from the section Things: select your device, then “LwM2M” tab of OneEdge IoT portal:

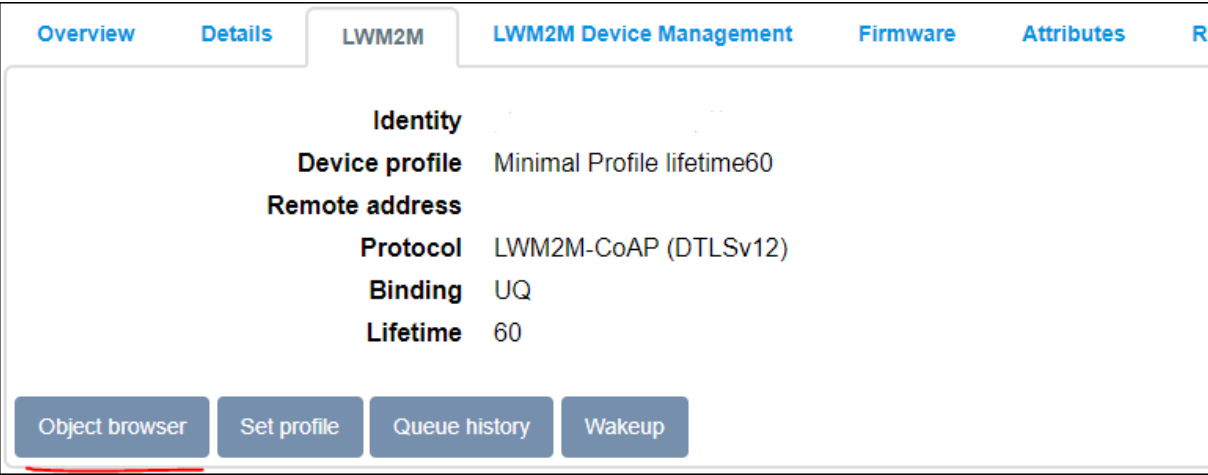


Figure 79

An instance of the object will be present and the resources can be modified.

m2mb lwm2m objset-get demo /32011						
Instance 0	/32011/0	Observe	Read	Write	Attributes	Delete
Read Write String	/32011/0/11	Observe	Read	Write	Attributes	hello world
Read Write Integer	/32011/0/12	Observe	Read	Write	Attributes	70
Read Write Float	/32011/0/13	Observe	Read	Write	Attributes	-12.5
Read Write Boolean	/32011/0/14	Observe	Read	Write	Attributes	false
Read Write Opaque	/32011/0/15	Observe	Read	Write	Attributes	view binary data
Read Write Time	/32011/0/16	Observe	Read	Write	Attributes	2021-05-12T14:11:42Z
Read Write Objlink	/32011/0/17	Observe	Read	Write	Attributes	instid=0, objid=0
Read Write Multiinstance String	/32011/0/31	Observe	Read	Write	Attributes	0=Greetings, 1=from, ... Read more
Read Write Multiinstance Boolean	/32011/0/34	Observe	Read	Write	Attributes	0=true, 1=false, 2=tr... Read more

Figure 80

3.4.31 LWM2M REG

Sample application showcasing LWM2M client registration management using M2MB API. Debug prints on **AUX UART**

Features

- Configure LWM2M client and enable it
- Get LWM2M server information using REG apis
- Deregister from LWM2M server using REG apis
- Register to the LWM2M server using REG apis
- Update registration on LWM2M server using REG apis
- Deregister from LWM2M server using REG apis

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6","<user apn>"
```

Depending on the Mobile Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.4.31.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask - Check registration status

- Initialize LWM2M client,
- Check for XML file for custom object
- Enable LWM2M client
- Wait for client to register to Portal
- Get the LWM2M server information
- Performs client portal deregistration
- Performs client portal registration

- Performs client portal registration Update
- Performs client portal deregistration

Lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)

3.4.31.2 Application execution example

```
Starting LWM2M REG demo. This is v1.0.00 built on Sep 11 2023 15:11:48.
On OneEdge portal, be sure that the 'lwm2m resources demo' Device Profile has been uploaded on the IoT Portal

Initializing resources...
LWM2M enable result OK
Waiting LWM2M Registering (120 seconds timeout)...
sshid[99] status[2] lastReg[1694446138] nextReg[1694532238] lastActivity[1694446138]
Deregistration action using LWM2M REG api!
Waiting deregistration event!
Deregistration event received!
Registration action using LWM2M REG api!
Waiting registration event!
Registration event received!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Deregistration action using LWM2M REG api!
Waiting deregistration event!
Deregistration event received!
Application complete.
```

Figure 81

3.4.31.3 Device Profile upload

LWM2M resources demo device profile must be imported to have a real-time update of resources values on the LWM2M browser.

To do so, import the file `json/lwm2m_resources_demo.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

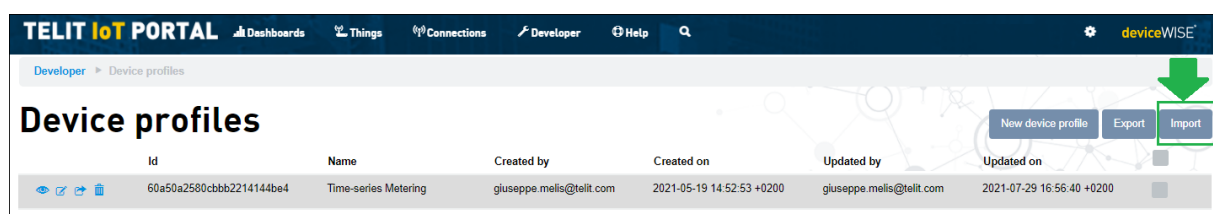


Figure 82

3.4.31.4 Custom Object configuration

The XML file content must be loaded on the Telit IoT Portal for the demo application to be fully executed.

First, enter Developer section from the top menu



Figure 83

Choose Object Registry

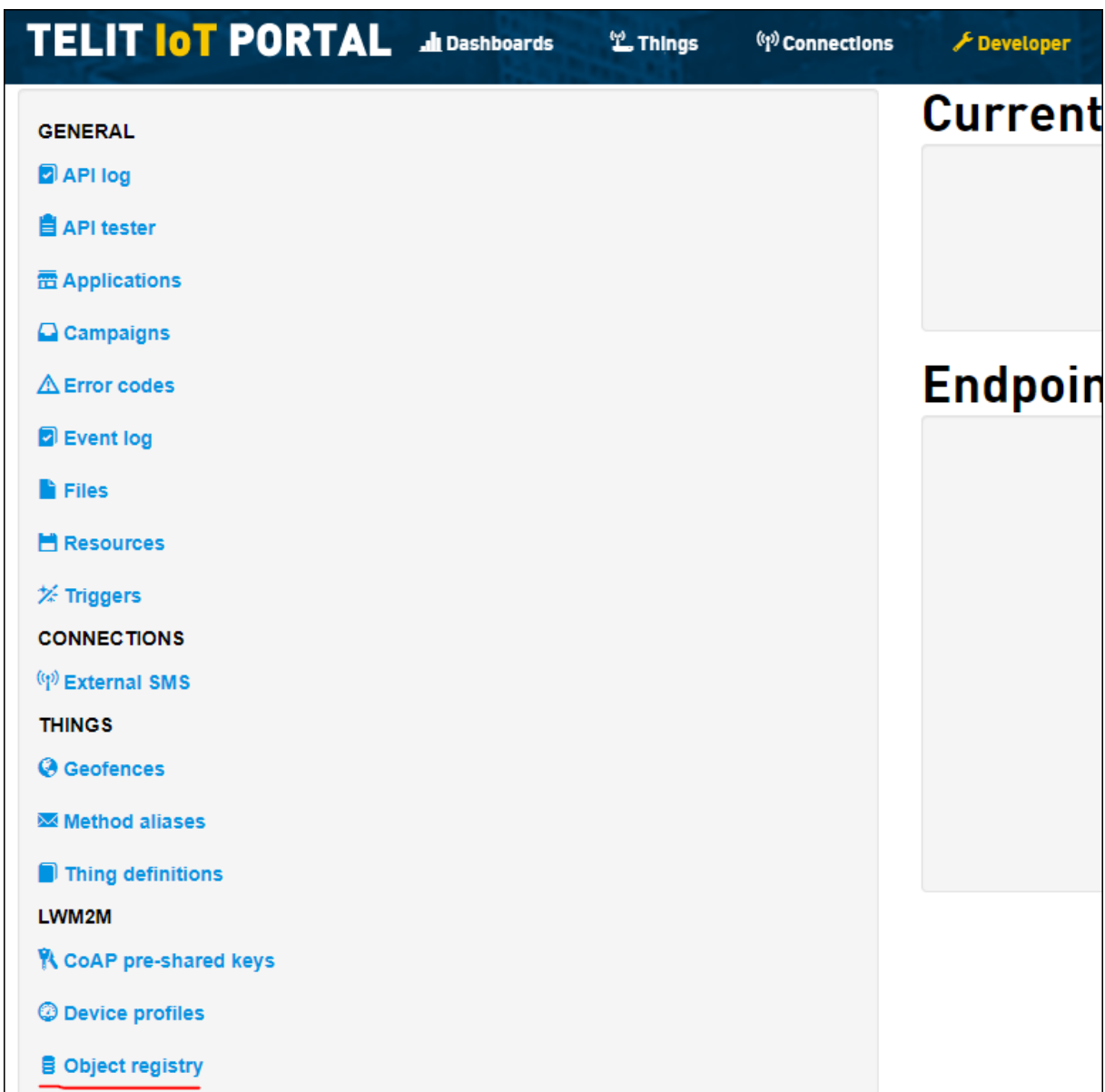


Figure 84

Create a New Object

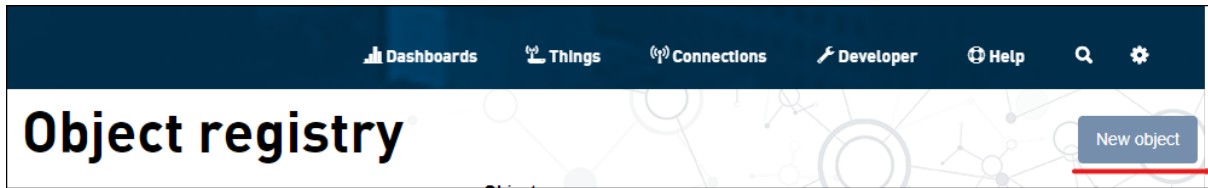


Figure 85

Copy the xml file content and paste it in the new Object form

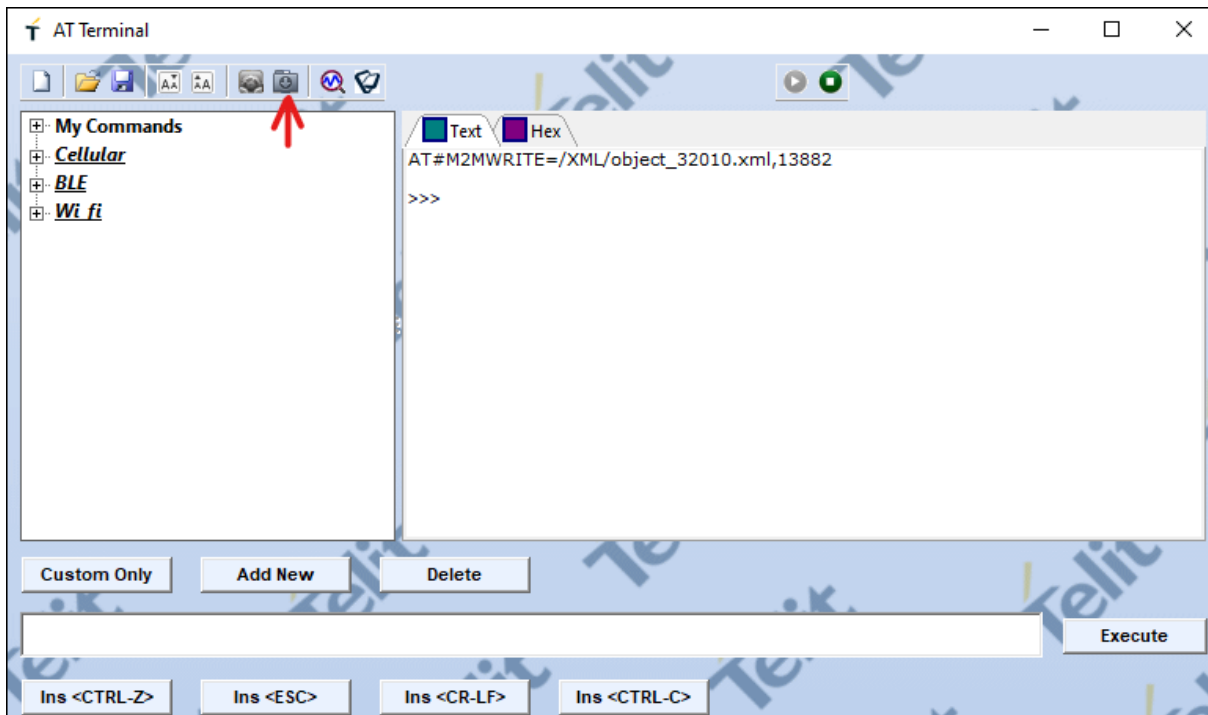
The screenshot shows a form titled "New object". Inside the form, there is a large text area with the placeholder text "Paste your LWM2M object definition here:". At the bottom of the form, there are two buttons: "Add" and "Cancel".

Figure 86

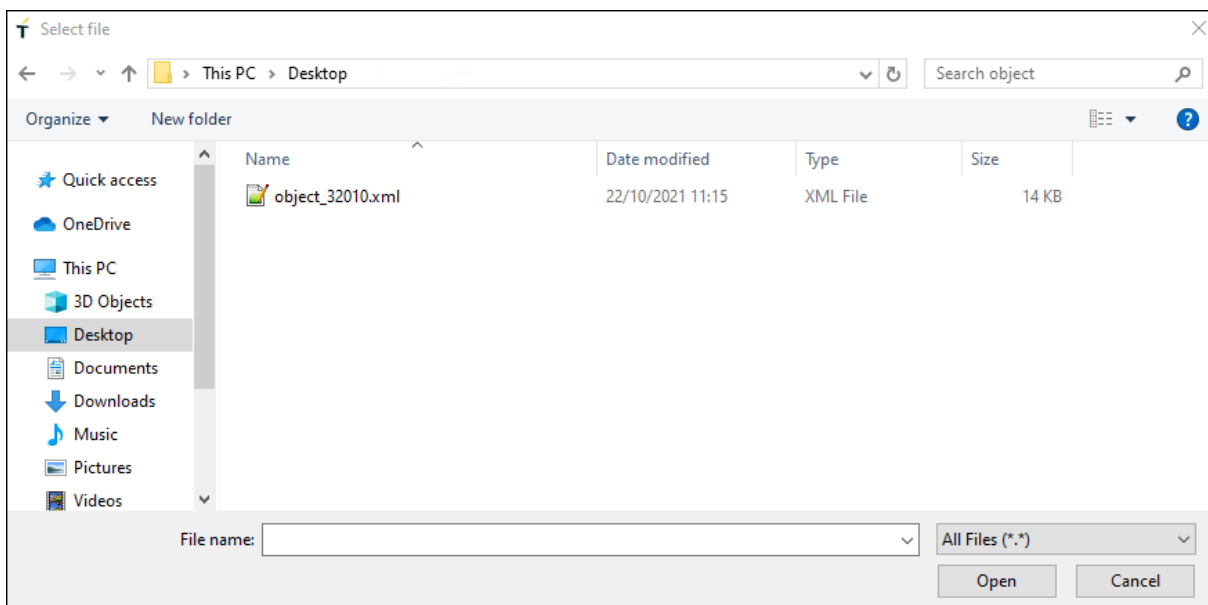
Also, the application requires the XML file `xml/object_32010.xml` (provided with the sample files) to be stored in module's `/XML/` folder. It can be done with

```
AT#M2MWRITE=/XML/object_32010.xml,<size_in_bytes>
```

To load the XML file in the module, Telit AT Controller (TATC) can be used. Once the command above is issued, press the load content button:

**Figure 87**

Select the file from your computer

**Figure 88**

The file is successfully loaded on the module

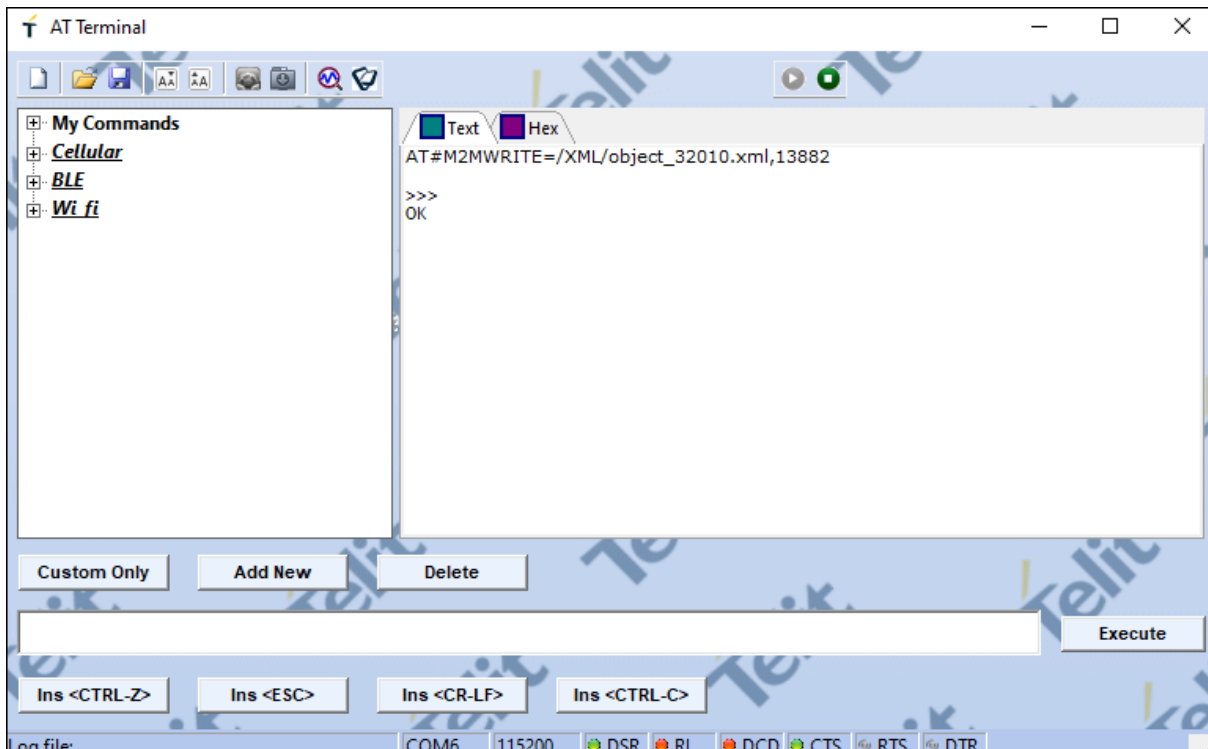


Figure 89

3.4.31.5 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue `AT#TID` to get the Telit ID. The command response will be similar to

`#TID: xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx,1 OK`

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on “**Things**” section, click “**New Thing**” button in the top right corner.

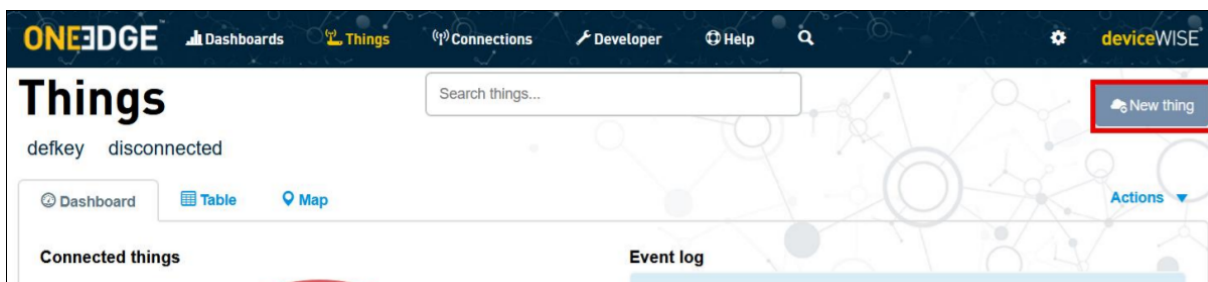


Figure 90

In the Create a new thing dialog, select “Telit Module”

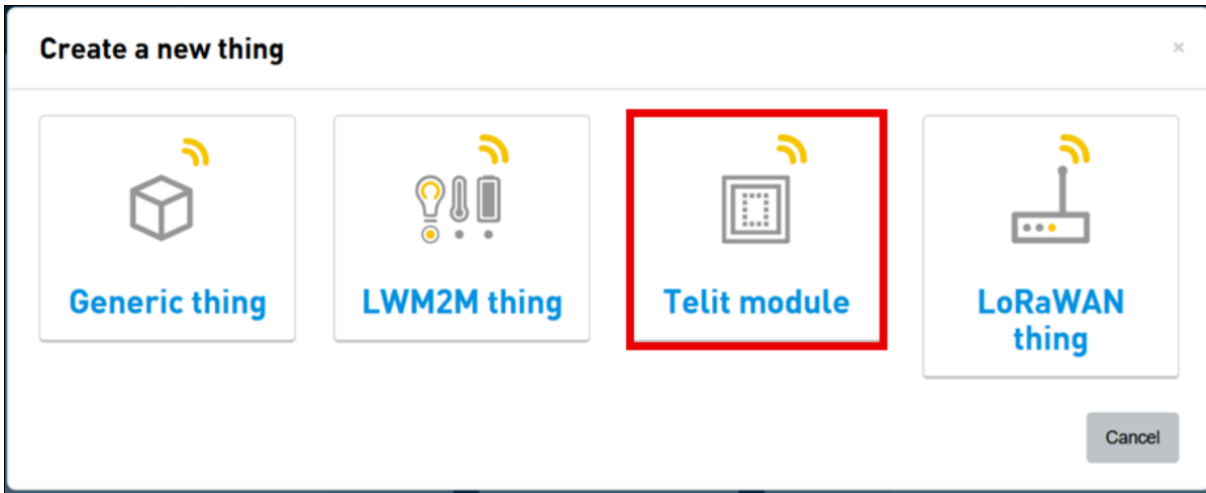


Figure 91

A dialog appears: select “Default” thing definition

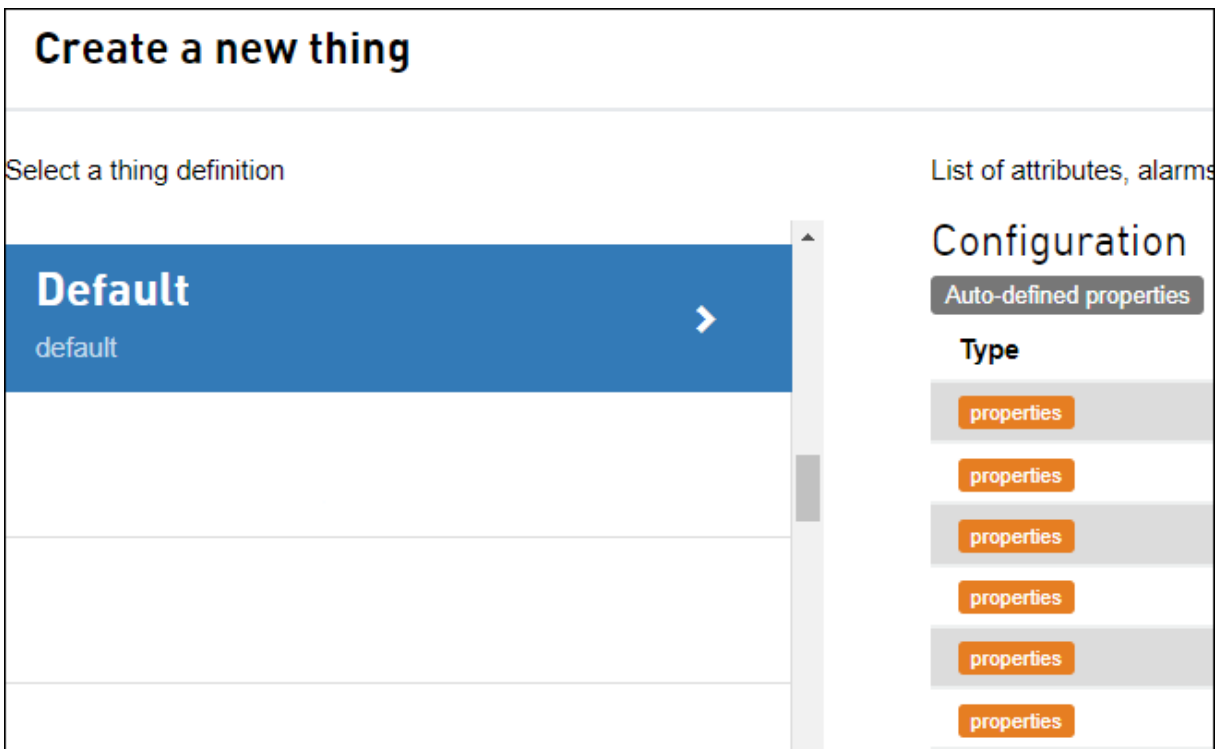


Figure 92

In the following screen, provide the Telit ID as “Identifier”. Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the screenshot below

Figure 93

Click **“Add”** to complete the new thing creation procedure.

If the Thing already exists, its device profile can be changed by following the steps shown in the picture below

Figure 94

3.4.32 Logging Demo

Sample application showing how to print on one of the available output interfaces.
Debug prints on **AUX UART**

Features

- How to open a logging channel
- How to set a logging level
- How to use different logging macros

3.4.32.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Print a message with every log level

```
Starting Logging demo app. This is v1.0.7 built on Mar 26 2020 13:57:06.  
[WARN ] 20.17 M2MB_main.c:74 - M2MB_main{M2M_DamsStart}$ This is a WARNING MESSAGE  
[ERROR] 20.18 M2MB_main.c:76 - M2MB_main{M2M_DamsStart}$ THIS IS AN ERROR MESSAGE  
[CRITICAL] 20.19 M2MB_main.c:78 - M2MB_main{M2M_DamsStart}$ THIS IS AN CRITICAL MESSAGE  
[DEBUG] 20.19 M2MB_main.c:80 - M2MB_main{M2M_DamsStart}$ This is a DEBUG message  
[TRACE] 20.20 M2MB_main.c:82 - M2MB_main{M2M_DamsStart}$ This is a TRACE message  
END.
```

Figure 95

3.4.33 MD5 example

Sample application showing how to compute MD5 hashes using m2mb crypto. Debug prints on **AUX UART**

Features

- Compute MD5 hash of a file
- Compute MD5 hash of a string

3.4.33.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a temporary file with the expected content
- Compute MD5 hash of the provided text file
- Compare the hash with the expected one
- Compute MD5 hash of a string
- Compare the hash with the expected one
- Delete test file

```
Starting MD5 demo app. This is v1.0.7 built on Apr 7 2020 10:19:54.  
Buffer written successfully into file. 45 bytes were written.  
  
Computing hash from file...  
Computed hash: bb0fa6eff92c305f166803b6938dd33a  
Expected hash: bb0fa6eff92c305f166803b6938dd33a  
Hashes are the same!  
  
Computing hash from string...  
Computed hash: bb0fa6eff92c305f166803b6938dd33a  
Expected hash: bb0fa6eff92c305f166803b6938dd33a  
Hashes are the same!
```

Figure 96

3.4.34 MQTT Client

Sample application showcasing MQTT client functionalities (with SSL). Debug prints on **AUX UART**

Features

- How to check module registration and enable PDP context
- How to configure MQTT client parameters
- How to connect to a broker with SSL and exchange data over a subscribed topic

3.4.34.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage MQTT client and start it

mqtt_demo.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Init MQTT client
- Configure it with all parameters (Client ID, username, password, PDP context ID, keepalive timeout...)
- Connect MQTT client to broker
- Subscribe to two topics
- Publish 10 messages with increasing counter. Even messages are sent to topic 1, odd messages on topic 2.
- Print received message in mqtt_topc_cb function
- Disconnect MQTT client and deinit it
- Disable PDP context

```

Starting MQTT demo app. This is v1.0.7 built on Apr 7 2020 10:34:08.
[DEBUG] 16.18 mqtt_demo.c:192 - MQTT_Task(MQTT_TASK)$ INIT
[DEBUG] 16.18 mqtt_demo.c:206 - MQTT_Task(MQTT_TASK)$ m2mb_os_ev_init success
[DEBUG] 16.19 mqtt_demo.c:214 - MQTT_Task(MQTT_TASK)$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 16.19 mqtt_demo.c:221 - MQTT_Task(MQTT_TASK)$ Waiting for registration...
[DEBUG] 16.20 mqtt_demo.c:131 - NetCallback{pubTspt_0}$ Module is registered
[DEBUG] 16.21 mqtt_demo.c:232 - MQTT_Task(MQTT_TASK)$ Pdp context activation
[DEBUG] 18.26 mqtt_demo.c:246 - MQTT_Task(MQTT_TASK)$ Activate PDP with APN web.omnitel.it on CID 3....
[DEBUG] 18.95 mqtt_demo.c:155 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 18.96 mqtt_demo.c:159 - PdpCallback{pubTspt_0}$ IP address: 37.118.201.56
[DEBUG] 18.96 mqtt_demo.c:268 - MQTT_Task(MQTT_TASK)$ Init MQTT
[DEBUG] 18.97 mqtt_demo.c:278 - MQTT_Task(MQTT_TASK)$ m2mb_mqtt_init succeeded

Connecting to broker <api-dev.devicewise.com>:1883...
Done.
Subscribing to test_topic and test_topic2..
[DEBUG] 20.35 mqtt_demo.c:367 - MQTT_Task(MQTT_TASK)$ Done.

[DEBUG] 20.36 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 2> to topic test_topic
[DEBUG] 20.37 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 20.71 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 27
[DEBUG] 20.72 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 2>
[DEBUG] 23.37 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 3> to topic test_topic2
[DEBUG] 23.38 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 23.92 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 27
[DEBUG] 23.93 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 3>
[DEBUG] 26.40 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 4> to topic test_topic
[DEBUG] 26.41 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 26.93 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 27
[DEBUG] 26.93 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 4>
[DEBUG] 29.42 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 5> to topic test_topic2
[DEBUG] 29.43 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 29.99 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 27
[DEBUG] 30.00 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 5>
[DEBUG] 32.46 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 6> to topic test_topic
[DEBUG] 32.48 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 33.00 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 27
[DEBUG] 33.01 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 6>
[DEBUG] 35.47 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 7> to topic test_topic2
[DEBUG] 35.48 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 36.01 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 27
[DEBUG] 36.02 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 7>
[DEBUG] 38.50 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 8> to topic test_topic
[DEBUG] 38.51 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 39.15 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 27
[DEBUG] 39.16 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 8>
[DEBUG] 41.52 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 9> to topic test_topic2
[DEBUG] 41.53 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 42.10 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 27
[DEBUG] 42.12 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 9>
[DEBUG] 44.56 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 10> to topic test_topic
[DEBUG] 44.57 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 45.09 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 28
[DEBUG] 45.11 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 10>
[DEBUG] 47.58 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 11> to topic test_topic2
[DEBUG] 47.59 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 48.12 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 28
[DEBUG] 48.13 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 11>

Disconnecting from MQTT broker..
[DEBUG] 50.60 mqtt_demo.c:414 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 50.61 mqtt_demo.c:443 - MQTT_Task(MQTT_TASK)$ application exit
[DEBUG] 50.62 mqtt_demo.c:453 - MQTT_Task(MQTT_TASK)$ m2mb_pdp_deactivate returned success
[DEBUG] 50.63 mqtt_demo.c:457 - MQTT_Task(MQTT_TASK)$ Application complete.
[DEBUG] 51.23 mqtt_demo.c:164 - PdpCallback{pubTspt_0}$ Context deactivated!

```

Figure 97

3.4.35 MultiTask

Sample application showcasing multi tasking functionalities with M2MB API. Debug prints on **AUX UART**

Features

- How to create tasks using azx utilities
- How to use send messages to tasks
- How to use a semaphore to synchronize two tasks

3.4.35.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create three tasks with the provided utility (this calls public m2mb APIs)
- Send a message to the task1, its callback function azx_msgTask1 will be called

azx_msgTask1

- Print received parameters from main
- Send modified parameters to task2 (its callback function azx_msgTask2 will be called)
- wait for an InterProcess Communication semaphore to be available (released by task3)
- Once the semaphore is available, print a message and return

azx_msgTask2

- Print received parameters from caller
- If first parameter is bigger than a certain value, Send modified parameters to task3
- Else, use the second parameter as a task handle and print the corresponding name plus the value of the first parameter

azx_msgTask3

- Print received parameters from task 2
- release IPC semaphore
- send message to task 2 with first parameter below the threshold and second parameter with task3 handle


```
Starting MultiTask demo app. This is v1.0.12-C1 built on Jun 23 2020 15:36:31.
Inside "myTask1" user callback function. Received parameters from MAIN: 3 4 5
Task1 - Sending a message to task 2 with modified parameters...
Task1 - Waiting for semaphore to be released by task 3 now...

Inside "myTask2" user callback function. Received parameters: 5 7 10
Task2 - Sending a message to task 3 with modified parameters...
Task2 - Done.

Inside "myTask3" user callback function. Received parameters from Task 2: 15 14 9
Task3 - Releasing IPC semaphore...

Task1 - After semaphore! return...

Task3 - IPC semaphore released.
Task3 - Sending a message to task 2 with specific 'type' parameter value of 0 and task 3 handle as param1...

Inside "myTask2" user callback function. Received parameters: 0 1073951320 9
Task3 - Done.
Task2 - Received type 0 from task "myTask3"
Task2 - Done.
```

Figure 98

3.4.36 MutEx

Sample application showing mutex usage, with ownership and prioritization usage.
Debug prints on **AUX UART**

Features

- How to create a mutex
- How to use the mutex with tasks having different priorities
- how to reorder the pending tasks queue for the mutex

3.4.36.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create four tasks with the provided utility (this calls public m2mb APIs). The first task is a “producer”, putting data on a shared buffer. The second is a “consumer” of said data, the other two are used for prioritization demo
- run producer and consumer tasks at the same pace. the shared buffer will stay empty, because the resource is consumed right after creation
- run producer twice as fast as consumer. The buffer is slowly filled
- run consumer twice as fast as publisher. The buffer is always empty.
- reserve the mutex in the main task and run producer, support and support2 tasks (in this order). Then release the mutex and check the execution order. It should be by arrival.
- reserve the mutex in the main task and run the same three task, but before releasing the mutex, call the prioritization API. the task with highest priority (producer) is put as first in the queue.

```

Starting Mutex app. This is v1.0.12-C1 built on Jul 1 2020 08:37:15.
[DEBUG] 14.50 M2MB_main:90 - mutex_init{M2M_DamsStart}$ [MUTEX] Mutex initialized

[CASE 1 ] Producer and consumer have same idle time

[DEBUG] 14.51 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 14.52 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 14.53 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 14.53 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 14.54 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 14.54 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 14.55 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 14.56 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 15.56 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 15.56 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 15.57 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 15.58 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 15.58 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 15.59 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 15.60 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 15.60 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 16.61 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 16.61 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 16.62 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 16.63 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 16.63 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 16.64 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 16.64 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 16.65 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released

```

Figure 99

```

[CASE 2 ] Producer has double idle time

[DEBUG] 17.56 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 17.56 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 17.57 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 17.58 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 17.58 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 17.59 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 17.59 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 17.60 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 18.63 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 18.64 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 0 items
[DEBUG] 18.64 M2MB_main:268 - msgConsumer{CONSUMER}$ Can't consume anything, buffer size is 0
[DEBUG] 18.65 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 19.62 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 19.62 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 19.63 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 19.64 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 19.68 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 19.69 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 19.69 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 19.70 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 20.73 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 20.74 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 0 items
[DEBUG] 20.75 M2MB_main:268 - msgConsumer{CONSUMER}$ Can't consume anything, buffer size is 0
[DEBUG] 20.75 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 21.67 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 21.68 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 21.68 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 21.69 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 21.77 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 21.79 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 21.80 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 21.80 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released

```

Figure 100

```
[CASE 3 ] Producer has half idle time

[DEBUG] 22.62 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 22.63 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 22.64 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 22.64 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 22.65 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 22.65 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 22.66 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 22.67 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 23.67 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 23.68 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 23.68 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 23.69 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 24.71 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 24.72 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 24.72 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 24.73 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 24.74 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 24.74 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 24.75 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 24.76 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 25.79 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 25.79 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 1 items
[DEBUG] 25.80 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 1
[DEBUG] 25.81 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 26.78 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 26.78 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 2 items
[DEBUG] 26.79 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 1
[DEBUG] 26.79 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 26.84 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 26.84 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 1 items
[DEBUG] 26.85 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 1
[DEBUG] 26.86 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
```

Figure 101

```
[CASE 4 ] NO HTPF

Reserve MUTEX so all tasks are enqueued
[DEBUG] 30.77 M2MB_main:387 - msgSupport{HPTF_SUPPORT}$ freepos = 0 | evaluate[freepos]= 3
[DEBUG] 30.78 M2MB_main:416 - msgSupport2{HPTF_SUPPORT2}$ freepos = 1 | evaluate[freepos]= 4
[DEBUG] 30.79 M2MB_main:223 - msgProducer{PRODUCER}$ producer: freepos = 2 | evaluate[freepos]= 1
[DEBUG] 35.85 M2MB_main:586 - M2MB_main{M2M_DamsStart}$ EVALUATE SEQUENCE IS 3 4 1. Expected: 3 4 1
NO HTPF OK

[CASE 4.1 ] HTPF USED

Reserve MUTEX so all tasks are enqueued
m2mb_os_mtx_hptf OK
[DEBUG] 41.98 M2MB_main:223 - msgProducer{PRODUCER}$ producer: freepos = 0 | evaluate[freepos]= 1
[DEBUG] 41.99 M2MB_main:387 - msgSupport{HPTF_SUPPORT}$ freepos = 1 | evaluate[freepos]= 3
[DEBUG] 42.00 M2MB_main:416 - msgSupport2{HPTF_SUPPORT2}$ freepos = 2 | evaluate[freepos]= 4
[DEBUG] 44.03 M2MB_main:650 - M2MB_main{M2M_DamsStart}$ EVALUATE SEQUENCE IS 1 3 4, expected 1 3 4
HTPF DEMO OK
The application has ended...
```

Figure 102

3.4.37 NTP example

The application connects to an NTP server, gets current date and time and updates module's internal clock. Debug prints on **AUX UART**

Features

- How to get current date and time from an NTP server
- How to set current date and time on module

3.4.37.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Send message to ntpTask

ntp_task.c

NTP_task() - Waits module registration - When module is registered, initializes ntp setting CID, server url and timeout - When PDP context is correctly opened, a query to NTP server is done to get current date and time - On SET_MODULE_RTC message type reception, module RTC is set with date time value got from NTP server.

m2mb_ntp_ind_callback() - As soon as M2MB_NTP_VALID_TIME event is received, current date and time is printed and a message (with SET_MODULE_RTC type) is sent to NTP_task

```
Start NTP demo application. This is v1.0 built on Apr 16 2021 09:36:12.
Waiting for registration...
Module is registered!

Activate PDP context with APN ibox.tim.it on CID 3
Context activated, IP address: 2.195.170.123
Get current time from server 0.pool.ntp.org, PORT: 123

Current time is: Friday 2021-04-16, 07:37:33

Current time correctly set on module
Module system time is: 2021-04-16, 07:37:33
```

Figure 103

3.4.38 ON OFF button management example

Sample application on how to handle ON/OFF button. Debug prints on **AUX UART**

Features

- How to enable ON/OFF button handling
- How to turn off module after ON/OFF button press/release

Application workflow

M2MB_main.c

- Open gpio_ON_OFF
- Configure the interrupt and the callback
- Init power functionality
- Wait for on/off button to be pressed
- If button has been pressed for more than 3 seconds (and then released) then module is turned off.

```
Starting ON OFF button handling demo app. This is v1.1.8 built on Feb  9 2023 11:56:05.  
Waiting for ON/OFF button to be pressed...  
onoff pressed!  
onoff released!  
ON/OFF pressed form more than 3 sec, turn off module!
```

Figure 104

3.4.39 RTC example

Sample application that shows RTC apis functionalities: how to get/set module system time and timestamp. Debug prints on **AUX UART**

Features

- How to read module timestamp
- How to read module system time
- How to set new system time

3.4.39.1 Application workflow

M2MB_main.c

- Init log azx and print a welcome message
- Init net functionality and wait for module registration
- Init RTC functionality and get module time in timestamp format (seconds from the epoch)
- Get module system time in date/time format
- Add 1 hour to timestamp, convert it to system time and set it to module

```
Start RTC demo application. This is v1.0 built on Oct  1 2021 15:01:40.
Waiting for registration...
Module is registered!

Current time in seconds from the epoch: 1633101266
Module system time is: 2021-10-01, 15:14:26

Get current time and add an hour

Current time in seconds from the epoch: 1633101266
New time to be set : 2021-10-01, 16:14:26, tz:4, dst:0

Set new time and check the setting
NEW module system time is: 2021-10-01, 16:14:26
```

Figure 105

3.4.40 SIM event handler example

Sim Event Demo application. Debug prints on **AUX UART**, using AZX log example functions

Features

- How to use ATI function for asynchronous management
- How to catch URC from an AppZone application
- How to catch SIM related events and handle them

Application workflow

M2MB_main.c

- Print welcome message
- Initialize AT interface
- Initialize AT URC manager task
- Initialize SIM event manager task
- Send "AT#SIMPR=1" to activate SIM URCs
- Insert SIM in SIM slot 1 and receive SIM inserted message
- Remove SIM from SIM slot 1 and receive SIM removed message

```
Starting SIM Presence Demo app. This is v1.1.17 built on Dec 16 2022 15:14:28.  
Please ensure SIM is not inserted before starting this sample application  
Sending command AT#SIMPR=1 to enable SIM presence URC messages...  
Command set.  
SIM 0 state changed to 1!  
  
SIM CARD HAS BEEN INSERTED!  
  
SIM 0 state changed to 0!  
  
SIM CARD HAS BEEN REMOVED!
```

Figure 106

3.4.41 SMS PDU

Sample application showcasing how to create and decode PDUs to be used with m2mb_sms_* API set. A SIM card and antenna must be present. Debug prints on **AUX UART**

Features

- How to enable SMS functionality
- How to use encode an SMS PDU to be sent with m2mb_api
- How to decode a received SMS response from PDU to ASCII mode.

3.4.41.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Init sms functionality
- Create PDU from text message
- Send message to destination number
- Wait for response
- When SMS PDU response is received, decode it and print information about it, plus the message content

```
m2mb_sms_init() succeeded

Sending message <How are you?>...
m2mb_sms_send() - succeeded
M2MB_SMS_SEND_RESP Callback
Send resp msg ID 10
SMS received!
SMS correctly received!

Reading SMS from memory...
m2mb_sms_read() request succeeded

--- SMS read ---
SMS tag M2MB_SMS_TAG_MT_NOT_READ
SMS format M2MB_SMS_FORMAT_3GPP
Code type: 0
Sender type: 145
Msg len: 12
Msg bytes: 11
Msg date 19/7/17 16:7:58 (timezone: 2)
Received SMS, content: <<Fine thanks >>
Sender: +[REDACTED]
```

Figure 107

3.4.42 SMS_atCmd example

Sample application showcasing how to receive an SMS containing an AT command, process the AT command and send its answer to sender (configurable in sms_config.txt). A SIM card and antenna must be present. Debug prints on **AUX UART**

Features

- How to receive an SMS with an AT command as text inside
- How to send AT command to parser and read the answer
- How to send the AT command answer back to sender via SMS

Optional configuration file to be put in /mod folder, copy sms_config.txt file into your module running the following AT command:

```
AT#M2MWRITE="/mod/sms_config.txt",138
>>> here receive the prompt; then type or send the file, sized 138 bytes
```

3.4.42.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Init SMS functionality
- Read configuration file sms_config.txt (send SMS with AT command answer back, delete SMS received)
- Init AT command parser
- Create a task to handle SMS parsing and AT command sending
- Wait for an incoming SMS

callbacks.c

msgSMSparse()

- When SMS has been received, content is decoded and printed. If there is an AT command inside, command is executed and answer printed and sent back to sender as an SMS (depending on sms_config.txt setting)

```

yStarting SMS with AT command demo app. This is v1.0.13-C1 built on Mar 18 2021 12:42:22.
[DEBUG] 16.61 M2MB_main:135 - M2MB_main{M2M_DamsStart}$ m2mb_os_ev_init success
m2mb_sms_init() succeeded
[DEBUG] 16.62 M2MB_main:168 - M2MB_main{M2M_DamsStart}$ M2MB_SMS_INCOMING_IND indication enabled
[DEBUG] 16.63 M2MB_main:179 - M2MB_main{M2M_DamsStart}$ M2MB_SMS_INCOMING_IND MEMORY FULL indication enabled
[DEBUG] 16.64 M2MB_main:196 - M2MB_main{M2M_DamsStart}$ Storage set to M2MB_SMS_STORAGE_SM
[DEBUG] 16.65 callbacks:114 - readConfigFromFile{M2M_DamsStart}$ Reading parameters from file
[DEBUG] 16.66 callbacks:116 - readConfigFromFile{M2M_DamsStart}$ Opening /mod/sms_config.txt in read mode..
Default: SMS with answer sending DISABLED, delete sms DISABLED
[DEBUG] 16.67 at_async:115 - at_cmd_async_init{M2M_DamsStart}$ m2mb_ati_init() on instance 0
Please send an SMS with a configuration as ("ATCMD: <atcmd>")...

```

Figure 108

3.4.43 SMTP Client

Sample application showing SMTP echo demo with M2MB API. Debug prints on **AUX UART**

Features

- How to check module registration and activate PDP context
- How to open a SMTP client
- How to send a mail

Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage SMTP client and start it

M2MB_main.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Initialize SMTP client and connect to SMTP server
- Prepare email and send it
- Close SMTP client
- Disable PDP context

```
Starting SMTP demo app. This is v1.1.17 built on Jan  9 2023 09:57:51.
Reading parameters from file...
Set APN to: <<nxt17.net>>
Set APN USER to: <<>>
Set APN PASS to: <<>>
Set PDP_CTX to: 1
Set MAIL SERVER to: <<          >>
Set MAIL PORT to: <<      >>
Set MAIL USER to: <<          >>
Set MAIL PASS to: <<          >>
Set MAIL FROM to: <<          >>
Set MAIL FROM NAME to: << >>
Set MAIL SUBJECT to: <<Hello from AppZone>>
Set MAIL BODY to: <<AppZone Test>>
Set MAIL TO to: <<          >>
Set MAIL TO NAME to: << >>
Waiting for registration...
Pdp context activation
Connecting to SMTP Server...
SMTP Client connected!
Preparing example's mail...
Sending example's mail...
Example's mail sent!
SMTP client closed
Application complete.
```

Figure 109

3.4.44 SW Timer (Software Timer)

The sample application shows how to use SW Timers M2MB API. Debug prints on **AUX UART**

Features

- How to open configure a SW timer
- How to use the timer to manage recurring events

3.4.44.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create sw timer structure
- Configure it with 4 seconds timeout, periodic timer (auto fires when expires)
- Init the timer with the parameters
- Start the timer
- Wait 10 seconds
- Stop the timer

timerCb

- Print a message with inside the callback

```
Starting SW Timers demo app. This is v1.0.7 built on Apr 7 2020 09:51:25.  
timer expired!  
[DEBUG] 21.41 M2MB_main.c:59 - timerCb{pubTspt_0}$ timer handle: 0x4002b004  
timer expired!  
[DEBUG] 25.47 M2MB_main.c:59 - timerCb{pubTspt_0}$ timer handle: 0x4002b004  
stopping the timer  
Stop a running timer: success  
Application end
```

Figure 110

3.4.45 Secure MicroService

Sample application showcasing how to manage secure microservice functionalities.
Debug prints on **AUX UART**

Features

- Write data in Secure Data Area (SDA), non protected
- Read the written data and compare with the original buffer
- Write a cripty key in Secure Data Area (SDA), non protected
- Perform a rotate of the written key data
- Perform MD5 sum of written data from TZ file
- Compare computed digest with expected one
- Write data in trust zone as a trusted object (it will not be possible to read it again but only use its content for crypto operations)
- Try to read the trusted object and verify it fails
- Rotate trusted item and verify retrieving the content fails
- compute MD5 sum of trusted item and compare with the expected one
- Try to pass data from a trusted item to a non trusted item using untrusted TZ buffers, and verify it fails

3.4.45.1 Application workflow

M2MB_main.c

- Write a buffer in a SDA item using `m2mb_secure_ms_write`
- Read the same item using `m2mb_secure_ms_read`
- Write a buffer containing some cripty key in a SDA item using `m2mb_secure_ms_write`
- Rotate the content of the key item
- Read it with `m2mb_secure_ms_read`
- Load the key content using `m2mb_secure_ms_crypto_alloc` and `m2mb_secure_crypto_add_it` in a SECURE_MS buffer
- Compute MD digest with `m2mb_secure_ms_crypto_md`
- Write a buffer containing some cripty key in a SDA item using `m2mb_secure_ms_write` but with **TRUSTED** option in `m2mb_secure_ms_open`
- Verify that `m2mb_secure_ms_read` on the trusted item fails
- Verify that `m2mb_secure_ms_crypto_rotate` fails for the trusted item
- Verify the MD5 digest
- Try to copy the trusted item data in a SECURE_MS buffer with `m2mb_secure_ms_crypto_alloc` and `m2mb_secure_crypto_add_item`, then load it in an untrusted object with `m2mb_secure_ms_crypto_write`, and verify it fails.

```

Starting secure ms demo app. This is v1.0.13-C1 built on Jul 30 2020 12:19:02.
Writing data in normal item
Stored input data in Secure Data Area
Reading data from normal item
Data length in SDA: 11 bytes
Securely loaded the data from the SDA
Read 11 bytes: <hello world>
original and retrieved strings are the same

Writing key in normal item
Stored input data in Secure Data Area

Rotate data in normal item
Original key: AA_THIS_IS_MY_SECRET_KEY_BB
Rotated key:

Compute MD5 of data in normal item
Data length in SDA: 27 bytes
MD5: 8EDAD26E26E1C74C7C02386C1C7F541D
hash is the expected one!

Writing data in trusted item
Stored input data in Secure Data Area
Reading data from trusted item (should fail!)
Data length in SDA: 27 bytes
m2mb_secure_ms_read() failed for trusted item, as expected!

Rotate data in trusted item
[ERROR] 17.01 M2MB_main:329 - read_rotate{M2M_DamsStart}$ Cannot read data from SECURE_MS_BUFFER to user buffer
Original key: AA_THIS_IS_MY_SECRET_KEY_BB
Rotated key:

Compute MD5 of data in trusted item
Data length in SDA: 27 bytes
MD5: 8EDAD26E26E1C74C7C02386C1C7F541D
Hash is the expected one!

Try to pass data from trusted to untrusted through TZ buffers
Cannot store data from SECURE_MS_BUFFER to SDA 'non-trusted', as expected

```

Figure 111

3.4.46 TCP IP

Sample application showcasing TCP echo demo with M2MB API. Debug prints on **AUX UART**

Features

- How to check module registration and activate PDP context
- How to open a TCP client socket
- How to communicate over the socket

3.4.46.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage socket and start it

m2m_tcp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and link it to the PDP context id
- Connect to the server
- Send data and receive response
- Close socket
- Disable PDP context

```

Starting TCP-IP demo app. This is v1.0.7 built on Mar 26 2020 16:20:30.
[DEBUG] 21.23 m2m_tcp_test.c:201 - M2M_msgTCPTask{TCP_TASK}$ INIT
[DEBUG] 21.25 m2m_tcp_test.c:217 - M2M_msgTCPTask{TCP_TASK}$ m2mb_os_ev_init success
[DEBUG] 21.26 m2m_tcp_test.c:223 - M2M_msgTCPTask{TCP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 21.26 m2m_tcp_test.c:231 - M2M_msgTCPTask{TCP_TASK}$ Waiting for registration...
[DEBUG] 21.28 m2m_tcp_test.c:128 - NetCallback{pubTspt_0}$ Module is registered to cell 0x816B!
[DEBUG] 21.29 m2m_tcp_test.c:244 - M2M_msgTCPTask{TCP_TASK}$ Pdp context activation
[DEBUG] 21.30 m2m_tcp_test.c:248 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.34 m2m_tcp_test.c:263 - M2M_msgTCPTask{TCP_TASK}$ Activate PDP with APN web.omnitel.it....
[DEBUG] 24.52 m2m_tcp_test.c:155 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 24.52 m2m_tcp_test.c:158 - PdpCallback{pubTspt_0}$ IP address: 83.225.44.56
[DEBUG] 24.54 m2m_tcp_test.c:273 - M2M_msgTCPTask{TCP_TASK}$ Creating Socket...
[DEBUG] 24.54 m2m_tcp_test.c:284 - M2M_msgTCPTask{TCP_TASK}$ Socket created
[DEBUG] 24.55 m2m_tcp_test.c:294 - M2M_msgTCPTask{TCP_TASK}$ Socket ctx set to 3
[DEBUG] 24.95 m2m_tcp_test.c:307 - M2M_msgTCPTask{TCP_TASK}$ Retrieved IP: 185.86.42.218
[DEBUG] 25.17 m2m_tcp_test.c:322 - M2M_msgTCPTask{TCP_TASK}$ Socket Connected!
[DEBUG] 25.18 m2m_tcp_test.c:329 - M2M_msgTCPTask{TCP_TASK}$ Sending data over socket..
[DEBUG] 25.19 m2m_tcp_test.c:342 - M2M_msgTCPTask{TCP_TASK}$ Data send successfully (16 bytes)
[DEBUG] 27.20 m2m_tcp_test.c:356 - M2M_msgTCPTask{TCP_TASK}$ trying to receive 16 bytes..
[DEBUG] 27.21 m2m_tcp_test.c:364 - M2M_msgTCPTask{TCP_TASK}$ Data received (16): <hello from m2mb!>
[DEBUG] 27.21 m2m_tcp_test.c:373 - M2M_msgTCPTask{TCP_TASK}$ application exit
[DEBUG] 27.22 m2m_tcp_test.c:385 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 27.24 m2m_tcp_test.c:388 - M2M_msgTCPTask{TCP_TASK}$ Application complete.
[DEBUG] 29.43 m2m_tcp_test.c:164 - PdpCallback{pubTspt_0}$ Context successfully deactivated!

```

Figure 112

3.4.47 TCP Socket status

Sample application showcasing how to check a TPC connected socket current status.
Debug prints on **AUX UART**

Features

- How to check module registration and activate PDP context
- How to open a TCP client socket
- How to check if the TCP socket is still valid

3.4.47.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage socket and start it

m2m_tcp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and link it to the PDP context id
- Connect to the server
- Check in a loop the current socket status using the adv_select function with a 2 seconds timeout
- Close socket when the remote host closes it
- Disable PDP context

```

Starting TCP socket status check demo app. This is v1.0.14-C1 built on Sep  8 2020 14:59:25.
[DEBUG] 21.33 m2m_tcp_tes:324 - M2M_msgTCPTask{TCP_TASK}$ INIT
[DEBUG] 21.34 m2m_tcp_tes:338 - M2M_msgTCPTask{TCP_TASK}$ m2mb_os_ev_init success
[DEBUG] 21.34 m2m_tcp_tes:344 - M2M_msgTCPTask{TCP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 21.35 m2m_tcp_tes:352 - M2M_msgTCPTask{TCP_TASK}$ Waiting for registration...
[DEBUG] 21.36 m2m_tcp_tes:365 - M2M_msgTCPTask{TCP_TASK}$ Pdp context activation
[DEBUG] 21.37 m2m_tcp_tes:369 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.41 m2m_tcp_tes:384 - M2M_msgTCPTask{TCP_TASK}$ Activate PDP with APN NXT17.NET....
[DEBUG] 24.09 m2m_tcp_tes:281 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 24.10 m2m_tcp_tes:284 - PdpCallback{pubTspt_0}$ IP address: 100.77.5.223
[DEBUG] 24.10 m2m_tcp_tes:394 - M2M_msgTCPTask{TCP_TASK}$ Creating Socket...
[DEBUG] 24.11 m2m_tcp_tes:405 - M2M_msgTCPTask{TCP_TASK}$ Socket created
[DEBUG] 24.11 m2m_tcp_tes:415 - M2M_msgTCPTask{TCP_TASK}$ Socket ctx set to 3
[DEBUG] 24.60 m2m_tcp_tes:428 - M2M_msgTCPTask{TCP_TASK}$ Retrieved IP: 185.86.42.218
[DEBUG] 24.93 m2m_tcp_tes:443 - M2M_msgTCPTask{TCP_TASK}$ Socket Connected!
[DEBUG] 26.98 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 29.03 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
...
[DEBUG] 82.18 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 84.23 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 86.28 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 88.31 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 88.90 m2m_tcp_tes:154 - adv_select{TCP_TASK}$ Data is available on socket <0x40032b3c>
[DEBUG] 88.92 m2m_tcp_tes:160 - adv_select{TCP_TASK}$ There are <0> pending bytes on socket
Socket was closed by remote!
[DEBUG] 88.92 m2m_tcp_tes:494 - M2M_msgTCPTask{TCP_TASK}$ application exit
[DEBUG] 88.94 m2m_tcp_tes:506 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 88.94 m2m_tcp_tes:509 - M2M_msgTCPTask{TCP_TASK}$ Application complete.
[DEBUG] 89.31 m2m_tcp_tes:290 - PdpCallback{pubTspt_0}$ Context successfully deactivated!

```

Figure 113

3.4.48 TCP Server

Sample application showcasing TCP listening socket demo with M2MB API. Debug prints on **AUX UART**

Features

- How to check module registration and activate PDP context
- How to open a TCP listening socket
- How to manage external hosts connection and exchange data

3.4.48.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage socket and start it

m2m_tcp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and set it in non-blocking mode
- Bind the socket to the listening port
- Start listening for incoming connection
- Check if a connection is incoming using m2mb_socket_bsd_select function
- If a client connects, perform accept on the child socket
- Send a "START" message to the client
- Send some data
- Wait for data from client and print it
- Close the child socket
- Start listening again, up to 3 times
- Close listening socket
- Disable PDP context

Debug Log

```

Starting TCP Server demo app. This is v1.0.7 built on Apr  7 2020 13:28:24.
[DEBUG] 14.55 m2m_tcp_test.c:220 - M2M_msgTCPTask(TCP_TASK)$ INIT
[DEBUG] 14.55 m2m_tcp_test.c:236 - M2M_msgTCPTask(TCP_TASK)$ m2mb_os_ev_init success
[DEBUG] 14.57 m2m_tcp_test.c:242 - M2M_msgTCPTask(TCP_TASK)$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 14.57 m2m_tcp_test.c:250 - M2M_msgTCPTask(TCP_TASK)$ Waiting for registration...
[DEBUG] 14.58 m2m_tcp_test.c:138 - NetCallback(pubTspt_0)$ Module is registered to cell 0x5222!
[DEBUG] 14.59 m2m_tcp_test.c:263 - M2M_msgTCPTask(TCP_TASK)$ Pdp context activation
[DEBUG] 14.60 m2m_tcp_test.c:267 - M2M_msgTCPTask(TCP_TASK)$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 16.57 m2m_tcp_test.c:282 - M2M_msgTCPTask(TCP_TASK)$ Activate PDP with APN ibox.tim.it...
[DEBUG] 17.16 m2m_tcp_test.c:165 - PdpCallback(pubTspt_0)$ Context activated!
[DEBUG] 17.17 m2m_tcp_test.c:168 - PdpCallback(pubTspt_0)$ IP address: 2.195.165.137

-----
| Start TCP server |
-----

[DEBUG] 19.15 m2m_tcp_test.c:301 - M2M_msgTCPTask(TCP_TASK)$ Creating Socket...
[DEBUG] 19.15 m2m_tcp_test.c:312 - M2M_msgTCPTask(TCP_TASK)$ Socket created
[DEBUG] 19.16 m2m_tcp_test.c:313 - M2M_msgTCPTask(TCP_TASK)$ m2mb_socket_bsd_socket(): valid socket ID [0x4002E79C] - PASS
[DEBUG] 20.16 m2m_tcp_test.c:319 - M2M_msgTCPTask(TCP_TASK)$ issuing m2m_socket_bsd_ioctl() to set non-blocking mode ...
[DEBUG] 20.17 m2m_tcp_test.c:331 - M2M_msgTCPTask(TCP_TASK)$ Binding Socket...
[DEBUG] 22.12 m2m_tcp_test.c:343 - M2M_msgTCPTask(TCP_TASK)$ Socket Bind Pass

Start TCP listening on port 6500...
[DEBUG] 24.13 m2m_tcp_test.c:368 - M2M_msgTCPTask(TCP_TASK)$ select...
Select result: 0
[DEBUG] 28.13 m2m_tcp_test.c:368 - M2M_msgTCPTask(TCP_TASK)$ select...
Select result: 1

TCP Server Coming Connection
--> Accept
[DEBUG] 30.52 m2m_tcp_test.c:397 - M2M_msgTCPTask(TCP_TASK)$ Socket Accept Pass

Connected! (socket dial n.1)

[DEBUG] 30.53 m2m_tcp_test.c:403 - M2M_msgTCPTask(TCP_TASK)$ Client Source Address: 185.86.42.254
[DEBUG] 30.54 m2m_tcp_test.c:404 - M2M_msgTCPTask(TCP_TASK)$ Client Port: 58658
[DEBUG] 30.54 m2m_tcp_test.c:405 - M2M_msgTCPTask(TCP_TASK)$ Client Family: 2
[DEBUG] 31.56 m2m_tcp_test.c:410 - M2M_msgTCPTask(TCP_TASK)$

-----
[DEBUG] 31.57 m2m_tcp_test.c:411 - M2M_msgTCPTask(TCP_TASK)$ | Send/receive data test |
[DEBUG] 31.57 m2m_tcp_test.c:412 - M2M_msgTCPTask(TCP_TASK)$ -----

[DEBUG] 32.58 m2m_tcp_test.c:416 - M2M_msgTCPTask(TCP_TASK)$
--> issuing m2mb_socket_bsd_send(): transmit "START" packet...
[DEBUG] 32.59 m2m_tcp_test.c:423 - M2M_msgTCPTask(TCP_TASK)$ --> done (11 have been transmitted)
[DEBUG] 32.60 m2m_tcp_test.c:425 - M2M_msgTCPTask(TCP_TASK)$ ALL data transmitted - PASS
[DEBUG] 32.61 m2m_tcp_test.c:430 - M2M_msgTCPTask(TCP_TASK)$
--> issuing m2mb_socket_bsd_send(): transmit 58 bytes...
[DEBUG] 32.62 m2m_tcp_test.c:437 - M2M_msgTCPTask(TCP_TASK)$ --> done (58 have been transmitted)
[DEBUG] 32.63 m2m_tcp_test.c:440 - M2M_msgTCPTask(TCP_TASK)$ ALL data transmitted - PASS
[DEBUG] 32.64 m2m_tcp_test.c:448 - M2M_msgTCPTask(TCP_TASK)$
Waiting for data...

[DEBUG] 39.64 m2m_tcp_test.c:457 - M2M_msgTCPTask(TCP_TASK)$ test
[DEBUG] 99.61 m2m_tcp_test.c:465 - M2M_msgTCPTask(TCP_TASK)$
m2mb_socket_bsd_recv() has received 6 bytes

[DEBUG] 102.60 m2m_tcp_test.c:469 - M2M_msgTCPTask(TCP_TASK)$
Server TCP is closing the current connection ...

```

Figure 114

Data on a PuTTY terminal

```
START
aaaaaaaaa-bbbbbbbbbb-ccccccccc-ddddddddd-eeeeeeeeee
test
█
```

Figure 115

3.4.49 TLS SSL Client

Sample application showcasing TLS/SSL with client certificates usage with M2MB API. Debug prints on **AUX UART**

Features

- How to check module registration and enable PDP context
- How to open a SSL client socket
- How to communicate over SSL socket

3.4.49.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the connection and start it

ssl_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and link it to the PDP context id
- Connect to the server over TCP socket
- Initialize the TLS parameters (TLS1.2) andh auth mode (server+client auth in the example)
- Create SSL context
- Read certificates files and store them
- Create secure socket and connect to the server using SSL
- Send data and receive response
- Close secure socket
- Close socket
- Delete SSL context
- Disable PDP context

The application requires the certificates to be stored in /mod/ssl_certs/ folder. It can be created with

AT#M2MMKDIR=/mod/ssl_certs

Certificates can then be loaded with

AT#M2MWRITE="/mod/ssl_certs/modulesCA.crt",1740

and providing the file content in RAW mode (for example using the "Transfer Data" button in Telit AT Controller)

For client certificates (if required), the commands will be

AT#M2MWRITE="/mod/ssl_certs/modulesClient.crt",1651

AT#M2MWRITE="/mod/ssl_certs/modulesClient_pkcs1.key",1679

PLEASE NOTE: always verify the file sizes to be used in the commands above as they might change

```
Starting TLS-SSL demo app. This is v1.1.2 built on Mar  3 2021 10:15:00.
[DEBUG] 10.85 ssl_test:252 - msgHTTPSTask{TLS_TASK}$ INIT
[DEBUG] 10.85 ssl_test:266 - msgHTTPSTask{TLS_TASK}$ m2mb_os_ev_init success
[DEBUG] 10.85 ssl_test:270 - msgHTTPSTask{TLS_TASK}$ Init SSL session test app
[DEBUG] 10.85 ssl_test:285 - msgHTTPSTask{TLS_TASK}$ m2mb_ssl_create_config sslConfigHnd1 = 0x40037958, sslRes= 0
[DEBUG] 10.85 ssl_test:294 - msgHTTPSTask{TLS_TASK}$ m2mb_ssl_create_config PASSED
[DEBUG] 10.85 ssl_test:306 - msgHTTPSTask{TLS_TASK}$ m2mb_ssl_create_ctxt PASSED
[DEBUG] 10.85 ssl_test:311 - msgHTTPSTask{TLS_TASK}$ loading CA CERT from file /mod/ssl_certs/modulesCA.crt
[DEBUG] 10.85 ssl_test:315 - msgHTTPSTask{TLS_TASK}$ file size: 1740
[DEBUG] 10.85 ssl_test:328 - msgHTTPSTask{TLS_TASK}$ Reading content from file. Size: 1740
Buffer successfully received from file. 1740 bytes were loaded.
Closing file.
[DEBUG] 10.85 ssl_test:361 - msgHTTPSTask{TLS_TASK}$ loading client CERT from file /mod/ssl_certs/modulesClient.crt
[DEBUG] 10.85 ssl_test:365 - msgHTTPSTask{TLS_TASK}$ file size: 1651
[DEBUG] 10.85 ssl_test:378 - msgHTTPSTask{TLS_TASK}$ Reading content from file. Size: 1651
Buffer successfully received from file. 1651 bytes were loaded.
Closing file.
[DEBUG] 10.85 ssl_test:401 - msgHTTPSTask{TLS_TASK}$ loading client KEY from file /mod/ssl_certs/modulesClient_pkcs1.key
[DEBUG] 10.85 ssl_test:405 - msgHTTPSTask{TLS_TASK}$ file size: 1679
[DEBUG] 10.85 ssl_test:418 - msgHTTPSTask{TLS_TASK}$ Reading content from file. Size: 1679
Buffer successfully received from file. 1679 bytes were loaded.
Closing file.
[DEBUG] 10.85 ssl_test:448 - msgHTTPSTask{TLS_TASK}$ certificates successfully stored!
[DEBUG] 10.85 ssl_test:457 - msgHTTPSTask{TLS_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 10.85 ssl_test:465 - msgHTTPSTask{TLS_TASK}$ Waiting for registration...
[DEBUG] 10.86 ssl_test:171 - NetCallback{pubTspt_0}$ Module is registered to cell 0x468E!
[DEBUG] 10.86 ssl_test:477 - msgHTTPSTask{TLS_TASK}$ Pdp context activation
[DEBUG] 10.86 ssl_test:481 - msgHTTPSTask{TLS_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 12.87 ssl_test:496 - msgHTTPSTask{TLS_TASK}$ Activate PDP with APN web.omnitel.it....
[DEBUG] 13.71 ssl_test:197 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 13.71 ssl_test:200 - PdpCallback{pubTspt_0}$ IP address: 2.41.76.63
[DEBUG] 13.71 ssl_test:514 - msgHTTPSTask{TLS_TASK}$ Creating Socket...
[DEBUG] 13.71 ssl_test:525 - msgHTTPSTask{TLS_TASK}$ Socket created
[DEBUG] 13.71 ssl_test:535 - msgHTTPSTask{TLS_TASK}$ Socket ctx set to 3
[DEBUG] 13.92 ssl_test:548 - msgHTTPSTask{TLS_TASK}$ Retrieved IP: 185.86.42.218
[DEBUG] 14.05 ssl_test:562 - msgHTTPSTask{TLS_TASK}$ Socket Connected!
[DEBUG] 15.97 ssl_test:587 - msgHTTPSTask{TLS_TASK}$ m2mb_ssl_connect ret 0
[DEBUG] 17.99 ssl_test:593 - msgHTTPSTask{TLS_TASK}$ Sending bytes..
```

```
[DEBUG] 17.99 ssl_test:593 - msgHTTPSTask{TLS_TASK}$ Sending bytes..
[DEBUG] 17.99 ssl_test:596 - msgHTTPSTask{TLS_TASK}$ SSL write result = 44
[DEBUG] 22.03 ssl_test:608 - msgHTTPSTask{TLS_TASK}$ pending bytes: 1087
[DEBUG] 22.03 ssl_test:612 - msgHTTPSTask{TLS_TASK}$ trying to receive 1087 bytes..
[DEBUG] 22.03 ssl_test:618 - msgHTTPSTask{TLS_TASK}$ Server response: (269)<HTTP/1.1 200 OK
Date: Wed, 03 Mar 2021 09:18:22 GMT
Server: Apache/2.2.15 (CentOS)
Last-Modified: Mon, 22 Jan 2018 10:57:39 GMT
ETag: "1fffc-27f-5635b4c6f12b3"
Accept-Ranges: bytes
Content-Length: 639
Connection: close
Content-Type: text/html; charset=UTF-8
>
[DEBUG] 22.03 ssl_test:634 - msgHTTPSTask{TLS_TASK}$ pending bytes: 762
[DEBUG] 22.03 ssl_test:638 - msgHTTPSTask{TLS_TASK}$ trying to receive remaining 762 bytes..
[DEBUG] 22.03 ssl_test:644 - msgHTTPSTask{TLS_TASK}$ Server response: (639)<<html>
<head>
  <title>module.telit.com</title>
  <meta content="text/html; charset=utf-8" />
</head>
<body>
  <table border=0 align=center>
    <tr>
      <td height="100" align=center><h2>modules.telit.com - Test HTML page</h2></td>
    </tr>
    <tr>
      <td align=center><img src=Telit.jpg alt="Telit logo" height="126" width="410"></img></td>
    </tr>
    <tr>
      <td height="200" align=center> This is a simple HTML page, </br>
      made with simple HTML code,</br>
      just for test!
    </td>
    </tr>
    <tr>
      <td height="100" align=center><font size="3">Telit &copy; 2015 - 2017 All rights reserved</font></td>
    </tr>
  </table>
</body>
</html>
>
[DEBUG] 22.03 ssl_test:662 - msgHTTPSTask{TLS_TASK}$ application exit
[DEBUG] 22.03 ssl_test:680 - msgHTTPSTask{TLS_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 22.03 ssl_test:683 - msgHTTPSTask{TLS_TASK}$ Application complete.
[DEBUG] 22.77 ssl_test:206 - PdpCallback{pubTspt_0}$ Context deactivated!
```

3.4.50 UDP client

Sample application showcasing UDP echo demo with M2MB API. Debug prints on **AUX UART**

Features

- How to check module registration and activate PDP context
- How to open a UDP client socket
- How to communicate over the socket

3.4.50.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task and start it

m2m_udp_test.c - Initialize Network structure and check registration - Initialize PDP structure and start PDP context - Create socket and link it to the PDP context id - Send data and receive response - Close socket - Disable PDP context

```
Starting UDP client demo app. This is v1.0.7 built on Apr 1 2020 14:57:13.
INIT
[DEBUG] 21.23 m2m_udp_test.c:223 - M2M_msgUDPTask{UDP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
Waiting for registration...
[DEBUG] 21.25 m2m_udp_test.c:131 - NetCallback{pubTspt_0}$ Module is registered to cell 0xC4CF!
[DEBUG] 21.26 m2m_udp_test.c:241 - M2M_msgUDPTask{UDP_TASK}$ Pdp context initialization
[DEBUG] 21.26 m2m_udp_test.c:245 - M2M_msgUDPTask{UDP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
Activate PDP with APN web.omnitel.it...
[DEBUG] 24.11 m2m_udp_test.c:157 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 24.11 m2m_udp_test.c:160 - PdpCallback{pubTspt_0}$ IP address: 109.113.222.12
[DEBUG] 24.12 m2m_udp_test.c:268 - M2M_msgUDPTask{UDP_TASK}$ Creating Socket...
[DEBUG] 24.13 m2m_udp_test.c:280 - M2M_msgUDPTask{UDP_TASK}$ Socket created
Socket ctx set to 3
[DEBUG] 24.41 m2m_udp_test.c:306 - M2M_msgUDPTask{UDP_TASK}$ Retrieved IP: 185.86.42.218
Socket ready.
Data successfully sent (16 bytes)
Socket rcv...
[DEBUG] 26.47 m2m_udp_test.c:352 - M2M_msgUDPTask{UDP_TASK}$ m2mb_socket_bsd_set_sock_opt() M2MB_SOCKET_BSD_SO_RCVTIMEO - success
trying to receive 16 bytes..
Data received (16): <hello from m2mb!>
[DEBUG] 26.48 m2m_udp_test.c:377 - M2M_msgUDPTask{UDP_TASK}$ application exit
Socket Closed
[DEBUG] 26.49 m2m_udp_test.c:399 - M2M_msgUDPTask{UDP_TASK}$ m2mb_pdp_deactivate returned success
Application complete.
[DEBUG] 27.04 m2m_udp_test.c:166 - PdpCallback{pubTspt_0}$ Context successfully deactivated!
```

Figure 116

3.4.51 USB Cable Check

Sample application showing how to check if USB cable is plugged in or not. Debug prints on **AUX UART**

Features

- How to open an USB channel and configure it with a callback function
- How to manage USB cable events in the callback function

3.4.51.1 Application workflow

M2MB_main.c

- Open UART/UART_AUX for debug
- open usb channel and set the callback
- Print greeting message
- Print current usb status

USB_Cb

- if the event is a connection/disconnection, show the current status

```
Starting USB cable check demo app. This is v1.0.0 built on Aug 19 2020 10:27:40.
m2mb_usb_open succeeded
m2mb_usb_ioctl: set usb callback
m2mb_usb_ioctl: got cable status
USB cable CONNECTED, status: 1

Waiting for USB cable to be plugged/unplugged...
Usb cable check event, USB status: 0
Usb cable check event, USB status: 1
Usb cable check event, USB status: 0
Usb cable check event, USB status: 1
```

Figure 117

3.4.52 Watchdog example

Sample application that shows how to set and implement a watchdog. Task has been locked waiting for an event with a timeout longer than wd inactivity timeout. If no wd kick or no actions

Features

- Enable watchdog on Task_1
- Handle watchdog kicks with a timer

3.4.52.1 Application workflow

M2MB_main.c

- Create Task_1 that will be put under watchdog control
- In Task_1 watchdog is enabled on it and a timer created to handle watchdog kicks. Then task enters in a loop where kicks are sent. After 5 kicks a task lock is simulated waiting for an event release (with a timeout longer than task watchdog inactivity timeout)
- As soon as watchdog inactivity timeout expired M2MB_WDOG_TIMEOUT_IND is generated and handled in WDcallback (in this case the event is released, no further actions are done)
- No further actions are performed by Task_1 (no kicks are sent) so after watchdog system timeout expiration app is rebooted.

```
Starting WATCHDOG demo app. This is v1.0.0 built on Nov  4 2021 14:34:06.
Event correctly initialized
Task started
Init WatchDog
m2mb_wDog_init OK
Tick duration: 1s
Adding Task under WD control with inactivity timeout of 60s
m2mb_wDog_addTask OK
Enabling the WatchDog
m2mb_wDog_enable OK
Timer creation OK

Start WD kick every 5s...
WD kick
WD kick done
```

Figure 118

3.4.53 ZLIB example

Sample application showing how to compress/uncompress with ZLIB. Debug prints on **AUX UART**

Features

- How to compress a file
- How to uncompress a file

In order to execute the entire test, copy test.gz file into your module running the following AT command:

```
AT#M2MWRITE="/mod/test.gz",138
```

>>> here receive the prompt; then type or send the file, sized 138 bytes

3.4.53.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Test the compression and decompression of a data string
- Test the decompression of a .gz file (test.gz), expected to be in /mod folder, into its content test.txt. The file must be uploaded by the user (see steps above).

```
Starting Logging demo app. This is v1.0.7 built on Apr  7 2020 09:02:35.
Starting TEST_COMPR_UNCOMPR.
len: 138; comprLen: 57
Compressed message:
W+EHU(,ILIVH^E/ISH^ PE*I-HMQE/K-R(Ï Êç$VU^#a$ê y4RI«¥1.
comprLen: 57; uncomprLen: 138
uncompress():
the quick brown fox jumped over the lazy dog. the quick brown fox jumped over the lazy dog. the quick brown fox jumped over the lazy dog.
Ending TEST_COMPR_UNCOMPR with SUCCESS.
Starting test_uncompress.
Data extracted correctly into the file ./mod/test.txt
test_uncompress finished correctly!
```

Figure 119

3.5 USB0

Applications that provide usage examples for various functionalities, log output on USB0

3.5.1 Alarm example

Sample application that shows how to set an alarm to wake-up module. Debug prints on **USB0**

Features

- How to set an alarm
- How to use it to turn on module

Application workflow

M2MB_main.c

- Init RTC
- Wait for registration
- Get current date and time
- Call function set_alarm
- Init Power and turn off module

```
12:34:48.532- Start Alarm demo application. This is v1.1.19 built on Sep 9 2024 12:34:02.
12:34:48.532-
12:34:48.532- Waiting for registration...
12:34:48.532- Module is registered!
12:34:48.532- NETWORK OPERATOR (mcc mnc): 222 01, Rat: 0
12:34:48.532- Module system time is: 2024-09-09, 12:33:35
12:34:48.532- Setting alarm in 2 minutes
12:34:48.532- Alarm will be set at: 2024-09-09, 12:35:35
12:34:48.741-
12:34:48.741- Wait 5 seconds and then Turn off module...
12:34:53.752- Power off moduley
```

Figure 120

3.5.2 ATI (AT Instance)

Sample application showing how to use AT Instance functionality (sending AT commands from code). The example supports both sync and async (using a callback) modes. Debug prints on **USB0**

Features

- How to open an AT interface from the application
- How to send AT commands and receive responses on the AT interface

3.5.2.1 Application workflow, sync mode

M2MB_main.c

- Open USB/UART/UART_AUX
- Init AT0 (first AT instance)
- Send AT+CGMR command
- Print response.
- Release AT0

at_sync.c

- Init ati functionality and take AT0
- Send AT+CGMR command, then read response after 2 seconds, then return it
- Deinit ati, releasing AT0

```
Starting AT demo app. This is v1.0.7 built on Apr  1 2020 15:12:58.
[DEBUG] 17.15  at_sync.c:53 - at_cmd_sync_init{M2M_DamsStart}$ m2mb_ati_init() on instance 0
Sending command AT+CGMR in sync mode
[DEBUG] 17.16  at_sync.c:79 - send_sync_at_command{M2M_DamsStart}$ Sending AT Command: AT+CGMR
Command response: <AT+CGMR
MOB.950004-B008

OK
>

[DEBUG] 19.21  at_sync.c:61 - at_cmd_sync_deinit{M2M_DamsStart}$ m2mb_ati_deinit() on instance 0
Application end
```

Figure 121

3.5.2.2 Application workflow, async mode

M2MB_main.c

- Open USB/UART/UART_AUX
- Init AT0 (first AT instance)
- Send AT+CGMR command
- Print response.

- Release AT0

at_async.c

- Init ati functionality and take AT0, register AT events callback
- Send AT+CGMR command, wait for response semaphore (released in callback), then read it and return it
- Deinit ati, releasing AT0

```
Starting AT demo app. This is v1.0.7 built on Apr 1 2020 15:07:45.
[DEBUG] 17.13 at_async.c:116 - at_cmd_async_init{M2M_DamsStart}$ m2mb_ati_init() on instance 0
Sending command AT+CGMR in async mode
[DEBUG] 17.15 at_async.c:153 - send_async_at_command{M2M_DamsStart}$ Sending AT Command: AT+CGMR
[DEBUG] 17.15 at_async.c:169 - send_async_at_command{M2M_DamsStart}$ waiting command response...
[DEBUG] 17.17 at_async.c:88 - at_cmd_async_callback{pubTspt_0}$ Callback - available bytes: 25
[DEBUG] 17.18 at_async.c:181 - send_async_at_command{M2M_DamsStart}$ Receive response...
Command response: <AT+CGMR
MOB.950004-B008

OK
>

[DEBUG] 17.19 at_async.c:136 - at_cmd_async_deinit{M2M_DamsStart}$ m2mb_ati_deinit() on instance 0
Application end
```

Figure 122

3.5.3 AWS demo

Sample application showcasing AWS IoT Core MQTT communication. Debug prints on **USB0**

Features

- How to check module registration and enable PDP context
- How to load certificates into device SSL session storage
- How to configure MQTT client parameters
- How to connect to AWS server with SSL and exchange data over a topic

3.5.3.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage MQTT client and start it

aws_demo.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Init MQTT client
- Configure it with all parameters (Client ID, PDP context ID, keepalive timeout...)
- Initialize the TLS parameters (TLS1.2) and auth mode (server+client auth in the example)
- Create SSL context
- Read certificates files and store them
- Connect MQTT client to broker
- Subscribe to topic
- Publish 10 messages with increasing counter
- Print received message in mqtt_topc_cb function
- Disconnect MQTT client and deinit it
- Disable PDP context

3.5.3.2 How to get started with AWS IoT

- Go to [AWS console](#) and create an account if one is not available yet.
- Go to **IoT Core** section
- Go to **Secure > Policies** section
- Create a new policy, which describes what the device will be allowed to do (e.g. subscribe, publish)
- Give it a name, then configure it using the configuration below (it is possible to copy/paste by clicking on **Add statements** section, then **Advanced mode**) :

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Action": [
        "iot:Publish",
        "iot:Subscribe",
        "iot:Connect",
        "iot:Receive"
      ],
      "Effect": "Allow",
      "Resource": [
        "*"
      ]
    }
  ]
}
```

- Click on create to complete the policy creation.
- Go to **Manage** section
- Press **Create**, then **Create a single thing**
- Give the new thing a name, then click on Next
- Select **One-click certificate creation (recommended)** by clicking on **Create certificate**
- Once presented with the **Certificate created** page, download all certificates and keys
- Click on the **Activate** button to enable the certificate authentication of the newly created device
- Click on **Attach a policy** and select the policy created in a previous step

For further information, please refer to the full [AWS IoT documentation](#)

3.5.3.3 Application setup

- Set **CLIENTCERTFILE** and **CLIENTKEYFILE** defines in **aws_demo.c** file in order to match the certificate and key created in the previous section.
- Set **AWS_BROKER_ADDRESS** to the correct AWS URL. It can be retrieved from AWS IoT Manage > Things > Interact in the HTTPS Rest API Endpoint URL.
- Set **CLIENT_ID** to the desired Client ID for your AWS device
- (Optional) if required, change **CACERTFILE** to match the one to be used.

3.5.3.4 Device setup

The application requires the certificates (provided in sample app **certs** subfolder) to be stored in **/mod/ssl_certs/** folder. It can be created with

```
AT#M2MMKDIR=/mod/ssl_certs
```

Certificates can then be loaded with

```
AT#M2MWRITE="/mod/ssl_certs/preload_CACert_01.crt",1468 AT#M2MWRITE="/mod/ssl_certs/IoT.crt",1646
```

providing the file content in RAW mode (for example using the “Transfer Data” button in Telit AT Controller)

For client certificates, the commands will be

```
AT#M2MWRITE="/mod/ssl_certs/xxxxx.crt",yyyy
```

```
AT#M2MWRITE="/mod/ssl_certs/xxxxx.key",zzzz
```

PLEASE NOTE: always verify the file sizes to be used in the commands above as they might change

```
Starting AWS IoT Core MQTT demo app. This is v1.1.5 built on Apr 30 2021 09:05:17.
[DEBUG] 15.51 aws_demo:607 - AWS_Task{MQTT_TASK}$ Init MQTT client for AWS
[DEBUG] 15.52 aws_demo:265 - PrepareSSLEnvironment{MQTT_TASK}$ m2mb_ssl_config SNI succeeded
[DEBUG] 15.52 aws_demo:271 - PrepareSSLEnvironment{MQTT_TASK}$ Root CA cert file /mod/ssl_certs/preload_CACert_01.crt
[DEBUG] 15.52 aws_demo:297 - PrepareSSLEnvironment{MQTT_TASK}$ Buffer successfully received from file. 1468 bytes were loaded.
[DEBUG] 15.52 aws_demo:308 - PrepareSSLEnvironment{MQTT_TASK}$ Cross Signed CA cert file /mod/ssl_certs/Amazon-IoT.crt
[DEBUG] 15.52 aws_demo:334 - PrepareSSLEnvironment{MQTT_TASK}$ Buffer successfully received from file. 1646 bytes were loaded.
[DEBUG] 15.52 aws_demo:360 - PrepareSSLEnvironment{MQTT_TASK}$ Client certificate file /mod/ssl_certs/ab71 -certificate.pem.crt
[DEBUG] 15.52 aws_demo:384 - PrepareSSLEnvironment{MQTT_TASK}$ Buffer successfully received from file. 1224 bytes were loaded.
[DEBUG] 15.52 aws_demo:396 - PrepareSSLEnvironment{MQTT_TASK}$ Client Key file /mod/ssl_certs/ab71 -private.pem.key
[DEBUG] 15.52 aws_demo:422 - PrepareSSLEnvironment{MQTT_TASK}$ Buffer successfully received from file. 1679 bytes were loaded.

SSL environment preparation completed
[DEBUG] 15.52 aws_demo:726 - AWS_Task{MQTT_TASK}$ Waiting for registration...
[DEBUG] 15.52 aws_demo:514 - NetCallback{pubTspt_0}$ Module is registered
[DEBUG] 15.52 aws_demo:738 - AWS_Task{MQTT_TASK}$ PDP context initialization
[DEBUG] 17.55 aws_demo:753 - AWS_Task{MQTT_TASK}$ Activate PDP with APN web.omnitel.it on CID 1....
[DEBUG] 18.37 aws_demo:557 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 18.37 aws_demo:561 - PdpCallback{pubTspt_0}$ IP address: 109.114.102.21

Connecting to Server <angy83rl5oizs-ats.iot.eu-west-2.amazonaws.com>:8883...
Done.
[DEBUG] 27.87 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 2, "data": {"RSSI": -72, "tm": "2021-04-30,09:05:50"}}> to topic
device/35308109f /updates
[DEBUG] 27.87 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 30.94 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 3, "data": {"RSSI": -72, "tm": "2021-04-30,09:05:53"}}> to topic
device/35308109f /updates
[DEBUG] 30.94 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 33.99 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 4, "data": {"RSSI": -72, "tm": "2021-04-30,09:05:56"}}> to topic
device/35308109f /updates
[DEBUG] 34.00 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 37.00 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 5, "data": {"RSSI": -72, "tm": "2021-04-30,09:05:59"}}> to topic
device/35308109f /updates
[DEBUG] 37.00 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 40.03 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 6, "data": {"RSSI": -72, "tm": "2021-04-30,09:06:02"}}> to topic
device/35308109f /updates
[DEBUG] 40.03 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 43.05 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 7, "data": {"RSSI": -72, "tm": "2021-04-30,09:06:05"}}> to topic
device/353081f /updates
[DEBUG] 43.05 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 46.13 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 8, "data": {"RSSI": -72, "tm": "2021-04-30,09:06:08"}}> to topic
device/3530810f /updates
[DEBUG] 46.13 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 49.15 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 9, "data": {"RSSI": -72, "tm": "2021-04-30,09:06:11"}}> to topic
device/3530810f /updates
[DEBUG] 49.15 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 52.19 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 10, "data": {"RSSI": -72, "tm": "2021-04-30,09:06:14"}}> to topic
device/3530810f /updates
[DEBUG] 52.19 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 55.22 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 11, "data": {"RSSI": -72, "tm": "2021-04-30,09:06:17"}}> to topic
device/353081f /updates
[DEBUG] 55.22 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.

Disconnecting from MQTT broker..
[DEBUG] 58.27 aws_demo:878 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 58.27 aws_demo:908 - AWS_Task{MQTT_TASK}$ application exit
[DEBUG] 58.27 aws_demo:918 - AWS_Task{MQTT_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 58.27 aws_demo:924 - AWS_Task{MQTT_TASK}$ Application complete.
```

Figure 123

Data received from a subscriber:

```
device/353081f /updates 1
QoS 0

device/3530810 /updates 1
30-04-2021 09:06:00.32760467 QoS 0
{ "ID": 2, "data": {"RSSI": -72, "tm": "2021-04-30,09:05:50"} }
```

Figure 124

3.5.4 App Manager

Sample application showing how to manage AppZone apps from m2mb code. Debug prints on **USB0**

Features

- How to get how many configured apps are available
- How to get the handle to manage the running app (change start delay, enable/disable)
- How to create the handle for a new binary app, enable it and set its parameters
- How to start the new app without rebooting the device, then stop it after a while.

3.5.4.1 Prerequisites

This app will try to manage another app called “second.bin”, which already exists in the module filesystem and can be anything (e.g. another sample app as GPIO toggle). the app must be built using the flag ROM_START=

in the Makefile to set a different starting address than the main app (by default, 0x40000000). For example, 0x41000000.

3.5.4.2 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- get a non existing app handle and verify it is NULL
- get the current app handle, then get the start delay **set in the INI file (so persistent)**
- change the current app delay value **in the INI file**
- verify that the change has been stored
- get current app state
- create an handle for a second application binary.
- add it to the INI file
- set its execution flag to 0
- get the delay time and the state from INI file for the new app
- get the current set address for the new app
- set the app delay **in RAM, INI will not be affected.**
- start the new app without reboot, using the right set delay
- wait some time, then get the app state and the used RAM amount
- wait 10 seconds, then stop the second app.
- set its execution flag to 1 so it will run at next boot.

```
Starting App Manager demo app. This is v1.0.14-C1 built on Sep 24 2020 12:33:25.  
There are 2 configured apps.  
Not existing app handle test (should be 0): 0x0  
Manager app handle: 0x809e20e0  
Manager app delay from nv memory: 5 seconds  
  
Changing Manager app delay time (on non volatile configuration) to 5 seconds..  
Manager app delay from nv memory is now 5 seconds  
Manager app state is M2MB_APPMNG_STATE_RUN  
  
Trying to get Second app handle...  
Second app handle is valid  
2nd app delay from nv memory is 1  
2nd app current state is M2MB_APPMNG_STATE_READY  
Second app current address is 0x41000000  
Setting volatile Second app delay (not stored in nvm) to 0 seconds...  
Starting Second app on the fly (without reboot)...  
Waiting 2 seconds...  
2nd app current state is M2MB_APPMNG_STATE_RUN  
Second app is running!  
Second App is using 475136 bytes of RAM  
Stopping Second app now...  
wait 10 seconds...  
2nd app current state is M2MB_APPMNG_STATE_STOP  
Set permanent run permission for Second app.  
Done. Second App will also run from next boot-up
```

Figure 125

3.5.5 App update OTA via FTP

Sample application showcasing Application OTA over FTP with AZX FTP. Debug prints on **USB0**

Features

- How to check module registration and activate PDP context
- How to connect to a FTP server
- How to download an application binary and update the local version

The app uses a predefined set of parameters. To load custom parameters, upload the `ota_config.txt` file (provided in project's `/src` folder) in module's `/mod` folder, for example with

```
AT#M2MWRITE="/mod/ota_config.txt",<filesize>
```

3.5.5.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage app OTA and start it

ftp_utils.c

- Set parameters to default
- Try to load parameters from `ota_config.txt` file
- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Initialize FTP client
- Connect to FTP server and log in
- Get new App binary file size on remote server
- Download the file in `/mod` folder, with the provided name
- Close FTP connection
- Disable PDP context
- Update applications configuration in **app_utils.c**

app_utils.c

- Set new application as default
- Delete old app binary
- Restart module

```
Starting FTP APP OTA demo app. This is v1.0.7 built on Apr 7 2020 17:04:05.
[DEBUG] 21.23 ftp_utils.c:447 - msgFTPTask{FTPOTA_TASK}$ INIT
[DEBUG] 21.25 ftp_utils.c:152 - readConfigFromFile{FTPOTA_TASK}$ Reading parameters from file
[DEBUG] 21.26 ftp_utils.c:154 - readConfigFromFile{FTPOTA_TASK}$ Opening /mod/ota_config.txt in read mode..
Set APN to: <<web.omnitel.it>>
Set FTP URL to: <<ftp.telit.com>>
Set FTP PORT to: 21
Set FTP USER to: <<_ _ _ _>>
Set FTP PASS to: <<_ _ _ _>>
Set FTP FILE URI to: <</samples/APP_OTA/helloworld.bin>>
Set LOCAL FINAL APP NAME to: <<helloworld.bin>>
Set LOCAL ORIGINAL APP NAME to: <<m2mapz.bin>>
[DEBUG] 23.53 ftp_utils.c:464 - msgFTPTask{FTPOTA_TASK}$ m2mb_os_ev_init success
[DEBUG] 23.54 ftp_utils.c:470 - msgFTPTask{FTPOTA_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.55 ftp_utils.c:478 - msgFTPTask{FTPOTA_TASK}$ Waiting for registration...
[DEBUG] 23.56 ftp_utils.c:371 - NetCallback{pubTspt_0}$ Module is registered to network
[DEBUG] 23.56 ftp_utils.c:491 - msgFTPTask{FTPOTA_TASK}$ Pdp context activation
[DEBUG] 23.57 ftp_utils.c:495 - msgFTPTask{FTPOTA_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 25.61 ftp_utils.c:504 - msgFTPTask{FTPOTA_TASK}$ Activate PDP with APN web.omnitel.it on cid 3....
[DEBUG] 26.30 ftp_utils.c:398 - PdpCallback{pubTspt_0}$ Context active
[DEBUG] 26.30 ftp_utils.c:401 - PdpCallback{pubTspt_0}$ IP address: 176.246.110.148
Start ftp client...
[DEBUG] 27.36 ftp_utils.c:533 - msgFTPTask{FTPOTA_TASK}$ Connected.
[DEBUG] 28.87 ftp_utils.c:546 - msgFTPTask{FTPOTA_TASK}$ FTP login successful.
Get remote file /samples/APP_OTA/helloworld.bin size
[DEBUG] 29.31 ftp_utils.c:568 - msgFTPTask{FTPOTA_TASK}$ Done. File size: 116224.
Starting download of remote file /samples/APP_OTA/helloworld.bin into local /mod/helloworld.bin
/samples/APP_OTA/helloworld.bin 4.68% 5440
/samples/APP_OTA/helloworld.bin 9.36% 10880
/samples/APP_OTA/helloworld.bin 14.04% 16320
/samples/APP_OTA/helloworld.bin 18.72% 21760
/samples/APP_OTA/helloworld.bin 23.40% 27200
/samples/APP_OTA/helloworld.bin 28.08% 32640
/samples/APP_OTA/helloworld.bin 32.76% 38080
/samples/APP_OTA/helloworld.bin 37.44% 43520
/samples/APP_OTA/helloworld.bin 42.13% 48960
/samples/APP_OTA/helloworld.bin 46.81% 54400
/samples/APP_OTA/helloworld.bin 51.49% 59840
/samples/APP_OTA/helloworld.bin 56.17% 65280
/samples/APP_OTA/helloworld.bin 60.85% 70720
/samples/APP_OTA/helloworld.bin 65.53% 76160
/samples/APP_OTA/helloworld.bin 70.21% 81600
/samples/APP_OTA/helloworld.bin 74.89% 87040
/samples/APP_OTA/helloworld.bin 79.57% 92480
/samples/APP_OTA/helloworld.bin 84.25% 97920
/samples/APP_OTA/helloworld.bin 88.93% 103360
/samples/APP_OTA/helloworld.bin 93.61% 108800
/samples/APP_OTA/helloworld.bin 97.42% 113220
[DEBUG] 43.54 ftp_utils.c:608 - msgFTPTask{FTPOTA_TASK}$ download successful.
FTP quit...
[DEBUG] 43.77 ftp_utils.c:632 - msgFTPTask{FTPOTA_TASK}$ Deactivating PDP
[DEBUG] 43.77 ftp_utils.c:642 - msgFTPTask{FTPOTA_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 44.20 ftp_utils.c:407 - PdpCallback{pubTspt_0}$ Context deactive
[DEBUG] 45.44 app_utils.c:76 - update_app{FTPOTA_TASK}$ Application successfully configured.
[DEBUG] 45.45 app_utils.c:82 - update_app{FTPOTA_TASK}$ Deleting old application /mod/m2mapz.bin
Starting. This is v1.0.7 built on Apr 7 2020 17:02:52. LEVEL: 2

Start Hello world Application [ version: 2.000000 ]

Hello world 2.0 [ 000001 ]
Hello world 2.0 [ 000002 ]
Hello world 2.0 [ 000003 ]
```

Figure 126

3.5.6 cJSON example:

Sample application showcasing how to manage JSON objects. Debug prints on **USB0**

Features

- How to read a JSON using cJSON library
- How to write a JSON
- How to manipulate JSON objects

3.5.6.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Parse an example string into a JSON object and print the result in a formatted string
- Print some test outcomes (e.g. non existing item correctly not found)
- Retrieve single elements from the parsed JSON object and use them to format a descriptive string
- Delete the JSON object
- Create a new JSON object appending elements to it
- Print the result JSON string from the object

```

Starting Logging demo app. This is v1.0.7 built on Apr  7 2020 08:33:03.
And here is what we got:
{
  "name":      "Atlantic Ocean",
  "format":    {
    "type":     "salt",
    "volume":   310410900,
    "depth":    -8486,
    "volume_percent": 23.300000,
    "tide":     -3.500000,
    "calm":     false,
    "life":     ["plankton", "corals", "fish", "mammals"]
  }
}
inexistent key not found
name found: Atlantic Ocean
format found (null)
Our JSON string contains info about an ocean named Atlantic Ocean,
has a volume of 310410900 km^3 of salt water with -8486 meters max depth,
represents 23.3% of total oceans volume,
has an average low tide of -3.5 meters,
hosts a huge number of living creatures such as plankton, corals, fish, mammals,
and is not always calm.

Let's build a TR50 command with a property.publish and an alarm.publish for MQTT (no auth).
And here is what we got:
{
  "1": {
    "command": "property.publish",
    "params": {
      "thingKey": "mything",
      "key": "mykey",
      "value": 123.144000
    }
  },
  "2": {
    "command": "alarm.publish",
    "params": {
      "thingKey": "mything",
      "key": "mykey",
      "state": 3,
      "msg": "Message."
    }
  }
}
END.

```

Figure 127

3.5.7 Crypto Elliptic Curve Cryptography (ECC) example

Sample application showcasing how to manage Elliptic Curve Cryptography functionalities. Debug prints on **USB0**

Features

- How to initialize ECC contexts A (Alice) and B (Bob). Alice is emulating a remote host, from which a public key is known.
- How to generate keypairs for contexts and export public keys
- how to export keyblobs from a context (a keyblob is encrypted with hw specific keys, and can only be used on the module where it was created)
- How to save a keyblob in secured TrustZone.
- How to reload a keyblob from the TrustZone into an initialized context
- How to sign a message with ECDSA from context B (Bob) and verify it from another context A (Alice) with the signature and public key of Bob.
- How to make Bob and Alice derive a shared session keys using each other's public key.
- How to make Bob and Alice create an AES context with the newly created shared keys, encode data and decode it on the other side

3.5.7.1 Application workflow

M2MB_main.c

- Create Bob ECC context, create a keypair and export it in a keyblob
- Open a file in secured Trust Zone, then store the keyblob in it.
- Destroy Bob ECC context
- Recreate Bob ECC context, open the file from Trust Zone and read the keyblob.
- Import the keyblob in Bob context.
- Export Bob public key
- Create Alice ECC context, to simulate an external host. Generate a keypair and export the public key.
- Sign a message with Bob context, generating a signature.
- Use Alice to verify the signed message using Bob's signature and public key
- Derive a shared key for Bob, using Alice's public key
- Create an AES context for Bob
- Import the shared key into the AES context
- Encrypt a message using Bob's AES context.

- Derive a shared key for Alice, using Bob's public key
- Create an AES context for Alice
- Import the shared key into the AES context
- Decrypt the message using Alice's AES context.
- Check the decrypted message and the original one match
- Clear all resources

```
Starting Crypto ECC demo app. This is v1.0.9-C1 built on May 11 2020 16:30:23.
Bob (local) and Alice (remote) scenario
Bob's keypair generated
Bob's keyblob length is 224
Bob exported the keyblob to be securely stored.

Bob already had an item in Secure Data Area, it was removed to create a new one
Bob securely saved the keyblob in Secure Data Area
Releasing resources

Close Bob's context...
Done. Now Bob context does not exist anymore.

Re-initialize Bob Context and load the keyblob from the secure zone
Bob securely loaded the keyblob from the SDA
Import keyblob in Bob's context..
Done. Now export Bob's public key...
Bob's public key successfully exported

Alice's keypair generated
Alice's public key successfully exported

Bob's message signed with ECDSA!
Alice verified bob's message with his pubkey and signature!

-----
Bob and Alice will now exchange a message with AES encrypt
-----

Bob retrieved the generated shared key size
Bob's shared keyblob length is: 32. Allocate the required memory to store it.
Bob created a shared key using Alice's public key!

Bob created an AEX context to exchange encrypted data with Alice
Bob's AES context imported the shared keyblob
Bob Encrypted the message using AES and the shared key!
Encrypted data:
 94EE531E3B84B2A4EF05502186BFF5DA

Alice retrieved the generated shared key size
Alice's shared keyblob length is: 32. Allocate the required memory to store it.
Alice created a shared key using Bob's public key!

Alice created an AEX context to exchange encrypted data with Bob
Alice's AES context imported the shared keyblob
Alice decrypted the message using AES and the shared key!
Decrypted:
 414094941E8942A4445548035BFAE943

Original, plain message:
 414094941E8942A4445548035BFAE943

Plain and decrypted messages match!
```

Figure 128

3.5.8 EEPROM 24AA256

Sample application showing how to communicate with a MicroChip 24AA256T I2C EEPROM chip using azx eeprom utility APIs. Debug prints on **USB0**

Setup

This demo application requires that: - A0, A1, and A2 pins (1,2,3 chip pins) are connected to ground (pin 4) for device address 0xA0 - Pin 7 (WP) is connected to ground - Pin 6 (SCL) is connected to module GPIO 3 - Pin 5 (SDA) is connected to module GPIO 2 - Pin 4 is connected to one of the ground pins of the module - Pin 8 is connected to 1v8 supply (e.g. VPWRMON pin on the module)

Features

- Initialize the logs on the output channel
- configure the EEPROM utility, setting the slave address and the memory parameters (page size, memory size)
- Write single bytes on a random address
- Read written bytes as a page
- Write data using pages
- Read the new data using pages
- Read again using sequential reading
- Read a single byte from a specific address
- Read next byte using read from current address
- Erase the EEPROM
- Deinit EEPROM utility

3.5.8.1 Application workflow

M2MB_main.c

- call `azx_eeprom_init()` to set the utility parameters (SDA and SCL pins, page and memory sizes)
- call `azx_eeprom_writeByte()` to store a single byte with value "5" at the address 0x0213
- call `azx_eeprom_writeByte()` to store a single byte with value "6" at the address 0x0214
- call `azx_eeprom_readPages()` from address 0x0213 to retrieve the 2 bytes from the EEPROM
- call `azx_eeprom_writePages` to write 1024 bytes from a buffer, starting from address 0x00
- call `azx_eeprom_readPages()` again, to read 256 bytes from address 0x00
- call `azx_eeprom_readSequentially()` to read 256 bytes from 0x00 by without pages (less overhead on I2C protocol)

- call `azx_eeprom_readByte()` to get a single byte from address 0x00
- call `azx_eeprom_readByteFromCurrentAddress()` to get a byte from next address (0x01)
- call `azx_eeprom_eraseAll()` to completely erase the EEPROM memory (this writes 0xFF in each byte)
- call `azx_eeprom_readPages` from address 0x0213 to get 2 bytes and verify the values have been written to 0xFF
- call `azx_eeprom_deinit` to close the eeprom handler and the I2C channel

```
Starting I2C EEPROM 24AA256T demo app. This is v1.0.13-C1 built on Nov  3 2020 16:28:23.
Configuring the I2C device...
Opening I2C channel /dev/I2C-160 ( device address is 0xA0 )
Writing 1 byte at address 0x0213...
Done.
Writing 1 byte at address 0x0214...
Done.
Reading the 2 bytes from address 0x0213...
Done. Data: [0xFF 0xFF]

Writing 1024 bytes at address 0x0000..
Done.

Reading 256 bytes from address 0x0000...
Done. Data:
<<ABCDEFGHIJKLMNOPQRSTUVWXYZ.....abcdefghijklmnopqrstuvwxyz.....

Reading 256 bytes sequentially from address 0x0000...
Done. Data:
<<ABCDEFGHIJKLMNOPQRSTUVWXYZ.....abcdefghijklmnopqrstuvwxyz.....

Reading 1 byte from address 0x0000...
Done. Data: 'A'

Reading 1 byte from current address (should be 0x0001)...
Done. Data: 'B'

[DEBUG] 17.47 M2MB_main:177 - run_I2C_EEPROM_Demo{M2M_DamsStart}$ Erasing all the eeprom...
[DEBUG] 28.05 M2MB_main:185 - run_I2C_EEPROM_Demo{M2M_DamsStart}$ Done

Reading the 2 bytes from address 0x0213...
Done. Data: [0xFF 0xFF]

Deinit EEPROM...
Done
```

Figure 129

3.5.9 Easy AT example

Sample application showcasing Easy AT functionalities. Debug prints on **USB0**

Features

- Shows how to register custom commands

The application adds two custom commands to the list of available ones:

- AT#MYCMD
- AT#MYINPUT

3.5.9.1 AT#MYCMD

This is a simple parameter-waiting command. It expects one string parameter, and will print it on the logging interface once received. The command simply returns OK

3.5.9.2 AT#MYINPUT

This command expects a numeric parameter, which indicates how many bytes will be received over the interface at most (the command will provide a prompt indicating it is waiting data). Then the data management callback will print when data is received, and if CTRL+Z (0x1A in hex) is received, it will complete the process, printing in the log interface what was received. sending ESC will terminate the process discarding any pending data.

3.5.10 Protect and AT command with a pwd using EasyAT functionality example

Sample application showing how to protect an AT command with a pwd using EasyAT functionality. Debug prints on **USB0**

Features

- Shows how to register a new custom command and overwrite an existing one
- AT#M2MADMIN (new command)
- AT#M2MWRITE (existing one)

Application workflow

M2MB_main.c

- Entry point that calls EasyAT initialization function

3.5.10.1 AT#M2MADMIN

This command allows to enter a pwd to lock/unlock #M2MWRITE command. In the example app the new pwd is saved in the trustzone

AT#M2MADMIN="pwd",["newpwd"]

where mode: 0 - unlock command #m2mwrite 1 - lock command #m2mwrite 2 - change pwd

```
13:40:22.898- Starting Easy AT admin demo app. This is v1.1.19 built on Jul 24 2024 13:39:26.
13:40:22.898- Easy_at_init start
13:40:22.898- [DEBUG] 20.19 at_common:375 - easy_at_init{M2M_DamsStart}$ m2mb_atp_init succeeded
13:40:22.898- Register AT#M2MWRITE
13:40:22.898- Register AT#M2MADMIN
13:40:32.444- [DEBUG] 29.90 at_common:187 - _easy_at_task_entry{M2M_AT}$ task entry, message event type: CMD
13:40:32.444- [ERROR] 29.91 at_hash_M2M:74 - M2MWRITE_AT_Callback{M2M_AT}$ Operation not allowed!
13:40:37.464- [DEBUG] 34.93 at_common:187 - _easy_at_task_entry{M2M_AT}$ task entry, message event type: CMD
13:40:37.464- [DEBUG] 34.94 at_hash_M2M:223 - M2MADMIN_AT_Callback{M2M_AT}$ admin callback
13:40:37.464- Read form secure item res: 0
13:40:37.464- parameter is <mypassword>
13:40:37.464- Operation is <0>
13:40:37.464- hash: <89E01536AC207279409D4DE1E5253E01F4A1769E696DB0D6062CA9B8F56767C8>
13:40:37.464- right password!
13:40:37.909- [DEBUG] 35.31 at_hash_M2M:309 - M2MADMIN_AT_Callback{M2M_AT}$ Release SUCCESS was OK
13:41:04.285- [DEBUG] 62.27 at_common:187 - _easy_at_task_entry{M2M_AT}$ task entry, message event type: CMD
13:41:04.285- [DEBUG] 62.28 at_hash_M2M:223 - M2MADMIN_AT_Callback{M2M_AT}$ admin callback
13:41:04.285- parameter is <mypassword>
13:41:04.285- Operation is <1>
13:41:04.285- hash: <89E01536AC207279409D4DE1E5253E01F4A1769E696DB0D6062CA9B8F56767C8>
13:41:04.285- right password!
13:41:04.285- [DEBUG] 62.30 at_hash_M2M:362 - M2MADMIN_AT_Callback{M2M_AT}$ Release SUCCESS was OK
13:41:11.611- [DEBUG] 69.69 at_common:187 - _easy_at_task_entry{M2M_AT}$ task entry, message event type: CMD
13:41:11.611- [ERROR] 69.71 at_hash_M2M:74 - M2MWRITE_AT_Callback{M2M_AT}$ Operation not allowed!
```

Figure 130

3.5.11 Events

Sample application showcasing events setup and usage. Debug prints on **USB0**

Features

- How to setup OS events with a custom bitmask
- How to wait for events and generate them in callback functions to synchronize blocks of code

3.5.11.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create an event handler
- Create a timer to generate an event, with a 2 seconds expiration time
- Wait for a specific event bit on the event handler
- At timer expiration, set the same event bit and verify that the code flow went through after the event.

```
Starting Events demo app. This is v1.0.7 built on Apr 7 2020 08:44:29.  
[DEBUG] 20.55 M2MB_main.c:171 - M2MB_main{M2M_DamsStart}$ m2mb_os_ev_init success  
Set the timer attributes structure success.  
Timer successfully created  
[DEBUG] 20.57 M2MB_main.c:125 - setup_timer{M2M_DamsStart}$ Start the timer, success.  
[DEBUG] 22.60 M2MB_main.c:60 - hwTimerCb{pubTspt_0}$ Timer Callback, generate event!  
[DEBUG] 22.61 M2MB_main.c:183 - M2MB_main{M2M_DamsStart}$ event occurred!
```

Figure 131

3.5.12 Events - Barrier (multi events)

Sample application showcasing how to setup and use multiple events to create a barrier. Debug prints on **USB0**

Features

- How to setup OS events to be used as a barrier
- How to wait for multiple events in the same point, and generate them in call-back functions to synchronize blocks of code

3.5.12.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create an event handler
- Create a timer to generate an event, with a 3 seconds expiration time
- Create another timer to generate an event, with a 6 seconds expiration time
- Start both timers
- Wait for both event bits on the event handler (each one will be set by one of the timers)
- At first timer expiration, set the first event bit and verify that the code flow does not procede.
- At second timer expiration, set the second event bit and verify that the code flow went through after the event (implementing a barrier).

```
Starting Barrier demo app. This is v1.0.7 built on Apr 7 2020 08:48:30.
[DEBUG] 20.01 M2MB_main.c:179 - M2MB_main{M2M_DamsStart}$ m2mb_os_ev_init success
Set the timer attributes structure success.
Timer successfully created with 3000 timeout (ms)
Set the timer attributes structure success.
Timer successfully created with 6000 timeout (ms)
[DEBUG] 23.08 M2MB_main.c:66 - hwTimerCb1{pubTspt_0}$ Timer Callback, generate event 1!
[DEBUG] 26.12 M2MB_main.c:75 - hwTimerCb2{pubTspt_0}$ Timer Callback, generate event 2!
[DEBUG] 26.13 M2MB_main.c:214 - M2MB_main{M2M_DamsStart}$ BOTH events occurred!
```

Figure 132

3.5.13 FOTA example

Sample application showcasing FOTA usage with M2MB API. Debug prints on **USB0**

Features

- How download a delta file from a remote server
- How to apply the delta and update the module firmware

3.5.13.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a main task to manage connectivity.
- create a fota task to manage FOTA and start it with INIT option

fota.c

fotaTask()

- Initialize FOTA system then reset parameters.
- Check current FOTA state, if not in IDLE, return error.
- Send a message to mainTask so networking is initialized.
- after PdPCallback() notifies the correct context activation, configure the fota client parameters such as FTP server URL, username and password
- get delta file from server. when it is completed, FOTADownloadCallback is called.
- If delta download went fine, check it.
- If delta file is correct, apply it. Once complete, restart the module.

mainTask()

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context. Event will be received on **PdP-Callback** function
- Disable PDP context when required to stop the app

PdpCallback()

- When PDP context is enabled, send a message to fotaTask to start the download

```

Starting FOTA demo app. This is v1.1.7 built on Jun 11 2021 12:20:43.
[DEBUG] 23.60 fota:187 - fotaTask{FOTA_TASK}$ Init FOTA...

Session file not present, procede with FOTA...
[DEBUG] 23.61 fota:236 - fotaTask{FOTA_TASK}$ m2mb_fota_reset PASS
[DEBUG] 23.61 fota:260 - fotaTask{FOTA_TASK}$ m2mb_fota_state_get M2MB_FOTA_STATE_IDLE
[DEBUG] 23.62 fota:379 - mainTask{MAIN_TASK}$ INIT
[DEBUG] 23.62 fota:392 - mainTask{MAIN_TASK}$ m2mb_os_ev_init success
[DEBUG] 23.63 fota:398 - mainTask{MAIN_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.63 fota:405 - mainTask{MAIN_TASK}$ Waiting for registration...
[DEBUG] 23.64 fota:131 - NetCallback{pubTspt_0}$ Module is registered to network
[DEBUG] 23.65 fota:418 - mainTask{MAIN_TASK}$ Pdp context initialization
[DEBUG] 25.70 fota:431 - mainTask{MAIN_TASK}$ Activate PDP with APN web.omnitel.it on cid 1....
[DEBUG] 35.42 fota:152 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 35.43 fota:155 - PdpCallback{pubTspt_0}$ IP address: 2.41.116.139

[DEBUG] 35.43 fota:285 - fotaTask{FOTA_TASK}$
Trying to download "samples/FOTA/37.00.003.3_to_37.00.003.1_ME310G1_NANVWWAU.bin" delta file...
[DEBUG] 35.45 fota:295 - fotaTask{FOTA_TASK}$ m2mb_fota_get_delta OK - Waiting for the completion callback
[DEBUG] 119.43 fota:96 - FOTADownloadCallBack{pubTspt_0}$ FOTA download Success - performing packet validation...
[DEBUG] 119.44 fota:301 - fotaTask{FOTA_TASK}$ Validating delta file...
[DEBUG] 156.36 fota:317 - fotaTask{FOTA_TASK}$ Packet is valid, start update...
[DEBUG] 156.40 fota:329 - fotaTask{FOTA_TASK}$ m2mb_fota_start PASS
[DEBUG] 158.36 fota:342 - fotaTask{FOTA_TASK}$
Rebooting...After reboot there will be the new FW running on module!

#OTAEV: Module Upgraded To New Fw
Starting FOTA demo app. This is v1.1.7 built on Jun 11 2021 12:20:43.
[DEBUG] 29.24 fota:187 - fotaTask{FOTA_TASK}$ Init FOTA...

Session file is already present, stop.

```

Figure 133

3.5.14 FOTA FTP_client example

Sample application that shows how to download a delta file from an FTP server, stores it in the FOTA partition and deploys it. Debug prints on **USB0**

Features

- How to download a delta file from FTP server using FTP client
- How to store directly delta file in the FOTA partition
- How to deploy delta file to upgrade module fw.

Application workflow

M2MB_main.c

- Print welcome message
- Create a main task to manage connectivity, delta download and deployment

ftp_test.c

msgFTPTask()

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context. Event will be received on
- Initialize FOTA system then reset parameters.
- After PDP context activation notified by PdPCallback() configure fota client parameters as FTP server url, username and password and SSL
- Get delta file from server and store it directly in the FOTA partition
- If delta download went fine, check it (m2mb_fota_update_package_check_setup) and if it's correct apply it (m2mb_fota_start).
- Once completed restart module.

PdpCallback()

- When PDP context is enabled, send a message to fotaTask to start the download

buf_data_cb_OTA()

- Handles data reception and writing in the FOTA partition (one block size at a time)


```

yStarting FOTA delta file FTP download and deploy demo app. This is v1.1.19 built on May 28 2024 14:41:14.
14:45:06.588- [DEBUG] 19.69 ftp_test:915 - msgFTPTask{FTP_TASK}$ INIT
14:45:06.588- Set APN to: <<ibox.tim.it>>
14:45:06.588- Set APN_USER to: <<>>
14:45:06.588- Set APN_PASS to: <<>>
14:45:06.588- Set PDP_CDX to: 1
14:45:06.588- Set FTP_ADDR to: <<ftp.telit.com>>
14:45:06.588- Set FTP_PORT to: 21
14:45:06.588- Set FTP_USER to: <<[REDACTED]>>
14:45:06.588- Set FTP_PASS to: <<[REDACTED]>>
14:45:06.588- Set ENABLE_TLS to: <<0>>
14:45:06.588- Set AUTH_TYPE to: <<2>>
14:45:06.588- Set CA_CERT_PATH to: <</mod>>
14:45:06.807- Set CLIENT_CERT_PATH to: <</mod>>
14:45:06.807- Set CLIENT_KEY_PATH to: <</mod>>
14:45:06.807- Set REMOTE_FOLDER to: <<robertaga>>
14:45:06.807- Set DLTOBUF_FILE to: <<delta_dummy_6_B710.bin>>
14:45:06.807- Set FOTA_STATUS_FILE to: <</mod/fota_stat.txt>>
14:45:06.807- [DEBUG] 19.78 ftp_test:932 - msgFTPTask{FTP_TASK}$ m2mb_os_ev_init success
14:45:06.807- [DEBUG] 19.79 ftp_test:938 - msgFTPTask{FTP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
14:45:06.807- [DEBUG] 19.80 ftp_test:946 - msgFTPTask{FTP_TASK}$ Waiting for registration...
14:45:06.807- [DEBUG] 19.82 ftp_test:840 - NetCallback{pubTspt_0}$ Module is registered to network
14:45:06.807-
14:45:06.807- Check FOTA upgrade status in file system
14:45:06.807-
14:45:06.807- Module current fw version is:
14:45:06.807- 37.00.216-B010-POC.210000
14:45:06.807- MOC.200006-B010
14:45:06.807- POC.210000
14:45:06.807- AOC.210000
14:45:07.248- [WARN ] 20.37 ftp_test:334 - FOTAStatusInit{FTP_TASK}$ File doesn't exist create it, first app execution
14:45:07.248- File created, store current fw version and fota upgrade flag=0
14:45:08.315-
14:45:08.315- Read stored data
14:45:08.315-
14:45:08.315- FOTA Flag: 0 Module previous fw version:
14:45:08.315- 37.00.216-B010-POC.210000
14:45:08.315- MOC.200006-B010
14:45:08.315- POC.210000
14:45:08.315- AOC.210000
14:45:08.315- Fw to be upgraded...
14:45:08.315- [DEBUG] 21.44 ftp_test:1009 - msgFTPTask{FTP_TASK}$ Pdp context activation
14:45:08.315- [DEBUG] 21.45 ftp_test:1013 - msgFTPTask{FTP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
14:45:10.426- [DEBUG] 23.50 ftp_test:1022 - msgFTPTask{FTP_TASK}$ Activate PDP with APN ibox.tim.it on cid 1....
14:45:10.855- [DEBUG] 24.05 ftp_test:867 - PdpCallback{pubTspt_0}$ Context active
14:45:10.855- [DEBUG] 24.07 ftp_test:870 - PdpCallback{pubTspt_0}$ IP address: 2.193.245.246
14:45:10.855- Start ftp client...
14:45:10.855- Connecting to host: <ftp.telit.com>
14:45:11.294- Retrieved address: 185.86.42.214
14:45:11.294- socket 0x4003d2a4 ok

```

Figure 134

3.5.15 FOTA from Local File example

Sample application that shows how perform FOTA upgrade using a delta file stored into file system. Debug prints on **USB0**

Features

- How to store and get FOTA upgrade information to/from a file
- How to get delta file from module file system
- How to apply the delta and update module firmware

Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Check if module has been already upgraded or needs to be upgraded reading FOTA upgrade status from a file
- Create a fota task to manage FOTA and start it with INIT option

smartFotaTask() - Initialize FOTA system then reset parameters. - Get FOTA partition size and block size - Copy delta file from file system to FOTA partition. when it is completed, FOTADownloadCallback is called. - If delta file is correct, apply it. Once complete, write FOTA status flag and current fw version to a file, restart the module.

```
Starting FOTA application from local delta file demo app. This is v1.0.14-C1 built on May 16 2022 15:50:38.

Check FOTA upgrade status in file system

Module current fw version is:
37.00.214-POC.210001
MOC.200003
POC.210001
AOC.210000
[WARN ] 13.83 M2MB_main:190 - FOTAUpgradeStatusInit{M2M_DamsStart}$ File doesn't exist create it, first app execution
File created, store current fw version and fota upgrade flag=0

Read stored data

FOTA Flag: 0 Module previous fw version:
37.00.214-POC.210001
MOC.200003
POC.210001
AOC.210000

Start FOTA process
[DEBUG] 15.42 M2MB_main:450 - smartFotaTask{FOTA_TASK}$ INIT
[DEBUG] 15.43 M2MB_main:461 - smartFotaTask{FOTA_TASK}$ m2mb_fota_init success
[DEBUG] 15.44 M2MB_main:464 - smartFotaTask{FOTA_TASK}$
Get block and FOTA partition size
OTA blockSize: 131072
OTA partitionSize: 11272192
[DEBUG] 15.47 M2MB_main:472 - smartFotaTask{FOTA_TASK}$
Copy delta file from File system to FOTA partition
[DEBUG] 15.47 M2MB_main:277 - copyFromFileSystemToDelta{FOTA_TASK}$ File size: 746552
[DEBUG] 15.87 M2MB_main:480 - smartFotaTask{FOTA_TASK}$ Delta file writing completed
[DEBUG] 15.88 M2MB_main:484 - smartFotaTask{FOTA_TASK}$
Delta file check...
[DEBUG] 15.89 M2MB_main:347 - check_fota_delta{FOTA_TASK}$ -- check_fota_delta...
[DEBUG] 54.73 M2MB_main:366 - check_fota_delta{FOTA_TASK}$ Fota check integrity PASS
[DEBUG] 54.74 M2MB_main:492 - smartFotaTask{FOTA_TASK}$ ...delta file OK
[DEBUG] 54.75 M2MB_main:496 - smartFotaTask{FOTA_TASK}$
--> Start update...
[DEBUG] 56.83 M2MB_main:403 - startUpdate{FOTA_TASK}$
Reboot module to start delta deployment
```

Figure 135

3.5.16 FTP

Sample application showcasing FTP client demo with AZX FTP. Debug prints on **USB0**

Features

- How to check module registration and activate PDP context
- How to connect to a FTP server
- How to exchange data with the server

3.5.16.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage FTP client and start it

ftp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Init FTP client and set the debug function for it
- Connect to the server
- Perform log in
- Check remote file size and last modification time
- Download file from server to local filesystem. A data callback is set to report periodic info about the download status
- Upload the same file to the server with a different name. A data callback is set to report periodic info about the upload status
- Download another file content in a buffer instead of a file. A data callback is set to report periodic info about the download status
- Close the connection with FTP server
- Disable PDP context

```

Starting FTP demo app. This is v1.0.7 built on Apr 7 2020 11:17:36.
[DEBUG] 21.23 ftp_test.c:290 - msgFTPTask{FTP_TASK}$ INIT
[DEBUG] 21.23 ftp_test.c:304 - msgFTPTask{FTP_TASK}$ m2mb_os_ev_init success
[DEBUG] 21.23 ftp_test.c:310 - msgFTPTask{FTP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 21.23 ftp_test.c:318 - msgFTPTask{FTP_TASK}$ Waiting for registration...
[DEBUG] 21.25 ftp_test.c:214 - NetCallback{pubTspt_0}$ Module is registered to network
[DEBUG] 21.26 ftp_test.c:331 - msgFTPTask{FTP_TASK}$ Pdp context activation
[DEBUG] 21.27 ftp_test.c:335 - msgFTPTask{FTP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.31 ftp_test.c:344 - msgFTPTask{FTP_TASK}$ Activate PDP with APN web.omnitel.it on cid 3...
[DEBUG] 24.09 ftp_test.c:241 - PdpCallback{pubTspt_0}$ Context active
[DEBUG] 24.10 ftp_test.c:244 - PdpCallback{pubTspt_0}$ IP address: 176.244.166.181
Start ftp client...
[DEBUG] 24.82 ftp_test.c:373 - msgFTPTask{FTP_TASK}$ Connected.
[DEBUG] 26.32 ftp_test.c:386 - msgFTPTask{FTP_TASK}$ FTP login successful.
Get remote file /samples/pattern_big.txt size
[DEBUG] 26.69 ftp_test.c:428 - msgFTPTask{FTP_TASK}$ Done. File size: 20026.
Get remote file /samples/pattern_big.txt last modification date
[DEBUG] 26.89 ftp_test.c:450 - msgFTPTask{FTP_TASK}$ Done. File last mod date: 20200407090654
.

Starting download of remote file /samples/pattern_big.txt into local /mod/_pattern_big.txt
/samples/pattern_big.txt 47.54% 9520
/samples/pattern_big.txt 100.00% 20026
[DEBUG] 29.75 ftp_test.c:488 - msgFTPTask{FTP_TASK}$ download successful.
[DEBUG] 29.76 ftp_test.c:522 - msgFTPTask{FTP_TASK}$
Local file /mod/_pattern_big.txt size: 20026

Starting upload of local file /mod/_pattern_big.txt
/mod/_pattern_big.txt 81.81% 16384
Upload successful.

Starting download of remote file /samples/pattern.txt into local buffer
Getting remote file /samples/pattern.txt size..
[DEBUG] 32.97 ftp_test.c:583 - msgFTPTask{FTP_TASK}$ Done. File size: 988.
Starting download of remote file /samples/pattern.txt to buffer
[DEBUG] 34.08 ftp_test.c:145 - buf_data_cb{FTP_TASK}$ Received START event
[DEBUG] 34.09 ftp_test.c:149 - buf_data_cb{FTP_TASK}$ Received DATA: 988 bytes on buffer 0x400399e0
[DEBUG] 34.26 ftp_test.c:153 - buf_data_cb{FTP_TASK}$ Received END event
[DEBUG] 34.26 ftp_test.c:623 - msgFTPTask{FTP_TASK}$ Download successful. Received 988 bytes<<<
0 |-----| |-----| |-----| |-----| |-----| *
1 | A | | A | | A | | A | | A | | *
2 | AAA | | AAA | | AAA | | AAA | | AAA | | *
3 | AAAAA | | AAAAA | | AAAAA | | AAAAA | | AAAAA | | *
4 | AAAAAAA | | AAAAAAA | | AAAAAAA | | AAAAAAA | | AAAAAAA | | *
5 | AAAAAAAA | | AAAAAAAA | | AAAAAAAA | | AAAAAAAA | | AAAAAAAA | | *
6 | AAAAAAA | | AAAAAAA | | AAAAAAA | | AAAAAAA | | AAAAAAA | | *
7 | AAAAA | | AAAAA | | AAAAA | | AAAAA | | AAAAA | | *
8 | AAA | | AAA | | AAA | | AAA | | AAA | | *
9 | A | | A | | A | | A | | A | | *
10 |-----| |-----| |-----| |-----| |-----| *
11 | | | | | | | *
12 |-----| |-----| |-----| |-----| |-----|

```

Figure 136

3.5.17 File System example

Sample application showcasing M2MB File system API usage. Debug prints on **USB0**

Features

- How to open a file in write mode and write data in it
- How to reopen the file in read mode and read data from it

3.5.17.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Open file in write mode
- Write data in file
- Close file
- Reopen file in read mode
- Read data from file and print it
- Close file and delete it

```
Starting FileSystem demo app. This is v1.0.7 build on Mar 26 2020 09:50:19. LEVEL: 2
Opening /my_text_file.txt in write mode..
Buffer written successfully into file. 15 bytes were written.
Closing file.
Opening /my_text_file.txt in read only mode..
Received 15 bytes from file:
<Hello from file>
Closing file.
Deleting File
File deleted
App Completed
```

Figure 137

3.5.18 GNSS example

Sample application showing how to use GNSS functionality. Debug prints on **USB0**

Features

- How to enable GNSS receiver on module
- How to collect location information from receiver

Note: on MEx10G1 product family both M2MB_GNSS_SERVICE_NMEA_REPORT and M2MB_GNSS_SERVICE_POSITION_REPORT services are available, while on ME910C1 product family only M2MB_GNSS_SERVICE_POSITION_REPORT is available

3.5.18.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print a welcome message
- Create GNSS task and send a message to it

gps_task.c - Init Info feature and get module type - Init gnss, enable position/NMEA report and start it. - When a fix or a NMEA sentence is available, a message will be printed by the GNSS callback function

```
Starting GNSS demo app. This is v1.1.4 built on Oct  1 2021 15:27:44.
Model: ME910C1-E2
m2mb_gnss_enable, POSITION OK
m2mb_gnss_start OK, waiting for position/nmea sentences...
latitude_valid: 1 - latitude: 45.713643
longitude_valid: 1 - longitude: 13.738041
altitude_valid: 1 - altitude: 195.000000
uncertainty_valid: 1 - uncertainty: 95.000000
velocity_valid: 1 - codingType: 0
speed_horizontal: 0.650000
bearing: 0.000000
timestamp_valid: 1 -timestamp: 1633095357439
speed_valid: 1 - speed: 1.471360

***** Wait 120 seconds and then stop GPS *****
```

Figure 138

3.5.19 GPIO interrupt example

Sample application showing how to use GPIOs and interrupts. Debug prints on **USB0**

Features

- How to open a GPIO in input mode with interrupt
- How to open a second GPIO in output mode to trigger the first one

3.5.19.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Open GPIO 4 as output
- Open GPIO 3 as input and set interrupt for any edge (rising and falling). **A jumper must be used to short GPIO 3 and 4 pins.**
- Toggle GPIO 4 status high and low every second
- An interrupt is generated on GPIO 3

```
Starting GPIO interrupt demo app. This is v1.0.7 built on Mar 26 2020 16:33:01.  
Setting gpio 3 interrupt...  
Setting GPIO 4 HIGH  
CALLBACK->Interrupt on GPIO 3! Value: 1  
Setting GPIO 4 LOW  
CALLBACK->Interrupt on GPIO 3! Value: 0  
Setting GPIO 4 HIGH  
CALLBACK->Interrupt on GPIO 3! Value: 1  
Setting GPIO 4 LOW  
CALLBACK->Interrupt on GPIO 3! Value: 0  
Setting GPIO 4 HIGH  
CALLBACK->Interrupt on GPIO 3! Value: 1  
Setting GPIO 4 LOW  
CALLBACK->Interrupt on GPIO 3! Value: 0
```

Figure 139

3.5.20 GTP example

Sample application that shows how to get the position using GTP feature. Debug prints on **USB0**

Features

- How to init and enable GTP feature
- How to get the position using GTP

Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Init NET functionality and wait for module to be registered
- Init PDP functionality and set APN on CID1
- Init GTP functionality
- Check if GTP is already enabled. If not enable it and reboot module
- If GTP is enabled, get position

```
Starting GTP demo app. This is v1.1.8 built on Dec 20 2022 15:29:19.
[DEBUG] 15.08 M2MB_main:237 - M2MB_main{M2M_DamsStart}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 15.09 M2MB_main:245 - M2MB_main{M2M_DamsStart}$ Waiting for registration...
[DEBUG] 15.10 M2MB_main:253 - M2MB_main{M2M_DamsStart}$ Pdp context setting
[DEBUG] 15.11 M2MB_main:82 - NetCallback{pubTspt_0}$ Module is registered to cell 0x201
[DEBUG] 15.11 M2MB_main:257 - M2MB_main{M2M_DamsStart}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 15.13 M2MB_main:268 - M2MB_main{M2M_DamsStart}$ m2mb_pdp_deinit returned M2MB_RESULT_SUCCESS

Check if GTP has been already enabled
[DEBUG] 16.19 M2MB_main:312 - M2MB_main{M2M_DamsStart}$ GTP status: 1 => Enabled
[DEBUG] 16.20 M2MB_main:317 - M2MB_main{M2M_DamsStart}$ Get the position...

m2mb_gnss_GTP OK, waiting for position...
[DEBUG] 107.96 M2MB_main:160 - gnssCallbackFN{pubTspt_0}$ gnssCallback event: 1
GTP position got
latitude: 41.900002
longitude: 12.500000
altitude: 0.000000
altitudeMeanSeaLevel: -0.000000
accuracy: 0.000000
speed: -0.000000
bearing: -0.000000
timestamp: 0
verticalAccuracy 0.000000
speedAccuracy -0.000000
bearingAccuracy: -0.000000
```

Figure 140

3.5.21 General_INFO example

Sample application prints some Module/SIM information as IMEI, fw version, IMSI and so on; it prints also some information about registration. Debug prints on **USB0**

Features

- How to print some Module information as IMEI, FW version etc
- How to print some SIM information as IMSI, ICCID
- How to get and print some informatio about Module registration as Netowrk Operator, AcT, RSSI, etc

3.5.21.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Init NET functionality
- Init INFO functionality
- Get and print Module and SIM info
- Wait form module to register to network
- Get and print registration INFO

```

Starting. This is v1.1.4 built on Mar 31 2021 09:56:03. LEVEL: 2

Start General INFO application [ version: 1.000000 ]

=====
MODULE ME910C1-E2 INFO
=====
MANUFACTURER: Telit
IMEI: 353080091125422
MODEM FIRMWARE VERSION: MOB.700005
PACKAGE VERSION:
30.00.709-B005-P0B.700100
MOB.700005
P0B.700100
A0B.700000

=====
SIM INFO
=====
IMSI: 222015602268648
ICCID: 89390100001138084906

=====
Waiting for registration...

=====
Module is registered to HOME network cellID 0x5221
NETWORK OPERATOR (mcc mnc): 222 01
Network Technology 2G (AcT: 0) RSSI: -81

```

Figure 141

3.5.22 HTTP Client

Sample application showing how to use HTTPs client functionalities. Debug prints on **USB0**

Features

- How to check module registration and activate PDP context
- How to initialize the http client, set the debug hook function and the data callback to manage incoming data
- How to perform GET, HEAD or POST operations (GET also with single range support)

NOTE: the sample app has an optional dependency on `azx_base64.h` if basic authentication is required (refer to `HTTP_BASIC_AUTH_GET` define in `M2MB_main.c` for further details)

3.5.22.1 Application workflow

`M2MB_main.c`

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage HTTP client and start it

`httpTaskCB`

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create HTTP client options and initialize its functionality
- Create HTTP SSL config and initialize the SSL options
- Configure data management options for HTTP client
- Apply all configurations to HTTP client
- Perform a GET request to a server
- Disable PDP context

`DATA_CB`

- Print incoming data
- Set the abort flag to 0 to keep going.

```
Starting HTTP(s) client demo app. This is v1.1.15 built on Nov 28 2022 12:12:27.
[DEBUG] 17.12 read_param:101 - readConfigFromFile(HttpClient)$ Reading parameters from file
[DEBUG] 17.13 read_param:103 - readConfigFromFile(HttpClient)$ Opening /mod/HTTP_Client_config.txt in read mode..
Set APN to: <web.omnitel.it>
Set APN USER to: <>
Set APN PASS to: <>
Set CID to: 3
Set CACERTFILE to: <>
Set CLIENTCERTFILE to: <>
Set CLIENTKEYFILE to: <>
Set REQUEST_TYPE to: 2
Set SERVER to: <http://linux-ip.net>
[DEBUG] 17.16 m2mb_HTTP_t:267 - activatePdp(HttpClient)$ m2mb_os_ev_init success
[DEBUG] 17.17 m2mb_HTTP_t:273 - activatePdp(HttpClient)$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 17.18 m2mb_HTTP_t:281 - activatePdp(HttpClient)$ Waiting for registration..
[DEBUG] 17.19 m2mb_HTTP_t:101 - NetCallback{pubTspt_0}$ Module is registered to cell 0x468E!
[DEBUG] 17.20 m2mb_HTTP_t:295 - activatePdp(HttpClient)$ Pdp context initialization
[DEBUG] 19.26 m2mb_HTTP_t:305 - activatePdp(HttpClient)$ Activate PDP with APN web.omnitel.it....
[DEBUG] 20.16 m2mb_HTTP_t:128 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 20.18 m2mb_HTTP_t:131 - PdpCallback{pubTspt_0}$ IP address: 176.243.212.114
Performing a GET request...
Connecting to linux-ip.net:80//
Socket connected!
<!DOCTYPE html>
<html>
<head>
  <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
  <meta name="author" content="Martin A. Brown" />
  <meta name="robots" content="index, follow"/>

  <meta property="og:title" content="http://linux-ip.net/" />
  <meta property="og:url" content="http://linux-ip.net/" />
  <meta property="og:site_name" content="http://linux-ip.net/" />
  <meta property="og:type" content="website"/>

  <link rel="canonical" href="http://linux-ip.net" />

  <title>http://linux-ip.net/</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <link rel="stylesheet" type="text/css" href="//netdna.bootstrapcdn.com/font-awesome/4.0.3/css/font-awesome.css" />
  <link rel="stylesheet" type="text/css" href="//netdna.bootstrapcdn.com/twitter-bootstrap/2.3.2/css/bootstrap-combined.min.css" />
  <link rel="stylesheet" type="text/css" href="http://linux-ip.net/theme/css/main.css" />
</head>
<body>
  <div id="site-footer">
    <div class="row-fluid">
      <div class="span10 offset1">
        <address>
          <p>
            Powered by <a href="http://getpelican.com/">Pelican</a>
            and <a href="http://python.org">Python</a>.
            Theme based on <a href="http://github.com/jsliang/pelican-fresh">Fresh</a>
            by <a href="http://jsliang.com/">jsliang</a>
          </p>
        </address>
      </div>
    </div>
  </div>
</body>
</html>
Result: 200
[DEBUG] 26.80 m2mb_HTTP_t:137 - PdpCallback{pubTspt_0}$ Context deactivated!
```

Figure 142

3.5.23 HW Timer (Hardware Timer)

The sample application shows how to use HW Timers M2MB API. Debug prints on **USB0**

Features

- How to open configure a HW timer
- How to use the timer to manage recurring events

3.5.23.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create hw timer structure
- Configure it with 100 ms timeout, periodic timer (auto fires when expires) and autostart
- Init the timer with the parameters
- Wait 10 seconds
- Stop the timer

TimerCb

- Print a message with an increasing counter

```
Starting HW Timers demo app. This is v1.0.7 built on Mar 26 2020 13:04:14.
[DEBUG] 14.06 M2MB_main.c:114 - M2MB_main{M2M_DamsStart}$ Set the timer attributes structure: success.
Timer successfully created
Start the timer, success.
[DEBUG] 14.18 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [0]
[DEBUG] 14.28 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [1]
[DEBUG] 14.38 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [2]
[DEBUG] 14.48 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [3]
[DEBUG] 14.58 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [4]
[DEBUG] 14.69 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [5]
[DEBUG] 14.79 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [6]
[DEBUG] 14.88 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [7]
[DEBUG] 14.98 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [8]
[DEBUG] 15.08 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [9]

[DEBUG] 23.90 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [96]
[DEBUG] 24.01 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [97]
[DEBUG] 24.11 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [98]
Stop a running timer: success
Application end
```

Figure 143

3.5.24 Hello World

The application prints “Hello World!” over selected output every two seconds. Debug prints on **USB0**, using AZX log example functions

Features

- How to open an output channel using AZX LOG sample functions
- How to print logging information on the channel using AZX LOG sample functions

3.5.24.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print “Hello World!” every 2 seconds in a while loop

```
Starting. This is v1.0.7 built on Mar 26 2020 09:34:16. LEVEL: 2
Start Hello world Application [ version: 2.000000 ]
Hello world 2.0 [ 000001 ]
Hello world 2.0 [ 000002 ]
Hello world 2.0 [ 000003 ]
Hello world 2.0 [ 000004 ]
Hello world 2.0 [ 000005 ]
Hello world 2.0 [ 000006 ]
Hello world 2.0 [ 000007 ]
Hello world 2.0 [ 000008 ]
Hello world 2.0 [ 000009 ]
```

Figure 144

3.5.25 I2C example

Sample application showing how to communicate with an I2C slave device. Debug prints on **USB0**

Features

- How to open a communication channel with an I2C slave device
- How to send and receive data to/from the slave device

Setup

- Connect sensor VDD to 1v8 supply (e.g. Vaux/PwrMon pin of the module)
- Connect sensor GND to a GND pin of the module
- Connect sensor SDA to module GPIO2
- Connect sensor SCL to module GPIO3

3.5.25.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Open I2C bus, setting SDA and SCL pins as 2 and 3 respectively
- Set registers to configure accelerometer -Read in a loop the 6 registers carrying the 3 axes values and show the g value for each of them

```
Starting I2C demo app. This is v1.0.7 built on Mar 26 2020 16:50:40.
Configuring the Kionix device...
opening channel /dev/I2C-30
[DEBUG] 20.18 M2MB_main.c:218 - test_I2C{M2M_DamsStart}$|-
WHOAMI content: 0x01
Configuring I2C Registers - Writing 0x4D into 0x1D register (CTRL_REG3)...
Write: success

I2C reading data from 0x1D register (CTRL_REG3)...
Read: success.
Accelerometer Enabled. ODR tilt: 12.5Hz, ODR directional tap: 400Hz, ORD Motion Wakeup: 50Hz
Configuring I2C Registers - Writing 0xC0 into 0x1B register (CTRL_REG1)...
Write: success

I2C reading data from 0x1B register (CTRL_REG1)...
Read: success.
Accelerometer Enabled. Operative mode, 12bit resolution
I2C read axes registers
-----
Reading Success.

X: -0.050 g
Y: -0.046 g
Z: 1.006 g
Reading Success.

X: -0.049 g
Y: -0.044 g
Z: 1.004 g
Reading Success.

X: -0.052 g
Y: -0.044 g
Z: 1.007 g
Reading Success.

X: -0.048 g
Y: -0.045 g
Z: 1.005 g
```

Figure 145

3.5.26 I2C Combined

Sample application showing how to communicate with an I2C slave device with I2C raw mode. Debug prints on MAIN UART

Features

- How to open a communication channel with an I2C slave device
- How to send and receive data to/from the slave device using raw mode API

Setup

- Connect sensor VDD to 1v8 supply (e.g. Vaux/PwrMon pin of the module)
- Connect sensor GND to a GND pin of the module
- Connect sensor SDA to module GPIO2
- Connect sensor SCL to module GPIO3

3.5.26.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Open I2C bus, setting SDA and SCL pins as 2 and 3 respectively
- Set registers to configure accelerometer - Read in a loop the 6 registers carrying the 3 axes values and show the g value for each of them

```
Starting I2C raw demo app. This is v1.0.13-C1 built on Jul 30 2020 11:28:18.
Configuring the I2C device...
Opening I2C channel /dev/I2C-30 ( device address is 0x0F << 1 )
Accelerometer Enabled. ODR tilt: 12.5Hz, ODR directional tap: 400Hz, ORD Motion Wakeup: 50Hz
Accelerometer Enabled. Operative mode, 12bit resolution
I2C read axes registers
-----
X: 0.000 g
Y: 0.000 g
Z: 0.000 g

X: -0.270 g
Y: 0.016 g
Z: 0.917 g

X: -0.268 g
Y: 0.013 g
Z: 0.925 g

X: -0.271 g
Y: 0.015 g
Z: 0.922 g

X: -0.267 g
Y: 0.016 g
Z: 0.918 g

X: -0.274 g
Y: 0.019 g
Z: 0.915 g
```

Figure 146

3.5.27 Little FileSystem 2

Sample application showing how use lfs2 porting with RAM disk and SPI data flash.
Debug prints on **USB0**

Features

- How to create and manage Ram Disk
- How to manage file-system in Ram disk partition
- How to create and manage SPI Flash memory partition
- How to manage file-system in SPI Flash memory partition

3.5.27.1 Application workflow

M2MB_main.c

- Init logging system
- Call Ram Disk tests
- Call Flash memory tests

ram_utils_usage.c

- Initialize Ram Disk
- Format and Mount partition
- List files
- Files creation and write content
- List files
- Read files
- Unmount and Release resources

spi_utils_usage.c - Initialize SPI Flash chip - Initialize SPI Flash Disk - Format and Mount partition - List files - Files creation and write content - List files - Read files - Delete files - Directories creation and deletion - Unmount and Release resources

Notes:

For SPI Flash a JSC memory is used with chip select pin connected to module GPIO2 pin. For better performances, a 33kOhm pull-down resistor on SPI clock is suggested. Please refer to SPI_echo sample app for SPI connection details.

For LE910Cx (both Linux and ThreadX based devices), AT#SPIEN=1 command must be sent once before running the app

```

Starting lfs2 demo app. This is v1.0.14-C1 built on Oct 22 2020 09:43:08.
>>>>>> Starting RAMDiskDemo ...
[DEBUG] 18.28 azx_lfs_uti:125 - azx_ram_initialize{M2M_DamsStart}$ Ram Memory allocated correctly from 0x40042228 to 0x40046228!!
Mounting partition...
Formatting...
Mounting...

Mounted partition...
<<<<<<fileListUtils
List:
.., 0, 2
.., 0, 2
file_name: file000.txt
size: 10
buffer: content000
mode: 0
RAM TYPE size: 10000

File created and closed: file000.txt

<<<<<<fileListUtils
___INSIDE --->file000.txt, 10, 1
List:
.., 0, 2
.., 0, 2
file000.txt, 10, 1
----->File reading
File: file000.txt, Size: 10, Buffer: content000
Nand released
Partition unmounted
[DEBUG] 20.31 azx_lfs_uti:165 - azx_ram_releaseResources{M2M_DamsStart}$ Ram Memory released correctly!!
>>>>>>> Starting FlashDiskDemo ...
Starting initialization...

table id[0] = 191
table id[1] = 1
table id[2] = 0

nandLFS_CALLBACK Callback event <1>
NAND Callback event: NAND_JSC_INITIALIZED <1>
nandLFS_CALLBACK Callback event <1>
NAND Callback event: NAND_JSC_INITIALIZED <1>

Mounting partition...
Formatting...
spiErase: address = 0, len = 131072
spiErase: address = 131072, len = 131072

Mounting...

Mounted partition...

<<<<<<fileListUtils
List:
.., 0, 2
.., 0, 2

Formatting...
spiErase: address = 0, len = 131072
spiErase: address = 131072, len = 131072

Mounting...

Mounted partition...

<<<<<<fileListUtils
|
List:
.., 0, 2
.., 0, 2
file_name: file000.txt
size: 10
buffer: content000
mode: 0

File created and closed: file000.txt

```

```

<><><>fileListUtils
List:
., 0, 2
., 0, 2
file000.txt, 10, 1
file001.txt, 10, 1
file002.txt, 10, 1
file003.txt, 10, 1
file004.txt, 10, 1
----->File reading
File: file000.txt, Size: 10, Buffer: content000

File: file004.txt, Size: 10, Buffer: content004

File: file002.txt, Size: 10, Buffer: content002
----->File removing
file001.txt<<<<<<<

File removed: file001.txt|
file000.txt<<<<<<<

File removed: file000.txt
file004.txt<<<<<<<

File removed: file004.txt

<><><><>fileListUtils
List:
., 0, 2
., 0, 2
file002.txt, 10, 1
file003.txt, 10, 1
spiErase: address = 59637760, len = 131072
[DEBUG] 58.61 azx_lfs_uti:648 - azx_lfsDirCreationByContext{M2M_DamsStart}$ Directory created: dir000!!
[DEBUG] 59.78 azx_lfs_uti:631 - azx_lfsDirCreationByContext{M2M_DamsStart}$ Directory already exists: dir000!!
spiErase: address = 59899904, len = 131072
[DEBUG] 61.70 azx_lfs_uti:648 - azx_lfsDirCreationByContext{M2M_DamsStart}$ Directory created: dir001!!
spiErase: address = 60162048, len = 131072
[DEBUG] 63.67 azx_lfs_uti:648 - azx_lfsDirCreationByContext{M2M_DamsStart}$ Directory created: dir002!!

<><><><>fileListUtils
List:
., 0, 2
., 0, 2
dir000, 0, 2
dir001, 0, 2
dir002, 0, 2
file002.txt, 10, 1
file003.txt, 10, 1

<><><><>fileListUtils
List:
., 0, 2|
., 0, 2
dir001, 0, 2
dir002, 0, 2
file002.txt, 10, 1
file003.txt, 10, 1
Nand released
Partition unmounted
Unmounted process ended...
testAllInOneFunction ended...

```

3.5.28 LWM2M

Sample application showcasing LWM2M client usage with M2MB API. Debug prints on **USB0**

Features

- Configure LWM2M client and enable it
- Create an instance of a custom object
- Set an integer value on a read only resource
- Set two integer values on a multi-instance read only resource
- write a string on a read/write resource
- Manage exec requests from the portal
- Manage write, read and monitoring resources changed from the portal

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6", "<user apn>"
```

Depending on the Mobiler Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.5.28.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask - Check registration status

- Initialize LWM2M client,
- Check for XML file fo custom object
- Enable unsolicited messages from client
- Create a task (lwm2m_taskCB is its callback function)to manage events from Portal
- Enable Lwm2M client

- Create a new instance for the custom object
- Wait for client to register to Portal
- Performs all operations (set, read, get, write) on the related resources
- Performs a set with notify ack enabled
- Wait for events from server

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)
- Run lwm2m_taskCB when a monitored resource changes, to manage the action to be done

3.5.28.2 Device Profile upload

LWM2M resources demo device profile must be imported to have a real-time update of resources values on the LWM2M browser.

To do so, import the file `json/lwm2m_resources_demo.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

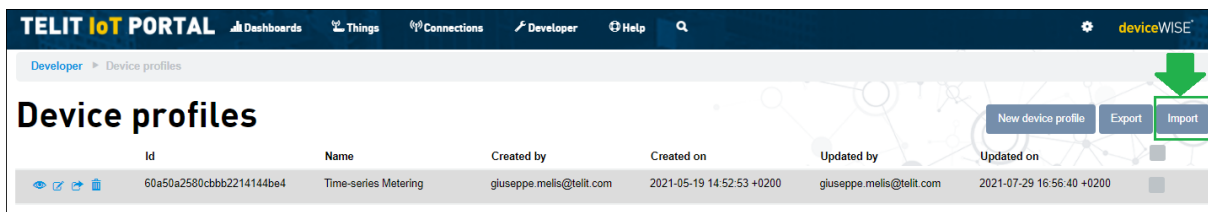


Figure 147

3.5.28.3 Custom Object configuration

The XML file content must be loaded on the Telit IoT Portal for the demo application to be fully executed.

First, enter Developer section from the top menu



Figure 148

Choose Object Registry

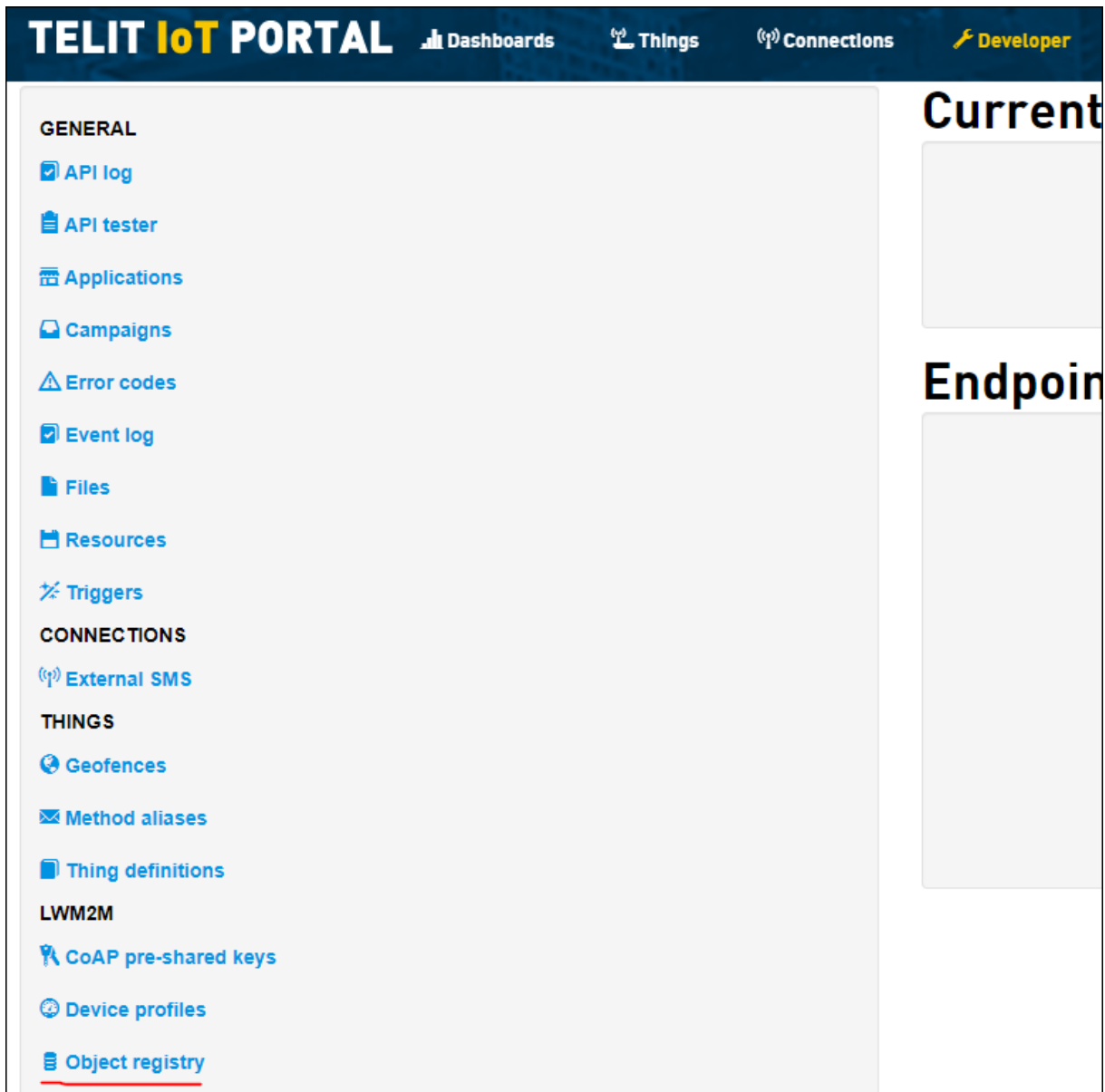


Figure 149

Create a New Object

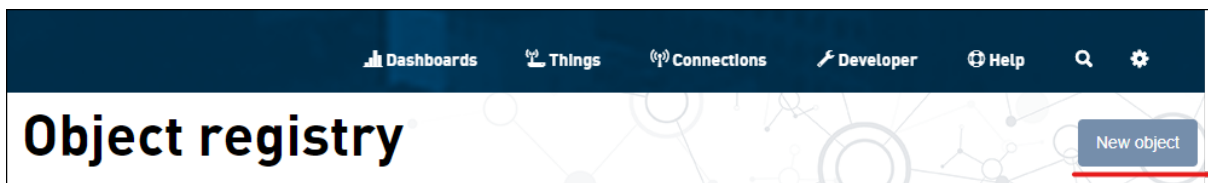
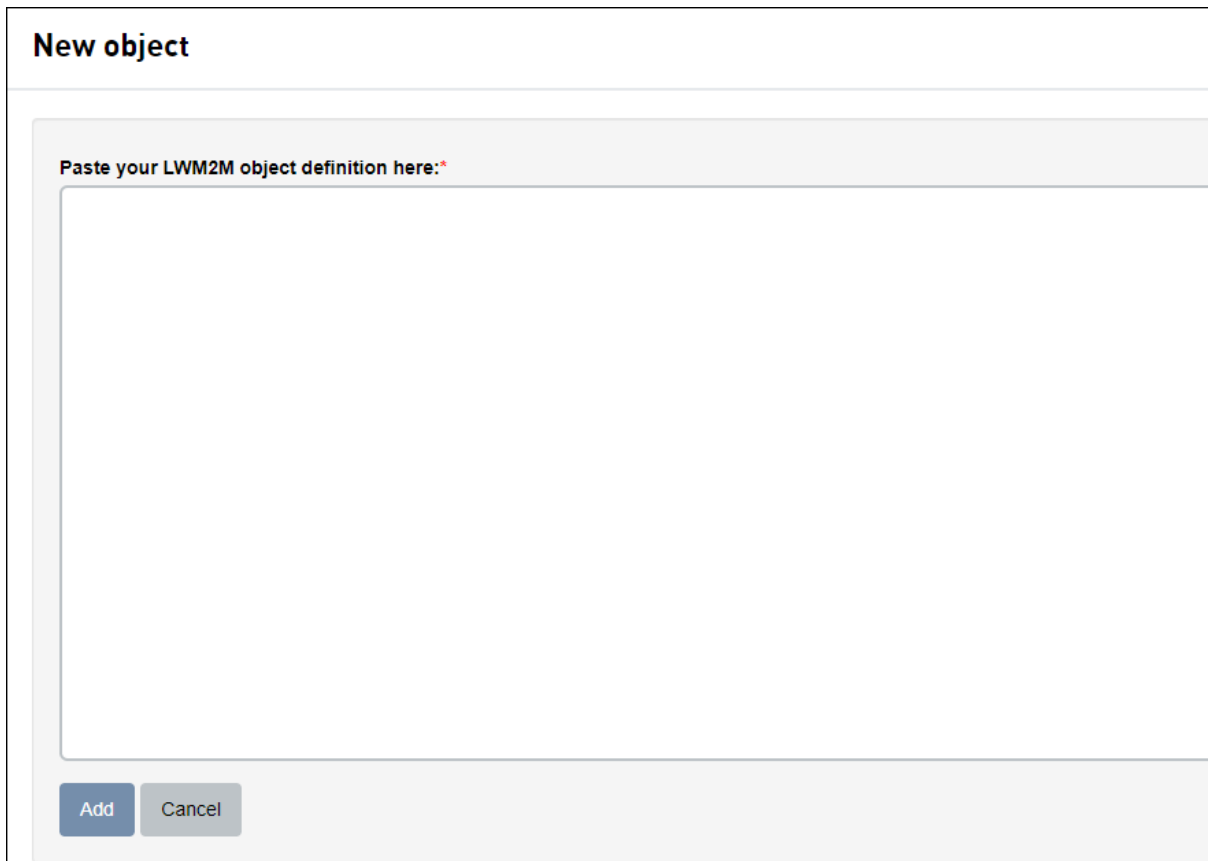


Figure 150

Copy the xml file content and paste it in the new Object form



New object

Paste your LWM2M object definition here:*

Add Cancel

Figure 151

Also, the application requires the XML file `xml/object_32010.xml` (provided with the sample files) to be stored in module's `/XML/` folder. It can be done with

```
AT#M2MWRITE=/XML/object_32010.xml,<size_in_bytes>
```

To load the XML file in the module, Telit AT Controller (TATC) can be used. Once the command above is issued, press the load content button:

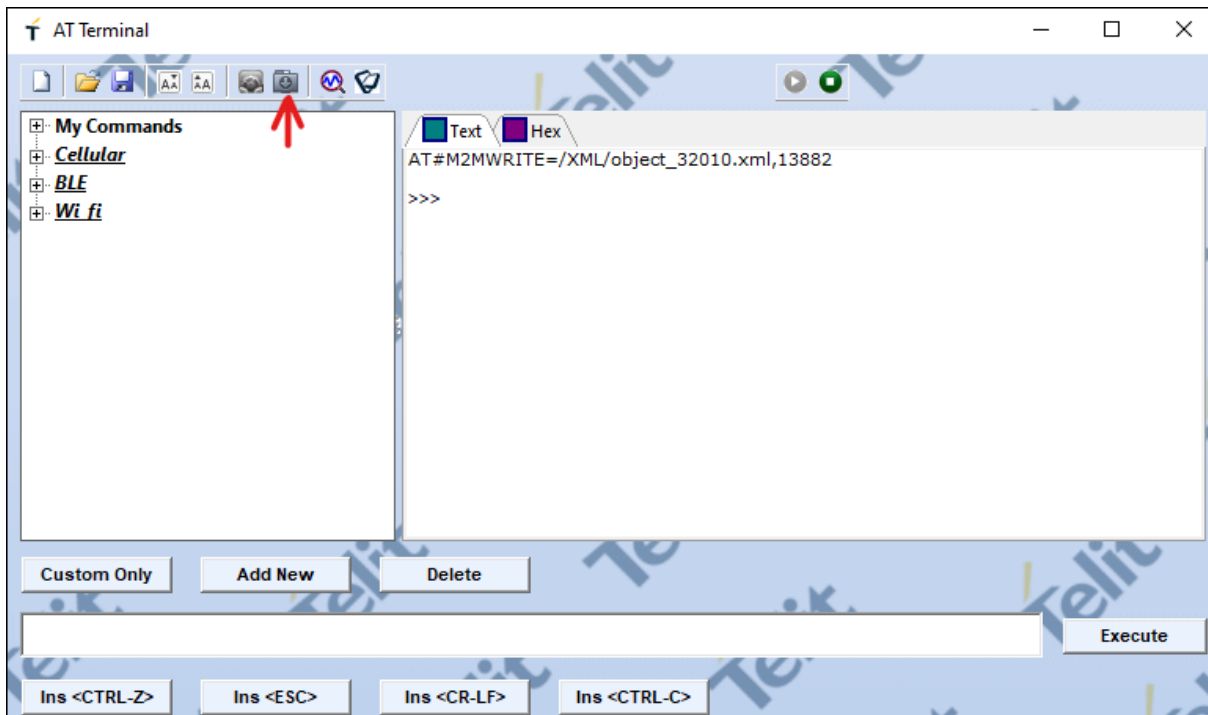


Figure 152

Select the file from your computer

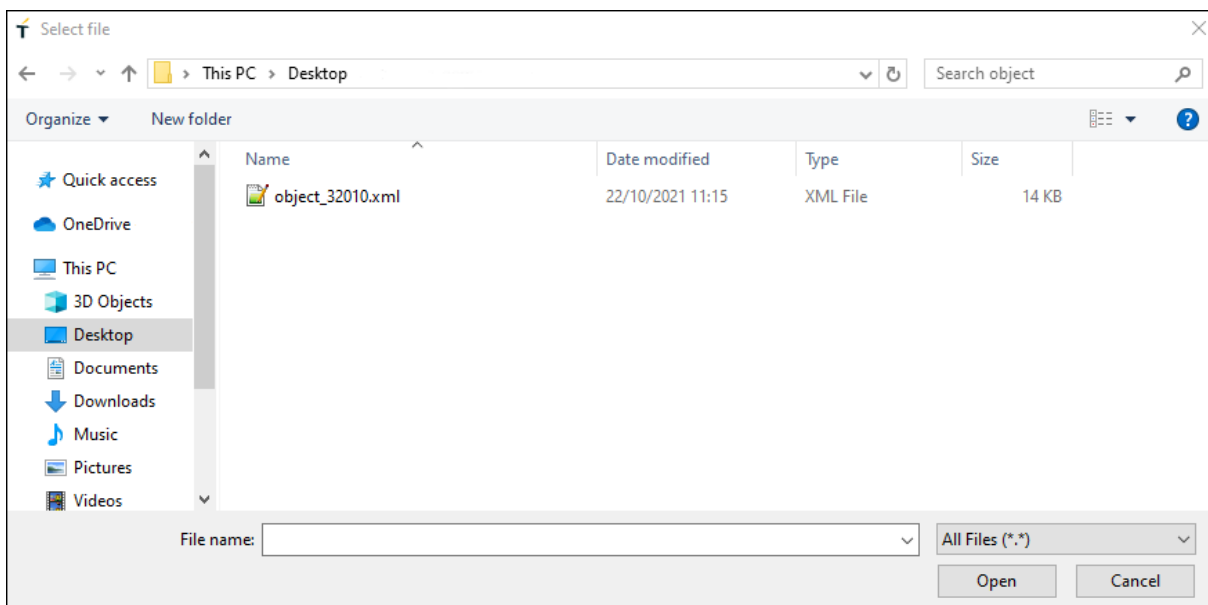


Figure 153

The file is successfully loaded on the module

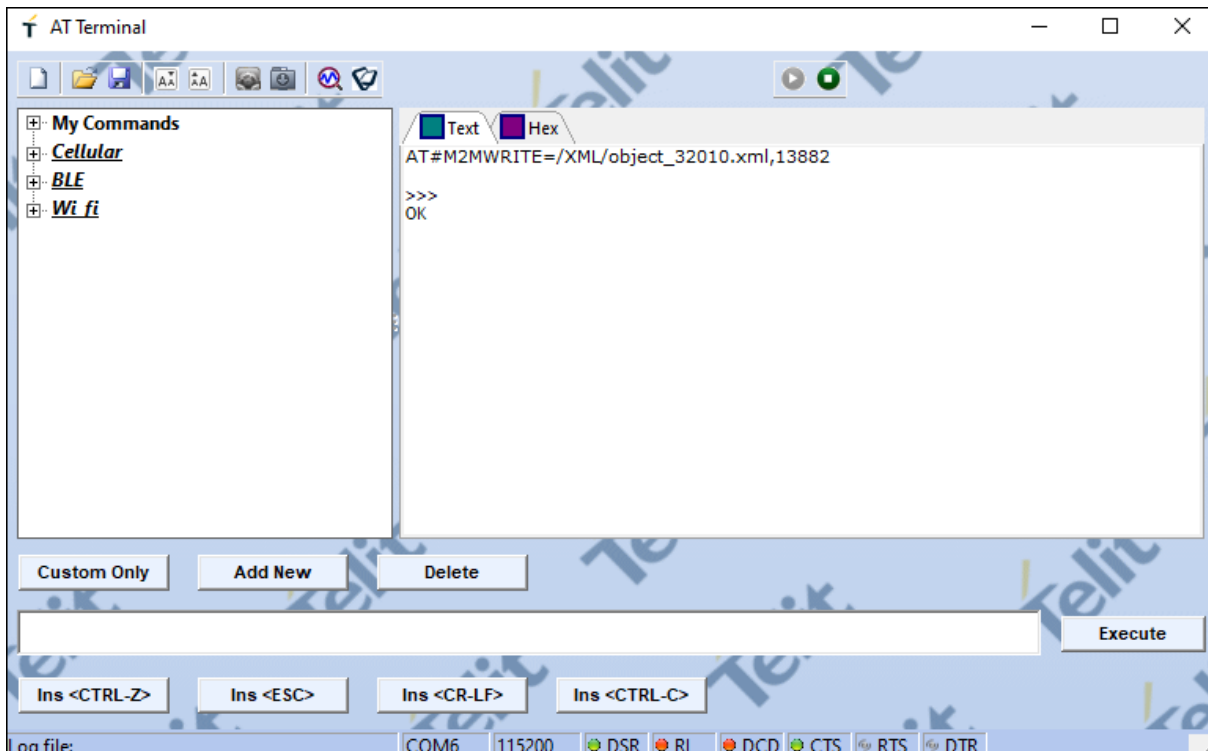


Figure 154

3.5.28.4 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on “**Things**” section, click “**New Thing**” button in the top right corner.

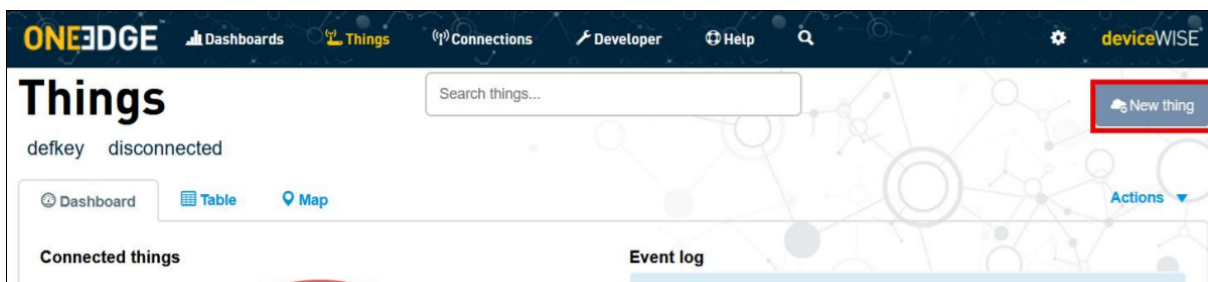


Figure 155

In the Create a new thing dialog, select “Telit Module”

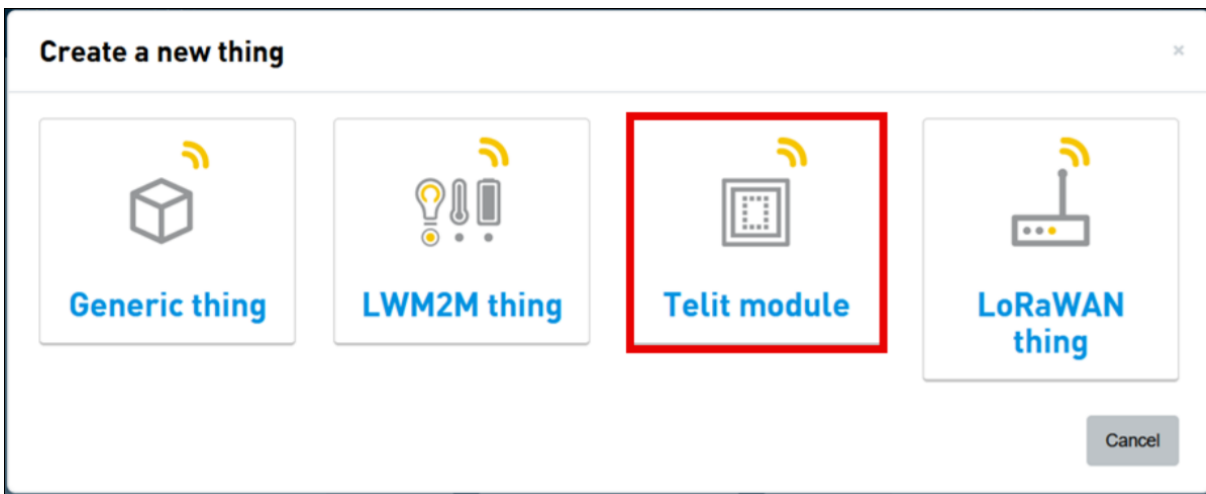


Figure 156

A dialog appears: select “Default” thing definition

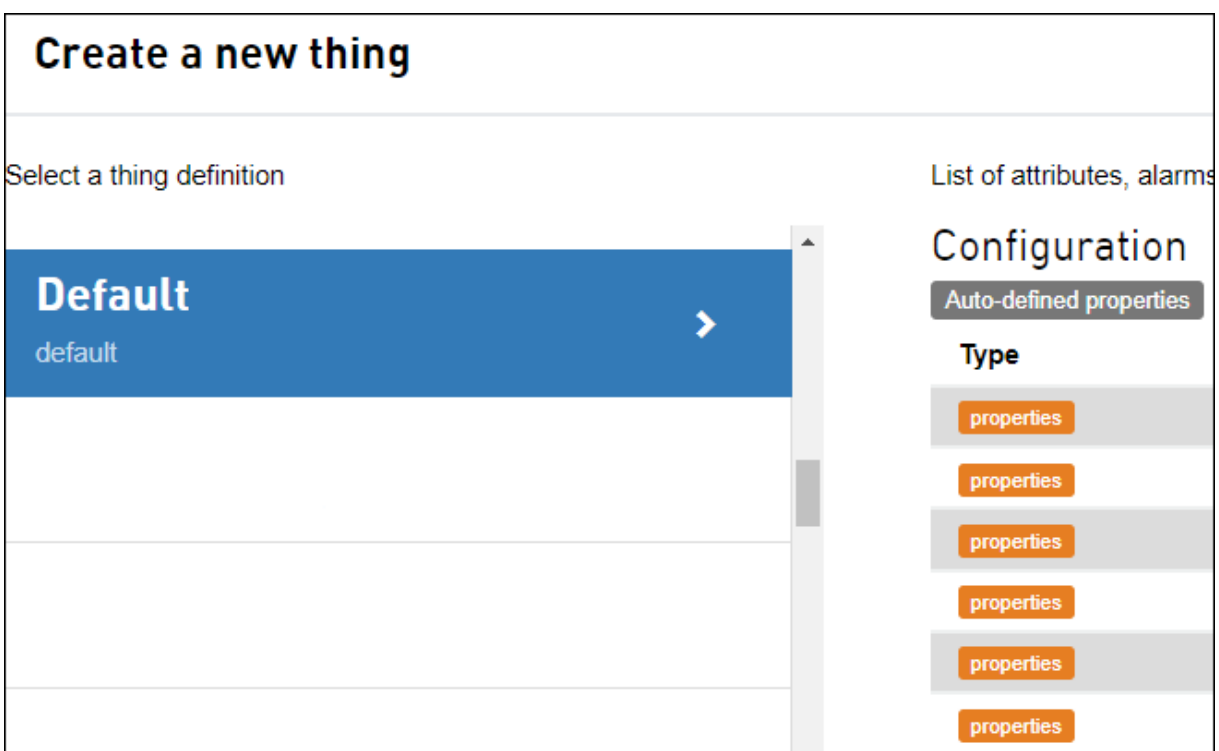


Figure 157

In the following screen, provide the Telit ID as “Identifier” Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the

screenshot below

Figure 158

Click **“Add”** to complete the new thing creation procedure.

If the Thing already exists, its device profile can be changed by following the steps shown in the picture below

Figure 159

3.5.28.5 Application execution example

```
Starting lwm2m demo. This is v1.1.10 built on Oct 22 2021 10:49:52.  
On OneEdge portal, be sure that the demo Device Profile has been uploaded on the IoT Portal  
Initializing resources...  
  
LWM2M enable result OK  
[DEBUG] 38.20 lwm2m_demo:998 - lwm2mIndicationCB{pubTspt_0}$ Monitoring enabled.  
  
Waiting LWM2M Registering (120 seconds timeout)...  
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING  
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED  
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING  
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED  
GET STATUS.  
IF Status: M2MB_LWM2M_IF_STATE_ACTIVE  
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED
```

Figure 160

```

=====
READ-ONLY RESOURCES
=====

Setting integer resource {32010/0/2} value to 50 on LWM2M client.
Reading integer resource {32010/0/2} value on LWM2M client.
Resource /32010/0/2/0 changed!
Reading integer resource {32010/0/2/0} value on LWM2M client.
---Integer value is now 50
Integer data in {32010/0/2/0} resource was updated to new value: 50
-----

Setting integer resource {32010/0/22/0} value to 10 on LWM2M client.
Resource /32010/0/22/0 changed!
Reading integer resource {32010/0/22/0} value on LWM2M client.
Integer data in {32010/0/22/0} resource was updated to new value: 10
Setting integer resource {32010/0/22/1} value to 11 on LWM2M client.
Resource /32010/0/22/1 changed!
Reading integer resource {32010/0/22/1} value on LWM2M client.
Integer data in {32010/0/22/1} resource was updated to new value: 11
-----

Setting double resource {32010/0/3} value to 20.500000 on LWM2M client.
Reading double resource {32010/0/3} value on LWM2M client.
Resource /32010/0/3/0 changed!
Reading double resource {32010/0/3/0} value on LWM2M client.
---Double value is now 20.500000
Float data in {32010/0/3/0} resource was updated to new value: 20.500000
-----

Setting boolean resource {32010/0/4} value to 1 on LWM2M client.
Reading boolean resource {32010/0/4} value on LWM2M client.
Resource /32010/0/4/0 changed!
Reading boolean resource {32010/0/4/0} value on LWM2M client.
---Boolean value is now true
Boolean data in {32010/0/4/0} resource was updated to new value: true
-----

```

```

Current time in seconds from the epoch: 1634900084
Setting timestamp resource {32010/0/6} value to 1634900084 on LWM2M client.
Reading timestamp resource {32010/0/6} value on LWM2M client.
Resource /32010/0/6/0 changed!
Reading timestamp resource {32010/0/6/0} value on LWM2M client.
---Timestamp value is now 1634900084
Time data in {32010/0/6/0} resource was updated to new value: 1634900084 (2021/10/22T10:54:44+00:00)
-----

Setting opaque resource {32010/0/5} on LWM2M client.
Resource /32010/0/5/0 changed!
Reading opaque resource {32010/0/5/0} on LWM2M client.
Opaque data in {32010/0/5/0} resource was updated to new content (10 bytes):
00 01 02 03 04 05 06 07 08 09
Reading opaque resource {32010/0/5} on LWM2M client.
---Opaque content is now (10 bytes): 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09
-----

Setting string resource {32010/0/1} value to Hello World! on LWM2M client.
Reading string resource {32010/0/1} value on LWM2M client.
Resource /32010/0/1/0 changed!
Reading integer resource {32010/0/1/0} value on LWM2M client.
---String content is now: <Hello World!>
String data in {32010/0/1/0} resource was updated to new content: <Hello World!>
-----

```



```
=====
READ-WRITE RESOURCES
=====

Writing integer resource {32010/0/12} value to 50 on LWM2M client.
Reading integer resource {32010/0/12} value on LWM2M client.
Resource /32010/0/12/0 changed!
Reading integer resource {32010/0/12/0} value on LWM2M client.
---Integer value is now 50
Integer data in {32010/0/12/0} resource was updated to new value: 50
-----

Writing double resource {32010/0/13} value to 20.500000 on LWM2M client.
Reading double resource {32010/0/13} value on LWM2M client.
Resource /32010/0/13/0 changed!
Reading double resource {32010/0/13/0} value on LWM2M client.
---Double value is now 20.500000
Float data in {32010/0/13/0} resource was updated to new value: 20.500000
-----

Writing boolean resource {32010/0/14} value to 1 on LWM2M client.
Reading boolean resource {32010/0/14} value on LWM2M client.
Resource /32010/0/14/0 changed!
Reading boolean resource {32010/0/14/0} value on LWM2M client.
---Boolean value is now true
Boolean data in {32010/0/14/0} resource was updated to new value: true
-----
```

```

Current time in seconds from the epoch: 1634900125
Writing timestamp resource {32010/0/16} value to 1634900125 on LWM2M client.
Reading timestamp resource {32010/0/16} value on LWM2M client.
Resource /32010/0/16/0 changed!
Reading timestamp resource {32010/0/16/0} value on LWM2M client.
---Timestamp value is now 1634900125
Time data in {32010/0/16/0} resource was updated to new value: 1634900125 (2021/10/22T10:55:25+00:00)
-----
Writing opaque resource {32010/0/15} on LWM2M client.
Resource /32010/0/15/0 changed!
Reading opaque resource {32010/0/15/0} on LWM2M client.
Opaque data in {32010/0/15/0} resource was updated to new content (10 bytes):
00 01 02 03 04 05 06 07 08 09
Reading opaque resource {32010/0/15} on LWM2M client.
---Opaque content is now (10 bytes): 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09
-----
Writing string resource {32010/0/11} value to <Hello World!> on LWM2M client.
Reading string resource {32010/0/11} value on LWM2M client.
Resource /32010/0/11/0 changed!
Reading integer resource {32010/0/11/0} value on LWM2M client.
---String content is now: <Hello World!>

```

```
=====
WRITE-ONLY RESOURCES
=====

Writing integer resource {32010/0/42} value to 50 on LWM2M client.
String data in {32010/0/11/0} resource was updated to new content: <Hello World!>
Getting integer resource {32010/0/42} value on LWM2M client.
Resource /32010/0/42/0 changed!
Getting integer resource {32010/0/42/0} value on LWM2M client.
---Integer value is now 50
Integer data in {32010/0/42/0} resource was updated to new value: 50
-----

Writing double resource {32010/0/43} value to 20.500000 on LWM2M client.
Getting double resource {32010/0/43} value on LWM2M client.
Resource /32010/0/43/0 changed!
Getting double resource {32010/0/43/0} value on LWM2M client.
---Double value is now 20.500000
Float data in {32010/0/43/0} resource was updated to new value: 20.500000
-----

Writing boolean resource {32010/0/44} value to 1 on LWM2M client.
Getting boolean resource {32010/0/44} value on LWM2M client.
Resource /32010/0/44/0 changed!
Getting boolean resource {32010/0/44/0} value on LWM2M client.
---Boolean value is now true
Boolean data in {32010/0/44/0} resource was updated to new value: true
-----
```

```

Current time in seconds from the epoch: 1634900163
Writing timestamp resource {32010/0/46} value to 1634900163 on LWM2M client.
Getting timestamp resource {32010/0/46} value on LWM2M client.
Resource /32010/0/46/0 changed!
Getting timestamp resource {32010/0/46/0} value on LWM2M client.
---Timestamp value is now 1634900163
Time data in {32010/0/46/0} resource was updated to new value: 1634900163 (2021/10/22T10:56:03+00:00)
-----
Writing opaque resource {32010/0/45} on LWM2M client.
Resource /32010/0/45/0 changed!
Getting opaque resource {32010/0/45/0} on LWM2M client.
Opaque data in {32010/0/45/0} resource was updated to new content (10 bytes):
00 01 02 03 04 05 06 07 08 09
Getting opaque resource {32010/0/45} on LWM2M client.
---Opaque content is now (10 bytes): 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09
-----
Writing string resource {32010/0/41} value to <Hello World!> on LWM2M client.
Getting string resource {32010/0/41} value on LWM2M client.
Resource /32010/0/41/0 changed!
Getting integer resource {32010/0/41/0} value on LWM2M client.
---String content is now: <Hello World!>
String data in {32010/0/41/0} resource was updated to new content: <Hello World!>
-----
Resources operations examples done.
-----
-----
Will perform a SET with notify ACK enabled.
-----
Enable notify ack
Setting integer resource {32010/0/2} value to 60 on LWM2M client.
Resource /32010/0/2/0 changed!
Reading integer resource {32010/0/2/0} value on LWM2M client.
Integer data in {32010/0/2/0} resource was updated to new value: 60
ACK received from server!
Reading integer resource {32010/0/2} value on LWM2M client.
---Integer value is now 60
Disable notify ack
Done.
Waiting for events from the OneEdge portal. Please write on monitored resources or call an 'exec' one.

```

After the Demo completes the initialization, it is possible to access the object resources from the section Things: select your device, then “LwM2M” tab of OneEdge IoT portal:

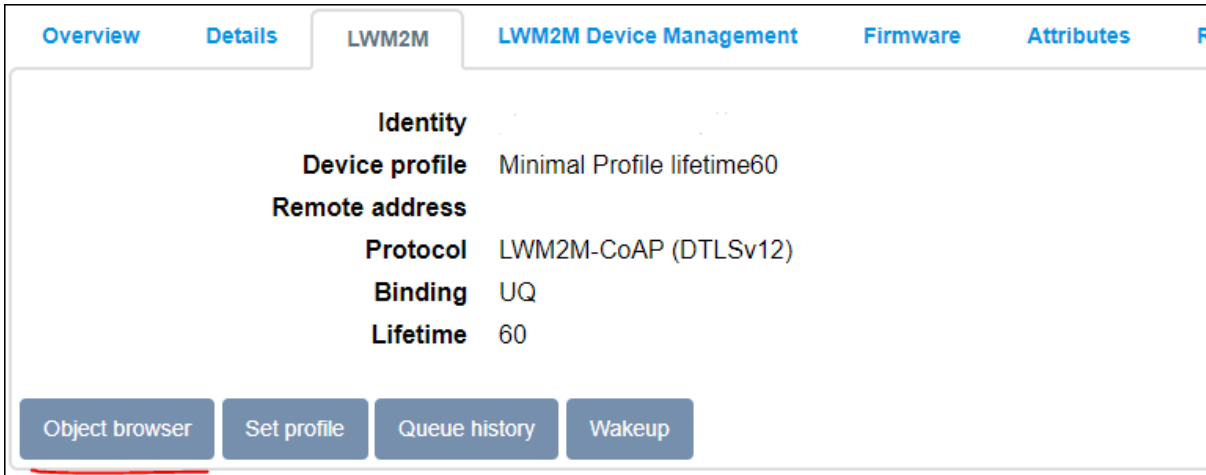


Figure 161

An instance of the object will be present and the resources can be modified.

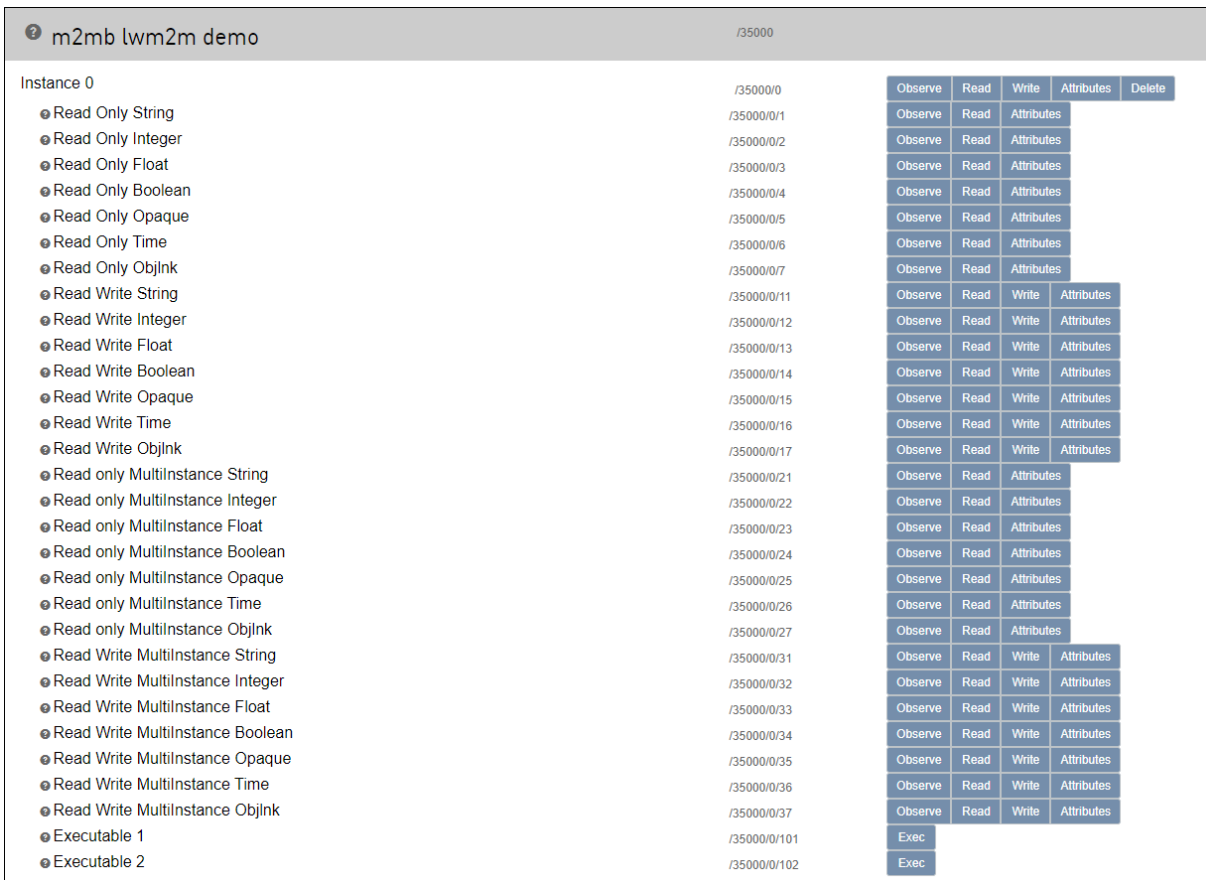


Figure 162

For example, executing the two Exec Resources at the bottom of the list, the application will react accordingly:

```
Info Exec Ind: 32010/0/101/0  
Asked to execute resource 101  
  
Resource /32010/0/11/0 changed!
```

Figure 163

Writing a string resource (id /32010/0/11), the application will notify the change

```
Reading integer resource {32010/0/11/0} value on LWM2M client.  
  
String data in {32010/0/11/0} resource was updated to new content: <Hello from the IoT Portal!>
```

Figure 164

3.5.29 LWM2M FOTA ACK management

Sample application showcasing LWM2M client FOTA events and ACKs management via APIs. Debug prints on **USB0**

Features

- Configure LWM2M client and enable it (if not already running)
- Enable monitoring on object 5 (Firmware Update)
- Configure lwm2m FOTA parameters to enable ACK requirement for each operation (Download, Update)
- Wait for an externally generated FOTA request to handle the steps by sending the acknowledgements when needed

Please refer to “80654NT11889A OneEdge Firmware Management Application Note” for further details about FOTA workflow. Get in touch with TS-OneEdge support to receive the document.

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6","<user apn>"
```

Depending on the Mobile Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.5.29.1 Device Profile upload

Minimal FOTA profile (short lifetime) device profile must be imported and selected to improve the responsiveness of the FOTA operations

To do so, import the file `json/lwm2m_fota_profile_short.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

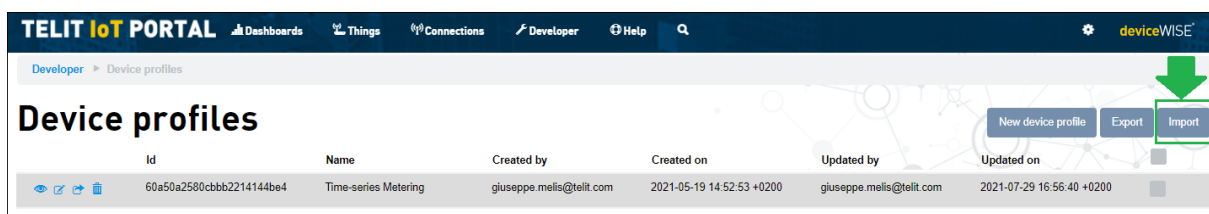


Figure 165

3.5.29.2 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on **“Things”** section, click **“New Thing”** button in the top right corner.

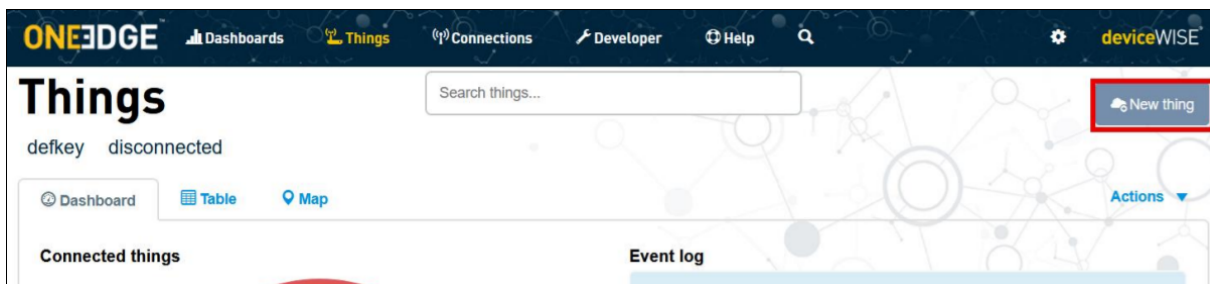


Figure 166

In the Create a new thing dialog, select “Telit Module”

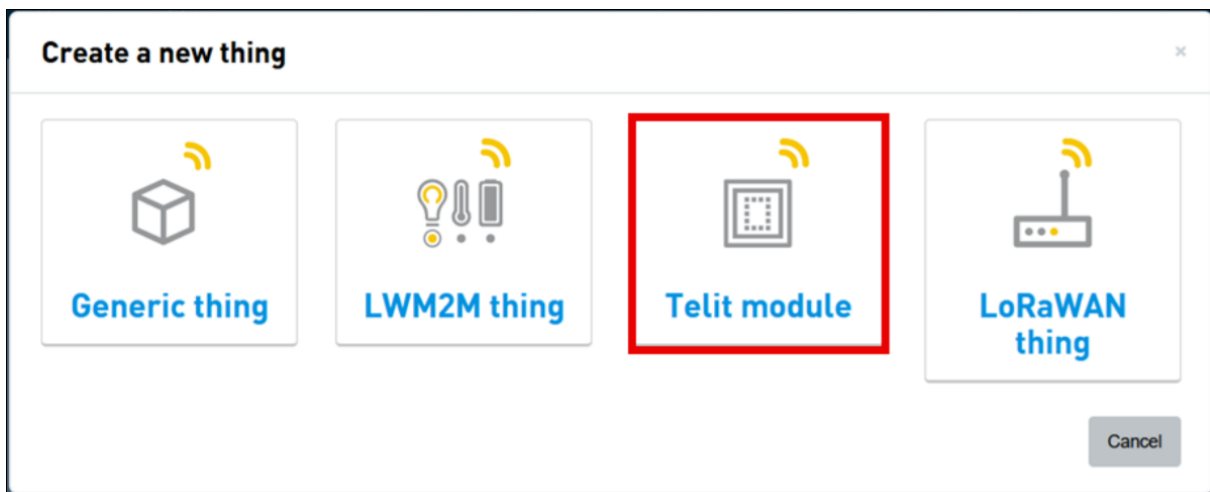


Figure 167

A dialog appears: select “Default” thing definition

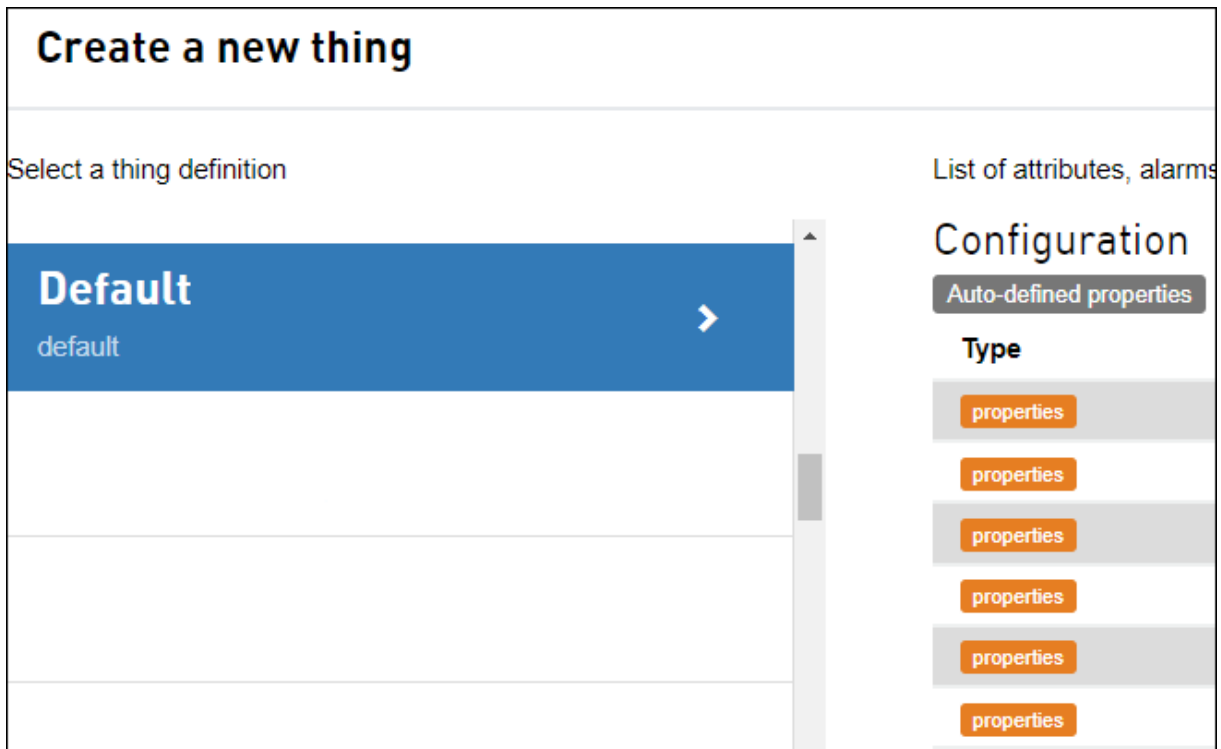


Figure 168

In the following screen, provide the Telit ID as “Identifier” Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the screenshot below

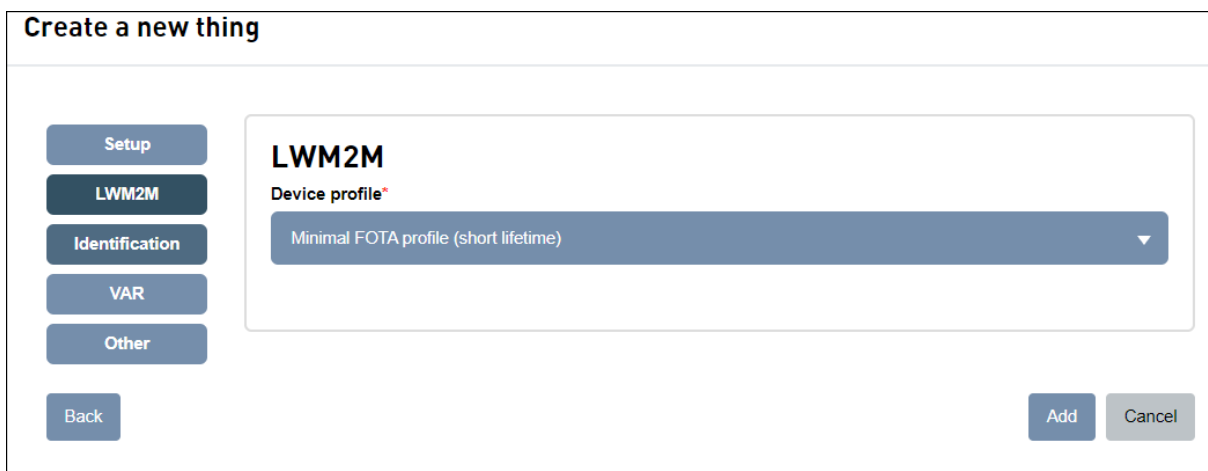


Figure 169

Click **“Add”** to complete the new thing creation procedure.

3.5.29.3 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask

- Initialize LWM2M client,
- Enable unsolicited messages from client
- Enable FOTA ACK configuration
- Create a task (lwm2m_taskCB is its callback function)to manage events from Portal
- Check if Lwm2M client is running, if not, enable it
- Wait for client to register to Portal
- Wait for FOTA events to arrive, and when the Update request arrives, notify the user that everything must stop before sending the last ACK (as the module will start updating and then reboot)

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)
- Run lwm2m_taskCB when a monitored resource changes, to manage the action to be done

3.5.29.4 Application execution example

```
Starting lwm2m FOTA management demo. This is v1.1.14 built on Jan 21 2022 14:39:47.
Initializing resources...
[DEBUG] 22.14 lwm2m_demo:283 - at_cmd_async_init{LWM2M_TASK}$ m2mb_ati_init() on instance 1

IF Status: M2MB_LWM2M_IF_STATE_DISABLED
Client Status: M2MB_LWM2M_CL_STATE_DISABLED
Enable lwm2m client
LWM2M enable result OK
Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

Waiting for events from the OneEdge portal. Please start a FOTA process.
```

Figure 170

```
Resource /5/0/1/0 changed!
[DEBUG] 108.42 lwm2m_demo:544 - lwm2m_taskCB{LWM2M_TASK}$ Firmware update event!
FOTA process is waiting for Download ACK, sending it!
Waiting for Update ACK!
Before sending the ACK to apply FOTA, stop any ongoing operation (filesystem, sockets)

Asked to perform UPDATE step, stopping all operations now.
Application complete.

----- Info, DEVICE REBOOT FOR FOTA -----
?????????
#OTAEV: Module Upgraded To New Fw
```

FOTA request arrives from server, ACK is sent for Download and Update. Device restarts with the new firmware version

```
Starting lwm2m FOTA management demo. This is v1.1.14 built on Jan 21 2022 14:39:47.
Initializing resources...
[DEBUG] 26.61 lwm2m_demo:283 - at_cmd_async_init{LWM2M_TASK}$ m2mb_ati_init() on instance 1
[WARN ] 34.98 lwm2m_demo:738 - lwm2mIndicationCB{pubTspt_0}$ write res 1

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
Enable already done, skip. State: M2MB_LWM2M_IF_STATE_ACTIVE
Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

Waiting for events from the OneEdge portal. Please start a FOTA process.
```

Application restarts and is ready for a new FOTA operation

3.5.30 LWM2M FOTA ACK management (AT URCs)

Sample application showcasing LWM2M client FOTA events and ACKs management via AT URCs. Debug prints on **USB0**

Features

- Configure LWM2M client and enable it (if not already running)
- Enable monitoring on object 5 (Firmware Update)
- Configure lwm2m FOTA parameters to enable ACK requirement for each operation (Download, Update)
- Wait for an externally generated FOTA request to handle the steps by sending the acknowledgements when needed

Please refer to “80654NT11889A OneEdge Firmware Management Application Note” for further details about FOTA workflow. Get in touch with TS-OneEdge support to receive the document.

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6","<user apn>"
```

Depending on the Mobile Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.5.30.1 Device Profile upload

Minimal FOTA profile (short lifetime) device profile must be imported and selected to improve the responsiveness of the FOTA operations

To do so, import the file `json/lwm2m_fota_profile_short.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

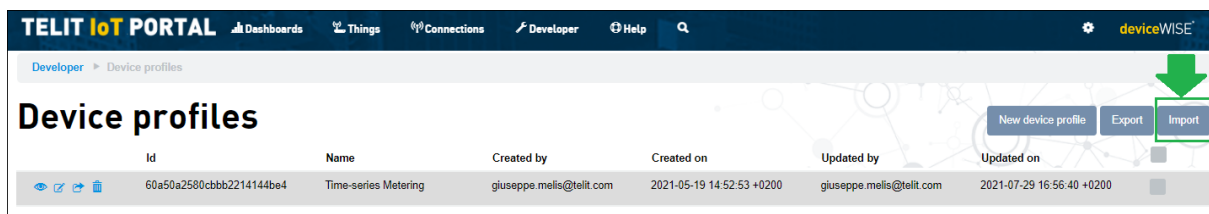


Figure 171

3.5.30.2 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on **“Things”** section, click **“New Thing”** button in the top right corner.

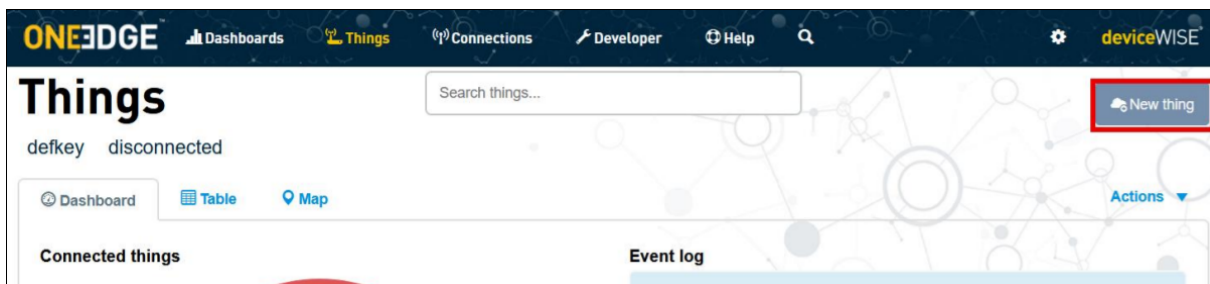


Figure 172

In the Create a new thing dialog, select “Telit Module”

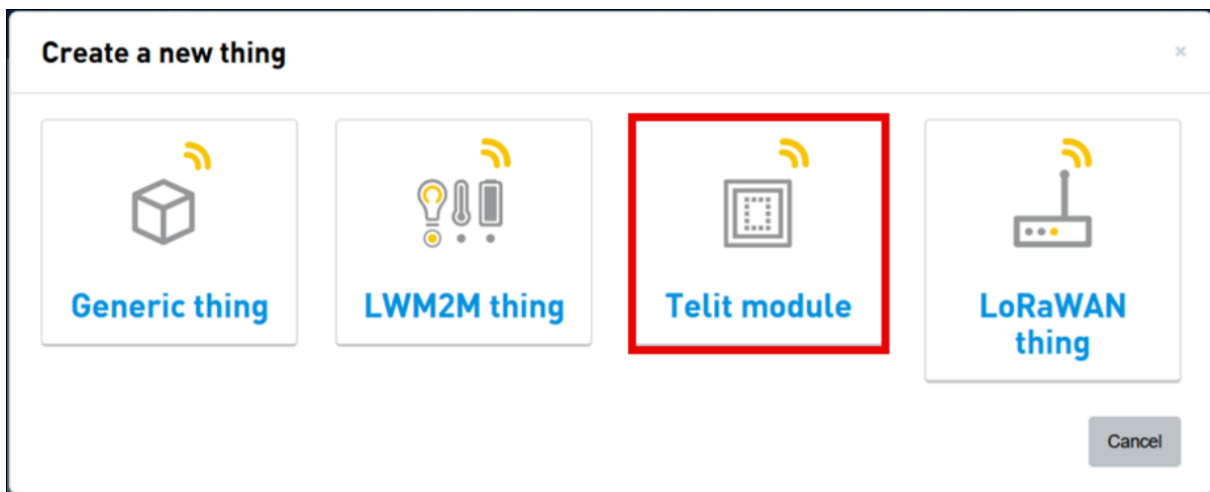


Figure 173

A dialog appears: select “Default” thing definition

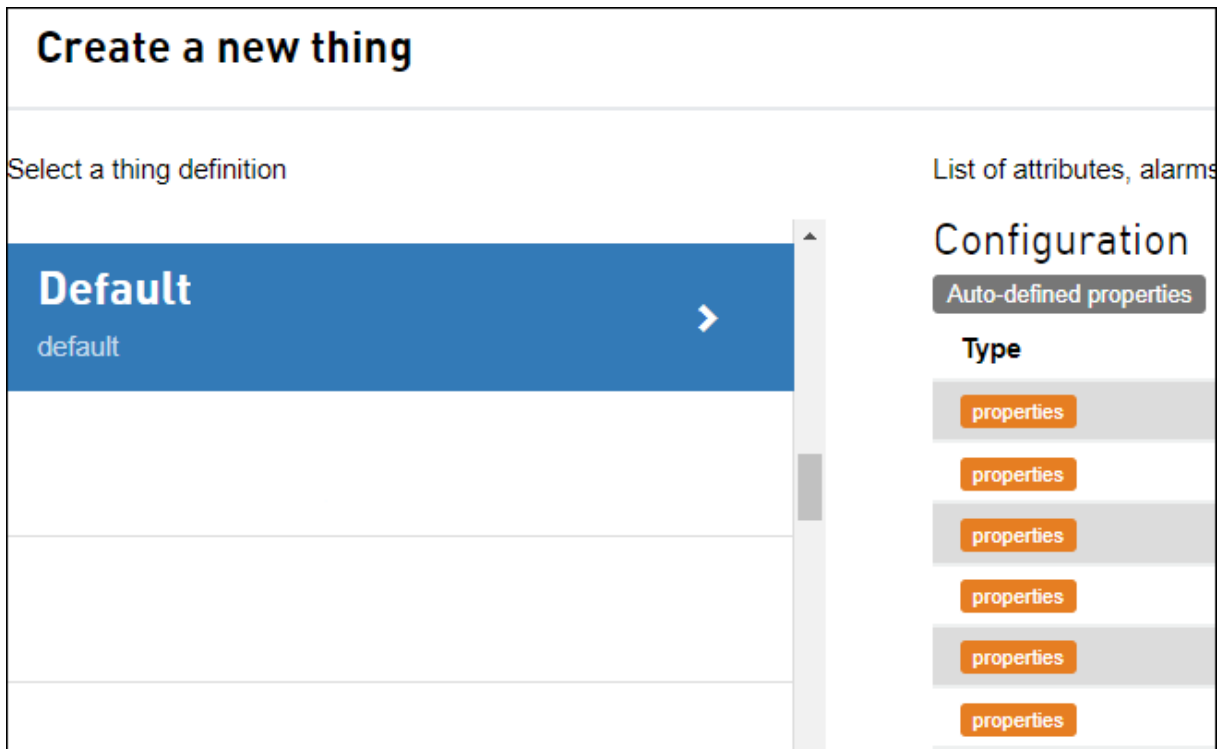


Figure 174

In the following screen, provide the Telit ID as “Identifier” Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the screenshot below

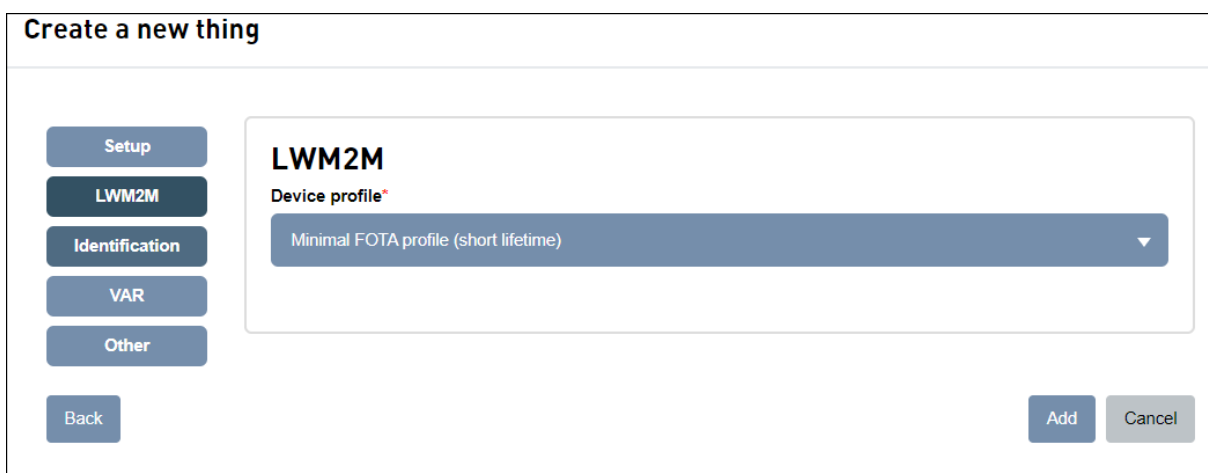


Figure 175

Click **“Add”** to complete the new thing creation procedure.

3.5.30.3 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask

- Initialize LWM2M client,
- Enable unsolicited messages from client
- Enable FOTA ACK configuration
- Create a task (lwm2m_taskCB is its callback function)to manage events from Portal
- Check if Lwm2M client is running, if not, enable it
- Wait for client to register to Portal
- Wait for FOTA events to arrive, and when the Update request arrives, notify the user that everything must stop before sending the last ACK (as the module will start updating and then reboot)

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)
- Run lwm2m_taskCB when a monitored resource changes, to manage the action to be done

3.5.30.4 Application execution example

```
Starting lwm2m FOTA management demo. This is v1.1.14 built on Jan 21 2022 14:39:47.
Initializing resources...
[DEBUG] 22.14 lwm2m_demo:283 - at_cmd_async_init{LWM2M_TASK}$ m2mb_ati_init() on instance 1

IF Status: M2MB_LWM2M_IF_STATE_DISABLED
Client Status: M2MB_LWM2M_CL_STATE_DISABLED
Enable lwm2m client
LWM2M enable result OK
Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

Waiting for events from the OneEdge portal. Please start a FOTA process.
```

Figure 176

```
Resource /5/0/1/0 changed!
[DEBUG] 108.42 lwm2m_demo:544 - lwm2m_taskCB{LWM2M_TASK}$ Firmware update event!
FOTA process is waiting for Download ACK, sending it!
Waiting for Update ACK!
Before sending the ACK to apply FOTA, stop any ongoing operation (filesystem, sockets)

Asked to perform UPDATE step, stopping all operations now.
Application complete.

----- Info, DEVICE REBOOT FOR FOTA -----
?????????
#OTAEV: Module Upgraded To New Fw
```

FOTA request arrives from server, ACK is sent for Download and Update. Device restarts with the new firmware version

```
Starting lwm2m FOTA management demo. This is v1.1.14 built on Jan 21 2022 14:39:47.
Initializing resources...
[DEBUG] 26.61 lwm2m_demo:283 - at_cmd_async_init{LWM2M_TASK}$ m2mb_ati_init() on instance 1
[WARN ] 34.98 lwm2m_demo:738 - lwm2mIndicationCB{pubTspt_0}$ write res 1

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
Enable already done, skip. State: M2MB_LWM2M_IF_STATE_ACTIVE
Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

Waiting for events from the OneEdge portal. Please start a FOTA process.
```

Application restarts and is ready for a new FOTA operation

3.5.31 LWM2M OBJ_GET AND OBJ_SET

Sample application showcasing LWM2M client m2mb_lwm2m_objget and m2mb_lwm2m_objset M2MB APIs usage. Debug prints on **USB0**

Features

- Configure LWM2M client and enable it
- Create an instance of a custom object
- Create a Json string
- Set string, integer, float, boolean, timestamp and opaque values with m2mb_lwm2m_objset
- Get all resources values with m2mb_lwm2m_objget
- Manage write, read and monitoring resources changed from the portal

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6", "<user apn>"
```

Depending on the Mobiler Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.5.31.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask - Check registration status

- Initialize LWM2M client,
- Check for XML file fo custom object
- Enable unsolicited messages from client
- Create a task (lwm2m_taskCB is its callback function)to manage events from Portal
- Enable Lwm2M client

- Create a new instance for the custom object
- Wait for client to register to Portal
- Performs obj_set and obj_get operations on the related resources
- Wait for events from server

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)
- Run lwm2m_taskCB when a monitored resource changes, to manage the action to be done

3.5.31.2 Device Profile upload

LWM2M resources demo device profile must be imported to have a real-time update of resources values on the LWM2M browser.

To do so, import the file `json/lwm2m_resources_demo.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

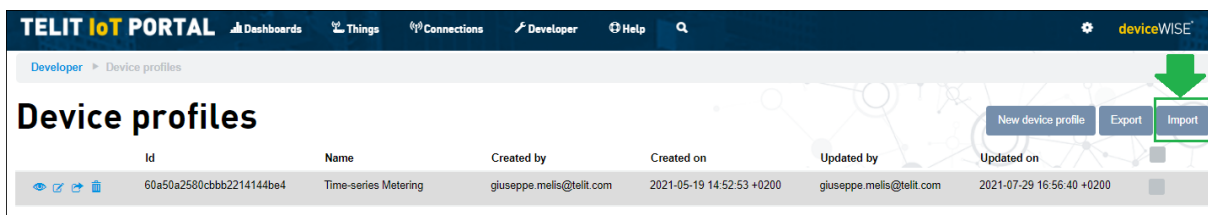


Figure 177

3.5.31.3 Custom Object configuration

The XML file content must be loaded on the Telit IoT Portal for the demo application to be fully executed.

First, enter Developer section from the top menu



Figure 178

Choose Object Registry

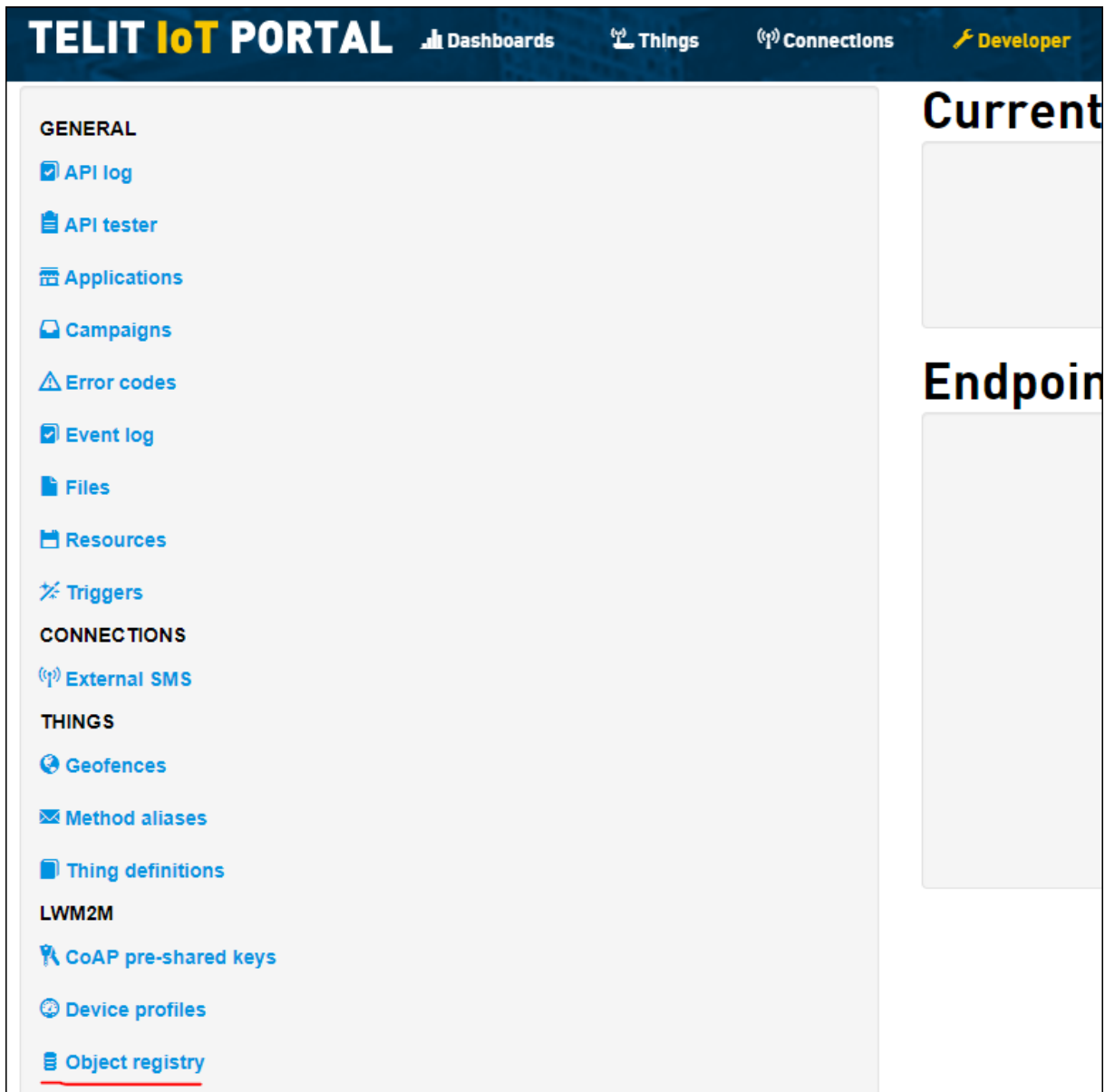


Figure 179

Create a New Object

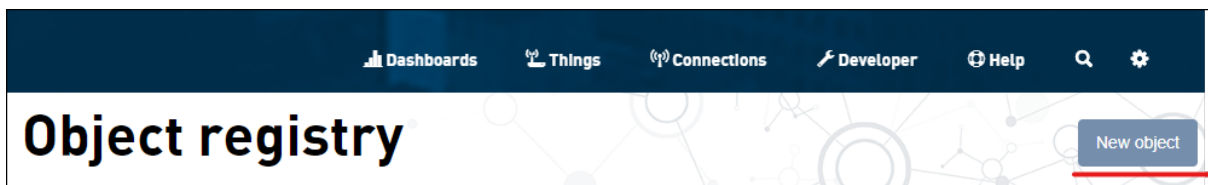
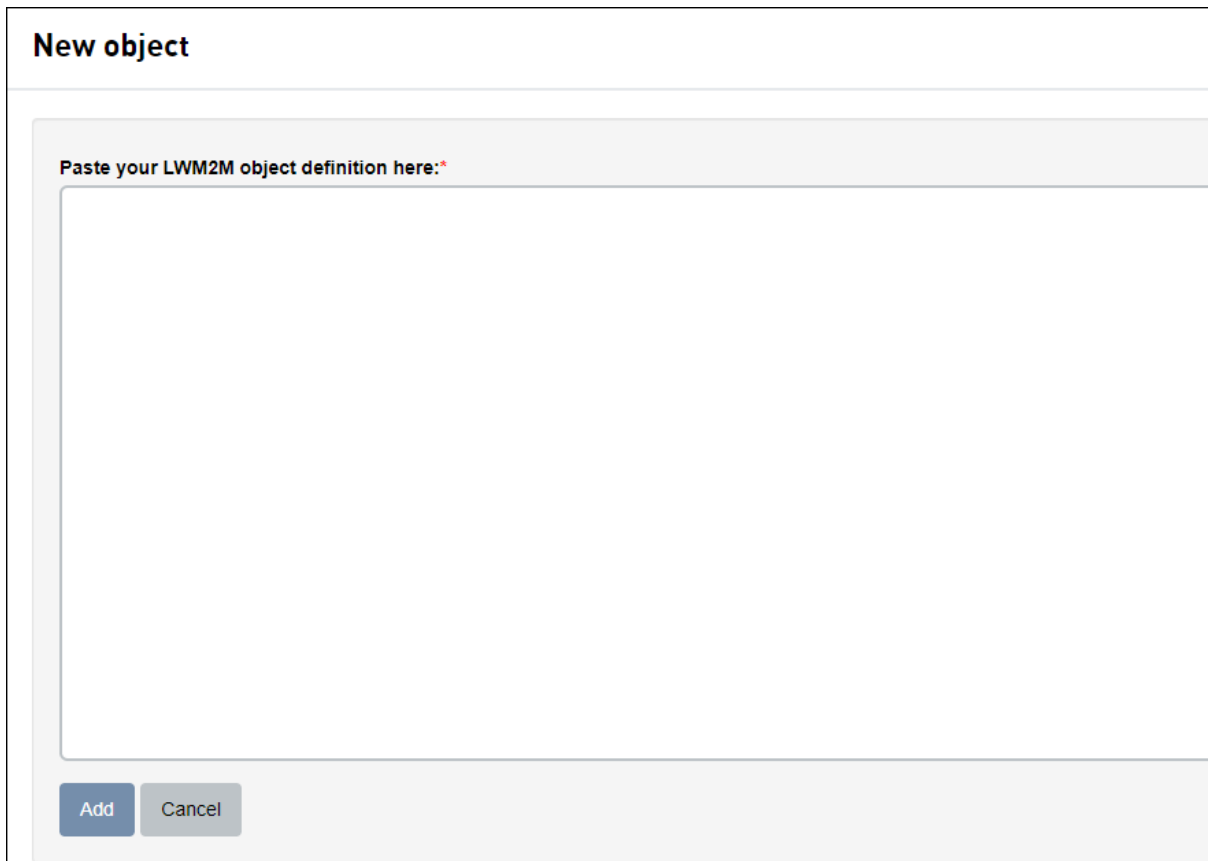


Figure 180

Copy the xml file content and paste it in the new Object form



New object

Paste your LWM2M object definition here:*

Add Cancel

Figure 181

Also, the application requires the XML file `xml/object_32011.xml` (provided with the sample files) to be stored in module's `/XML/` folder. It can be done with

```
AT#M2MWRITE=/XML/object_32011.xml,<size_in_bytes>
```

To load the XML file in the module, Telit AT Controller (TATC) can be used. Once the command above is issued, press the load content button:

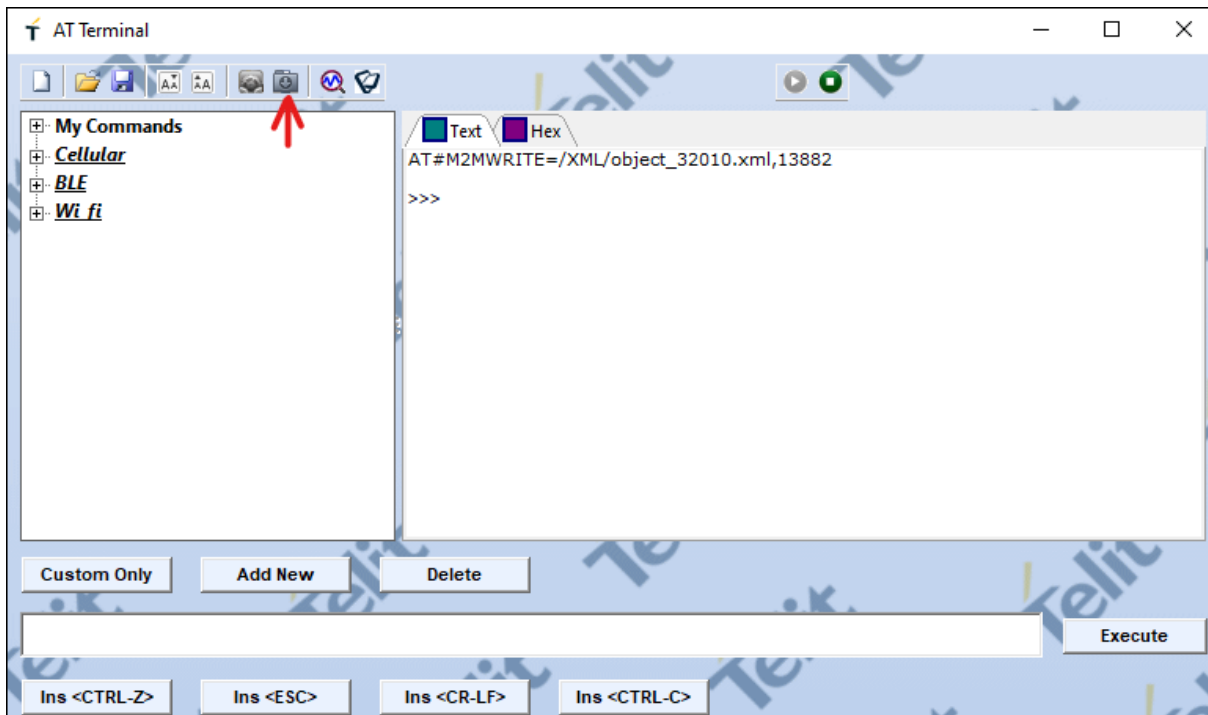


Figure 182

Select the file from your computer

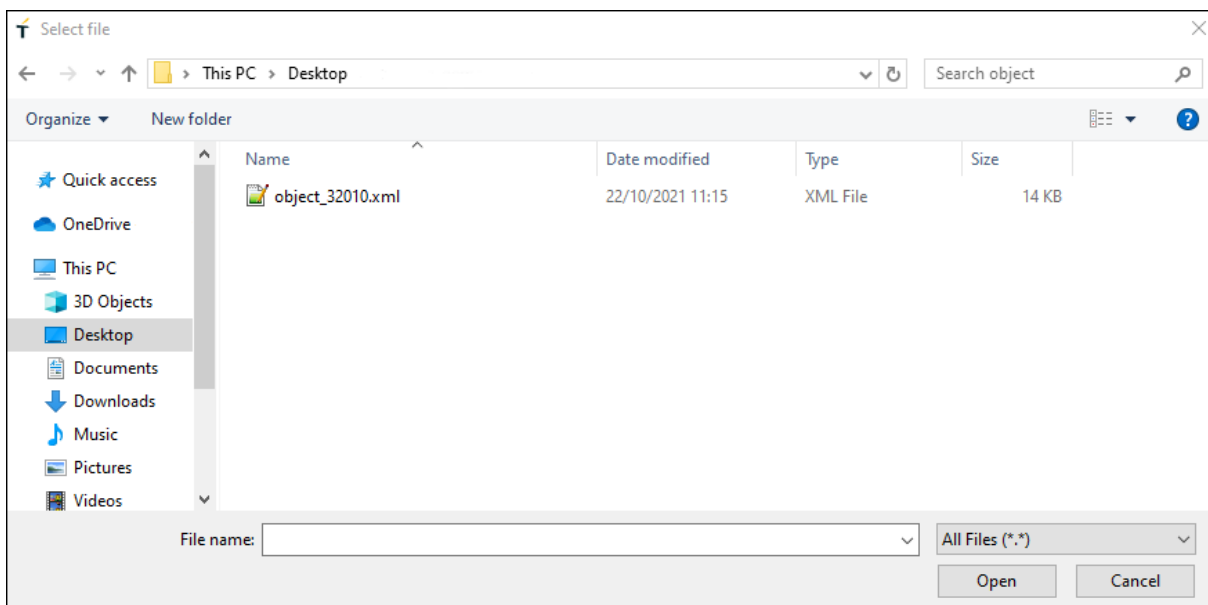


Figure 183

The file is successfully loaded on the module

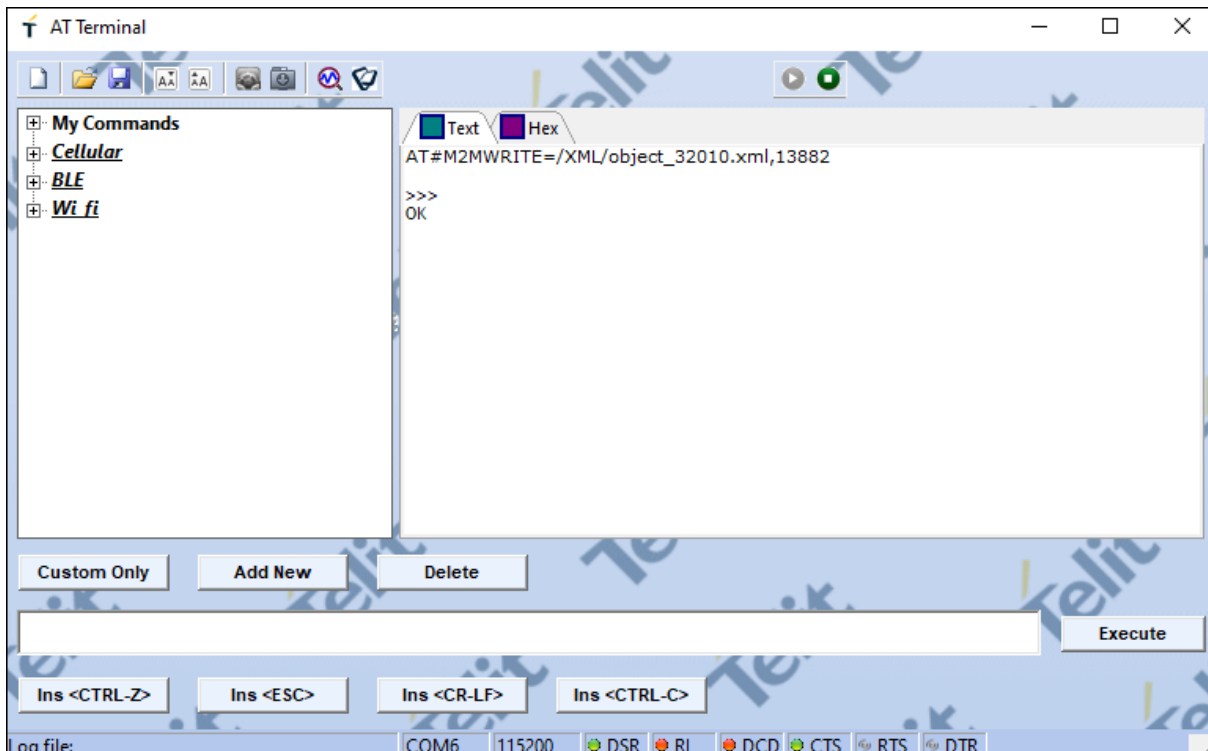


Figure 184

3.5.31.4 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on “**Things**” section, click “**New Thing**” button in the top right corner.

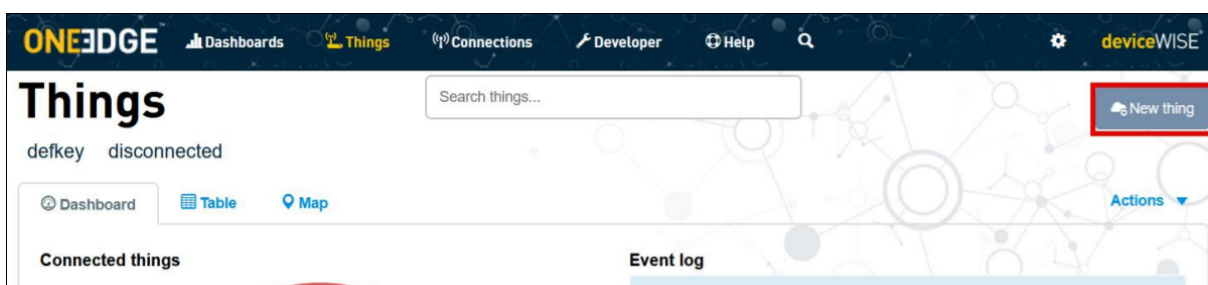


Figure 185

In the Create a new thing dialog, select “Telit Module”

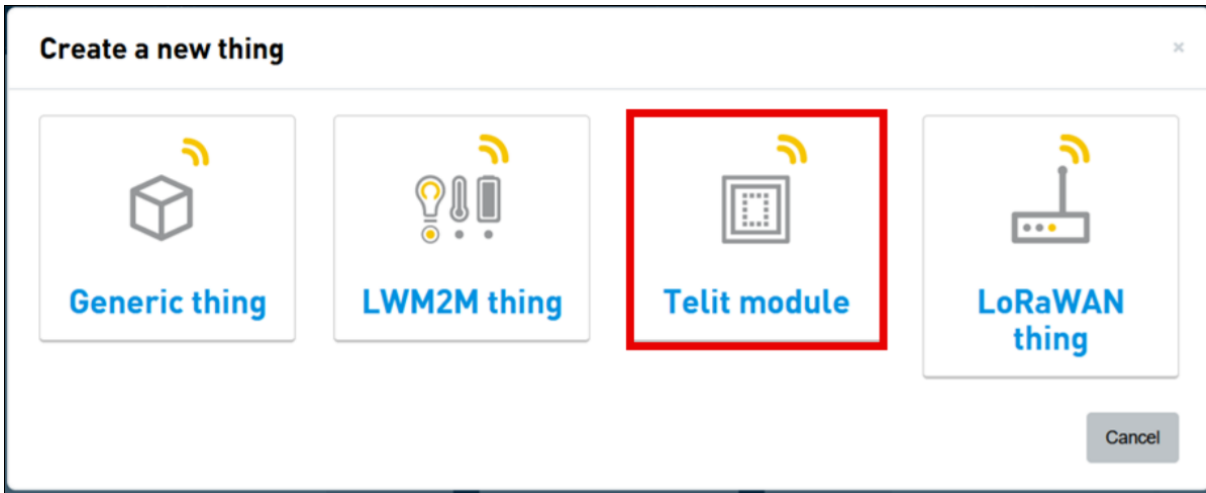


Figure 186

A dialog appears: select “Default” thing definition

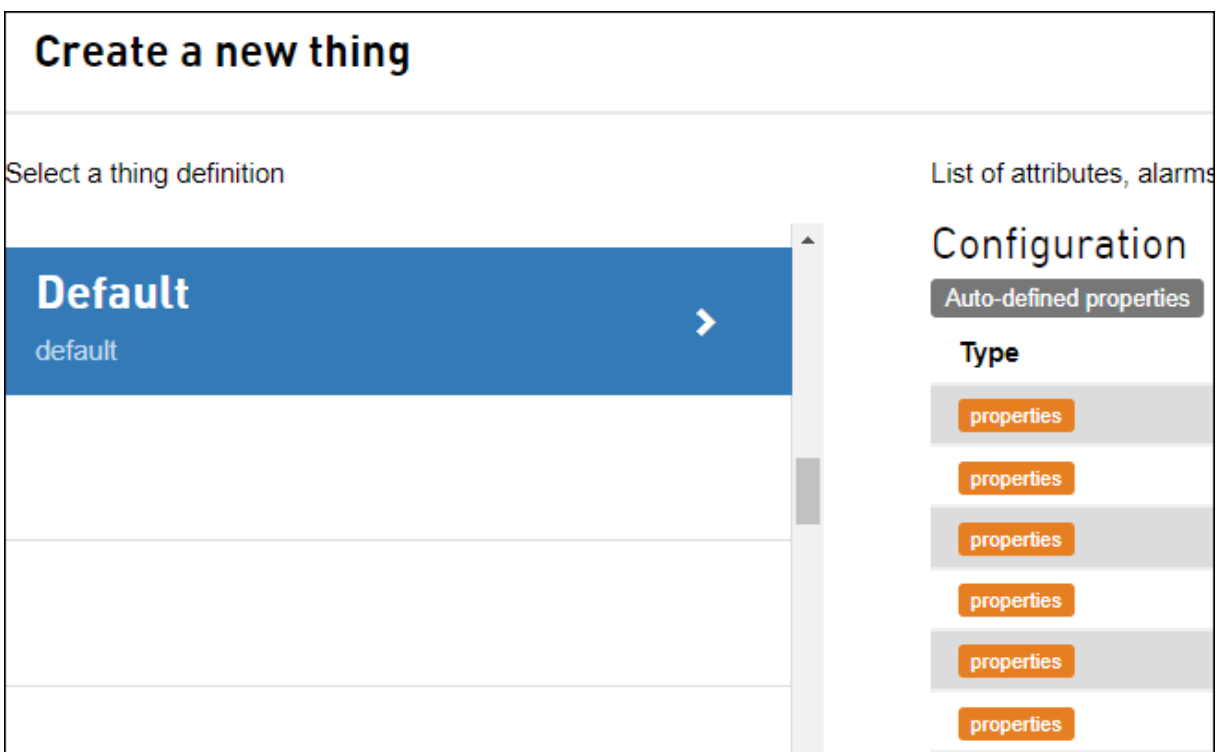


Figure 187

In the following screen, provide the Telit ID as “Identifier” Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the

screenshot below

Figure 188

Click **“Add”** to complete the new thing creation procedure.

If the Thing already exists, its device profile can be changed by following the steps shown in the picture below

Figure 189

3.5.31.5 Application execution example


```
Starting lwm2m demo. This is v1.1.10 built on Oct 22 2021 10:49:52.
On OneEdge portal, be sure that the demo Device Profile has been uploaded on the IoT Portal
Initializing resources...

LWM2M enable result OK
[DEBUG] 38.20 lwm2m_demo:998 - lwm2mIndicationCB{pubTspt_0}$ Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED
GET STATUS.
IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED
```

Figure 190

```
Starting lwm2m demo. This is v1.1.11 built on Nov 24 2021 13:32:52.
On OneEdge portal, be sure that the 'lwm2m resources objset - objget demo' Device Profile has been uploaded on the IoT Portal

Initializing resources...
LWM2M enable result OK
[DEBUG] 29.42 lwm2m_demo:593 - lwm2mIndicationCB{pubTspt_0}$ Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED
GET STATUS.
IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

m2mb_lwm2m_objset SUCCESS

Resource /32011/0/11/0 changed!

Reading string resource {32011/0/11/0} value on LWM2M client.

Resource /32011/0/12/0 changed!
Resource /32011/0/13/0 changed!
Resource /32011/0/15/0 changed!
Resource /32011/0/16/0 changed!
Resource /32011/0/31/0 changed!
Resource /32011/0/31/1 changed!
Resource /32011/0/31/2 changed!
Resource /32011/0/34/0 changed!
Resource /32011/0/34/1 changed!
Resource /32011/0/34/2 changed!
Resource /32011/0/34/3 changed!
```

```
String data in {32011/0/11/0} resource was updated to new content: <hello world>
Reading integer resource {32011/0/12/0} value on LWM2M client.
Integer data in {32011/0/12/0} resource was updated to new value: 70
Reading double resource {32011/0/13/0} value on LWM2M client.
Float data in {32011/0/13/0} resource was updated to new value: -12.500000
Reading opaque resource {32011/0/15/0} on LWM2M client.
Opaque data in {32011/0/15/0} resource was updated to new content (11 bytes):
68 65 6C 6C 6F 20 77 6F 72 6C 64
Reading timestamp resource {32011/0/16/0} value on LWM2M client.
Time data in {32011/0/16/0} resource was updated to new value: 1620828702 (2021/05/12T14:11:42+00:00)
Reading string resource {32011/0/31/0} value on LWM2M client.
String data in {32011/0/31/0} resource was updated to new content: <Greetings>
Reading string resource {32011/0/31/1} value on LWM2M client.
String data in {32011/0/31/1} resource was updated to new content: <from>
Reading string resource {32011/0/31/2} value on LWM2M client.
String data in {32011/0/31/2} resource was updated to new content: <AppZone!>
Reading boolean resource {32011/0/34/0} value on LWM2M client.
Boolean data in {32011/0/34/0} resource was updated to new value: true
Reading boolean resource {32011/0/34/1} value on LWM2M client.
Boolean data in {32011/0/34/1} resource was updated to new value: false
Reading boolean resource {32011/0/34/2} value on LWM2M client.
Boolean data in {32011/0/34/2} resource was updated to new value: true
Reading boolean resource {32011/0/34/3} value on LWM2M client.
Boolean data in {32011/0/34/3} resource was updated to new value: false
Waiting 5 seconds before the m2mb_lwm2m_objget
Waiting: 5
Waiting: 4
Waiting: 3
Waiting: 2
Waiting: 1
m2mb_lwm2m_objget SUCCESS
Content:
<{"bn":"/32011/0/",
"e":[
{"n":"11","sv":"hello world"},
{"n":"12","v":70},
{"n":"13","v":-12.500000},
{"n":"14","bv":false},
{"n":"15","sv":"aGVsbG8gd29ybGQ="},
{"n":"16","v":1620828702},
{"n":"17","ov":"0:0"},
{"n":"31/0","sv":"Greetings"},
{"n":"31/1","sv":"from"},
{"n":"31/2","sv":"AppZone!"},
{"n":"31/3","sv":""},
{"n":"31/4","sv":""},
{"n":"31/5","sv":""},
{"n":"31/6","sv":""},
{"n":"31/7","sv":""},
{"n":"31/8","sv":""},
{"n":"31/9","sv":""},
{"n":"34/0","bv":true},
{"n":"34/1","bv":false},
{"n":"34/2","bv":true},
{"n":"34/3","bv":false},
{"n":"34/4","bv":false},
{"n":"34/5","bv":false},
{"n":"34/6","bv":false},
{"n":"34/7","bv":false},
{"n":"34/8","bv":false},
{"n":"34/9","bv":false}]
}>
```

After the Demo completes the initialization, it is possible to access the object re-

sources from the section Things: select your device, then “LwM2M” tab of OneEdge IoT portal:

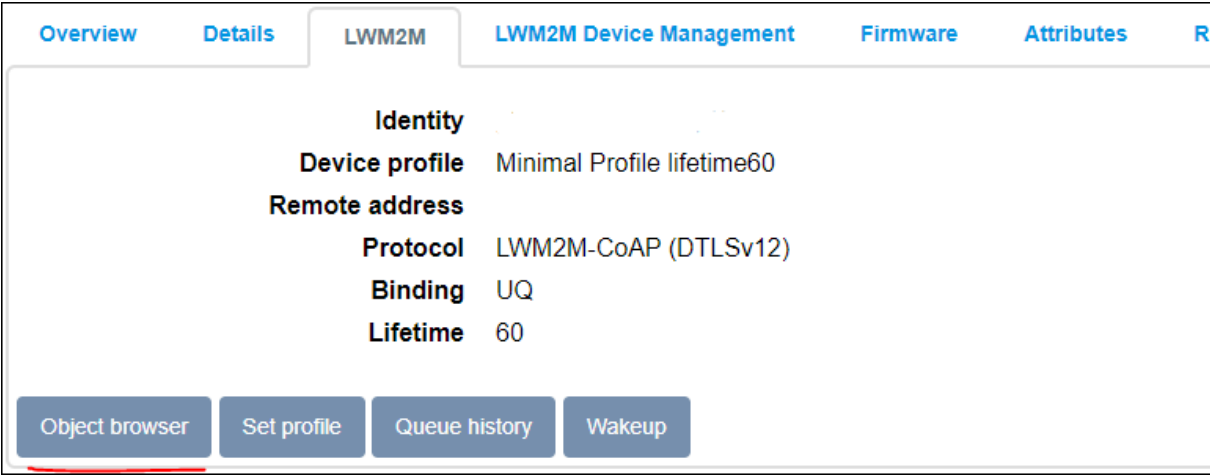


Figure 191

An instance of the object will be present and the resources can be modified.

m2mb lwm2m objset-get demo		/32011					
Instance 0	/32011/0	Observe	Read	Write	Attributes	Delete	
Read Write String	/32011/0/11	Observe	Read	Write	Attributes		hello world
Read Write Integer	/32011/0/12	Observe	Read	Write	Attributes		70
Read Write Float	/32011/0/13	Observe	Read	Write	Attributes		-12.5
Read Write Boolean	/32011/0/14	Observe	Read	Write	Attributes		false
Read Write Opaque	/32011/0/15	Observe	Read	Write	Attributes		view binary data
Read Write Time	/32011/0/16	Observe	Read	Write	Attributes		2021-05-12T14:11:42Z
Read Write Objlink	/32011/0/17	Observe	Read	Write	Attributes		instid=0, objid=0
Read Write Multiinstance String	/32011/0/31	Observe	Read	Write	Attributes		0=Greetings, 1=from, ... Read more
Read Write Multiinstance Boolean	/32011/0/34	Observe	Read	Write	Attributes		0=true, 1=false, 2=tr... Read more

Figure 192

3.5.32 LWM2M REG

Sample application showcasing LWM2M client registration management using M2MB API. Debug prints on **USB0**

Features

- Configure LWM2M client and enable it
- Get LWM2M server information using REG apis
- Deregister from LWM2M server using REG apis
- Register to the LWM2M server using REG apis
- Update registration on LWM2M server using REG apis
- Deregister from LWM2M server using REG apis

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6","<user apn>"
```

Depending on the Mobile Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.5.32.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask - Check registration status

- Initialize LWM2M client,
- Check for XML file for custom object
- Enable LWM2M client
- Wait for client to register to Portal
- Get the LWM2M server information
- Performs client portal deregistration
- Performs client portal registration

- Performs client portal registration Update
- Performs client portal deregistration

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)

3.5.32.2 Application execution example

```
Starting LWM2M REG demo. This is v1.0.00 built on Sep 11 2023 15:11:48.
On OneEdge portal, be sure that the 'lwm2m resources demo' Device Profile has been uploaded on the IoT Portal

Initializing resources...
LWM2M enable result OK
Waiting LWM2M Registering (120 seconds timeout)...
sshd[99] status[2] lastReg[1694446138] nextReg[1694532238] lastActivity[1694446138]
Deregistration action using LWM2M REG api!
Waiting deregistration event!
Deregistration event received!
Registration action using LWM2M REG api!
Waiting registration event!
Registration event received!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Deregistration action using LWM2M REG api!
Waiting deregistration event!
Deregistration event received!
Application complete.
```

Figure 193

3.5.32.3 Device Profile upload

LWM2M resources demo device profile must be imported to have a real-time update of resources values on the LWM2M browser.

To do so, import the file `json/lwm2m_resources_demo.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

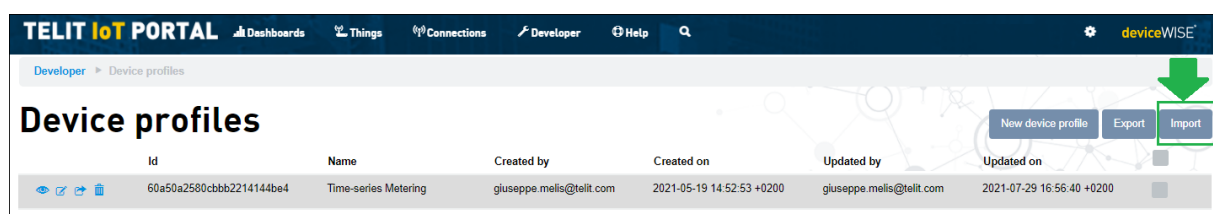


Figure 194

3.5.32.4 Custom Object configuration

The XML file content must be loaded on the Telit IoT Portal for the demo application to be fully executed.

First, enter Developer section from the top menu



Figure 195

Choose Object Registry

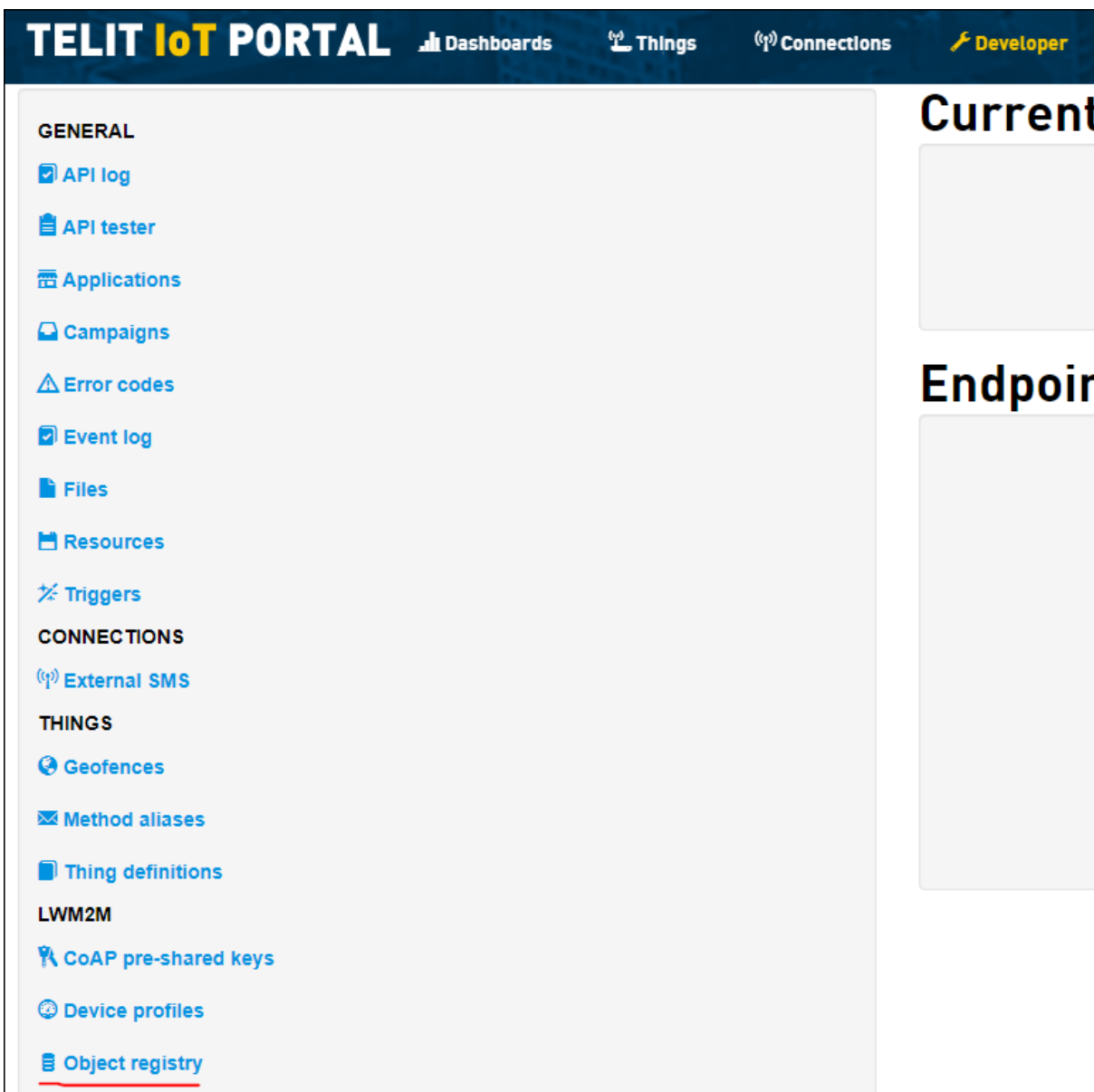


Figure 196

Create a New Object

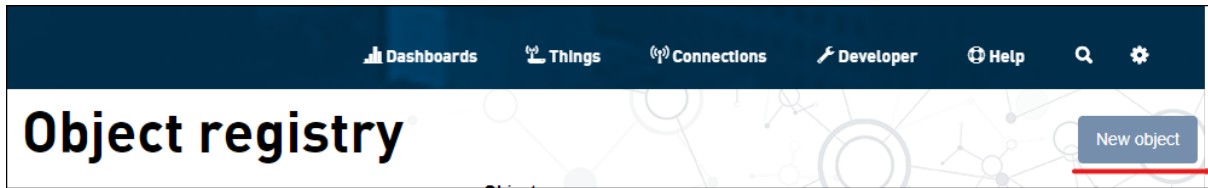


Figure 197

Copy the xml file content and paste it in the new Object form

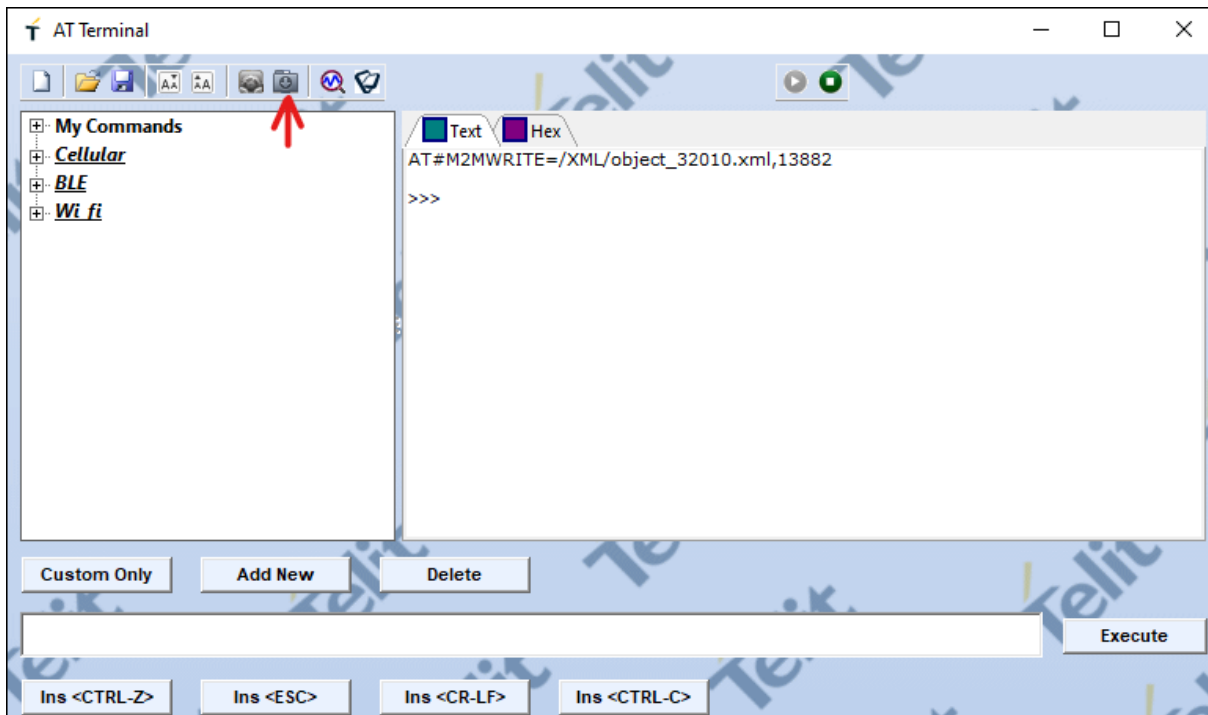
The screenshot shows a form titled "New object". Inside the form, there is a large text area with the placeholder text "Paste your LWM2M object definition here:". At the bottom of the form, there are two buttons: "Add" and "Cancel".

Figure 198

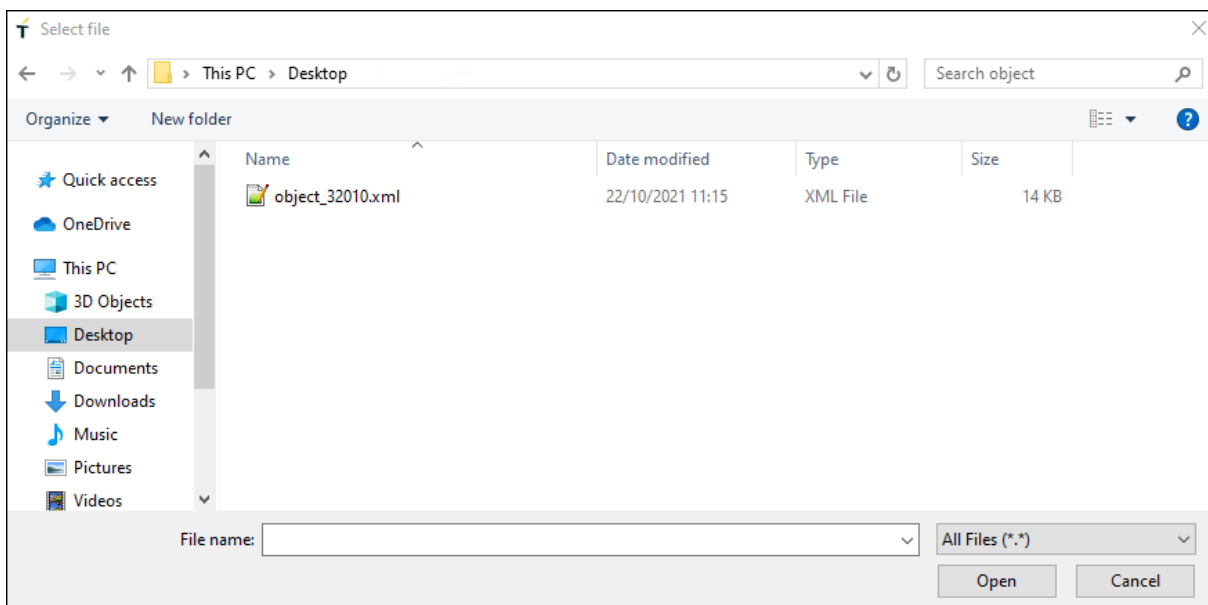
Also, the application requires the XML file `xml/object_32010.xml` (provided with the sample files) to be stored in module's `/XML/` folder. It can be done with

```
AT#M2MWRITE=/XML/object_32010.xml,<size_in_bytes>
```

To load the XML file in the module, Telit AT Controller (TATC) can be used. Once the command above is issued, press the load content button:

**Figure 199**

Select the file from your computer

**Figure 200**

The file is successfully loaded on the module

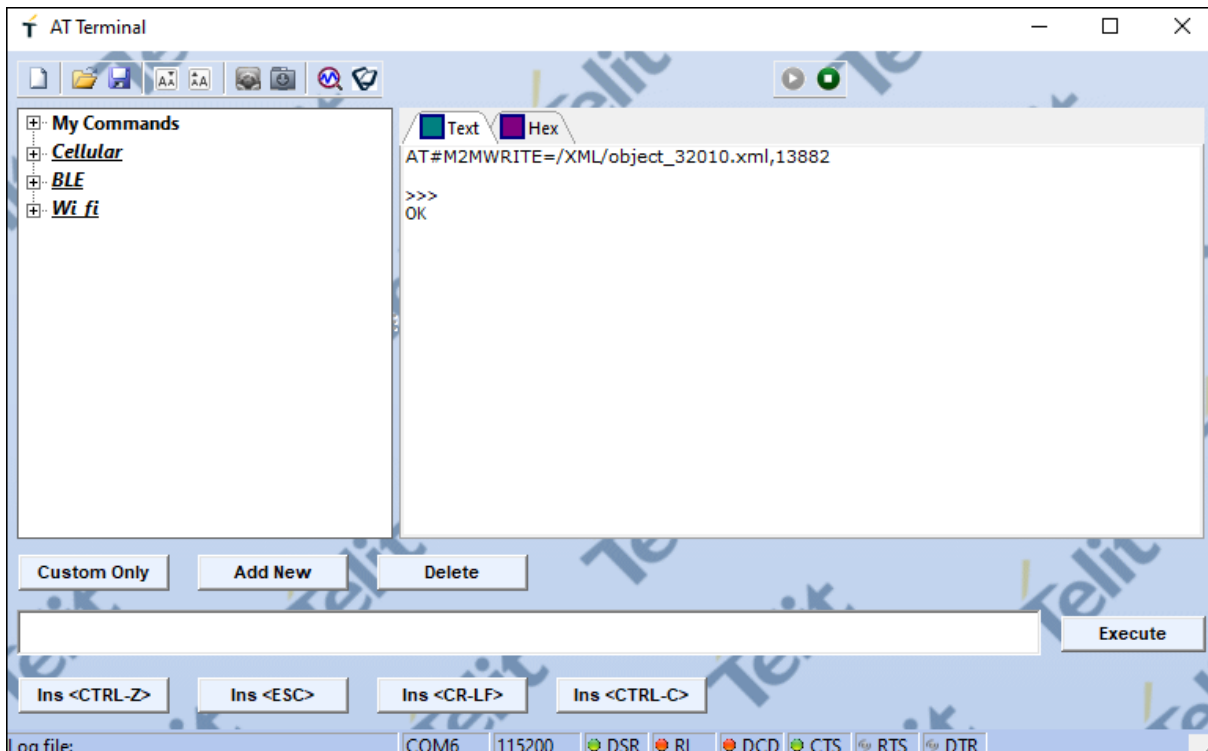


Figure 201

3.5.32.5 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on “**Things**” section, click “**New Thing**” button in the top right corner.

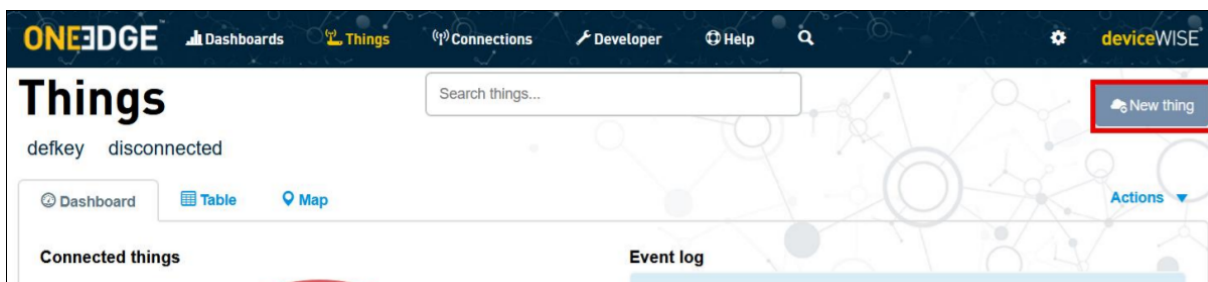


Figure 202

In the Create a new thing dialog, select “Telit Module”

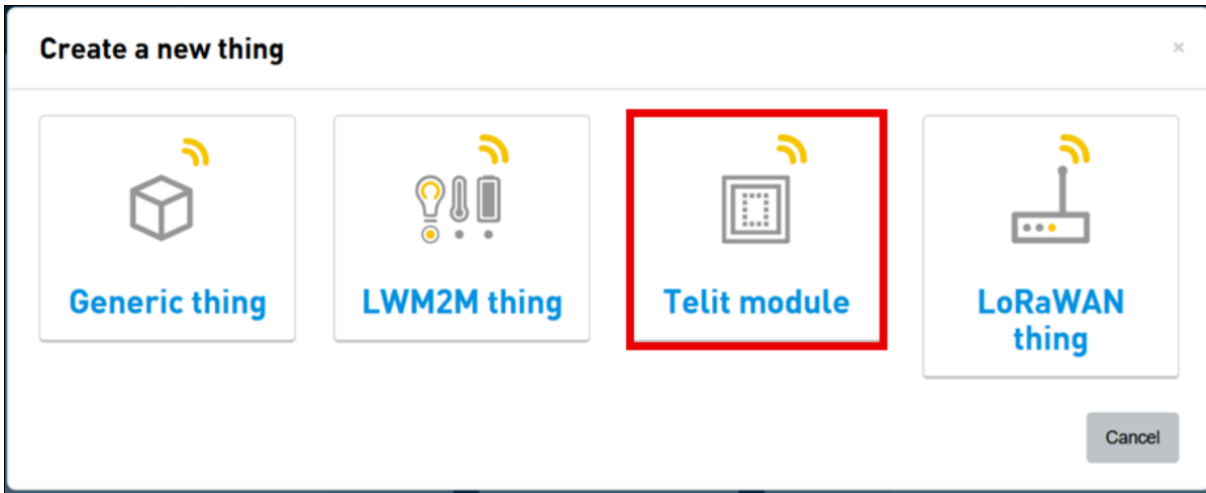


Figure 203

A dialog appears: select “Default” thing definition

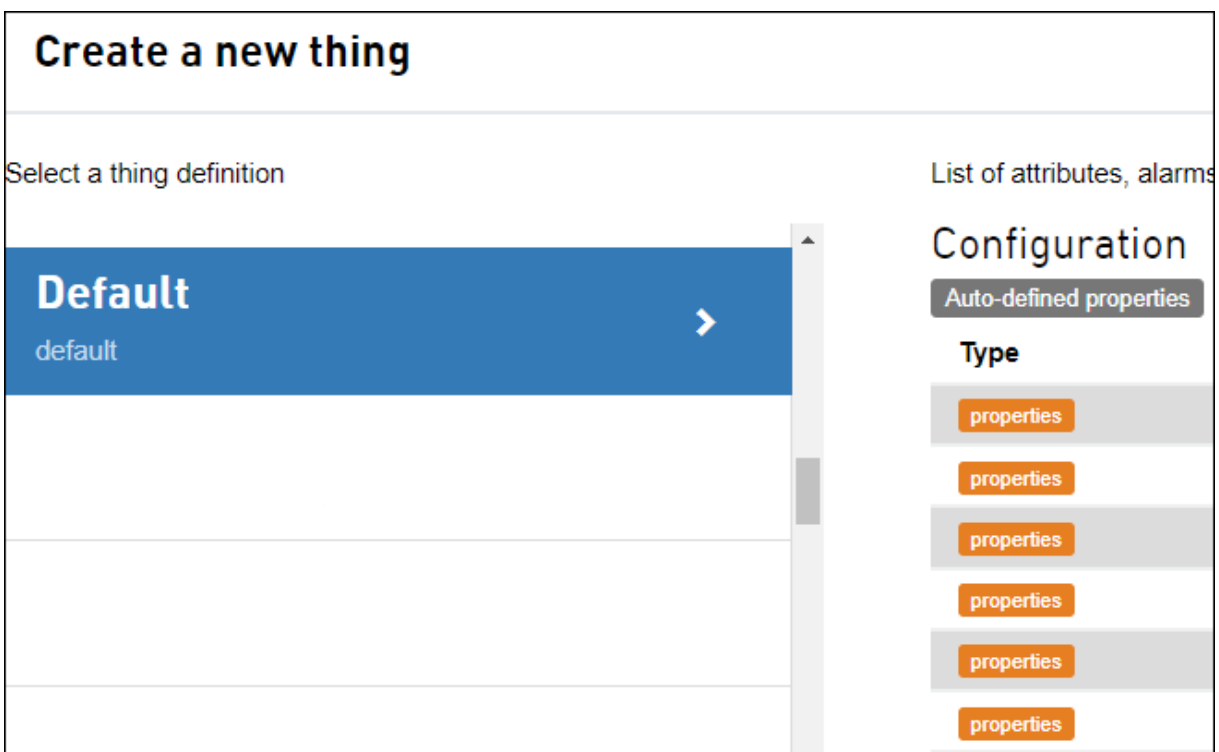


Figure 204

In the following screen, provide the Telit ID as “Identifier”. Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the screenshot below

Create a new thing

Setup
LWM2M
 Identification
 VAR
 Other
 Back

LWM2M
 Device profile*
 Lwm2m resources demo

Add Cancel

Figure 205

Click **“Add”** to complete the new thing creation procedure.

If the Thing already exists, its device profile can be changed by following the steps shown in the picture below

TELIT IoT PORTAL Dashboards Things Connections Developer Help deviceWISE

Things > ep_353080091125356

ep_353080091125356

Overview Details **LWM2M** LWM2M Device Management Firmware Attributes Remote AT Events Files Tunnels API usage Actions

Identity NDWfxhWyZGatw114
 Device profile Device Management Profile - Troubleshooting
 Remote address 141.226.117.241:3702
 Protocol LWM2M-CoAP (DTLSv12)
 Binding UQ
 Lifetime 630

Object browser **Set profile** Queue history Wakeup

Set LWM2M profile

Change profile*
 Lwm2m resources demo

Set Cancel

Figure 206

3.5.33 Logging Demo

Sample application showing how to print on one of the available output interfaces.
Debug prints on **USB0**

Features

- How to open a logging channel
- How to set a logging level
- How to use different logging macros

3.5.33.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Print a message with every log level

```
Starting Logging demo app. This is v1.0.7 built on Mar 26 2020 13:57:06.  
[WARN ] 20.17 M2MB_main.c:74 - M2MB_main{M2M_DamsStart}$ This is a WARNING MESSAGE  
[ERROR] 20.18 M2MB_main.c:76 - M2MB_main{M2M_DamsStart}$ THIS IS AN ERROR MESSAGE  
[CRITICAL] 20.19 M2MB_main.c:78 - M2MB_main{M2M_DamsStart}$ THIS IS AN CRITICAL MESSAGE  
[DEBUG] 20.19 M2MB_main.c:80 - M2MB_main{M2M_DamsStart}$ This is a DEBUG message  
[TRACE] 20.20 M2MB_main.c:82 - M2MB_main{M2M_DamsStart}$ This is a TRACE message  
END.
```

Figure 207

3.5.34 MD5 example

Sample application showing how to compute MD5 hashes using m2mb crypto. Debug prints on **USB0**

Features

- Compute MD5 hash of a file
- Compute MD5 hash of a string

3.5.34.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a temporary file with the expected content
- Compute MD5 hash of the provided text file
- Compare the hash with the expected one
- Compute MD5 hash of a string
- Compare the hash with the expected one
- Delete test file

```
Starting MD5 demo app. This is v1.0.7 built on Apr 7 2020 10:19:54.  
Buffer written successfully into file. 45 bytes were written.  
  
Computing hash from file...  
Computed hash: bb0fa6eff92c305f166803b6938dd33a  
Expected hash: bb0fa6eff92c305f166803b6938dd33a  
Hashes are the same!  
  
Computing hash from string...  
Computed hash: bb0fa6eff92c305f166803b6938dd33a  
Expected hash: bb0fa6eff92c305f166803b6938dd33a  
Hashes are the same!
```

Figure 208

3.5.35 MQTT Client

Sample application showcasing MQTT client functionalities (with SSL). Debug prints on **USB0**

Features

- How to check module registration and enable PDP context
- How to configure MQTT client parameters
- How to connect to a broker with SSL and exchange data over a subscribed topic

3.5.35.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage MQTT client and start it

mqtt_demo.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Init MQTT client
- Configure it with all parameters (Client ID, username, password, PDP context ID, keepalive timeout...)
- Connect MQTT client to broker
- Subscribe to two topics
- Publish 10 messages with increasing counter. Even messages are sent to topic 1, odd messages on topic 2.
- Print received message in mqtt_topc_cb function
- Disconnect MQTT client and deinit it
- Disable PDP context

```

Starting MQTT demo app. This is v1.0.7 built on Apr 7 2020 10:34:08.
[DEBUG] 16.18 mqtt_demo.c:192 - MQTT_Task(MQTT_TASK)$ INIT
[DEBUG] 16.18 mqtt_demo.c:206 - MQTT_Task(MQTT_TASK)$ m2mb_os_ev_init success
[DEBUG] 16.19 mqtt_demo.c:214 - MQTT_Task(MQTT_TASK)$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 16.19 mqtt_demo.c:221 - MQTT_Task(MQTT_TASK)$ Waiting for registration...
[DEBUG] 16.20 mqtt_demo.c:131 - NetCallback{pubTspt_0}$ Module is registered
[DEBUG] 16.21 mqtt_demo.c:232 - MQTT_Task(MQTT_TASK)$ Pdp context activation
[DEBUG] 18.26 mqtt_demo.c:246 - MQTT_Task(MQTT_TASK)$ Activate PDP with APN web.omnitel.it on CID 3....
[DEBUG] 18.95 mqtt_demo.c:155 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 18.96 mqtt_demo.c:159 - PdpCallback{pubTspt_0}$ IP address: 37.118.201.56
[DEBUG] 18.96 mqtt_demo.c:268 - MQTT_Task(MQTT_TASK)$ Init MQTT
[DEBUG] 18.97 mqtt_demo.c:278 - MQTT_Task(MQTT_TASK)$ m2mb_mqtt_init succeeded

Connecting to broker <api-dev.devicewise.com>:1883...
Done.
Subscribing to test_topic and test_topic2..
[DEBUG] 20.35 mqtt_demo.c:367 - MQTT_Task(MQTT_TASK)$ Done.

[DEBUG] 20.36 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 2> to topic test_topic
[DEBUG] 20.37 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 20.71 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 27
[DEBUG] 20.72 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 2>
[DEBUG] 23.37 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 3> to topic test_topic2
[DEBUG] 23.38 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 23.92 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 27
[DEBUG] 23.93 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 3>
[DEBUG] 26.40 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 4> to topic test_topic
[DEBUG] 26.41 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 26.93 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 27
[DEBUG] 26.93 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 4>
[DEBUG] 29.42 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 5> to topic test_topic2
[DEBUG] 29.43 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 29.99 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 27
[DEBUG] 30.00 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 5>
[DEBUG] 32.46 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 6> to topic test_topic
[DEBUG] 32.48 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 33.00 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 27
[DEBUG] 33.01 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 6>
[DEBUG] 35.47 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 7> to topic test_topic2
[DEBUG] 35.48 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 36.01 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 27
[DEBUG] 36.02 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 7>
[DEBUG] 38.50 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 8> to topic test_topic
[DEBUG] 38.51 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 39.15 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 27
[DEBUG] 39.16 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 8>
[DEBUG] 41.52 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 9> to topic test_topic2
[DEBUG] 41.53 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 42.10 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 27
[DEBUG] 42.12 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 9>
[DEBUG] 44.56 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 10> to topic test_topic
[DEBUG] 44.57 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 45.09 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 28
[DEBUG] 45.11 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 10>
[DEBUG] 47.58 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 11> to topic test_topic2
[DEBUG] 47.59 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 48.12 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 28
[DEBUG] 48.13 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 11>

Disconnecting from MQTT broker..
[DEBUG] 50.60 mqtt_demo.c:414 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 50.61 mqtt_demo.c:443 - MQTT_Task(MQTT_TASK)$ application exit
[DEBUG] 50.62 mqtt_demo.c:453 - MQTT_Task(MQTT_TASK)$ m2mb_pdp_deactivate returned success
[DEBUG] 50.63 mqtt_demo.c:457 - MQTT_Task(MQTT_TASK)$ Application complete.
[DEBUG] 51.23 mqtt_demo.c:164 - PdpCallback{pubTspt_0}$ Context deactivated!

```

Figure 209

3.5.36 MultiTask

Sample application showcasing multi tasking functionalities with M2MB API. Debug prints on **USB0**

Features

- How to create tasks using azx utilities
- How to use send messages to tasks
- How to use a semaphore to synchronize two tasks

3.5.36.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create three tasks with the provided utility (this calls public m2mb APIs)
- Send a message to the task1, its callback function azx_msgTask1 will be called

azx_msgTask1

- Print received parameters from main
- Send modified parameters to task2 (its callback function azx_msgTask2 will be called)
- wait for an InterProcess Communication semaphore to be available (released by task3)
- Once the semaphore is available, print a message and return

azx_msgTask2

- Print received parameters from caller
- If first parameter is bigger than a certain value, Send modified parameters to task3
- Else, use the second parameter as a task handle and print the corresponding name plus the value of the first parameter

azx_msgTask3

- Print received parameters from task 2
- release IPC semaphore
- send message to task 2 with first parameter below the threshold and second parameter with task3 handle


```
Starting MultiTask demo app. This is v1.0.12-C1 built on Jun 23 2020 15:36:31.
Inside "myTask1" user callback function. Received parameters from MAIN: 3 4 5
Task1 - Sending a message to task 2 with modified parameters...
Task1 - Waiting for semaphore to be released by task 3 now...
Inside "myTask2" user callback function. Received parameters: 5 7 10
Task2 - Sending a message to task 3 with modified parameters...
Task2 - Done.
Inside "myTask3" user callback function. Received parameters from Task 2: 15 14 9
Task3 - Releasing IPC semaphore...
Task1 - After semaphore! return...
Task3 - IPC semaphore released.
Task3 - Sending a message to task 2 with specific 'type' parameter value of 0 and task 3 handle as param1...
Inside "myTask2" user callback function. Received parameters: 0 1073951320 9
Task3 - Done.
Task2 - Received type 0 from task "myTask3"
Task2 - Done.
```

Figure 210

3.5.37 Mutex

Sample application showing mutex usage, with ownership and prioritization usage.
Debug prints on **USB0**

Features

- How to create a mutex
- How to use the mutex with tasks having different priorities
- how to reorder the pending tasks queue for the mutex

3.5.37.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create four tasks with the provided utility (this calls public m2mb APIs). The first task is a “producer”, putting data on a shared buffer. The second is a “consumer” of said data, the other two are used for prioritization demo
- run producer and consumer tasks at the same pace. the shared buffer will stay empty, because the resource is consumed right after creation
- run producer twice as fast as consumer. The buffer is slowly filled
- run consumer twice as fast as publisher. The buffer is always empty.
- reserve the mutex in the main task and run producer, support and support2 tasks (in this order). Then release the mutex and check the execution order. It should be by arrival.
- reserve the mutex in the main task and run the same three task, but before releasing the mutex, call the prioritization API. the task with highest priority (producer) is put as first in the queue.

```

Starting MutEx app. This is v1.0.12-C1 built on Jul 1 2020 08:37:15.
[DEBUG] 14.50 M2MB_main:90 - mutex_init{M2M_DamsStart}$ [MUTEX] Mutex initialized

[CASE 1 ] Producer and consumer have same idle time

[DEBUG] 14.51 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 14.52 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 14.53 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 14.53 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 14.54 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 14.54 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 14.55 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 14.56 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 15.56 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 15.56 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 15.57 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 15.58 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 15.58 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 15.59 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 15.60 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 15.60 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 16.61 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 16.61 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 16.62 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 16.63 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 16.63 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 16.64 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 16.64 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 16.65 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released

```

Figure 211

```

[CASE 2 ] Producer has double idle time

[DEBUG] 17.56 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 17.56 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 17.57 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 17.58 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 17.58 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 17.59 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 17.59 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 17.60 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 18.63 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 18.64 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 0 items
[DEBUG] 18.64 M2MB_main:268 - msgConsumer{CONSUMER}$ Can't consume anything, buffer size is 0
[DEBUG] 18.65 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 19.62 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 19.62 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 19.63 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 19.64 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 19.68 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 19.69 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 19.69 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 19.70 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 20.73 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 20.74 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 0 items
[DEBUG] 20.75 M2MB_main:268 - msgConsumer{CONSUMER}$ Can't consume anything, buffer size is 0
[DEBUG] 20.75 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 21.67 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 21.68 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 21.68 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 21.69 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 21.77 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 21.79 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 21.80 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 21.80 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released

```

Figure 212

```
[CASE 3 ] Producer has half idle time

[DEBUG] 22.62 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 22.63 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 22.64 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 22.64 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 22.65 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 22.65 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 22.66 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 22.67 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 23.67 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 23.68 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 23.68 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 23.69 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 24.71 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 24.72 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 24.72 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 24.73 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 24.74 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 24.74 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 24.75 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 24.76 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 25.79 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 25.79 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 1 items
[DEBUG] 25.80 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 1
[DEBUG] 25.81 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 26.78 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 26.78 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 2 items
[DEBUG] 26.79 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 1
[DEBUG] 26.79 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 26.84 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 26.84 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 1 items
[DEBUG] 26.85 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 1
[DEBUG] 26.86 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
```

Figure 213

```
[CASE 4 ] NO HTPF

Reserve MUTEX so all tasks are enqueued
[DEBUG] 30.77 M2MB_main:387 - msgSupport{HPTF_SUPPORT}$ freepos = 0 | evaluate[freepos]= 3
[DEBUG] 30.78 M2MB_main:416 - msgSupport2{HPTF_SUPPORT2}$ freepos = 1 | evaluate[freepos]= 4
[DEBUG] 30.79 M2MB_main:223 - msgProducer{PRODUCER}$ producer: freepos = 2 | evaluate[freepos]= 1
[DEBUG] 35.85 M2MB_main:586 - M2MB_main{M2M_DamsStart}$ EVALUATE SEQUENCE IS 3 4 1. Expected: 3 4 1
NO HTPF OK

[CASE 4.1 ] HTPF USED

Reserve MUTEX so all tasks are enqueued
m2mb_os_mtx_hptf OK
[DEBUG] 41.98 M2MB_main:223 - msgProducer{PRODUCER}$ producer: freepos = 0 | evaluate[freepos]= 1
[DEBUG] 41.99 M2MB_main:387 - msgSupport{HPTF_SUPPORT}$ freepos = 1 | evaluate[freepos]= 3
[DEBUG] 42.00 M2MB_main:416 - msgSupport2{HPTF_SUPPORT2}$ freepos = 2 | evaluate[freepos]= 4
[DEBUG] 44.03 M2MB_main:650 - M2MB_main{M2M_DamsStart}$ EVALUATE SEQUENCE IS 1 3 4, expected 1 3 4
HTPF DEMO OK
The application has ended...
```

Figure 214

3.5.38 NTP example

The application connects to an NTP server, gets current date and time and updates module's internal clock. Debug prints on **USB0**

Features

- How to get current date and time from an NTP server
- How to set current date and time on module

3.5.38.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Send message to ntpTask

ntp_task.c

NTP_task() - Waits module registration - When module is registered, initializes ntp setting CID, server url and timeout - When PDP context is correctly opened, a query to NTP server is done to get current date and time - On SET_MODULE_RTC message type reception, module RTC is set with date time value got from NTP server.

m2mb_ntp_ind_callback() - As soon as M2MB_NTP_VALID_TIME event is received, current date and time is printed and a message (with SET_MODULE_RTC type) is sent to NTP_task

```
Start NTP demo application. This is v1.0 built on Apr 16 2021 09:36:12.
Waiting for registration...
Module is registered!

Activate PDP context with APN ibox.tim.it on CID 3
Context activated, IP address: 2.195.170.123
Get current time from server 0.pool.ntp.org, PORT: 123

Current time is: Friday 2021-04-16, 07:37:33

Current time correctly set on module
Module system time is: 2021-04-16, 07:37:33
```

Figure 215

3.5.39 ON OFF button management example

Sample application on how to handle ON/OFF button. Debug prints on **USB0**

Features

- How to enable ON/OFF button handling
- How to turn off module after ON/OFF button press/release

Application workflow

M2MB_main.c

- Open gpio_ON_OFF
- Configure the interrupt and the callback
- Init power functionality
- Wait for on/off button to be pressed
- If button has been pressed for more than 3 seconds (and then released) then module is turned off.

```
Starting ON OFF button handling demo app. This is v1.1.8 built on Feb  9 2023 11:56:05.  
Waiting for ON/OFF button to be pressed...  
onoff pressed!  
onoff released!  
ON/OFF pressed form more than 3 sec, turn off module!
```

Figure 216

3.5.40 RTC example

Sample application that shows RTC apis functionalities: how to get/set module system time and timestamp. Debug prints on **USB0**

Features

- How to read module timestamp
- How to read module system time
- How to set new system time

3.5.40.1 Application workflow

M2MB_main.c

- Init log azx and print a welcome message
- Init net functionality and wait for module registration
- Init RTC functionality and get module time in timestamp format (seconds from the epoch)
- Get module system time in date/time format
- Add 1 hour to timestamp, convert it to system time and set it to module

```
Start RTC demo application. This is v1.0 built on Oct  1 2021 15:01:40.
Waiting for registration...
Module is registered!

Current time in seconds from the epoch: 1633101266
Module system time is: 2021-10-01, 15:14:26

Get current time and add an hour

Current time in seconds from the epoch: 1633101266
New time to be set : 2021-10-01, 16:14:26, tz:4, d1st:0

Set new time and check the setting
NEW module system time is: 2021-10-01, 16:14:26
```

Figure 217

3.5.41 SIM event handler example

Sim Event Demo application. Debug prints on **USB0**, using AZX log example functions

Features

- How to use ATI function for asynchronous management
- How to catch URC from an AppZone application
- How to catch SIM related events and handle them

Application workflow

M2MB_main.c

- Print welcome message
- Initialize AT interface
- Initialize AT URC manager task
- Initialize SIM event manager task
- Send "AT#SIMPR=1" to activate SIM URCs
- Insert SIM in SIM slot 1 and receive SIM inserted message
- Remove SIM from SIM slot 1 and receive SIM removed message

```
Starting SIM Presence Demo app. This is v1.1.17 built on Dec 16 2022 15:14:28.
Please ensure SIM is not inserted before starting this sample application
Sending command AT#SIMPR=1 to enable SIM presence URC messages...
Command set.
SIM 0 state changed to 1!

SIM CARD HAS BEEN INSERTED!

SIM 0 state changed to 0!

SIM CARD HAS BEEN REMOVED!
```

Figure 218

3.5.42 SMS PDU

Sample application showcasing how to create and decode PDUs to be used with m2mb_sms_* API set. A SIM card and antenna must be present. Debug prints on **USB0**

Features

- How to enable SMS functionality
- How to use encode an SMS PDU to be sent with m2mb_api
- How to decode a received SMS response from PDU to ASCII mode.

3.5.42.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Init sms functionality
- Create PDU from text message
- Send message to destination number
- Wait for response
- When SMS PDU response is received, decode it and print information about it, plus the message content

```
m2mb_sms_init() succeeded

Sending message <How are you?>...
m2mb_sms_send() - succeeded
M2MB_SMS_SEND_RESP Callback
Send resp msg ID 10
SMS received!
SMS correctly received!

Reading SMS from memory...
m2mb_sms_read() request succeeded

--- SMS read ---
SMS tag M2MB_SMS_TAG_MT_NOT_READ
SMS format M2MB_SMS_FORMAT_3GPP
Code type: 0
Sender type: 145
Msg len: 12
Msg bytes: 11
Msg date 19/7/17 16:7:58 (timezone: 2)
Received SMS, content: <<Fine thanks >>
Sender: +[REDACTED]
```

Figure 219

3.5.43 SMS_atCmd example

Sample application showcasing how to receive an SMS containing an AT command, process the AT command and send its answer to sender (configurable in sms_config.txt). A SIM card and antenna must be present. Debug prints on **USB0**

Features

- How to receive an SMS with an AT command as text inside
- How to send AT command to parser and read the answer
- How to send the AT command answer back to sender via SMS

Optional configuration file to be put in /mod folder, copy sms_config.txt file into your module running the following AT command:

```
AT#M2MWRITE="/mod/sms_config.txt",138
```

>>> here receive the prompt; then type or send the file, sized 138 bytes

3.5.43.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Init SMS functionality
- Read configuration file sms_config.txt (send SMS with AT command answer back, delete SMS received)
- Init AT command parser
- Create a task to handle SMS parsing and AT command sending
- Wait for an incoming SMS

callbacks.c

msgSMSparse()

- When SMS has been received, content is decoded and printed. If there is an AT command inside, command is executed and answer printed and sent back to sender as an SMS (depending on sms_config.txt setting)

```
yStarting SMS with AT command demo app. This is v1.0.13-C1 built on Mar 18 2021 12:42:22.
[DEBUG] 16.61 M2MB_main:135 - M2MB_main{M2M_DamsStart}$ m2mb_os_ev_init success
m2mb_sms_init() succeeded
[DEBUG] 16.62 M2MB_main:168 - M2MB_main{M2M_DamsStart}$ M2MB_SMS_INCOMING_IND indication enabled
[DEBUG] 16.63 M2MB_main:179 - M2MB_main{M2M_DamsStart}$ M2MB_SMS_INCOMING_IND MEMORY FULL indication enabled
[DEBUG] 16.64 M2MB_main:196 - M2MB_main{M2M_DamsStart}$ Storage set to M2MB_SMS_STORAGE_SM
[DEBUG] 16.65 callbacks:114 - readConfigFromFile{M2M_DamsStart}$ Reading parameters from file
[DEBUG] 16.66 callbacks:116 - readConfigFromFile{M2M_DamsStart}$ Opening /mod/sms_config.txt in read mode..
Default: SMS with answer sending DISABLED, delete sms DISABLED
[DEBUG] 16.67 at_async:115 - at_cmd_async_init{M2M_DamsStart}$ m2mb_ati_init() on instance 0
Please send an SMS with a configuration as ("ATCMD: <atcmd>")...
```

Figure 220

3.5.44 SMTP Client

Sample application showing SMTP echo demo with M2MB API. Debug prints on **USB0**

Features

- How to check module registration and activate PDP context
- How to open a SMTP client
- How to send a mail

Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage SMTP client and start it

M2MB_main.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Initialize SMTP client and connect to SMTP server
- Prepare email and send it
- Close SMTP client
- Disable PDP context

```
Starting SMTP demo app. This is v1.1.17 built on Jan  9 2023 09:57:51.
Reading parameters from file...
Set APN to: <<nxt17.net>>
Set APN USER to: <<>>
Set APN PASS to: <<>>
Set PDP_CTX to: 1
Set MAIL SERVER to: <<          >>
Set MAIL PORT to: <<      >>
Set MAIL USER to: <<          >>
Set MAIL PASS to: <<          >>
Set MAIL FROM to: <<          >>
Set MAIL FROM NAME to: << >>
Set MAIL SUBJECT to: <<Hello from AppZone>>
Set MAIL BODY to: <<AppZone Test>>
Set MAIL TO to: <<          >>
Set MAIL TO NAME to: << >>
Waiting for registration...
Pdp context activation
Connecting to SMTP Server...
SMTP Client connected!
Preparing example's mail...
Sending example's mail...
Example's mail sent!
SMTP client closed
Application complete.
```

Figure 221

3.5.45 SPI Echo

Sample application showing how to communicate over SPI with m2mb API. Debug prints on **USB0**

Features

- How to open an SPI bus. MOSI and MISO will be shorted, to have an echo.
- How to communicate over SPI bus

3.5.45.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Open SPI bus, set parameters
- Send data on MOSI and read the same in MISO

Notes:

For LE910Cx (both Linux and ThreadX based devices), AT#SPIEN=1 command must be sent once before running the app

```
Starting SPI demo app. This is v1.0.7 built on Apr  1 2020 13:48:05.  
Transfer successful. Received: hello from spi echo
```

Figure 222

3.5.46 SPI sensors

Sample application showing SPI usage, configuring two ST devices: a magnetometer (ST LIS3MDL) and a gyroscope (ST L3G4200D). The application will read values from both devices using GPIO4 and 3 (respectively) as magnetometer CS and gyro CS. Debug prints on **USB0**

Features

- How to open an SPI bus with a slave device
- How to communicate with the device over the SPI bus

Setup

- Connect sensor VDD to 3v8 supply (e.g. Vbatt on the module)
- Connect sensor GND to a GND pin of the module
- Connect sensors MOSI to module SPI_MOSI
- Connect sensors MISO to module SPI_MISO
- Connect sensors CLK to module SPI_CLK
- Connect magnetometer CS to module GPIO 2
- Connect gyroscope CS to module GPIO 3

3.5.46.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Open SPI bus, set parameters
- Configure GPIO 2 and GPIO 3 as output, set them high (idle)
- Set registers to configure magnetometer
- Read in a loop (10 iterations) the registers carrying the 3 axes values and show the gauss value for each of them. A metal object is put close to the sensor to change the read values.
- Set registers to configure gyroscope
- Read in a loop (10 iterations) the registers carrying the 3 axes values and show the degrees per second value for each of them. The board is rotated to change the read values.

Notes:

For LE910Cx (both Linux and ThreadX based devices), AT#SPIEN=1 command must be sent once before running the app

```
Starting SPI demo app. This is v1.0.7 built on Apr 1 2020 13:58:25.
SPI start

Magnetometer SPI Demo start
Reading Magnetometer WHOAMI. Expected: 0x3D
Expected response received!
Setting continuous conversion mode...
Continuous conversion mode successfully set.
Setting 10 Hz Output Data Rate, Medium performance mode X Y axis...
Magnetometer Enabled. 10Hz ODR, Medium Perf. Mode (X,Y).
Setting Medium performance for Z axis, little endian...
Medium Perf. Mode (Z), little endian.
Setting complete, starting reading loop...

X: 0.204 gauss
Y: -0.321 gauss
Z: 0.305 gauss

X: 0.290 gauss
Y: -0.103 gauss
Z: 0.043 gauss

X: -2.513 gauss
Y: -0.353 gauss
Z: -4.000 gauss

X: 1.980 gauss
Y: 0.174 gauss
Z: -1.945 gauss

X: 4.000 gauss
Y: -0.090 gauss
Z: -4.000 gauss

X: -0.605 gauss
Y: -0.154 gauss
Z: 0.210 gauss

X: -0.580 gauss
Y: 2.004 gauss
Z: -0.047 gauss

X: 0.177 gauss
Y: -0.359 gauss
Z: 0.295 gauss

X: 0.173 gauss
Y: -0.356 gauss
Z: 0.301 gauss

X: 0.174 gauss
Y: -0.356 gauss
Z: 0.298 gauss
Reading complete.
```

Figure 223

3.5.47 SW Timer (Software Timer)

The sample application shows how to use SW Timers M2MB API. Debug prints on **USB0**

Features

- How to open configure a SW timer
- How to use the timer to manage recurring events

3.5.47.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create sw timer structure
- Configure it with 4 seconds timeout, periodic timer (auto fires when expires)
- Init the timer with the parameters
- Start the timer
- Wait 10 seconds
- Stop the timer

timerCb

- Print a message with inside the callback

```
Starting SW Timers demo app. This is v1.0.7 built on Apr 7 2020 09:51:25.  
timer expired!  
[DEBUG] 21.41 M2MB_main.c:59 - timerCb{pubTspt_0}$ timer handle: 0x4002b004  
timer expired!  
[DEBUG] 25.47 M2MB_main.c:59 - timerCb{pubTspt_0}$ timer handle: 0x4002b004  
stopping the timer  
Stop a running timer: success  
Application end
```

Figure 224

3.5.48 Secure MicroService

Sample application showcasing how to manage secure microservice functionalities.
Debug prints on **USB0**

Features

- Write data in Secure Data Area (SDA), non protected
- Read the written data and compare with the original buffer
- Write a cripty key in Secure Data Area (SDA), non protected
- Perform a rotate of the written key data
- Perform MD5 sum of written data from TZ file
- Compare computed digest with expected one
- Write data in trust zone as a trusted object (it will not be possible to read it again but only use its content for crypto operations)
- Try to read the trusted object and verify it fails
- Rotate trusted item and verify retrieving the content fails
- compute MD5 sum of trusted item and compare with the expected one
- Try to pass data from a trusted item to a non trusted item using untrusted TZ buffers, and verify it fails

3.5.48.1 Application workflow

M2MB_main.c

- Write a buffer in a SDA item using `m2mb_secure_ms_write`
- Read the same item using `m2mb_secure_ms_read`
- Write a buffer containing some cripty key in a SDA item using `m2mb_secure_ms_write`
- Rotate the content of the key item
- Read it with `m2mb_secure_ms_read`
- Load the key content using `m2mb_secure_ms_crypto_alloc` and `m2mb_secure_crypto_add_it` in a SECURE_MS buffer
- Compute MD digest with `m2mb_secure_ms_crypto_md`
- Write a buffer containing some cripty key in a SDA item using `m2mb_secure_ms_write` but with **TRUSTED** option in `m2mb_secure_ms_open`
- Verify that `m2mb_secure_ms_read` on the trusted item fails
- Verify that `m2mb_secure_ms_crypto_rotate` fails for the trusted item
- Verify the MD5 digest
- Try to copy the trusted item data in a SECURE_MS buffer with `m2mb_secure_ms_crypto_alloc` and `m2mb_secure_crypto_add_item`, then load it in an untrusted object with `m2mb_secure_ms_crypto_write`, and verify it fails.

```

Starting secure ms demo app. This is v1.0.13-C1 built on Jul 30 2020 12:19:02.
Writing data in normal item
Stored input data in Secure Data Area
Reading data from normal item
Data length in SDA: 11 bytes
Securely loaded the data from the SDA
Read 11 bytes: <hello world>
original and retrieved strings are the same

Writing key in normal item
Stored input data in Secure Data Area

Rotate data in normal item
Original key: AA_THIS_IS_MY_SECRET_KEY_BB
Rotated key:

Compute MD5 of data in normal item
Data length in SDA: 27 bytes
MD5: 8EDAD26E26E1C74C7C02386C1C7F541D
hash is the expected one!

Writing data in trusted item
Stored input data in Secure Data Area
Reading data from trusted item (should fail!)
Data length in SDA: 27 bytes
m2mb_secure_ms_read() failed for trusted item, as expected!

Rotate data in trusted item
[ERROR] 17.01 M2MB_main:329 - read_rotate{M2M_DamsStart}$ Cannot read data from SECURE_MS_BUFFER to user buffer
Original key: AA_THIS_IS_MY_SECRET_KEY_BB
Rotated key:

Compute MD5 of data in trusted item
Data length in SDA: 27 bytes
MD5: 8EDAD26E26E1C74C7C02386C1C7F541D
Hash is the expected one!

Try to pass data from trusted to untrusted through TZ buffers
Cannot store data from SECURE_MS_BUFFER to SDA 'non-trusted', as expected

```

Figure 225

3.5.49 TCP IP

Sample application showcasing TCP echo demo with M2MB API. Debug prints on **USB0**

Features

- How to check module registration and activate PDP context
- How to open a TCP client socket
- How to communicate over the socket

3.5.49.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage socket and start it

m2m_tcp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and link it to the PDP context id
- Connect to the server
- Send data and receive response
- Close socket
- Disable PDP context

```
Starting TCP-IP demo app. This is v1.0.7 built on Mar 26 2020 16:20:30.
[DEBUG] 21.23 m2m_tcp_test.c:201 - M2M_msgTCPTask{TCP_TASK}$ INIT
[DEBUG] 21.25 m2m_tcp_test.c:217 - M2M_msgTCPTask{TCP_TASK}$ m2mb_os_ev_init success
[DEBUG] 21.26 m2m_tcp_test.c:223 - M2M_msgTCPTask{TCP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 21.26 m2m_tcp_test.c:231 - M2M_msgTCPTask{TCP_TASK}$ Waiting for registration...
[DEBUG] 21.28 m2m_tcp_test.c:128 - NetCallback{pubTspt_0}$ Module is registered to cell 0x816B!
[DEBUG] 21.29 m2m_tcp_test.c:244 - M2M_msgTCPTask{TCP_TASK}$ Pdp context activation
[DEBUG] 21.30 m2m_tcp_test.c:248 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.34 m2m_tcp_test.c:263 - M2M_msgTCPTask{TCP_TASK}$ Activate PDP with APN web.omnitel.it....
[DEBUG] 24.52 m2m_tcp_test.c:155 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 24.52 m2m_tcp_test.c:158 - PdpCallback{pubTspt_0}$ IP address: 83.225.44.56
[DEBUG] 24.54 m2m_tcp_test.c:273 - M2M_msgTCPTask{TCP_TASK}$ Creating Socket...
[DEBUG] 24.54 m2m_tcp_test.c:284 - M2M_msgTCPTask{TCP_TASK}$ Socket created
[DEBUG] 24.55 m2m_tcp_test.c:294 - M2M_msgTCPTask{TCP_TASK}$ Socket ctx set to 3
[DEBUG] 24.95 m2m_tcp_test.c:307 - M2M_msgTCPTask{TCP_TASK}$ Retrieved IP: 185.86.42.218
[DEBUG] 25.17 m2m_tcp_test.c:322 - M2M_msgTCPTask{TCP_TASK}$ Socket Connected!
[DEBUG] 25.18 m2m_tcp_test.c:329 - M2M_msgTCPTask{TCP_TASK}$ Sending data over socket..
[DEBUG] 25.19 m2m_tcp_test.c:342 - M2M_msgTCPTask{TCP_TASK}$ Data send successfully (16 bytes)
[DEBUG] 27.20 m2m_tcp_test.c:356 - M2M_msgTCPTask{TCP_TASK}$ trying to receive 16 bytes..
[DEBUG] 27.21 m2m_tcp_test.c:364 - M2M_msgTCPTask{TCP_TASK}$ Data received (16): <hello from m2mb!>
[DEBUG] 27.21 m2m_tcp_test.c:373 - M2M_msgTCPTask{TCP_TASK}$ application exit
[DEBUG] 27.22 m2m_tcp_test.c:385 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 27.24 m2m_tcp_test.c:388 - M2M_msgTCPTask{TCP_TASK}$ Application complete.
[DEBUG] 29.43 m2m_tcp_test.c:164 - PdpCallback{pubTspt_0}$ Context successfully deactivated!
```

Figure 226

3.5.50 TCP non blocking example

Sample application that shows how to configure and connect a TCP-IP non blocking socket. Debug prints on **USB0**

Features

- How to check module registration and activate PDP context
- How to open a TCP client non Blocking socket
- How to communicate over the socket

Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage socket and start it

m2m_tcp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and link it to the PDP context id
- Set the socket as non Blocking and connect to server. Uses `m2mb_socket_bsd_select`, `m2mb_socket_bsd_fd_isset_func` to check when socket is connected.
- Send data and receive response
- Close socket
- Disable PDP context

```

Starting TCP-IP non blocking demo app. This is v1.1.9 built on Mar 22 2022 15:35:56.
[DEBUG] 19.24 m2mb_tcp_te:215 - M2M_msgTCPTask{TCP_TASK}$ INIT
[DEBUG] 19.25 m2mb_tcp_te:231 - M2M_msgTCPTask{TCP_TASK}$ m2mb_os_ev_init success
[DEBUG] 19.25 m2mb_tcp_te:237 - M2M_msgTCPTask{TCP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 19.26 m2mb_tcp_te:245 - M2M_msgTCPTask{TCP_TASK}$ Waiting for registration...
[DEBUG] 19.27 m2mb_tcp_te:142 - NetCallback{pubTspt_0}$ Module is registered to cell 0x20!
[DEBUG] 19.28 m2mb_tcp_te:258 - M2M_msgTCPTask{TCP_TASK}$ Pdp context activation
[DEBUG] 19.29 m2mb_tcp_te:262 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 21.34 m2mb_tcp_te:277 - M2M_msgTCPTask{TCP_TASK}$ Activate PDP with APN ibox.tim.it....
[DEBUG] 21.89 m2mb_tcp_te:169 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 21.90 m2mb_tcp_te:172 - PdpCallback{pubTspt_0}$ IP address: 2.195.160.174
[DEBUG] 21.90 m2mb_tcp_te:287 - M2M_msgTCPTask{TCP_TASK}$ Creating Socket...
[DEBUG] 21.91 m2mb_tcp_te:299 - M2M_msgTCPTask{TCP_TASK}$ Socket created
[DEBUG] 21.92 m2mb_tcp_te:309 - M2M_msgTCPTask{TCP_TASK}$ Socket ctx set to 1
[DEBUG] 22.10 m2mb_tcp_te:322 - M2M_msgTCPTask{TCP_TASK}$ Retrieved IP: 185.86.42.218
[DEBUG] 22.11 m2mb_tcp_te:332 - M2M_msgTCPTask{TCP_TASK}$ Socket set to nonBlocking
[DEBUG] 22.11 m2mb_tcp_te:351 - M2M_msgTCPTask{TCP_TASK}$ Connection in progress...
[DEBUG] 22.29 m2mb_tcp_te:377 - M2M_msgTCPTask{TCP_TASK}$ Socket <0x400343a0> is connected!
[DEBUG] 22.30 m2mb_tcp_te:398 - M2M_msgTCPTask{TCP_TASK}$ Socket Connected!
[DEBUG] 22.32 m2mb_tcp_te:411 - M2M_msgTCPTask{TCP_TASK}$ Sending data over socket..
[DEBUG] 22.32 m2mb_tcp_te:424 - M2M_msgTCPTask{TCP_TASK}$ Data send successfully (16 bytes)
[DEBUG] 24.37 m2mb_tcp_te:438 - M2M_msgTCPTask{TCP_TASK}$ trying to receive 16 bytes..
[DEBUG] 24.38 m2mb_tcp_te:446 - M2M_msgTCPTask{TCP_TASK}$ Data received (16): <hello from m2mb!>
[DEBUG] 24.39 m2mb_tcp_te:456 - M2M_msgTCPTask{TCP_TASK}$ application exit
[DEBUG] 24.43 m2mb_tcp_te:468 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 24.45 m2mb_tcp_te:471 - M2M_msgTCPTask{TCP_TASK}$ Application complete.
[DEBUG] 24.79 m2mb_tcp_te:178 - PdpCallback{pubTspt_0}$ Context successfully deactivated!

```

Figure 227

3.5.51 TCP Socket status

Sample application showcasing how to check a TPC connected socket current status.
Debug prints on **USB0**

Features

- How to check module registration and activate PDP context
- How to open a TCP client socket
- How to check if the TCP socket is still valid

3.5.51.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage socket and start it

m2m_tcp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and link it to the PDP context id
- Connect to the server
- Check in a loop the current socket status using the adv_select function with a 2 seconds timeout
- Close socket when the remote host closes it
- Disable PDP context


```

Starting TCP socket status check demo app. This is v1.0.14-C1 built on Sep  8 2020 14:59:25.
[DEBUG] 21.33 m2m_tcp_tes:324 - M2M_msgTCPTask{TCP_TASK}$ INIT
[DEBUG] 21.34 m2m_tcp_tes:338 - M2M_msgTCPTask{TCP_TASK}$ m2mb_os_ev_init success
[DEBUG] 21.34 m2m_tcp_tes:344 - M2M_msgTCPTask{TCP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 21.35 m2m_tcp_tes:352 - M2M_msgTCPTask{TCP_TASK}$ Waiting for registration...
[DEBUG] 21.36 m2m_tcp_tes:365 - M2M_msgTCPTask{TCP_TASK}$ Pdp context activation
[DEBUG] 21.37 m2m_tcp_tes:369 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.41 m2m_tcp_tes:384 - M2M_msgTCPTask{TCP_TASK}$ Activate PDP with APN NXT17.NET....
[DEBUG] 24.09 m2m_tcp_tes:281 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 24.10 m2m_tcp_tes:284 - PdpCallback{pubTspt_0}$ IP address: 100.77.5.223
[DEBUG] 24.10 m2m_tcp_tes:394 - M2M_msgTCPTask{TCP_TASK}$ Creating Socket...
[DEBUG] 24.11 m2m_tcp_tes:405 - M2M_msgTCPTask{TCP_TASK}$ Socket created
[DEBUG] 24.11 m2m_tcp_tes:415 - M2M_msgTCPTask{TCP_TASK}$ Socket ctx set to 3
[DEBUG] 24.60 m2m_tcp_tes:428 - M2M_msgTCPTask{TCP_TASK}$ Retrieved IP: 185.86.42.218
[DEBUG] 24.93 m2m_tcp_tes:443 - M2M_msgTCPTask{TCP_TASK}$ Socket Connected!
[DEBUG] 26.98 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 29.03 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
...
[DEBUG] 82.18 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 84.23 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 86.28 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 88.31 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 88.90 m2m_tcp_tes:154 - adv_select{TCP_TASK}$ Data is available on socket <0x40032b3c>
[DEBUG] 88.92 m2m_tcp_tes:160 - adv_select{TCP_TASK}$ There are <0> pending bytes on socket
Socket was closed by remote!
[DEBUG] 88.92 m2m_tcp_tes:494 - M2M_msgTCPTask{TCP_TASK}$ application exit
[DEBUG] 88.94 m2m_tcp_tes:506 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 88.94 m2m_tcp_tes:509 - M2M_msgTCPTask{TCP_TASK}$ Application complete.
[DEBUG] 89.31 m2m_tcp_tes:290 - PdpCallback{pubTspt_0}$ Context successfully deactivated!

```

Figure 228

3.5.52 TCP Server

Sample application showcasing TCP listening socket demo with M2MB API. Debug prints on **USB0**

Features

- How to check module registration and activate PDP context
- How to open a TCP listening socket
- How to manage external hosts connection and exchange data

3.5.52.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage socket and start it

m2m_tcp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and set it in non-blocking mode
- Bind the socket to the listening port
- Start listening for incoming connection
- Check if a connection is incoming using m2mb_socket_bsd_select function
- If a client connects, perform accept on the child socket
- Send a "START" message to the client
- Send some data
- Wait for data from client and print it
- Close the child socket
- Start listening again, up to 3 times
- Close listening socket
- Disable PDP context

Debug Log

```

Starting TCP Server demo app. This is v1.0.7 built on Apr  7 2020 13:28:24.
[DEBUG] 14.55 m2m_tcp_test.c:220 - M2M_msgTCPTask(TCP_TASK)$ INIT
[DEBUG] 14.55 m2m_tcp_test.c:236 - M2M_msgTCPTask(TCP_TASK)$ m2mb_os_ev_init success
[DEBUG] 14.57 m2m_tcp_test.c:242 - M2M_msgTCPTask(TCP_TASK)$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 14.57 m2m_tcp_test.c:250 - M2M_msgTCPTask(TCP_TASK)$ Waiting for registration...
[DEBUG] 14.58 m2m_tcp_test.c:138 - NetCallback(pubTspt_0)$ Module is registered to cell 0x5222!
[DEBUG] 14.59 m2m_tcp_test.c:263 - M2M_msgTCPTask(TCP_TASK)$ Pdp context activation
[DEBUG] 14.60 m2m_tcp_test.c:267 - M2M_msgTCPTask(TCP_TASK)$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 16.57 m2m_tcp_test.c:282 - M2M_msgTCPTask(TCP_TASK)$ Activate PDP with APN ibox.tim.it...
[DEBUG] 17.16 m2m_tcp_test.c:165 - PdpCallback(pubTspt_0)$ Context activated!
[DEBUG] 17.17 m2m_tcp_test.c:168 - PdpCallback(pubTspt_0)$ IP address: 2.195.165.137

-----
| Start TCP server |
|-----|

[DEBUG] 19.15 m2m_tcp_test.c:301 - M2M_msgTCPTask(TCP_TASK)$ Creating Socket...
[DEBUG] 19.15 m2m_tcp_test.c:312 - M2M_msgTCPTask(TCP_TASK)$ Socket created
[DEBUG] 19.16 m2m_tcp_test.c:313 - M2M_msgTCPTask(TCP_TASK)$ m2mb_socket_bsd_socket(): valid socket ID [0x4002E79C] - PASS
[DEBUG] 20.16 m2m_tcp_test.c:319 - M2M_msgTCPTask(TCP_TASK)$ issuing m2m_socket_bsd_ioctl() to set non-blocking mode ...
[DEBUG] 20.17 m2m_tcp_test.c:331 - M2M_msgTCPTask(TCP_TASK)$ Binding Socket...
[DEBUG] 22.12 m2m_tcp_test.c:343 - M2M_msgTCPTask(TCP_TASK)$ Socket Bind Pass

Start TCP listening on port 6500...
[DEBUG] 24.13 m2m_tcp_test.c:368 - M2M_msgTCPTask(TCP_TASK)$ select...
Select result: 0
[DEBUG] 28.13 m2m_tcp_test.c:368 - M2M_msgTCPTask(TCP_TASK)$ select...
Select result: 1

TCP Server Coming Connection
--> Accept
[DEBUG] 30.52 m2m_tcp_test.c:397 - M2M_msgTCPTask(TCP_TASK)$ Socket Accept Pass

Connected! (socket dial n.1)
[DEBUG] 30.53 m2m_tcp_test.c:403 - M2M_msgTCPTask(TCP_TASK)$ Client Source Address: 185.86.42.254
[DEBUG] 30.54 m2m_tcp_test.c:404 - M2M_msgTCPTask(TCP_TASK)$ Client Port: 58658
[DEBUG] 30.54 m2m_tcp_test.c:405 - M2M_msgTCPTask(TCP_TASK)$ Client Family: 2
[DEBUG] 31.56 m2m_tcp_test.c:410 - M2M_msgTCPTask(TCP_TASK)$

[DEBUG] 31.57 m2m_tcp_test.c:411 - M2M_msgTCPTask(TCP_TASK)$ | Send/receive data test |
[DEBUG] 31.57 m2m_tcp_test.c:412 - M2M_msgTCPTask(TCP_TASK)$ -----
[DEBUG] 32.58 m2m_tcp_test.c:416 - M2M_msgTCPTask(TCP_TASK)$
--> issuing m2mb_socket_bsd_send(): transmit "START" packet...
[DEBUG] 32.59 m2m_tcp_test.c:423 - M2M_msgTCPTask(TCP_TASK)$ --> done (11 have been transmitted)
[DEBUG] 32.60 m2m_tcp_test.c:425 - M2M_msgTCPTask(TCP_TASK)$ ALL data transmitted - PASS
[DEBUG] 32.61 m2m_tcp_test.c:430 - M2M_msgTCPTask(TCP_TASK)$
--> issuing m2mb_socket_bsd_send(): transmit 58 bytes...
[DEBUG] 32.62 m2m_tcp_test.c:437 - M2M_msgTCPTask(TCP_TASK)$ --> done (58 have been transmitted)
[DEBUG] 32.63 m2m_tcp_test.c:440 - M2M_msgTCPTask(TCP_TASK)$ ALL data transmitted - PASS
[DEBUG] 32.64 m2m_tcp_test.c:448 - M2M_msgTCPTask(TCP_TASK)$
Waiting for data...

[DEBUG] 39.64 m2m_tcp_test.c:457 - M2M_msgTCPTask(TCP_TASK)$ test
[DEBUG] 99.61 m2m_tcp_test.c:465 - M2M_msgTCPTask(TCP_TASK)$
m2mb_socket_bsd_recv() has received 6 bytes
[DEBUG] 102.60 m2m_tcp_test.c:469 - M2M_msgTCPTask(TCP_TASK)$
Server TCP is closing the current connection ...

```

Figure 229

Data on a PuTTY terminal

```
START  
aaaaaaaaaa-bbbbbbbbbb-ccccccccc-ddddddddd-eeeeeeeeee  
test  
█
```

Figure 230

3.5.53 TLS SSL Client

Sample application showcasing TLS/SSL with client certificates usage with M2MB API. Debug prints on **USB0**

Features

- How to check module registration and enable PDP context
- How to open a SSL client socket
- How to communicate over SSL socket

3.5.53.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the connection and start it

ssl_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and link it to the PDP context id
- Connect to the server over TCP socket
- Initialize the TLS parameters (TLS1.2) andh auth mode (server+client auth in the example)
- Create SSL context
- Read certificates files and store them
- Create secure socket and connect to the server using SSL
- Send data and receive response
- Close secure socket
- Close socket
- Delete SSL context
- Disable PDP context

The application requires the certificates to be stored in /mod/ssl_certs/ folder. It can be created with

AT#M2MMKDIR=/mod/ssl_certs

Certificates can then be loaded with

AT#M2MWRITE="/mod/ssl_certs/modulesCA.crt",1740

and providing the file content in RAW mode (for example using the "Transfer Data" button in Telit AT Controller)

For client certificates (if required), the commands will be

AT#M2MWRITE="/mod/ssl_certs/modulesClient.crt",1651

AT#M2MWRITE="/mod/ssl_certs/modulesClient_pkcs1.key",1679

PLEASE NOTE: always verify the file sizes to be used in the commands above as they might change

```
Starting TLS-SSL demo app. This is v1.1.2 built on Mar  3 2021 10:15:00.
[DEBUG] 10.85 ssl_test:252 - msgHTTPSTask{TLS_TASK}$ INIT
[DEBUG] 10.85 ssl_test:266 - msgHTTPSTask{TLS_TASK}$ m2mb_os_ev_init success
[DEBUG] 10.85 ssl_test:270 - msgHTTPSTask{TLS_TASK}$ Init SSL session test app
[DEBUG] 10.85 ssl_test:285 - msgHTTPSTask{TLS_TASK}$ m2mb_ssl_create_config sslConfigHnd1 = 0x40037958, sslRes= 0
[DEBUG] 10.85 ssl_test:294 - msgHTTPSTask{TLS_TASK}$ m2mb_ssl_create_config PASSED
[DEBUG] 10.85 ssl_test:306 - msgHTTPSTask{TLS_TASK}$ m2mb_ssl_create_ctxt PASSED
[DEBUG] 10.85 ssl_test:311 - msgHTTPSTask{TLS_TASK}$ loading CA CERT from file /mod/ssl_certs/modulesCA.crt
[DEBUG] 10.85 ssl_test:315 - msgHTTPSTask{TLS_TASK}$ file size: 1740
[DEBUG] 10.85 ssl_test:328 - msgHTTPSTask{TLS_TASK}$ Reading content from file. Size: 1740
Buffer successfully received from file. 1740 bytes were loaded.
Closing file.
[DEBUG] 10.85 ssl_test:361 - msgHTTPSTask{TLS_TASK}$ loading client CERT from file /mod/ssl_certs/modulesClient.crt
[DEBUG] 10.85 ssl_test:365 - msgHTTPSTask{TLS_TASK}$ file size: 1651
[DEBUG] 10.85 ssl_test:378 - msgHTTPSTask{TLS_TASK}$ Reading content from file. Size: 1651
Buffer successfully received from file. 1651 bytes were loaded.
Closing file.
[DEBUG] 10.85 ssl_test:401 - msgHTTPSTask{TLS_TASK}$ loading client KEY from file /mod/ssl_certs/modulesClient_pkcs1.key
[DEBUG] 10.85 ssl_test:405 - msgHTTPSTask{TLS_TASK}$ file size: 1679
[DEBUG] 10.85 ssl_test:418 - msgHTTPSTask{TLS_TASK}$ Reading content from file. Size: 1679
Buffer successfully received from file. 1679 bytes were loaded.
Closing file.
[DEBUG] 10.85 ssl_test:448 - msgHTTPSTask{TLS_TASK}$ certificates successfully stored!
[DEBUG] 10.85 ssl_test:457 - msgHTTPSTask{TLS_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 10.85 ssl_test:465 - msgHTTPSTask{TLS_TASK}$ Waiting for registration...
[DEBUG] 10.86 ssl_test:171 - NetCallback{pubTspt_0}$ Module is registered to cell 0x468E!
[DEBUG] 10.86 ssl_test:477 - msgHTTPSTask{TLS_TASK}$ Pdp context activation
[DEBUG] 10.86 ssl_test:481 - msgHTTPSTask{TLS_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 12.87 ssl_test:496 - msgHTTPSTask{TLS_TASK}$ Activate PDP with APN web.omnitel.it...
[DEBUG] 13.71 ssl_test:197 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 13.71 ssl_test:200 - PdpCallback{pubTspt_0}$ IP address: 2.41.76.63
[DEBUG] 13.71 ssl_test:514 - msgHTTPSTask{TLS_TASK}$ Creating Socket...
[DEBUG] 13.71 ssl_test:525 - msgHTTPSTask{TLS_TASK}$ Socket created
[DEBUG] 13.71 ssl_test:535 - msgHTTPSTask{TLS_TASK}$ Socket ctx set to 3
[DEBUG] 13.92 ssl_test:548 - msgHTTPSTask{TLS_TASK}$ Retrieved IP: 185.86.42.218
[DEBUG] 14.05 ssl_test:562 - msgHTTPSTask{TLS_TASK}$ Socket Connected!
[DEBUG] 15.97 ssl_test:587 - msgHTTPSTask{TLS_TASK}$ m2mb_ssl_connect ret 0
[DEBUG] 17.99 ssl_test:593 - msgHTTPSTask{TLS_TASK}$ Sending bytes..
```

```
[DEBUG] 17.99 ssl_test:593 - msgHTTPSTask{TLS_TASK}$ Sending bytes..
[DEBUG] 17.99 ssl_test:596 - msgHTTPSTask{TLS_TASK}$ SSL write result = 44
[DEBUG] 22.03 ssl_test:608 - msgHTTPSTask{TLS_TASK}$ pending bytes: 1087
[DEBUG] 22.03 ssl_test:612 - msgHTTPSTask{TLS_TASK}$ trying to receive 1087 bytes..
[DEBUG] 22.03 ssl_test:618 - msgHTTPSTask{TLS_TASK}$ Server response: (269)<HTTP/1.1 200 OK
Date: Wed, 03 Mar 2021 09:18:22 GMT
Server: Apache/2.2.15 (CentOS)
Last-Modified: Mon, 22 Jan 2018 10:57:39 GMT
ETag: "1fffc-27f-5635b4c6f12b3"
Accept-Ranges: bytes
Content-Length: 639
Connection: close
Content-Type: text/html; charset=UTF-8
>
[DEBUG] 22.03 ssl_test:634 - msgHTTPSTask{TLS_TASK}$ pending bytes: 762
[DEBUG] 22.03 ssl_test:638 - msgHTTPSTask{TLS_TASK}$ trying to receive remaining 762 bytes..
[DEBUG] 22.03 ssl_test:644 - msgHTTPSTask{TLS_TASK}$ Server response: (639)<<html>
<head>
  <title>module.telit.com</title>
  <meta content="text/html; charset=utf-8" />
</head>
<body>
  <table border=0 align=center>
    <tr>
      <td height="100" align=center><h2>modules.telit.com - Test HTML page</h2></td>
    </tr>
    <tr>
      <td align=center><img src=Telit.jpg alt="Telit logo" height="126" width="410"></img></td>
    </tr>
    <tr>
      <td height="200" align=center> This is a simple HTML page, </br>
      made with simple HTML code,</br>
      just for test!
    </td>
    </tr>
    <tr>
      <td height="100" align=center><font size="3">Telit &copy; 2015 - 2017 All rights reserved</font></td>
    </tr>
  </table>
</body>
</html>
>
[DEBUG] 22.03 ssl_test:662 - msgHTTPSTask{TLS_TASK}$ application exit
[DEBUG] 22.03 ssl_test:680 - msgHTTPSTask{TLS_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 22.03 ssl_test:683 - msgHTTPSTask{TLS_TASK}$ Application complete.
[DEBUG] 22.77 ssl_test:206 - PdpCallback{pubTspt_0}$ Context deactivated!
```

3.5.54 UDP client

Sample application showcasing UDP echo demo with M2MB API. Debug prints on **USB0**

Features

- How to check module registration and activate PDP context
- How to open a UDP client socket
- How to communicate over the socket

3.5.54.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task and start it

m2m_udp_test.c - Initialize Network structure and check registration - Initialize PDP structure and start PDP context - Create socket and link it to the PDP context id - Send data and receive response - Close socket - Disable PDP context

```
Starting UDP client demo app. This is v1.0.7 built on Apr 1 2020 14:57:13.
INIT
[DEBUG] 21.23 m2m_udp_test.c:223 - M2M_msgUDPTask{UDP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
Waiting for registration...
[DEBUG] 21.25 m2m_udp_test.c:131 - NetCallback{pubTspt_0}$ Module is registered to cell 0xC4CF!
[DEBUG] 21.26 m2m_udp_test.c:241 - M2M_msgUDPTask{UDP_TASK}$ Pdp context initialization
[DEBUG] 21.26 m2m_udp_test.c:245 - M2M_msgUDPTask{UDP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
Activate PDP with APN web.omnitel.it...
[DEBUG] 24.11 m2m_udp_test.c:157 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 24.11 m2m_udp_test.c:160 - PdpCallback{pubTspt_0}$ IP address: 109.113.222.12
[DEBUG] 24.12 m2m_udp_test.c:268 - M2M_msgUDPTask{UDP_TASK}$ Creating Socket...
[DEBUG] 24.13 m2m_udp_test.c:280 - M2M_msgUDPTask{UDP_TASK}$ Socket created
Socket ctx set to 3
[DEBUG] 24.41 m2m_udp_test.c:306 - M2M_msgUDPTask{UDP_TASK}$ Retrieved IP: 185.86.42.218
Socket ready.
Data successfully sent (16 bytes)
Socket rcv...
[DEBUG] 26.47 m2m_udp_test.c:352 - M2M_msgUDPTask{UDP_TASK}$ m2mb_socket_bsd_set_sock_opt() M2MB_SOCKET_BSD_SO_RCVTIMEO - success
trying to receive 16 bytes..
Data received (16): <hello from m2mb!>
[DEBUG] 26.48 m2m_udp_test.c:377 - M2M_msgUDPTask{UDP_TASK}$ application exit
Socket Closed
[DEBUG] 26.49 m2m_udp_test.c:399 - M2M_msgUDPTask{UDP_TASK}$ m2mb_pdp_deactivate returned success
Application complete.
[DEBUG] 27.04 m2m_udp_test.c:166 - PdpCallback{pubTspt_0}$ Context successfully deactivated!
```

Figure 231

3.5.55 UDP_Server example

Sample application that shows UDP listening socket demo with m2mb apis. Debug prints on **USB0**

Features

- How to configure an UDP socket into listen mode
- How to receive data using m2mb_socket_bsd_select
- How to read data received and send data to client

Application workflow

M2MB_main.c

- Print welcome message
- Init task apis and create M2M_msgUDPTask to handle UDP socket

m2mb_udp_test.c

M2M_msgUDPTask

- Wait for module registration
- Activate PDP context
- Create UDP listen socket
- Wait for incoming data from client using m2mb_socket_bsd_select
- When there are data on socket, read them and send some data back to client

```

Starting UDP Server demo app. This is v1.1.10 built on Mar  3 2022 15:42:39.

[DEBUG] 19.28 m2mb_udp_te:221 - M2M_msgUDPTask{UDP_TASK}$ INIT
[DEBUG] 19.29 m2mb_udp_te:237 - M2M_msgUDPTask{UDP_TASK}$ m2mb_os_ev_init success
[DEBUG] 19.30 m2mb_udp_te:243 - M2M_msgUDPTask{UDP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 19.31 m2mb_udp_te:251 - M2M_msgUDPTask{UDP_TASK}$ Waiting for registration...
[DEBUG] 19.32 m2mb_udp_te:140 - NetCallback{pubTspt_0}$ Module is registered to cell 0x5222!
[DEBUG] 19.33 m2mb_udp_te:264 - M2M_msgUDPTask{UDP_TASK}$ Pdp context activation
[DEBUG] 19.33 m2mb_udp_te:268 - M2M_msgUDPTask{UDP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 21.39 m2mb_udp_te:283 - M2M_msgUDPTask{UDP_TASK}$ Activate PDP with APN ibox.tim.it...
[DEBUG] 22.00 m2mb_udp_te:167 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 22.01 m2mb_udp_te:170 - PdpCallback{pubTspt_0}$ IP address: 2.195.163.102

---- Start UDP server Listen IP: 2.195.163.102 PORT: 6500 ----

[DEBUG] 24.07 m2mb_udp_te:299 - M2M_msgUDPTask{UDP_TASK}$ Creating Socket...
[DEBUG] 24.08 m2mb_udp_te:310 - M2M_msgUDPTask{UDP_TASK}$ Socket created
[DEBUG] 24.08 m2mb_udp_te:311 - M2M_msgUDPTask{UDP_TASK}$ m2mb_socket_bsd_socket(): valid socket ID [0x40035490] - PASS
[DEBUG] 25.12 m2mb_udp_te:317 - M2M_msgUDPTask{UDP_TASK}$ issuing m2m_socket_bsd_ioctl() to set non-blocking mode ...
[DEBUG] 25.13 m2mb_udp_te:329 - M2M_msgUDPTask{UDP_TASK}$ Binding Socket...
[DEBUG] 27.18 m2mb_udp_te:341 - M2M_msgUDPTask{UDP_TASK}$ Socket Bind Pass

Waiting for Incoming UDP data, Listen IP: 2.195.163.102 PORT: 6500

[DEBUG] 29.25 m2mb_udp_te:359 - M2M_msgUDPTask{UDP_TASK}$ select...
[DEBUG] 32.32 m2mb_udp_te:362 - M2M_msgUDPTask{UDP_TASK}$ Select result: 0
[DEBUG] 33.34 m2mb_udp_te:359 - M2M_msgUDPTask{UDP_TASK}$ select...
[DEBUG] 36.41 m2mb_udp_te:362 - M2M_msgUDPTask{UDP_TASK}$ Select result: 0
[DEBUG] 37.45 m2mb_udp_te:359 - M2M_msgUDPTask{UDP_TASK}$ select...
[DEBUG] 40.52 m2mb_udp_te:362 - M2M_msgUDPTask{UDP_TASK}$ Select result: 0
[DEBUG] 41.54 m2mb_udp_te:359 - M2M_msgUDPTask{UDP_TASK}$ select...
[DEBUG] 44.61 m2mb_udp_te:362 - M2M_msgUDPTask{UDP_TASK}$ Select result: 0
[DEBUG] 45.64 m2mb_udp_te:359 - M2M_msgUDPTask{UDP_TASK}$ select...
[DEBUG] 45.94 m2mb_udp_te:362 - M2M_msgUDPTask{UDP_TASK}$ Select result: 1
[DEBUG] 45.95 m2mb_udp_te:366 - M2M_msgUDPTask{UDP_TASK}$
Incoming UDP data available
---- Receive/send data test ----

trying to receive bytes..
[DEBUG] 46.99 m2mb_udp_te:399 - M2M_msgUDPTask{UDP_TASK}$ Client Source Address: 151.99.1.17
[DEBUG] 47.00 m2mb_udp_te:400 - M2M_msgUDPTask{UDP_TASK}$ Client Port: 36368
[DEBUG] 47.01 m2mb_udp_te:401 - M2M_msgUDPTask{UDP_TASK}$ Client Family: 2
Data received (4): <test>
Data successfully sent (16 bytes)

```

Figure 232

3.5.56 Watchdog example

Sample application that shows how to set and implement a watchdog. Task has been locked waiting for an event with a timeout longer than wd inactivity timeout. If no wd kick or no actions

Features

- Enable watchdog on Task_1
- Handle watchdog kicks with a timer

3.5.56.1 Application workflow

M2MB_main.c

- Create Task_1 that will be put under watchdog control
- In Task_1 watchdog is enabled on it and a timer created to handle watchdog kicks. Then task enters in a loop where kicks are sent. After 5 kicks a task lock is simulated waiting for an event release (with a timeout longer than task watchdog inactivity timeout)
- As soon as watchdog inactivity timeout expired M2MB_WDOG_TIMEOUT_IND is generated and handled in WDcallback (in this case the event is released, no further actions are done)
- No further actions are performed by Task_1 (no kicks are sent) so after watchdog system timeout expiration app is rebooted.

```
Starting WATCHDOG demo app. This is v1.0.0 built on Nov  4 2021 14:34:06.
Event correctly initialized
Task started
Init WatchDog
m2mb_wDog_init OK
Tick duration: 1s
Adding Task under WD control with inactivity timeout of 60s
m2mb_wDog_addTask OK
Enabling the WatchDog
m2mb_wDog_enable OK
Timer creation OK

Start WD kick every 5s...
WD kick
WD kick done
```

Figure 233

3.5.57 ZLIB example

Sample application showing how to compress/uncompress with ZLIB. Debug prints on **USB0**

Features

- How to compress a file
- How to uncompress a file

In order to execute the entire test, copy test.gz file into your module running the following AT command:

```
AT#M2MWRITE="/mod/test.gz",138
```

>>> here receive the prompt; then type or send the file, sized 138 bytes

3.5.57.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Test the compression and decompression of a data string
- Test the decompression of a .gz file (test.gz), expected to be in /mod folder, into its content test.txt. The file must be uploaded by the user (see steps above).

```
Starting Logging demo app. This is v1.0.7 built on Apr 7 2020 09:02:35.
Starting TEST_COMPR_UNCOMPR.
len: 138; comprLen: 57
Compressed message:
W+EHU(,ILIVH^E/ISHE`PE*I-HMQE/K-R(Ï Êç$VU^#a$ê y4RI«¥1.
comprLen: 57; uncomprLen: 138
uncompress():
the quick brown fox jumped over the lazy dog. the quick brown fox jumped over the lazy dog. the quick brown fox jumped over the lazy dog.
Ending TEST_COMPR_UNCOMPR with SUCCESS.
Starting test_uncompress.
Data extracted correctly into the file ./mod/test.txt
test_uncompress finished correctly!
```

Figure 234

3.6 MAIN UART

Applications that provide usage examples for various functionalities, log output on MAIN UART

3.6.1 Alarm example

Sample application that shows how to set an alarm to wake-up module. Debug prints on **MAIN UART**

Features

- How to set an alarm
- How to use it to turn on module

Application workflow

M2MB_main.c

- Init RTC
- Wait for registration
- Get current date and time
- Call function set_alarm
- Init Power and turn off module

```
12:34:48.532- Start Alarm demo application. This is v1.1.19 built on Sep 9 2024 12:34:02.
12:34:48.532-
12:34:48.532- Waiting for registration...
12:34:48.532- Module is registered!
12:34:48.532- NETWORK OPERATOR (mcc mnc): 222 01, Rat: 0
12:34:48.532- Module system time is: 2024-09-09, 12:33:35
12:34:48.532- Setting alarm in 2 minutes
12:34:48.532- Alarm will be set at: 2024-09-09, 12:35:35
12:34:48.741-
12:34:48.741- Wait 5 seconds and then Turn off module...
12:34:53.752- Power off moduley
```

Figure 235

3.6.2 ATI (AT Instance)

Sample application showing how to use AT Instance functionality (sending AT commands from code). The example supports both sync and async (using a callback) modes. Debug prints on **MAIN UART**

Features

- How to open an AT interface from the application
- How to send AT commands and receive responses on the AT interface

3.6.2.1 Application workflow, sync mode

M2MB_main.c

- Open USB/UART/UART_AUX
- Init AT0 (first AT instance)
- Send AT+CGMR command
- Print response.
- Release AT0

at_sync.c

- Init ati functionality and take AT0
- Send AT+CGMR command, then read response after 2 seconds, then return it
- Deinit ati, releasing AT0

```
Starting AT demo app. This is v1.0.7 built on Apr  1 2020 15:12:58.
[DEBUG] 17.15  at_sync.c:53 - at_cmd_sync_init{M2M_DamsStart}$ m2mb_ati_init() on instance 0
Sending command AT+CGMR in sync mode
[DEBUG] 17.16  at_sync.c:79 - send_sync_at_command{M2M_DamsStart}$ Sending AT Command: AT+CGMR
Command response: <AT+CGMR
MOB.950004-B008

OK
>

[DEBUG] 19.21  at_sync.c:61 - at_cmd_sync_deinit{M2M_DamsStart}$ m2mb_ati_deinit() on instance 0
Application end
```

Figure 236

3.6.2.2 Application workflow, async mode

M2MB_main.c

- Open USB/UART/UART_AUX
- Init AT0 (first AT instance)
- Send AT+CGMR command
- Print response.

- Release AT0

at_async.c

- Init ati functionality and take AT0, register AT events callback
- Send AT+CGMR command, wait for response semaphore (released in callback), then read it and return it
- Deinit ati, releasing AT0

```
Starting AT demo app. This is v1.0.7 built on Apr 1 2020 15:07:45.
[DEBUG] 17.13 at_async.c:116 - at_cmd_async_init{M2M_DamsStart}$ m2mb_ati_init() on instance 0
Sending command AT+CGMR in async mode
[DEBUG] 17.15 at_async.c:153 - send_async_at_command{M2M_DamsStart}$ Sending AT Command: AT+CGMR
[DEBUG] 17.15 at_async.c:169 - send_async_at_command{M2M_DamsStart}$ waiting command response...
[DEBUG] 17.17 at_async.c:88 - at_cmd_async_callback{pubTspt_0}$ Callback - available bytes: 25
[DEBUG] 17.18 at_async.c:181 - send_async_at_command{M2M_DamsStart}$ Receive response...
Command response: <AT+CGMR
MOB.950004-B008

OK
>

[DEBUG] 17.19 at_async.c:136 - at_cmd_async_deinit{M2M_DamsStart}$ m2mb_ati_deinit() on instance 0
Application end
```

Figure 237

3.6.3 AT Tunnel

Sample application showcasing how to perform an AT tunnel from Main UART to an AT instance. Debug prints on **USB1**.

Features

- How to open an AT interface from the application
- How to receive data from main UART and tunnel it to the AT interface, then report back to UART the AT response

3.6.3.1 Application workflow

M2MB_main.c

- Open USB1 for debug
- Initialize UART with callback function to manage input data
- Initialize AT system to manage AT commands from UART
- wait 5 minutes then deinit AT system

Main UART:

```
Starting AT tunnel demo app. Waiting for AT commands...
AT+CGMM
ME910C1-P2

OK
AT+CGREG?
+CGREG: 0,1

OK
```

Figure 238

USB1 debug log:


```

Starting AT tunnel demo app. This is v1.0.7 built on Apr  7 2020 08:21:41.
Uart opened, setting callback for data..
[DEBUG] 17.21 M2MB_main.c:183 - at_cmd_async_init{M2M_DamsStart}$ m2mb_ati_init() on instance 0
[DEBUG] 20.43 M2MB_main.c:144 - UART_Cb{pubTspt_0}$ Received 8 bytes
[DEBUG] 20.43 M2MB_main.c:84 - msgUARTTask{uart_task}$ Received data on uart, read it and send on ATI
UART IN: <AT+CGMM
>. Sending to ATI...
[DEBUG] 20.43 M2MB_main.c:171 - at_cmd_async_callback{pubTspt_0}$ Callback - available bytes: 8
[DEBUG] 20.43 M2MB_main.c:107 - msgUARTTask{uart_task}$ Received data on ATI, read it and send on UART
[DEBUG] 20.43 M2MB_main.c:116 - msgUARTTask{uart_task}$ Received: <AT+CGMM
>
[DEBUG] 20.43 M2MB_main.c:171 - at_cmd_async_callback{pubTspt_0}$ Callback - available bytes: 20
[DEBUG] 20.43 M2MB_main.c:107 - msgUARTTask{uart_task}$ Received data on ATI, read it and send on UART
[DEBUG] 20.43 M2MB_main.c:116 - msgUARTTask{uart_task}$ Received: <
ME910C1-P2
OK
>
[DEBUG] 32.82 M2MB_main.c:144 - UART_Cb{pubTspt_0}$ Received 10 bytes
[DEBUG] 32.82 M2MB_main.c:84 - msgUARTTask{uart_task}$ Received data on uart, read it and send on ATI
UART IN: <AT+CGREG?
>. Sending to ATI...
[DEBUG] 32.82 M2MB_main.c:171 - at_cmd_async_callback{pubTspt_0}$ Callback - available bytes: 10
[DEBUG] 32.82 M2MB_main.c:107 - msgUARTTask{uart_task}$ Received data on ATI, read it and send on UART
[DEBUG] 32.82 M2MB_main.c:116 - msgUARTTask{uart_task}$ Received: <AT+CGREG?
>
[DEBUG] 32.83 M2MB_main.c:171 - at_cmd_async_callback{pubTspt_0}$ Callback - available bytes: 21
[DEBUG] 32.83 M2MB_main.c:107 - msgUARTTask{uart_task}$ Received data on ATI, read it and send on UART
[DEBUG] 32.83 M2MB_main.c:116 - msgUARTTask{uart_task}$ Received: <
+CGREG: 0,1
OK
>

```

Figure 239

3.6.4 AWS demo

Sample application showcasing AWS IoT Core MQTT communication. Debug prints on **MAIN UART**

Features

- How to check module registration and enable PDP context
- How to load certificates into device SSL session storage
- How to configure MQTT client parameters
- How to connect to AWS server with SSL and exchange data over a topic

3.6.4.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage MQTT client and start it

aws_demo.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Init MQTT client
- Configure it with all parameters (Client ID, PDP context ID, keepalive timeout...)
- Initialize the TLS parameters (TLS1.2) and auth mode (server+client auth in the example)
- Create SSL context
- Read certificates files and store them
- Connect MQTT client to broker
- Subscribe to topic
- Publish 10 messages with increasing counter
- Print received message in mqtt_topc_cb function
- Disconnect MQTT client and deinit it
- Disable PDP context

3.6.4.2 How to get started with AWS IoT

- Go to [AWS console](#) and create an account if one is not available yet.
- Go to **IoT Core** section
- Go to **Secure > Policies** section
- Create a new policy, which describes what the device will be allowed to do (e.g. subscribe, publish)
- Give it a name, then configure it using the configuration below (it is possible to copy/paste by clicking on **Add statements** section, then **Advanced mode**) :

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Action": [
        "iot:Publish",
        "iot:Subscribe",
        "iot:Connect",
        "iot:Receive"
      ],
      "Effect": "Allow",
      "Resource": [
        "*"
      ]
    }
  ]
}
```

- Click on create to complete the policy creation.
- Go to **Manage** section
- Press **Create**, then **Create a single thing**
- Give the new thing a name, then click on Next
- Select **One-click certificate creation (recommended)** by clicking on **Create certificate**
- Once presented with the **Certificate created** page, download all certificates and keys
- Click on the **Activate** button to enable the certificate authentication of the newly created device
- Click on **Attach a policy** and select the policy created in a previous step

For further information, please refer to the full [AWS IoT documentation](#)

3.6.4.3 Application setup

- Set **CLIENTCERTFILE** and **CLIENTKEYFILE** defines in **aws_demo.c** file in order to match the certificate and key created in the previous section.
- Set **AWS_BROKER_ADDRESS** to the correct AWS URL. It can be retrieved from AWS IoT **Manage > Things > Interact** in the **HTTPS Rest API Endpoint** URL.
- Set **CLIENT_ID** to the desired Client ID for your AWS device
- (Optional) if required, change **CACERTFILE** to match the one to be used.

3.6.4.4 Device setup

The application requires the certificates (provided in sample app **certs** subfolder) to be stored in **/mod/ssl_certs/** folder. It can be created with

```
AT#M2MMKDIR=/mod/ssl_certs
```

Certificates can then be loaded with

```
AT#M2MWRITE="/mod/ssl_certs/preload_CACert_01.crt",1468 AT#M2MWRITE="/mod/ssl_certs/IoT.crt",1646
```

providing the file content in RAW mode (for example using the “Transfer Data” button in Telit AT Controller)

For client certificates, the commands will be

```
AT#M2MWRITE="/mod/ssl_certs/xxxxx.crt",yyyy
```

```
AT#M2MWRITE="/mod/ssl_certs/xxxxx.key",zzzz
```

PLEASE NOTE: always verify the file sizes to be used in the commands above as they might change

```

Starting AWS IoT Core MQTT demo app. This is v1.1.5 built on Apr 30 2021 09:05:17.
[DEBUG] 15.51 aws_demo:607 - AWS_Task{MQTT_TASK}$ Init MQTT client for AWS
[DEBUG] 15.52 aws_demo:265 - PrepareSSLEnvironment{MQTT_TASK}$ m2mb_ssl_config SNI succeeded
[DEBUG] 15.52 aws_demo:271 - PrepareSSLEnvironment{MQTT_TASK}$ Root CA cert file /mod/ssl_certs/preload_CACert_01.crt
[DEBUG] 15.52 aws_demo:297 - PrepareSSLEnvironment{MQTT_TASK}$ Buffer successfully received from file. 1468 bytes were loaded.
[DEBUG] 15.52 aws_demo:308 - PrepareSSLEnvironment{MQTT_TASK}$ Cross Signed CA cert file /mod/ssl_certs/Amazon-IoT.crt
[DEBUG] 15.52 aws_demo:334 - PrepareSSLEnvironment{MQTT_TASK}$ Buffer successfully received from file. 1646 bytes were loaded.
[DEBUG] 15.52 aws_demo:360 - PrepareSSLEnvironment{MQTT_TASK}$ Client certificate file /mod/ssl_certs/ab71 -certificate.pem.crt
[DEBUG] 15.52 aws_demo:384 - PrepareSSLEnvironment{MQTT_TASK}$ Buffer successfully received from file. 1224 bytes were loaded.
[DEBUG] 15.52 aws_demo:396 - PrepareSSLEnvironment{MQTT_TASK}$ Client Key file /mod/ssl_certs/ab71 -private.pem.key
[DEBUG] 15.52 aws_demo:422 - PrepareSSLEnvironment{MQTT_TASK}$ Buffer successfully received from file. 1679 bytes were loaded.

SSL environment preparation completed
[DEBUG] 15.52 aws_demo:726 - AWS_Task{MQTT_TASK}$ Waiting for registration...
[DEBUG] 15.52 aws_demo:514 - NetCallback{pubTspt_0}$ Module is registered
[DEBUG] 15.52 aws_demo:738 - AWS_Task{MQTT_TASK}$ PDP context initialization
[DEBUG] 17.55 aws_demo:753 - AWS_Task{MQTT_TASK}$ Activate PDP with APN web.omnitel.it on CID 1....
[DEBUG] 18.37 aws_demo:557 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 18.37 aws_demo:561 - PdpCallback{pubTspt_0}$ IP address: 109.114.102.21

Connecting to Server <angy83rl5oizs-ats.iot.eu-west-2.amazonaws.com>:8883...
Done.
[DEBUG] 27.87 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 2, "data": {"RSSI": -72, "tm": "2021-04-30,09:05:50"}}> to topic
device/353081090 /updates
[DEBUG] 27.87 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 30.94 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 3, "data": {"RSSI": -72, "tm": "2021-04-30,09:05:53"}}> to topic
device/353081090 /updates
[DEBUG] 30.94 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 33.99 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 4, "data": {"RSSI": -72, "tm": "2021-04-30,09:05:56"}}> to topic
device/353081090 /updates
[DEBUG] 34.00 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 37.00 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 5, "data": {"RSSI": -72, "tm": "2021-04-30,09:05:59"}}> to topic
device/353081090 /updates
[DEBUG] 37.00 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 40.03 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 6, "data": {"RSSI": -72, "tm": "2021-04-30,09:06:02"}}> to topic
device/353081090 /updates
[DEBUG] 40.03 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 43.05 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 7, "data": {"RSSI": -72, "tm": "2021-04-30,09:06:05"}}> to topic
device/353081090 /updates
[DEBUG] 43.05 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 46.13 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 8, "data": {"RSSI": -72, "tm": "2021-04-30,09:06:08"}}> to topic
device/353081090 /updates
[DEBUG] 46.13 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 49.15 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 9, "data": {"RSSI": -72, "tm": "2021-04-30,09:06:11"}}> to topic
device/353081090 /updates
[DEBUG] 49.15 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 52.19 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 10, "data": {"RSSI": -72, "tm": "2021-04-30,09:06:14"}}> to topic
device/353081090 /updates
[DEBUG] 52.19 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 55.22 aws_demo:852 - AWS_Task{MQTT_TASK}$ PUBLISHING <{"ID": 11, "data": {"RSSI": -72, "tm": "2021-04-30,09:06:17"}}> to topic
device/353081090 /updates
[DEBUG] 55.22 aws_demo:857 - AWS_Task{MQTT_TASK}$ Done.

Disconnecting from MQTT broker..
[DEBUG] 58.27 aws_demo:878 - AWS_Task{MQTT_TASK}$ Done.
[DEBUG] 58.27 aws_demo:908 - AWS_Task{MQTT_TASK}$ application exit
[DEBUG] 58.27 aws_demo:918 - AWS_Task{MQTT_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 58.27 aws_demo:924 - AWS_Task{MQTT_TASK}$ Application complete.

```

Figure 240

Data received from a subscriber:

```

device/353081090 /updates 1
QoS 0

device/353081090 /updates 1
30-04-2021 09:06:00.32760467 QoS 0
{ "ID": 2, "data": {"RSSI": -72, "tm": "2021-04-30,09:05:50"} }

```

Figure 241

3.6.5 App Manager

Sample application showing how to manage AppZone apps from m2mb code. Debug prints on **MAIN UART**

Features

- How to get how many configured apps are available
- How to get the handle to manage the running app (change start delay, enable/disable)
- How to create the handle for a new binary app, enable it and set its parameters
- How to start the new app without rebooting the device, then stop it after a while.

3.6.5.1 Prerequisites

This app will try to manage another app called “second.bin”, which already exists in the module filesystem and can be anything (e.g. another sample app as GPIO toggle). the app must be built using the flag ROM_START=

in the Makefile to set a different starting address than the main app (by default, 0x40000000). For example, 0x41000000.

3.6.5.2 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- get a non existing app handle and verify it is NULL
- get the current app handle, then get the start delay **set in the INI file (so persistent)**
- change the current app delay value **in the INI file**
- verify that the change has been stored
- get current app state
- create an handle for a second application binary.
- add it to the INI file
- set its execution flag to 0
- get the delay time and the state from INI file for the new app
- get the current set address for the new app
- set the app delay **in RAM, INI will not be affected.**
- start the new app without reboot, using the right set delay
- wait some time, then get the app state and the used RAM amount
- wait 10 seconds, then stop the second app.
- set its execution flag to 1 so it will run at next boot.

```
Starting App Manager demo app. This is v1.0.14-C1 built on Sep 24 2020 12:33:25.  
There are 2 configured apps.  
Not existing app handle test (should be 0): 0x0  
Manager app handle: 0x809e20e0  
Manager app delay from nv memory: 5 seconds  
  
Changing Manager app delay time (on non volatile configuration) to 5 seconds..  
Manager app delay from nv memory is now 5 seconds  
Manager app state is M2MB_APPMNG_STATE_RUN  
  
Trying to get Second app handle...  
Second app handle is valid  
2nd app delay from nv memory is 1  
2nd app current state is M2MB_APPMNG_STATE_READY  
Second app current address is 0x41000000  
Setting volatile Second app delay (not stored in nvm) to 0 seconds...  
Starting Second app on the fly (without reboot)...  
Waiting 2 seconds...  
2nd app current state is M2MB_APPMNG_STATE_RUN  
Second app is running!  
Second App is using 475136 bytes of RAM  
Stopping Second app now...  
wait 10 seconds...  
2nd app current state is M2MB_APPMNG_STATE_STOP  
Set permanent run permission for Second app.  
Done. Second App will also run from next boot-up
```

Figure 242

3.6.6 App update OTA via FTP

Sample application showcasing Application OTA over FTP with AZX FTP. Debug prints on **MAIN UART**

Features

- How to check module registration and activate PDP context
- How to connect to a FTP server
- How to download an application binary and update the local version

The app uses a predefined set of parameters. To load custom parameters, upload the `ota_config.txt` file (provided in project's `/src` folder) in module's `/mod` folder, for example with

```
AT#M2MWRITE="/mod/ota_config.txt",<filesize>
```

3.6.6.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage app OTA and start it

ftp_utils.c

- Set parameters to default
- Try to load parameters from `ota_config.txt` file
- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Initialize FTP client
- Connect to FTP server and log in
- Get new App binary file size on remote server
- Download the file in `/mod` folder, with the provided name
- Close FTP connection
- Disable PDP context
- Update applications configuration in **app_utils.c**

app_utils.c

- Set new application as default
- Delete old app binary
- Restart module

```

Starting FTP APP OTA demo app. This is v1.0.7 built on Apr 7 2020 17:04:05.
[DEBUG] 21.23 ftp_utils.c:447 - msgFTPTask{FTPOTA_TASK}$ INIT
[DEBUG] 21.25 ftp_utils.c:152 - readConfigFromFile{FTPOTA_TASK}$ Reading parameters from file
[DEBUG] 21.26 ftp_utils.c:154 - readConfigFromFile{FTPOTA_TASK}$ Opening /mod/ota_config.txt in read mode..
Set APN to: <<web.omnitel.it>>
Set FTP URL to: <<ftp.telit.com>>
Set FTP PORT to: 21
Set FTP USER to: <<_ _ _ _ _>>
Set FTP PASS to: <<_ _ _ _ _>>
Set FTP FILE URI to: <</samples/APP_OTA/helloworld.bin>>
Set LOCAL FINAL APP NAME to: <<helloworld.bin>>
Set LOCAL ORIGINAL APP NAME to: <<m2mapz.bin>>
[DEBUG] 23.53 ftp_utils.c:464 - msgFTPTask{FTPOTA_TASK}$ m2mb_os_ev_init success
[DEBUG] 23.54 ftp_utils.c:470 - msgFTPTask{FTPOTA_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.55 ftp_utils.c:478 - msgFTPTask{FTPOTA_TASK}$ Waiting for registration...
[DEBUG] 23.56 ftp_utils.c:371 - NetCallback{pubTspt_0}$ Module is registered to network
[DEBUG] 23.56 ftp_utils.c:491 - msgFTPTask{FTPOTA_TASK}$ Pdp context activation
[DEBUG] 23.57 ftp_utils.c:495 - msgFTPTask{FTPOTA_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 25.61 ftp_utils.c:504 - msgFTPTask{FTPOTA_TASK}$ Activate PDP with APN web.omnitel.it on cid 3....
[DEBUG] 26.30 ftp_utils.c:398 - PdpCallback{pubTspt_0}$ Context active
[DEBUG] 26.30 ftp_utils.c:401 - PdpCallback{pubTspt_0}$ IP address: 176.246.110.148
Start ftp client...
[DEBUG] 27.36 ftp_utils.c:533 - msgFTPTask{FTPOTA_TASK}$ Connected.
[DEBUG] 28.87 ftp_utils.c:546 - msgFTPTask{FTPOTA_TASK}$ FTP login successful.
Get remote file /samples/APP_OTA/helloworld.bin size
[DEBUG] 29.31 ftp_utils.c:568 - msgFTPTask{FTPOTA_TASK}$ Done. File size: 116224.
Starting download of remote file /samples/APP_OTA/helloworld.bin into local /mod/helloworld.bin
/samples/APP_OTA/helloworld.bin 4.68% 5440
/samples/APP_OTA/helloworld.bin 9.36% 10880
/samples/APP_OTA/helloworld.bin 14.04% 16320
/samples/APP_OTA/helloworld.bin 18.72% 21760
/samples/APP_OTA/helloworld.bin 23.40% 27200
/samples/APP_OTA/helloworld.bin 28.08% 32640
/samples/APP_OTA/helloworld.bin 32.76% 38080
/samples/APP_OTA/helloworld.bin 37.44% 43520
/samples/APP_OTA/helloworld.bin 42.13% 48960
/samples/APP_OTA/helloworld.bin 46.81% 54400
/samples/APP_OTA/helloworld.bin 51.49% 59840
/samples/APP_OTA/helloworld.bin 56.17% 65280
/samples/APP_OTA/helloworld.bin 60.85% 70720
/samples/APP_OTA/helloworld.bin 65.53% 76160
/samples/APP_OTA/helloworld.bin 70.21% 81600
/samples/APP_OTA/helloworld.bin 74.89% 87040
/samples/APP_OTA/helloworld.bin 79.57% 92480
/samples/APP_OTA/helloworld.bin 84.25% 97920
/samples/APP_OTA/helloworld.bin 88.93% 103360
/samples/APP_OTA/helloworld.bin 93.61% 108800
/samples/APP_OTA/helloworld.bin 97.42% 113220
[DEBUG] 43.54 ftp_utils.c:608 - msgFTPTask{FTPOTA_TASK}$ download successful.
FTP quit...
[DEBUG] 43.77 ftp_utils.c:632 - msgFTPTask{FTPOTA_TASK}$ Deactivating PDP
[DEBUG] 43.77 ftp_utils.c:642 - msgFTPTask{FTPOTA_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 44.20 ftp_utils.c:407 - PdpCallback{pubTspt_0}$ Context deactive
[DEBUG] 45.44 app_utils.c:76 - update_app{FTPOTA_TASK}$ Application successfully configured.
[DEBUG] 45.45 app_utils.c:82 - update_app{FTPOTA_TASK}$ Deleting old application /mod/m2mapz.bin
Starting. This is v1.0.7 built on Apr 7 2020 17:02:52. LEVEL: 2

Start Hello world Application [ version: 2.000000 ]

Hello world 2.0 [ 000001 ]
Hello world 2.0 [ 000002 ]
Hello world 2.0 [ 000003 ]

```

Figure 243

3.6.7 CJSON example:

Sample application showcasing how to manage JSON objects. Debug prints on **MAIN UART**

Features

- How to read a JSON using cJSON library
- How to write a JSON
- How to manipulate JSON objects

3.6.7.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Parse an example string into a JSON object and print the result in a formatted string
- Print some test outcomes (e.g. non existing item correctly not found)
- Retrieve single elements from the parsed JSON object and use them to format a descriptive string
- Delete the JSON object
- Create a new JSON object appending elements to it
- Print the result JSON string from the object

```

Starting Logging demo app. This is v1.0.7 built on Apr  7 2020 08:33:03.
And here is what we got:
{
  "name":      "Atlantic Ocean",
  "format":    {
    "type":     "salt",
    "volume":   310410900,
    "depth":    -8486,
    "volume_percent": 23.300000,
    "tide":     -3.500000,
    "calm":     false,
    "life":     ["plankton", "corals", "fish", "mammals"]
  }
}
inexistent key not found
name found: Atlantic Ocean
format found (null)
Our JSON string contains info about an ocean named Atlantic Ocean,
has a volume of 310410900 km^3 of salt water with -8486 meters max depth,
represents 23.3% of total oceans volume,
has an average low tide of -3.5 meters,
hosts a huge number of living creatures such as plankton, corals, fish, mammals,
and is not always calm.

Let's build a TR50 command with a property.publish and an alarm.publish for MQTT (no auth).
And here is what we got:
{
  "1": {
    "command": "property.publish",
    "params": {
      "thingKey": "mything",
      "key": "mykey",
      "value": 123.144000
    }
  },
  "2": {
    "command": "alarm.publish",
    "params": {
      "thingKey": "mything",
      "key": "mykey",
      "state": 3,
      "msg": "Message."
    }
  }
}
END.

```

Figure 244

3.6.8 Crypto Elliptic Curve Cryptography (ECC) example

Sample application showcasing how to manage Elliptic Curve Cryptography functionalities. Debug prints on **MAIN UART**

Features

- How to initialize ECC contexts A (Alice) and B (Bob). Alice is emulating a remote host, from which a public key is known.
- How to generate keypairs for contexts and export public keys
- how to export keyblobs from a context (a keyblob is encrypted with hw specific keys, and can only be used on the module where it was created)
- How to save a keyblob in secured TrustZone.
- How to reload a keyblob from the TrustZone into an initialized context
- How to sign a message with ECDSA from context B (Bob) and verify it from another context A (Alice) with the signature and public key of Bob.
- How to make Bob and Alice derive a shared session keys using each other's public key.
- How to make Bob and Alice create an AES context with the newly created shared keys, encode data and decode it on the other side

3.6.8.1 Application workflow

M2MB_main.c

- Create Bob ECC context, create a keypair and export it in a keyblob
- Open a file in secured Trust Zone, then store the keyblob in it.
- Destroy Bob ECC context
- Recreate Bob ECC context, open the file from Trust Zone and read the keyblob.
- Import the keyblob in Bob context.
- Export Bob public key
- Create Alice ECC context, to simulate an external host. Generate a keypair and export the public key.
- Sign a message with Bob context, generating a signature.
- Use Alice to verify the signed message using Bob's signature and public key
- Derive a shared key for Bob, using Alice's public key
- Create an AES context for Bob
- Import the shared key into the AES context
- Encrypt a message using Bob's AES context.

- Derive a shared key for Alice, using Bob's public key
- Create an AES context for Alice
- Import the shared key into the AES context
- Decrypt the message using Alice's AES context.
- Check the decrypted message and the original one match
- Clear all resources

```
Starting Crypto ECC demo app. This is v1.0.9-C1 built on May 11 2020 16:30:23.
Bob (local) and Alice (remote) scenario
Bob's keypair generated
Bob's keyblob length is 224
Bob exported the keyblob to be securely stored.

Bob already had an item in Secure Data Area, it was removed to create a new one
Bob securely saved the keyblob in Secure Data Area
Releasing resources

Close Bob's context...
Done. Now Bob context does not exist anymore.

Re-initialize Bob Context and load the keyblob from the secure zone
Bob securely loaded the keyblob from the SDA
Import keyblob in Bob's context..
Done. Now export Bob's public key...
Bob's public key successfully exported

Alice's keypair generated
Alice's public key successfully exported

Bob's message signed with ECDSA!
Alice verified bob's message with his pubkey and signature!

-----
Bob and Alice will now exchange a message with AES encrypt
-----

Bob retrieved the generated shared key size
Bob's shared keyblob length is: 32. Allocate the required memory to store it.
Bob created a shared key using Alice's public key!

Bob created an AEX context to exchange encrypted data with Alice
Bob's AES context imported the shared keyblob
Bob Encrypted the message using AES and the shared key!
Encrypted data:
 94EE531E3B84B2A4EF05502186BFF5DA

Alice retrieved the generated shared key size
Alice's shared keyblob length is: 32. Allocate the required memory to store it.
Alice created a shared key using Bob's public key!

Alice created an AEX context to exchange encrypted data with Bob
Alice's AES context imported the shared keyblob
Alice decrypted the message using AES and the shared key!
Decrypted:
 414094941E8942A4445548035BFAE943

Original, plain message:
 414094941E8942A4445548035BFAE943

Plain and decrypted messages match!
```

Figure 245

3.6.9 EEPROM 24AA256

Sample application showing how to communicate with a MicroChip 24AA256T I2C EEPROM chip using azx eeprom utility APIs. Debug prints on **MAIN UART**

Setup

This demo application requires that: - A0, A1, and A2 pins (1,2,3 chip pins) are connected to ground (pin 4) for device address 0xA0 - Pin 7 (WP) is connected to ground - Pin 6 (SCL) is connected to module GPIO 3 - Pin 5 (SDA) is connected to module GPIO 2 - Pin 4 is connected to one of the ground pins of the module - Pin 8 is connected to 1v8 supply (e.g. VPWRMON pin on the module)

Features

- Initialize the logs on the output channel
- configure the EEPROM utility, setting the slave address and the memory parameters (page size, memory size)
- Write single bytes on a random address
- Read written bytes as a page
- Write data using pages
- Read the new data using pages
- Read again using sequential reading
- Read a single byte from a specific address
- Read next byte using read from current address
- Erase the EEPROM
- Deinit EEPROM utility

3.6.9.1 Application workflow

M2MB_main.c

- call `azx_eeprom_init()` to set the utility parameters (SDA and SCL pins, page and memory sizes)
- call `azx_eeprom_writeByte()` to store a single byte with value "5" at the address 0x0213
- call `azx_eeprom_writeByte()` to store a single byte with value "6" at the address 0x0214
- call `azx_eeprom_readPages()` from address 0x0213 to retrieve the 2 bytes from the EEPROM
- call `azx_eeprom_writePages` to write 1024 bytes from a buffer, starting from address 0x00
- call `azx_eeprom_readPages()` again, to read 256 bytes from address 0x00
- call `azx_eeprom_readSequentially()` to read 256 bytes from 0x00 by without pages (less overhead on I2C protocol)

- call `azx_eeprom_readByte()` to get a single byte from address 0x00
- call `azx_eeprom_readByteFromCurrentAddress()` to get a byte from next address (0x01)
- call `azx_eeprom_eraseAll()` to completely erase the EEPROM memory (this writes 0xFF in each byte)
- call `azx_eeprom_readPages` from address 0x0213 to get 2 bytes and verify the values have been written to 0xFF
- call `azx_eeprom_deinit` to close the eeprom handler and the I2C channel

```
Starting I2C EEPROM 24AA256T demo app. This is v1.0.13-C1 built on Nov  3 2020 16:28:23.
Configuring the I2C device...
Opening I2C channel /dev/I2C-160 ( device address is 0xA0 )
Writing 1 byte at address 0x0213...
Done.
Writing 1 byte at address 0x0214...
Done.
Reading the 2 bytes from address 0x0213...
Done. Data: [0xFF 0xFF]

Writing 1024 bytes at address 0x0000..
Done.

Reading 256 bytes from address 0x0000...
Done. Data:
<<ABCDEFGHIJKLMNOPQRSTUVWXYZ.....abcdefghijklmnopqrstuvwxyz.....

Reading 256 bytes sequentially from address 0x0000...
Done. Data:
<<ABCDEFGHIJKLMNOPQRSTUVWXYZ.....abcdefghijklmnopqrstuvwxyz.....

Reading 1 byte from address 0x0000...
Done. Data: 'A'

Reading 1 byte from current address (should be 0x0001)...
Done. Data: 'B'

[DEBUG] 17.47 M2MB_main:177 - run_I2C_EEPROM_Demo{M2M_DamsStart}$ Erasing all the eeprom...
[DEBUG] 28.05 M2MB_main:185 - run_I2C_EEPROM_Demo{M2M_DamsStart}$ Done

Reading the 2 bytes from address 0x0213...
Done. Data: [0xFF 0xFF]

Deinit EEPROM...
Done
```

Figure 246

3.6.10 Easy AT example

Sample application showcasing Easy AT functionalities. Debug prints on **MAIN UART**

Features

- Shows how to register custom commands

The application adds two custom commands to the list of available ones:

- AT#MYCMD
- AT#MYINPUT

3.6.10.1 AT#MYCMD

This is a simple parameter-waiting command. It expects one string parameter, and will print it on the logging interface once received. The command simply returns OK

3.6.10.2 AT#MYINPUT

This command expects a numeric parameter, which indicates how many bytes will be received over the interface at most (the command will provide a prompt indicating it is waiting data). Then the data management callback will print when data is received, and if CTRL+Z (0x1A in hex) is received, it will complete the process, printing in the log interface what was received. sending ESC will terminate the process discarding any pending data.

3.6.11 Protect and AT command with a pwd using EasyAT functionality example

Sample application showing how to protect an AT command with a pwd using EasyAT functionality. Debug prints on **MAIN UART**

Features

- Shows how to register a new custom command and overwrite an existing one
- AT#M2MADMIN (new command)
- AT#M2MWRITE (existing one)

Application workflow

M2MB_main.c

- Entry point that calls EasyAT initialization function

3.6.11.1 AT#M2MADMIN

This command allows to enter a pwd to lock/unlock #M2MWRITE command. In the example app the new pwd is saved in the trustzone

AT#M2MADMIN="pwd",["newpwd"]

where mode: 0 - unlock command #m2mwrite 1 - lock command #m2mwrite 2 - change pwd

```
13:40:22.898- Starting Easy AT admin demo app. This is v1.1.19 built on Jul 24 2024 13:39:26.
13:40:22.898- Easy_at_init start
13:40:22.898- [DEBUG] 20.19 at_common:375 - easy_at_init{M2M_DamsStart}$ m2mb_atp_init succeeded
13:40:22.898- Register AT#M2MWRITE
13:40:22.898- Register AT#M2MADMIN
13:40:32.444- [DEBUG] 29.90 at_common:187 - _easy_at_task_entry{M2M_AT}$ task entry, message event type: CMD
13:40:32.444- [ERROR] 29.91 at_hash_M2M:74 - M2MWRITE_AT_Callback{M2M_AT}$ Operation not allowed!
13:40:37.464- [DEBUG] 34.93 at_common:187 - _easy_at_task_entry{M2M_AT}$ task entry, message event type: CMD
13:40:37.464- [DEBUG] 34.94 at_hash_M2M:223 - M2MADMIN_AT_Callback{M2M_AT}$ admin callback
13:40:37.464- Read form secure item res: 0
13:40:37.464- parameter is <mypassword>
13:40:37.464- Operation is <0>
13:40:37.464- hash: <89E01536AC207279409D4DE1E5253E01F4A1769E696DB0D6062CA9B8F56767C8>
13:40:37.464- right password!
13:40:37.909- [DEBUG] 35.31 at_hash_M2M:309 - M2MADMIN_AT_Callback{M2M_AT}$ Release SUCCESS was OK
13:41:04.285- [DEBUG] 62.27 at_common:187 - _easy_at_task_entry{M2M_AT}$ task entry, message event type: CMD
13:41:04.285- [DEBUG] 62.28 at_hash_M2M:223 - M2MADMIN_AT_Callback{M2M_AT}$ admin callback
13:41:04.285- parameter is <mypassword>
13:41:04.285- Operation is <1>
13:41:04.285- hash: <89E01536AC207279409D4DE1E5253E01F4A1769E696DB0D6062CA9B8F56767C8>
13:41:04.285- right password!
13:41:04.285- [DEBUG] 62.30 at_hash_M2M:362 - M2MADMIN_AT_Callback{M2M_AT}$ Release SUCCESS was OK
13:41:11.611- [DEBUG] 69.69 at_common:187 - _easy_at_task_entry{M2M_AT}$ task entry, message event type: CMD
13:41:11.611- [ERROR] 69.71 at_hash_M2M:74 - M2MWRITE_AT_Callback{M2M_AT}$ Operation not allowed!
```

Figure 247

3.6.12 Events

Sample application showcasing events setup and usage. Debug prints on **MAIN UART**

Features

- How to setup OS events with a custom bitmask
- How to wait for events and generate them in callback functions to synchronize blocks of code

3.6.12.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create an event handler
- Create a timer to generate an event, with a 2 seconds expiration time
- Wait for a specific event bit on the event handler
- At timer expiration, set the same event bit and verify that the code flow went through after the event.

```
Starting Events demo app. This is v1.0.7 built on Apr  7 2020 08:44:29.  
[DEBUG] 20.55 M2MB_main.c:171 - M2MB_main{M2M_DamsStart}$ m2mb_os_ev_init success  
Set the timer attributes structure success.  
Timer successfully created  
[DEBUG] 20.57 M2MB_main.c:125 - setup_timer{M2M_DamsStart}$ Start the timer, success.  
[DEBUG] 22.60 M2MB_main.c:60 - hwTimerCb{pubTspt_0}$ Timer Callback, generate event!  
[DEBUG] 22.61 M2MB_main.c:183 - M2MB_main{M2M_DamsStart}$ event occurred!
```

Figure 248

3.6.13 Events - Barrier (multi events)

Sample application showcasing how to setup and use multiple events to create a barrier. Debug prints on **MAIN UART**

Features

- How to setup OS events to be used as a barrier
- How to wait for multiple events in the same point, and generate them in call-back functions to synchronize blocks of code

3.6.13.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create an event handler
- Create a timer to generate an event, with a 3 seconds expiration time
- Create another timer to generate an event, with a 6 seconds expiration time
- Start both timers
- Wait for both event bits on the event handler (each one will be set by one of the timers)
- At first timer expiration, set the first event bit and verify that the code flow does not procede.
- At second timer expiration, set the second event bit and verify that the code flow went through after the event (implementing a barrier).

```
Starting Barrier demo app. This is v1.0.7 built on Apr 7 2020 08:48:30.
[DEBUG] 20.01 M2MB_main.c:179 - M2MB_main{M2M_DamsStart}$ m2mb_os_ev_init success
Set the timer attributes structure success.
Timer successfully created with 3000 timeout (ms)
Set the timer attributes structure success.
Timer successfully created with 6000 timeout (ms)
[DEBUG] 23.08 M2MB_main.c:66 - hwTimerCb1{pubTspt_0}$ Timer Callback, generate event 1!
[DEBUG] 26.12 M2MB_main.c:75 - hwTimerCb2{pubTspt_0}$ Timer Callback, generate event 2!
[DEBUG] 26.13 M2MB_main.c:214 - M2MB_main{M2M_DamsStart}$ BOTH events occurred!
```

Figure 249

3.6.14 FOTA example

Sample application showcasing FOTA usage with M2MB API. Debug prints on **MAIN UART**

Features

- How download a delta file from a remote server
- How to apply the delta and update the module firmware

3.6.14.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a main task to manage connectivity.
- create a fota task to manage FOTA and start it with INIT option

fota.c

fotaTask()

- Initialize FOTA system then reset parameters.
- Check current FOTA state, if not in IDLE, return error.
- Send a message to mainTask so networking is initialized.
- after PdPCallback() notifies the correct context activation, configure the fota client parameters such as FTP server URL, username and password
- get delta file from server. when it is completed, FOTADownloadCallback is called.
- If delta download went fine, check it.
- If delta file is correct, apply it. Once complete, restart the module.

mainTask()

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context. Event will be received on **PdP-Callback** function
- Disable PDP context when required to stop the app

PdpCallback()

- When PDP context is enabled, send a message to fotaTask to start the download

```

Starting FOTA demo app. This is v1.1.7 built on Jun 11 2021 12:20:43.
[DEBUG] 23.60 fota:187 - fotaTask{FOTA_TASK}$ Init FOTA...

Session file not present, procede with FOTA...
[DEBUG] 23.61 fota:236 - fotaTask{FOTA_TASK}$ m2mb_fota_reset PASS
[DEBUG] 23.61 fota:260 - fotaTask{FOTA_TASK}$ m2mb_fota_state_get M2MB_FOTA_STATE_IDLE
[DEBUG] 23.62 fota:379 - mainTask{MAIN_TASK}$ INIT
[DEBUG] 23.62 fota:392 - mainTask{MAIN_TASK}$ m2mb_os_ev_init success
[DEBUG] 23.63 fota:398 - mainTask{MAIN_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.63 fota:405 - mainTask{MAIN_TASK}$ Waiting for registration...
[DEBUG] 23.64 fota:131 - NetCallback{pubTspt_0}$ Module is registered to network
[DEBUG] 23.65 fota:418 - mainTask{MAIN_TASK}$ Pdp context initialization
[DEBUG] 25.70 fota:431 - mainTask{MAIN_TASK}$ Activate PDP with APN web.omnitel.it on cid 1....
[DEBUG] 35.42 fota:152 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 35.43 fota:155 - PdpCallback{pubTspt_0}$ IP address: 2.41.116.139

[DEBUG] 35.43 fota:285 - fotaTask{FOTA_TASK}$
Trying to download "samples/FOTA/37.00.003.3_to_37.00.003.1_ME310G1_NANVWWAU.bin" delta file...
[DEBUG] 35.45 fota:295 - fotaTask{FOTA_TASK}$ m2mb_fota_get_delta OK - Waiting for the completion callback
[DEBUG] 119.43 fota:96 - FOTADownloadCallBack{pubTspt_0}$ FOTA download Success - performing packet validation...
[DEBUG] 119.44 fota:301 - fotaTask{FOTA_TASK}$ Validating delta file...
[DEBUG] 156.36 fota:317 - fotaTask{FOTA_TASK}$ Packet is valid, start update...
[DEBUG] 156.40 fota:329 - fotaTask{FOTA_TASK}$ m2mb_fota_start PASS
[DEBUG] 158.36 fota:342 - fotaTask{FOTA_TASK}$
Rebooting...After reboot there will be the new FW running on module!

#OTAEV: Module Upgraded To New Fw
Starting FOTA demo app. This is v1.1.7 built on Jun 11 2021 12:20:43.
[DEBUG] 29.24 fota:187 - fotaTask{FOTA_TASK}$ Init FOTA...

Session file is already present, stop.

```

Figure 250

3.6.15 FOTA_FTP_client example

Sample application that shows how to download a delta file from an FTP server, stores it in the FOTA partition and deploys it. Debug prints on **MAIN UART**

Features

- How to download a delta file from FTP server using FTP client
- How to store directly delta file in the FOTA partition
- How to deploy delta file to upgrade module fw.

Application workflow

M2MB_main.c

- Print welcome message
- Create a main task to manage connectivity, delta download and deployment

ftp_test.c

msgFTPTask()

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context. Event will be received on
- Initialize FOTA system then reset parameters.
- After PDP context activation notified by PdPCallback() configure fota client parameters as FTP server url, username and password and SSL
- Get delta file from server and store it directly in the FOTA partition
- If delta download went fine, check it (m2mb_fota_update_package_check_setup) and if it's correct apply it (m2mb_fota_start).
- Once completed restart module.

PdpCallback()

- When PDP context is enabled, send a message to fotaTask to start the download

buf_data_cb_OTA()

- Handles data reception and writing in the FOTA partition (one block size at a time)


```

yStarting FOTA delta file FTP download and deploy demo app. This is v1.1.19 built on May 28 2024 14:41:14.
14:45:06.588- [DEBUG] 19.69 ftp_test:915 - msgFTPTask{FTP_TASK}$ INIT
14:45:06.588- Set APN to: <<ibox.tim.it>>
14:45:06.588- Set APN USER to: <<>>
14:45:06.588- Set APN PASS to: <<>>
14:45:06.588- Set PDP_CDX to: 1
14:45:06.588- Set FTP_ADDR to: <<ftp.telit.com>>
14:45:06.588- Set FTP_PORT to: 21
14:45:06.588- Set FTP_USER to: <<[REDACTED]>>
14:45:06.588- Set FTP_PASS to: <<[REDACTED]>>
14:45:06.588- Set ENABLE_TLS to: <<0>>
14:45:06.588- Set AUTH_TYPE to: <<2>>
14:45:06.588- Set CA_CERT_PATH to: <</mod>>
14:45:06.807- Set CLIENT_CERT_PATH to: <</mod>>
14:45:06.807- Set CLIENT_KEY_PATH to: <</mod>>
14:45:06.807- Set REMOTE_FOLDER to: <<robertaga>>
14:45:06.807- Set DLTOBUF_FILE to: <<delta_dummy_6_B710.bin>>
14:45:06.807- Set FOTA_STATUS_FILE to: <</mod/fota_stat.txt>>
14:45:06.807- [DEBUG] 19.78 ftp_test:932 - msgFTPTask{FTP_TASK}$ m2mb_os_ev_init success
14:45:06.807- [DEBUG] 19.79 ftp_test:938 - msgFTPTask{FTP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
14:45:06.807- [DEBUG] 19.80 ftp_test:946 - msgFTPTask{FTP_TASK}$ Waiting for registration...
14:45:06.807- [DEBUG] 19.82 ftp_test:840 - NetCallback{pubTspt_0}$ Module is registered to network
14:45:06.807-
14:45:06.807- Check FOTA upgrade status in file system
14:45:06.807-
14:45:06.807- Module current fw version is:
14:45:06.807- 37.00.216-B010-POC.210000
14:45:06.807- MOC.200006-B010
14:45:06.807- POC.210000
14:45:06.807- AOC.210000
14:45:07.248- [WARN ] 20.37 ftp_test:334 - FOTAStatusInit{FTP_TASK}$ File doesn't exist create it, first app execution
14:45:07.248- File created, store current fw version and fota upgrade flag=0
14:45:08.315-
14:45:08.315- Read stored data
14:45:08.315-
14:45:08.315- FOTA Flag: 0 Module previous fw version:
14:45:08.315- 37.00.216-B010-POC.210000
14:45:08.315- MOC.200006-B010
14:45:08.315- POC.210000
14:45:08.315- AOC.210000
14:45:08.315- Fw to be upgraded...
14:45:08.315- [DEBUG] 21.44 ftp_test:1009 - msgFTPTask{FTP_TASK}$ Pdp context activation
14:45:08.315- [DEBUG] 21.45 ftp_test:1013 - msgFTPTask{FTP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
14:45:10.426- [DEBUG] 23.50 ftp_test:1022 - msgFTPTask{FTP_TASK}$ Activate PDP with APN ibox.tim.it on cid 1....
14:45:10.855- [DEBUG] 24.05 ftp_test:867 - PdpCallback{pubTspt_0}$ Context active
14:45:10.855- [DEBUG] 24.07 ftp_test:870 - PdpCallback{pubTspt_0}$ IP address: 2.193.245.246
14:45:10.855- Start ftp client...
14:45:10.855- Connecting to host: <ftp.telit.com>
14:45:11.294- Retrieved address: 185.86.42.214
14:45:11.294- socket 0x4003d2a4 ok

```

Figure 251

3.6.16 FOTA from Local File example

Sample application that shows how perform FOTA upgrade using a delta file stored into file system. Debug prints on **MAIN UART**

Features

- How to store and get FOTA upgrade information to/from a file
- How to get delta file from module file system
- How to apply the delta and update module firmware

Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Check if module has been already upgraded or needs to be upgraded reading FOTA upgrade status from a file
- Create a fota task to manage FOTA and start it with INIT option

smartFotaTask() - Initialize FOTA system then reset parameters. - Get FOTA partition size and block size - Copy delta file from file system to FOTA partition. when it is completed, FOTADownloadCallback is called. - If delta file is correct, apply it. Once complete, write FOTA status flag and current fw version to a file, restart the module.

```

Starting FOTA application from local delta file demo app. This is v1.0.14-C1 built on May 16 2022 15:50:38.

Check FOTA upgrade status in file system

Module current fw version is:
37.00.214-POC.210001
MOC.200003
POC.210001
AOC.210000
[WARN ] 13.83 M2MB_main:190 - FOTAUpgradeStatusInit{M2M_DamsStart}$ File doesn't exist create it, first app execution
File created, store current fw version and fota upgrade flag=0

Read stored data

FOTA Flag: 0 Module previous fw version:
37.00.214-POC.210001
MOC.200003
POC.210001
AOC.210000

Start FOTA process
[DEBUG] 15.42 M2MB_main:450 - smartFotaTask{FOTA_TASK}$ INIT
[DEBUG] 15.43 M2MB_main:461 - smartFotaTask{FOTA_TASK}$ m2mb_fota_init success
[DEBUG] 15.44 M2MB_main:464 - smartFotaTask{FOTA_TASK}$
Get block and FOTA partition size
OTA blockSize: 131072
OTA partitionSize: 11272192
[DEBUG] 15.47 M2MB_main:472 - smartFotaTask{FOTA_TASK}$
Copy delta file from File system to FOTA partition
[DEBUG] 15.47 M2MB_main:277 - copyFromFileSystemToDelta{FOTA_TASK}$ File size: 746552
[DEBUG] 15.87 M2MB_main:480 - smartFotaTask{FOTA_TASK}$ Delta file writing completed
[DEBUG] 15.88 M2MB_main:484 - smartFotaTask{FOTA_TASK}$
Delta file check...
[DEBUG] 15.89 M2MB_main:347 - check_fota_delta{FOTA_TASK}$ -- check_fota_delta...
[DEBUG] 54.73 M2MB_main:366 - check_fota_delta{FOTA_TASK}$ Fota check integrity PASS
[DEBUG] 54.74 M2MB_main:492 - smartFotaTask{FOTA_TASK}$ ...delta file OK
[DEBUG] 54.75 M2MB_main:496 - smartFotaTask{FOTA_TASK}$
--> Start update...
[DEBUG] 56.83 M2MB_main:403 - startUpdate{FOTA_TASK}$
Reboot module to start delta deployment

```

Figure 252

3.6.17 FTP

Sample application showcasing FTP client demo with AZX FTP. Debug prints on **MAIN UART**

Features

- How to check module registration and activate PDP context
- How to connect to a FTP server
- How to exchange data with the server

3.6.17.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage FTP client and start it

ftp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Init FTP client and set the debug function for it
- Connect to the server
- Perform log in
- Check remote file size and last modification time
- Download file from server to local filesystem. A data callback is set to report periodic info about the download status
- Upload the same file to the server with a different name. A data callback is set to report periodic info about the upload status
- Download another file content in a buffer instead of a file. A data callback is set to report periodic info about the download status
- Close the connection with FTP server
- Disable PDP context

```

Starting FTP demo app. This is v1.0.7 built on Apr 7 2020 11:17:36.
[DEBUG] 21.23 ftp_test.c:290 - msgFTPTask{FTP_TASK}$ INIT
[DEBUG] 21.23 ftp_test.c:304 - msgFTPTask{FTP_TASK}$ m2mb_os_ev_init success
[DEBUG] 21.23 ftp_test.c:310 - msgFTPTask{FTP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 21.23 ftp_test.c:318 - msgFTPTask{FTP_TASK}$ Waiting for registration...
[DEBUG] 21.25 ftp_test.c:214 - NetCallback{pubTspt_0}$ Module is registered to network
[DEBUG] 21.26 ftp_test.c:331 - msgFTPTask{FTP_TASK}$ Pdp context activation
[DEBUG] 21.27 ftp_test.c:335 - msgFTPTask{FTP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.31 ftp_test.c:344 - msgFTPTask{FTP_TASK}$ Activate PDP with APN web.omnitel.it on cid 3...
[DEBUG] 24.09 ftp_test.c:241 - PdpCallback{pubTspt_0}$ Context active
[DEBUG] 24.10 ftp_test.c:244 - PdpCallback{pubTspt_0}$ IP address: 176.244.166.181
Start ftp client...
[DEBUG] 24.82 ftp_test.c:373 - msgFTPTask{FTP_TASK}$ Connected.
[DEBUG] 26.32 ftp_test.c:386 - msgFTPTask{FTP_TASK}$ FTP login successful.
Get remote file /samples/pattern_big.txt size
[DEBUG] 26.69 ftp_test.c:428 - msgFTPTask{FTP_TASK}$ Done. File size: 20026.
Get remote file /samples/pattern_big.txt last modification date
[DEBUG] 26.89 ftp_test.c:450 - msgFTPTask{FTP_TASK}$ Done. File last mod date: 20200407090654
.

Starting download of remote file /samples/pattern_big.txt into local /mod/_pattern_big.txt
/samples/pattern_big.txt 47.54% 9520
/samples/pattern_big.txt 100.00% 20026
[DEBUG] 29.75 ftp_test.c:488 - msgFTPTask{FTP_TASK}$ download successful.
[DEBUG] 29.76 ftp_test.c:522 - msgFTPTask{FTP_TASK}$
Local file /mod/_pattern_big.txt size: 20026

Starting upload of local file /mod/_pattern_big.txt
/mod/_pattern_big.txt 81.81% 16384
Upload successful.

Starting download of remote file /samples/pattern.txt into local buffer
Getting remote file /samples/pattern.txt size..
[DEBUG] 32.97 ftp_test.c:583 - msgFTPTask{FTP_TASK}$ Done. File size: 988.
Starting download of remote file /samples/pattern.txt to buffer
[DEBUG] 34.08 ftp_test.c:145 - buf_data_cb{FTP_TASK}$ Received START event
[DEBUG] 34.09 ftp_test.c:149 - buf_data_cb{FTP_TASK}$ Received DATA: 988 bytes on buffer 0x400399e0
[DEBUG] 34.26 ftp_test.c:153 - buf_data_cb{FTP_TASK}$ Received END event
[DEBUG] 34.26 ftp_test.c:623 - msgFTPTask{FTP_TASK}$ Download successful. Received 988 bytes<<<
0 |-----| |-----| |-----| |-----| |-----| *
1 | A | | A | | A | | A | | A | | *
2 | AAA | | AAA | | AAA | | AAA | | AAA | | *
3 | AAAAA | | AAAAA | | AAAAA | | AAAAA | | AAAAA | | *
4 | AAAAAAA | | AAAAAAA | | AAAAAAA | | AAAAAAA | | AAAAAAA | | *
5 | AAAAAAAA | | AAAAAAAA | | AAAAAAAA | | AAAAAAAA | | AAAAAAAA | | *
6 | AAAAAAA | | AAAAAAA | | AAAAAAA | | AAAAAAA | | AAAAAAA | | *
7 | AAAAA | | AAAAA | | AAAAA | | AAAAA | | AAAAA | | *
8 | AAA | | AAA | | AAA | | AAA | | AAA | | *
9 | A | | A | | A | | A | | A | | *
10 |-----| |-----| |-----| |-----| |-----| *
11 | | | | | | | *
12 |-----| |-----| |-----| |-----| |-----|

```

Figure 253

3.6.18 File System example

Sample application showcasing M2MB File system API usage. Debug prints on **MAIN UART**

Features

- How to open a file in write mode and write data in it
- How to reopen the file in read mode and read data from it

3.6.18.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Open file in write mode
- Write data in file
- Close file
- Reopen file in read mode
- Read data from file and print it
- Close file and delete it

```
Starting FileSystem demo app. This is v1.0.7 build on Mar 26 2020 09:50:19. LEVEL: 2
Opening /my_text_file.txt in write mode..
Buffer written successfully into file. 15 bytes were written.
Closing file.
Opening /my_text_file.txt in read only mode..
Received 15 bytes from file:
<Hello from file>
Closing file.
Deleting File
File deleted
App Completed
```

Figure 254

3.6.19 GNSS example

Sample application showing how to use GNSS functionality. Debug prints on **MAIN UART**

Features

- How to enable GNSS receiver on module
- How to collect location information from receiver

Note: on MEx10G1 product family both M2MB_GNSS_SERVICE_NMEA_REPORT and M2MB_GNSS_SERVICE_POSITION_REPORT services are available, while on ME910C1 product family only M2MB_GNSS_SERVICE_POSITION_REPORT is available

3.6.19.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print a welcome message
- Create GNSS task and send a message to it

gps_task.c - Init Info feature and get module type - Init gnss, enable position/NMEA report and start it. - When a fix or a NMEA sentence is available, a message will be printed by the GNSS callback function

```
Starting GNSS demo app. This is v1.1.4 built on Oct 1 2021 15:27:44.
Model: ME910C1-E2
m2mb_gnss_enable, POSITION OK
m2mb_gnss_start OK, waiting for position/nmea sentences...
latitude_valid: 1 - latitude: 45.713643
longitude_valid: 1 - longitude: 13.738041
altitude_valid: 1 - altitude: 195.000000
uncertainty_valid: 1 - uncertainty: 95.000000
velocity_valid: 1 - codingType: 0
speed_horizontal: 0.650000
bearing: 0.000000
timestamp_valid: 1 -timestamp: 1633095357439
speed_valid: 1 - speed: 1.471360

***** Wait 120 seconds and then stop GPS *****
```

Figure 255

3.6.20 GPIO interrupt example

Sample application showing how to use GPIOs and interrupts. Debug prints on **MAIN UART**

Features

- How to open a GPIO in input mode with interrupt
- How to open a second GPIO in output mode to trigger the first one

3.6.20.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Open GPIO 4 as output
- Open GPIO 3 as input and set interrupt for any edge (rising and falling). **A jumper must be used to short GPIO 3 and 4 pins.**
- Toggle GPIO 4 status high and low every second
- An interrupt is generated on GPIO 3

```
Starting GPIO interrupt demo app. This is v1.0.7 built on Mar 26 2020 16:33:01.
Setting gpio 3 interrupt...
Setting GPIO 4 HIGH
CALLBACK->Interrupt on GPIO 3! Value: 1
Setting GPIO 4 LOW
CALLBACK->Interrupt on GPIO 3! Value: 0
Setting GPIO 4 HIGH
CALLBACK->Interrupt on GPIO 3! Value: 1
Setting GPIO 4 LOW
CALLBACK->Interrupt on GPIO 3! Value: 0
Setting GPIO 4 HIGH
CALLBACK->Interrupt on GPIO 3! Value: 1
Setting GPIO 4 LOW
CALLBACK->Interrupt on GPIO 3! Value: 0
```

Figure 256

3.6.21 GTP example

Sample application that shows how to get the position using GTP feature. Debug prints on **MAIN UART**

Features

- How to init and enable GTP feature
- How to get the position using GTP

Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Init NET functionality and wait for module to be registered
- Init PDP functionality and set APN on CID1
- Init GTP functionality
- Check if GTP is already enabled. If not enable it and reboot module
- If GTP is enabled, get position

```
Starting GTP demo app. This is v1.1.8 built on Dec 20 2022 15:29:19.
[DEBUG] 15.08 M2MB_main:237 - M2MB_main{M2M_DamsStart}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 15.09 M2MB_main:245 - M2MB_main{M2M_DamsStart}$ Waiting for registration...
[DEBUG] 15.10 M2MB_main:253 - M2MB_main{M2M_DamsStart}$ Pdp context setting
[DEBUG] 15.11 M2MB_main:82 - NetCallback{pubTspt_0}$ Module is registered to cell 0x201
[DEBUG] 15.11 M2MB_main:257 - M2MB_main{M2M_DamsStart}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 15.13 M2MB_main:268 - M2MB_main{M2M_DamsStart}$ m2mb_pdp_deinit returned M2MB_RESULT_SUCCESS

Check if GTP has been already enabled
[DEBUG] 16.19 M2MB_main:312 - M2MB_main{M2M_DamsStart}$ GTP status: 1 => Enabled
[DEBUG] 16.20 M2MB_main:317 - M2MB_main{M2M_DamsStart}$ Get the position...

m2mb_gnss_GTP OK, waiting for position...
[DEBUG] 107.96 M2MB_main:160 - gnssCallbackFN{pubTspt_0}$ gnssCallback event: 1
GTP position got
latitude: 41.900002
longitude: 12.500000
altitude: 0.000000
altitudeMeanSeaLevel: -0.000000
accuracy: 0.000000
speed: -0.000000
bearing: -0.000000
timestamp: 0
verticalAccuracy 0.000000
speedAccuracy -0.000000
bearingAccuracy: -0.000000
```

Figure 257

3.6.22 General_INFO example

Sample application prints some Module/SIM information as IMEI, fw version, IMSI and so on; it prints also some information about registration. Debug prints on **MAIN UART**

Features

- How to print some Module information as IMEI, FW version etc
- How to print some SIM information as IMSI, ICCID
- How to get and print some informatio about Module registration as Netowrk Operator, AcT, RSSI, etc

3.6.22.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Init NET functionality
- Init INFO functionality
- Get and print Module and SIM info
- Wait form module to register to network
- Get and print registration INFO

```

Starting. This is v1.1.4 built on Mar 31 2021 09:56:03. LEVEL: 2

Start General INFO application [ version: 1.000000 ]

=====
MODULE ME910C1-E2 INFO
=====
MANUFACTURER: Telit
IMEI: 353080091125422
MODEM FIRMWARE VERSION: MOB.700005
PACKAGE VERSION:
30.00.709-B005-P0B.700100
MOB.700005
P0B.700100
A0B.700000

=====
SIM INFO
=====
IMSI: 222015602268648
ICCID: 89390100001138084906

=====
Waiting for registration...

=====
Module is registered to HOME network cellID 0x5221
NETWORK OPERATOR (mcc mnc): 222 01
Network Technology 2G (AcT: 0) RSSI: -81

```

Figure 258

3.6.23 HTTP Client

Sample application showing how to use HTTPs client functionalities. Debug prints on **MAIN UART**

Features

- How to check module registration and activate PDP context
- How to initialize the http client, set the debug hook function and the data callback to manage incoming data
- How to perform GET, HEAD or POST operations (GET also with single range support)

NOTE: the sample app has an optional dependency on `azx_base64.h` if basic authentication is required (refer to `HTTP_BASIC_AUTH_GET` define in `M2MB_main.c` for further details)

3.6.23.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage HTTP client and start it

httpTaskCB

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create HTTP client options and initialize its functionality
- Create HTTP SSL config and initialize the SSL options
- Configure data management options for HTTP client
- Apply all configurations to HTTP client
- Perform a GET request to a server
- Disable PDP context

DATA_CB

- Print incoming data
- Set the abort flag to 0 to keep going.

```

Starting HTTP(s) client demo app. This is v1.1.15 built on Nov 28 2022 12:12:27.
[DEBUG] 17.12 read_param:101 - readConfigFromFile(HttpClient)$ Reading parameters from file
[DEBUG] 17.13 read_param:103 - readConfigFromFile(HttpClient)$ Opening /mod/HTTP_Client_config.txt in read mode..
Set APN to: <web.omnitel.it>
Set APN USER to: <>
Set APN PASS to: <>
Set CID to: 3
Set CACERTFILE to: <>
Set CLIENTCERTFILE to: <>
Set CLIENTKEYFILE to: <>
Set REQUEST_TYPE to: 2
Set SERVER to: <http://linux-ip.net>
[DEBUG] 17.16 m2mb_HTTP_t:267 - activatePdp(HttpClient)$ m2mb_os_ev_init success
[DEBUG] 17.17 m2mb_HTTP_t:273 - activatePdp(HttpClient)$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 17.18 m2mb_HTTP_t:281 - activatePdp(HttpClient)$ Waiting for registration..
[DEBUG] 17.19 m2mb_HTTP_t:101 - NetCallback{pubTspt_0}$ Module is registered to cell 0x468E!
[DEBUG] 17.20 m2mb_HTTP_t:295 - activatePdp(HttpClient)$ Pdp context initialization
[DEBUG] 19.26 m2mb_HTTP_t:305 - activatePdp(HttpClient)$ Activate PDP with APN web.omnitel.it....
[DEBUG] 20.16 m2mb_HTTP_t:128 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 20.18 m2mb_HTTP_t:131 - PdpCallback{pubTspt_0}$ IP address: 176.243.212.114
Performing a GET request...
Connecting to linux-ip.net:80//
Socket connected!
<!DOCTYPE html>
<html>
<head>
  <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
  <meta name="author" content="Martin A. Brown" />
  <meta name="robots" content="index, follow"/>

  <meta property="og:title" content="http://linux-ip.net/" />
  <meta property="og:url" content="http://linux-ip.net/" />
  <meta property="og:site_name" content="http://linux-ip.net/" />
  <meta property="og:type" content="website"/>

  <link rel="canonical" href="http://linux-ip.net" />

  <title>http://linux-ip.net/</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <link rel="stylesheet" type="text/css" href="//netdna.bootstrapcdn.com/font-awesome/4.0.3/css/font-awesome.css" />
  <link rel="stylesheet" type="text/css" href="//netdna.bootstrapcdn.com/twitter-bootstrap/2.3.2/css/bootstrap-combined.min.css" />
  <link rel="stylesheet" type="text/css" href="http://linux-ip.net/theme/css/main.css" />
</head>
<body>
  <div id="site-footer">
    <div class="row-fluid">
      <div class="span10 offset1">
        <address>
          <p>
            Powered by <a href="http://getpelican.com/">Pelican</a>
            and <a href="http://python.org">Python</a>.
            Theme based on <a href="http://github.com/jsliang/pelican-fresh">Fresh</a>
            by <a href="http://jsliang.com/">jsliang</a>
          </p>
        </address>
      </div>
    </div>
  </div>
</body>
</html>
Result: 200
[DEBUG] 26.80 m2mb_HTTP_t:137 - PdpCallback{pubTspt_0}$ Context deactivated!

```

Figure 259

3.6.24 HW Timer (Hardware Timer)

The sample application shows how to use HW Timers M2MB API. Debug prints on **MAIN UART**

Features

- How to open configure a HW timer
- How to use the timer to manage recurring events

3.6.24.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create hw timer structure
- Configure it with 100 ms timeout, periodic timer (auto fires when expires) and autostart
- Init the timer with the parameters
- Wait 10 seconds
- Stop the timer

TimerCb

- Print a message with an increasing counter

```
Starting HW Timers demo app. This is v1.0.7 built on Mar 26 2020 13:04:14.
[DEBUG] 14.06 M2MB_main.c:114 - M2MB_main{M2M_DamsStart}$ Set the timer attributes structure: success.
Timer successfully created
Start the timer, success.
[DEBUG] 14.18 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [0]
[DEBUG] 14.28 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [1]
[DEBUG] 14.38 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [2]
[DEBUG] 14.48 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [3]
[DEBUG] 14.58 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [4]
[DEBUG] 14.69 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [5]
[DEBUG] 14.79 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [6]
[DEBUG] 14.88 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [7]
[DEBUG] 14.98 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [8]
[DEBUG] 15.08 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [9]

[DEBUG] 23.90 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [96]
[DEBUG] 24.01 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [97]
[DEBUG] 24.11 M2MB_main.c:55 - TimerCb{pubTspt_0}$ Callback Count: [98]
Stop a running timer: success
Application end
```

Figure 260

3.6.25 Hello World

The application prints “Hello World!” over selected output every two seconds. Debug prints on **MAIN UART**, using AZX log example functions

Features

- How to open an output channel using AZX LOG sample functions
- How to print logging information on the channel using AZX LOG sample functions

3.6.25.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print “Hello World!” every 2 seconds in a while loop

```
Starting. This is v1.0.7 built on Mar 26 2020 09:34:16. LEVEL: 2
Start Hello world Application [ version: 2.000000 ]
Hello world 2.0 [ 000001 ]
Hello world 2.0 [ 000002 ]
Hello world 2.0 [ 000003 ]
Hello world 2.0 [ 000004 ]
Hello world 2.0 [ 000005 ]
Hello world 2.0 [ 000006 ]
Hello world 2.0 [ 000007 ]
Hello world 2.0 [ 000008 ]
Hello world 2.0 [ 000009 ]
```

Figure 261

3.6.26 I2C example

Sample application showing how to communicate with an I2C slave device. Debug prints on **MAIN UART**

Features

- How to open a communication channel with an I2C slave device
- How to send and receive data to/from the slave device

Setup

- Connect sensor VDD to 1v8 supply (e.g. Vaux/PwrMon pin of the module)
- Connect sensor GND to a GND pin of the module
- Connect sensor SDA to module GPIO2
- Connect sensor SCL to module GPIO3

3.6.26.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Open I2C bus, setting SDA and SCL pins as 2 and 3 respectively
- Set registers to configure accelerometer -Read in a loop the 6 registers carrying the 3 axes values and show the g value for each of them

```
Starting I2C demo app. This is v1.0.7 built on Mar 26 2020 16:50:40.
Configuring the Kionix device...
opening channel /dev/I2C-30
[DEBUG] 20.18 M2MB_main.c:218 - test_I2C{M2M_DamsStart}$|-
WHOAMI content: 0x01
Configuring I2C Registers - Writing 0x4D into 0x1D register (CTRL_REG3)...
Write: success

I2C reading data from 0x1D register (CTRL_REG3)...
Read: success.
Accelerometer Enabled. ODR tilt: 12.5Hz, ODR directional tap: 400Hz, ORD Motion Wakeup: 50Hz
Configuring I2C Registers - Writing 0xC0 into 0x1B register (CTRL_REG1)...
Write: success

I2C reading data from 0x1B register (CTRL_REG1)...
Read: success.
Accelerometer Enabled. Operative mode, 12bit resolution
I2C read axes registers
-----
Reading Success.

X: -0.050 g
Y: -0.046 g
Z: 1.006 g
Reading Success.

X: -0.049 g
Y: -0.044 g
Z: 1.004 g
Reading Success.

X: -0.052 g
Y: -0.044 g
Z: 1.007 g
Reading Success.

X: -0.048 g
Y: -0.045 g
Z: 1.005 g
```

Figure 262

3.6.27 I2C Combined

Sample application showing how to communicate with an I2C slave device with I2C raw mode. Debug prints on MAIN UART

Features

- How to open a communication channel with an I2C slave device
- How to send and receive data to/from the slave device using raw mode API

Setup

- Connect sensor VDD to 1v8 supply (e.g. Vaux/PwrMon pin of the module)
- Connect sensor GND to a GND pin of the module
- Connect sensor SDA to module GPIO2
- Connect sensor SCL to module GPIO3

3.6.27.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Open I2C bus, setting SDA and SCL pins as 2 and 3 respectively
- Set registers to configure accelerometer - Read in a loop the 6 registers carrying the 3 axes values and show the g value for each of them

```
Starting I2C raw demo app. This is v1.0.13-C1 built on Jul 30 2020 11:28:18.
Configuring the I2C device...
Opening I2C channel /dev/I2C-30 ( device address is 0x0F << 1 )
Accelerometer Enabled. ODR tilt: 12.5Hz, ODR directional tap: 400Hz, ORD Motion Wakeup: 50Hz
Accelerometer Enabled. Operative mode, 12bit resolution
I2C read axes registers
-----
X: 0.000 g
Y: 0.000 g
Z: 0.000 g

X: -0.270 g
Y: 0.016 g
Z: 0.917 g

X: -0.268 g
Y: 0.013 g
Z: 0.925 g

X: -0.271 g
Y: 0.015 g
Z: 0.922 g

X: -0.267 g
Y: 0.016 g
Z: 0.918 g

X: -0.274 g
Y: 0.019 g
Z: 0.915 g
```

Figure 263

3.6.28 Little FileSystem 2

Sample application showing how use lfs2 porting with RAM disk and SPI data flash.
Debug prints on **MAIN UART**

Features

- How to create and manage Ram Disk
- How to manage file-system in Ram disk partition
- How to create and manage SPI Flash memory partition
- How to manage file-system in SPI Flash memory partition

3.6.28.1 Application workflow

M2MB_main.c

- Init logging system
- Call Ram Disk tests
- Call Flash memory tests

ram_utils_usage.c

- Initialize Ram Disk
- Format and Mount partition
- List files
- Files creation and write content
- List files
- Read files
- Unmount and Release resources

spi_utils_usage.c - Initialize SPI Flash chip - Initialize SPI Flash Disk - Format and Mount partition - List files - Files creation and write content - List files - Read files - Delete files - Directories creation and deletion - Unmount and Release resources

Notes:

For SPI Flash a JSC memory is used with chip select pin connected to module GPIO2 pin. For better performances, a 33kOhm pull-down resistor on SPI clock is suggested. Please refer to SPI_echo sample app for SPI connection details.

For LE910Cx (both Linux and ThreadX based devices), AT#SPIEN=1 command must be sent once before running the app

```

Starting lfs2 demo app. This is v1.0.14-C1 built on Oct 22 2020 09:43:08.
>>>>>> Starting RAMDiskDemo ...
[DEBUG] 18.28 azx_lfs_uti:125 - azx_ram_initialize{M2M_DamsStart}$ Ram Memory allocated correctly from 0x40042228 to 0x40046228!!
Mounting partition...
Formatting...
Mounting...

Mounted partition...
<<<<<<fileListUtils
List:
.., 0, 2
.., 0, 2
file_name: file000.txt
size: 10
buffer: content000
mode: 0
RAM TYPE size: 10000

File created and closed: file000.txt

<<<<<<fileListUtils
___INSIDE --->file000.txt, 10, 1
List:
.., 0, 2
.., 0, 2
file000.txt, 10, 1
----->File reading
File: file000.txt, Size: 10, Buffer: content000
Nand released
Partition unmounted
[DEBUG] 20.31 azx_lfs_uti:165 - azx_ram_releaseResources{M2M_DamsStart}$ Ram Memory released correctly!!

>>>>>>> Starting FlashDiskDemo ...
Starting initialization...

table id[0] = 191
table id[1] = 1
table id[2] = 0

nandLFS_CALLBACK Callback event <1>
NAND Callback event: NAND_JSC_INITIALIZED <1>
nandLFS_CALLBACK Callback event <1>
NAND Callback event: NAND_JSC_INITIALIZED <1>

Mounting partition...
Formatting...
spiErase: address = 0, len = 131072
spiErase: address = 131072, len = 131072

Mounting...

Mounted partition...

<<<<<<fileListUtils
List:
.., 0, 2
.., 0, 2

Formatting...
spiErase: address = 0, len = 131072
spiErase: address = 131072, len = 131072

Mounting...

Mounted partition...

<<<<<<fileListUtils
|
List:
.., 0, 2
.., 0, 2
file_name: file000.txt
size: 10
buffer: content000
mode: 0

File created and closed: file000.txt

```

```

<><><>fileListUtils
List:
., 0, 2
., 0, 2
file000.txt, 10, 1
file001.txt, 10, 1
file002.txt, 10, 1
file003.txt, 10, 1
file004.txt, 10, 1
----->File reading
File: file000.txt, Size: 10, Buffer: content000

File: file004.txt, Size: 10, Buffer: content004

File: file002.txt, Size: 10, Buffer: content002
----->File removing
file001.txt<<<<<<<

File removed: file001.txt|
file000.txt<<<<<<<

File removed: file000.txt
file004.txt<<<<<<<

File removed: file004.txt

<><><>fileListUtils
List:
., 0, 2
., 0, 2
file002.txt, 10, 1
file003.txt, 10, 1
spiErase: address = 59637760, len = 131072
[DEBUG] 58.61 azx_lfs_uti:648 - azx_lfsDirCreationByContext{M2M_DamsStart}$ Directory created: dir000!!
[DEBUG] 59.78 azx_lfs_uti:631 - azx_lfsDirCreationByContext{M2M_DamsStart}$ Directory already exists: dir000!!
spiErase: address = 59899904, len = 131072
[DEBUG] 61.70 azx_lfs_uti:648 - azx_lfsDirCreationByContext{M2M_DamsStart}$ Directory created: dir001!!
spiErase: address = 60162048, len = 131072
[DEBUG] 63.67 azx_lfs_uti:648 - azx_lfsDirCreationByContext{M2M_DamsStart}$ Directory created: dir002!!

<><><>fileListUtils
List:
., 0, 2
., 0, 2
dir000, 0, 2
dir001, 0, 2
dir002, 0, 2
file002.txt, 10, 1
file003.txt, 10, 1

<><><>fileListUtils
List:
., 0, 2|
., 0, 2
dir001, 0, 2
dir002, 0, 2
file002.txt, 10, 1
file003.txt, 10, 1

Nand released
Partition unmounted
Unmounted process ended...
testAllInOneFunction ended...

```

3.6.29 LWM2M

Sample application showcasing LWM2M client usage with M2MB API. Debug prints on **MAIN UART**

Features

- Configure LWM2M client and enable it
- Create an instance of a custom object
- Set an integer value on a read only resource
- Set two integer values on a multi-instance read only resource
- write a string on a read/write resource
- Manage exec requests from the portal
- Manage write, read and monitoring resources changed from the portal

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6", "<user apn>"
```

Depending on the Mobiler Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.6.29.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask - Check registration status

- Initialize LWM2M client,
- Check for XML file fo custom object
- Enable unsolicited messages from client
- Create a task (lwm2m_taskCB is its callback function)to manage events from Portal
- Enable Lwm2M client

- Create a new instance for the custom object
- Wait for client to register to Portal
- Performs all operations (set, read, get, write) on the related resources
- Performs a set with notify ack enabled
- Wait for events from server

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)
- Run lwm2m_taskCB when a monitored resource changes, to manage the action to be done

3.6.29.2 Device Profile upload

LWM2M resources demo device profile must be imported to have a real-time update of resources values on the LWM2M browser.

To do so, import the file `json/lwm2m_resources_demo.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

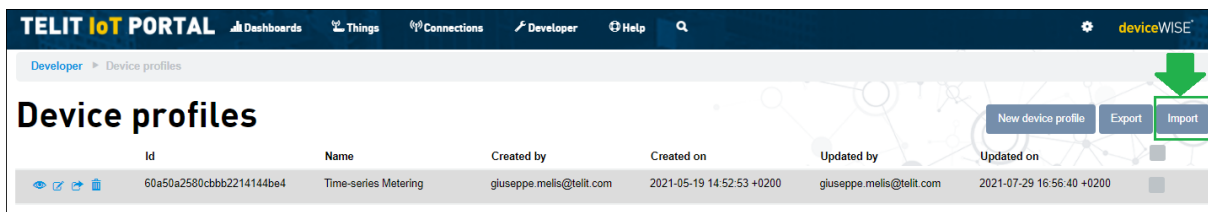


Figure 264

3.6.29.3 Custom Object configuration

The XML file content must be loaded on the Telit IoT Portal for the demo application to be fully executed.

First, enter Developer section from the top menu



Figure 265

Choose Object Registry

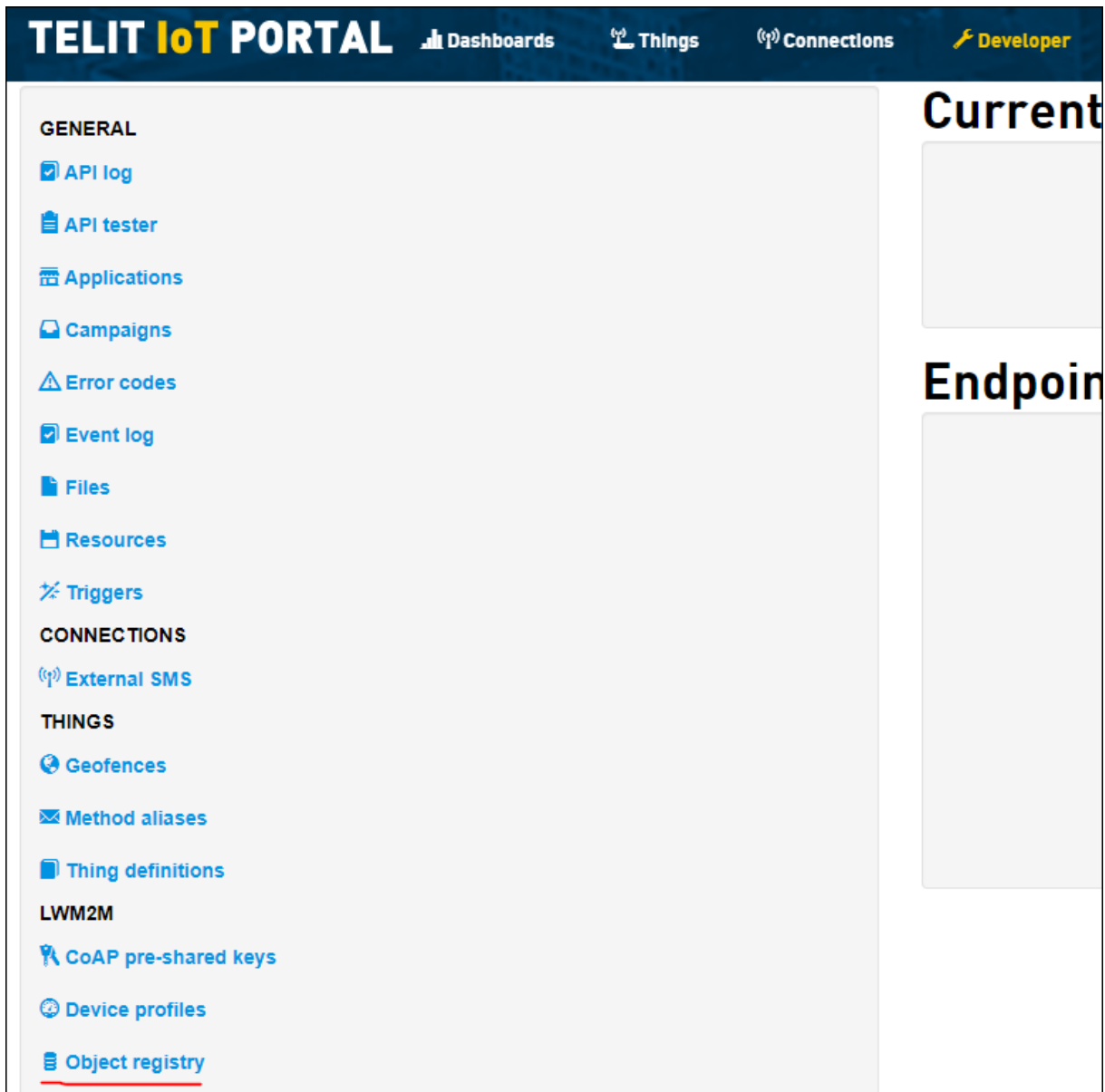


Figure 266

Create a New Object

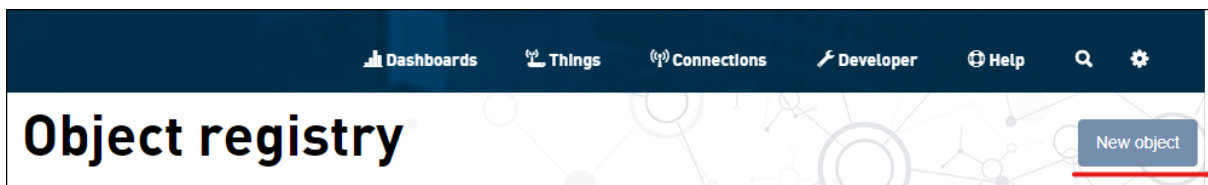
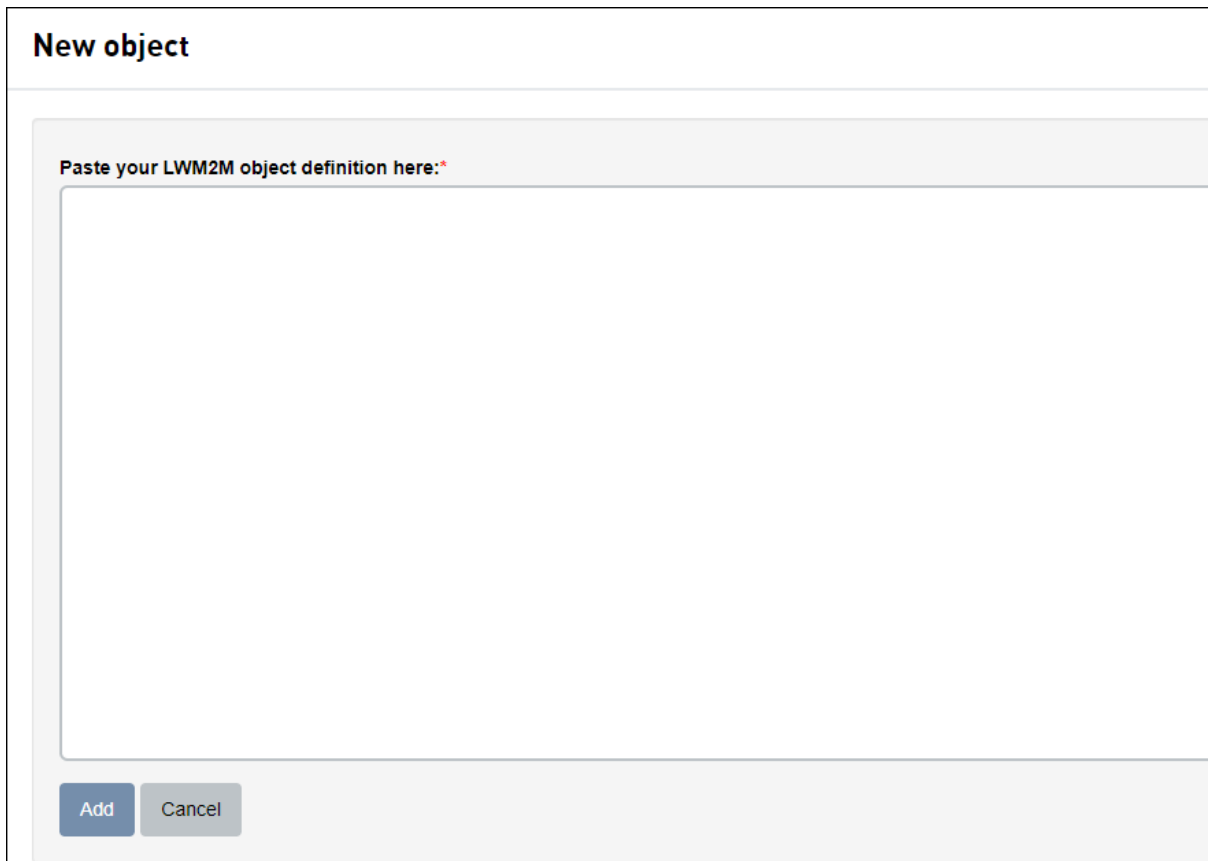


Figure 267

Copy the xml file content and paste it in the new Object form



New object

Paste your LWM2M object definition here:*

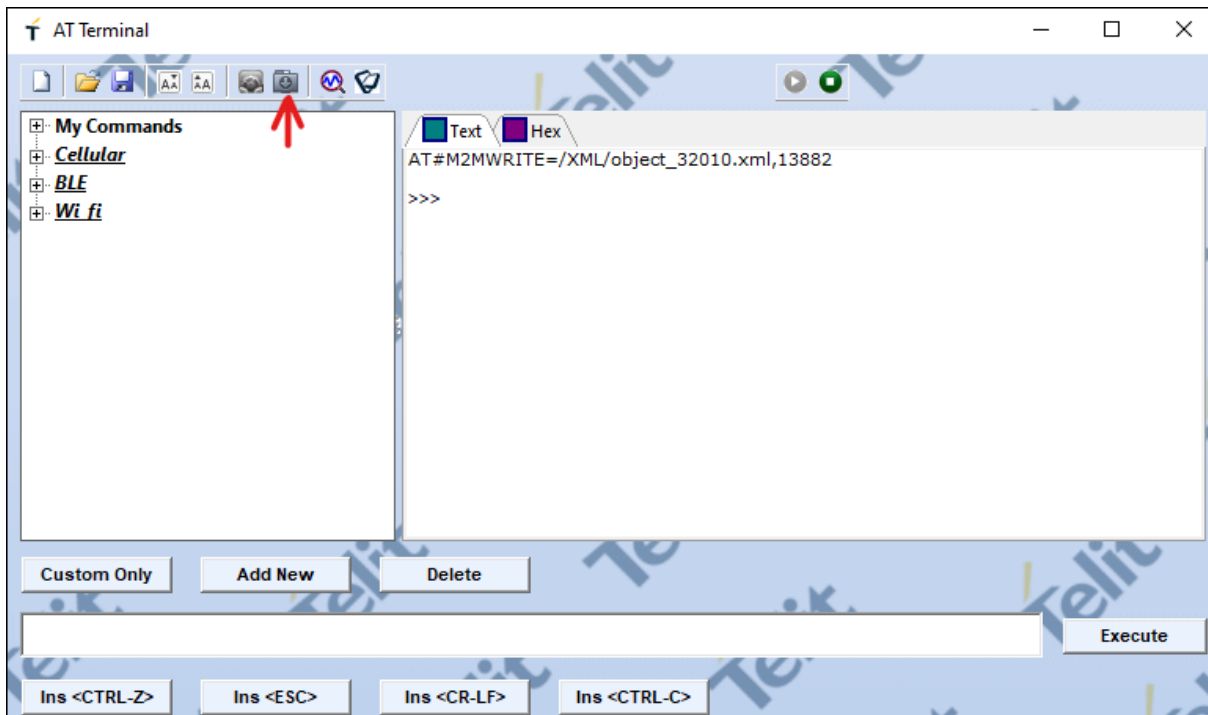
Add Cancel

Figure 268

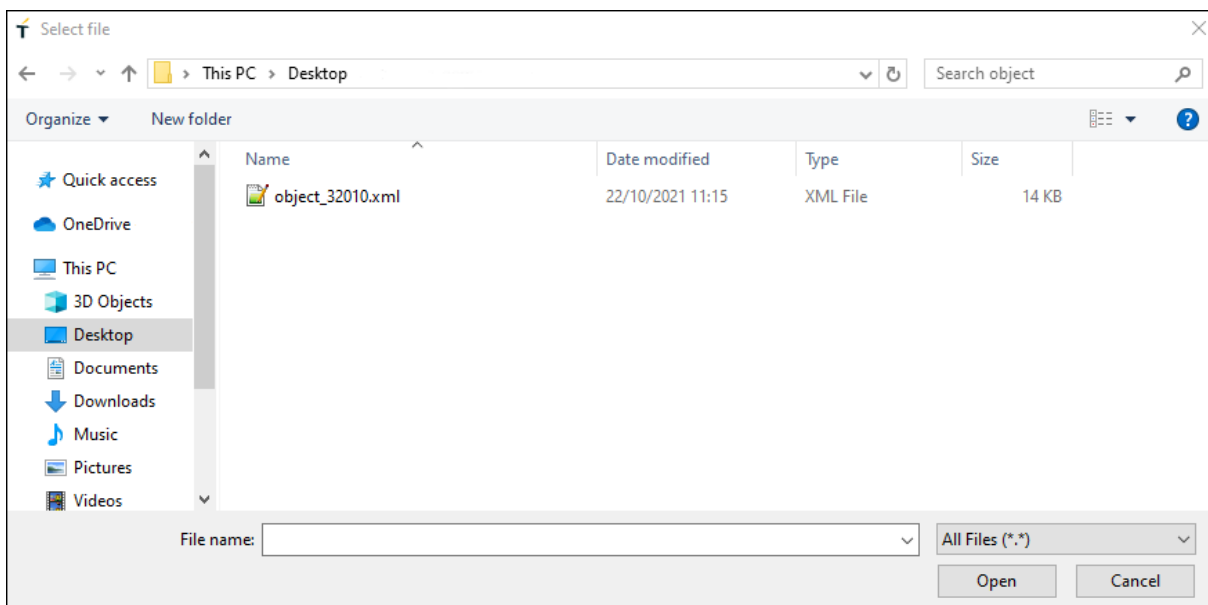
Also, the application requires the XML file `xml/object_32010.xml` (provided with the sample files) to be stored in module's `/XML/` folder. It can be done with

```
AT#M2MWRITE=/XML/object_32010.xml,<size_in_bytes>
```

To load the XML file in the module, Telit AT Controller (TATC) can be used. Once the command above is issued, press the load content button:

**Figure 269**

Select the file from your computer

**Figure 270**

The file is successfully loaded on the module

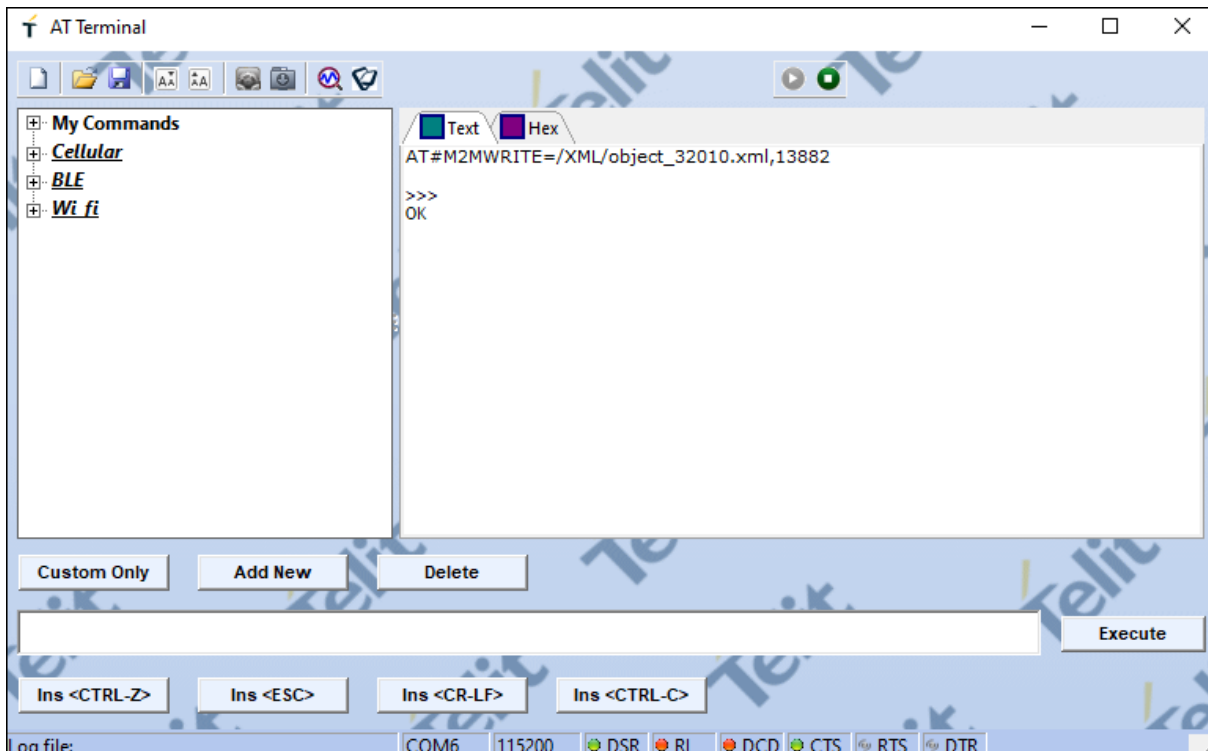


Figure 271

3.6.29.4 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on “**Things**” section, click “**New Thing**” button in the top right corner.

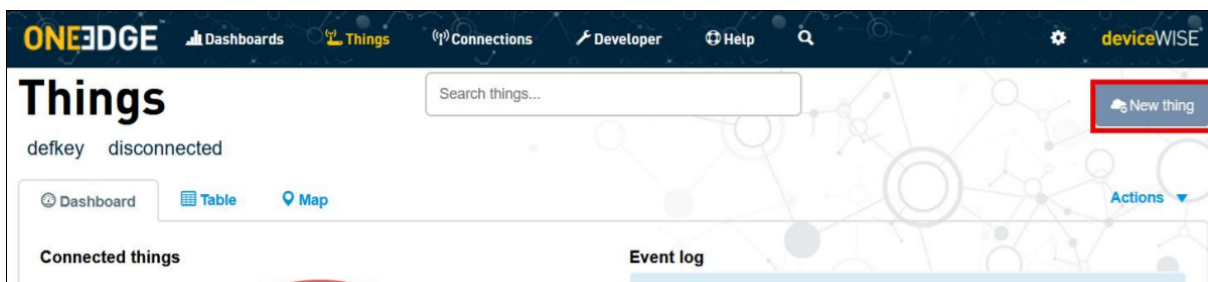


Figure 272

In the Create a new thing dialog, select “Telit Module”

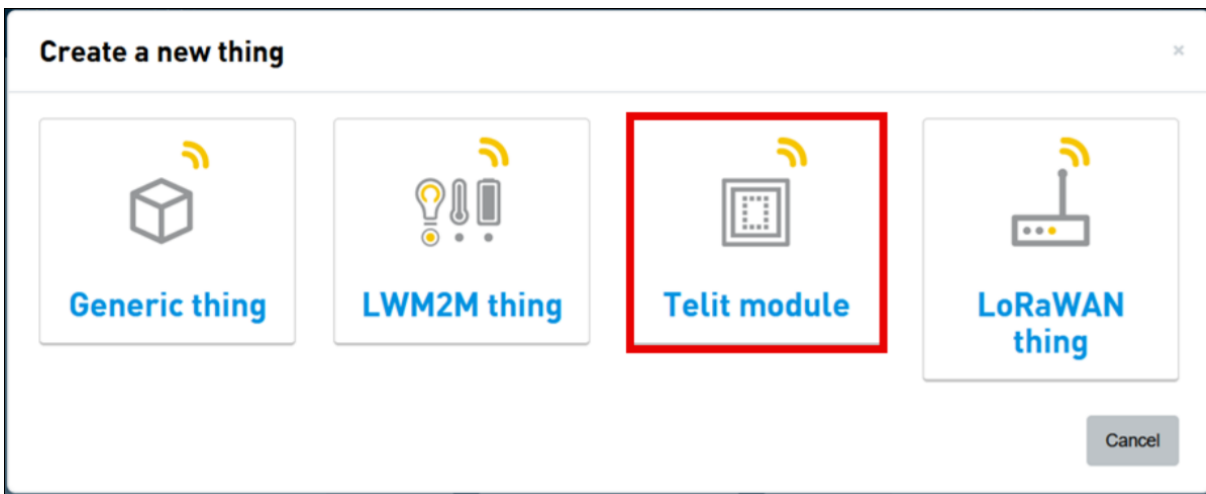


Figure 273

A dialog appears: select “Default” thing definition

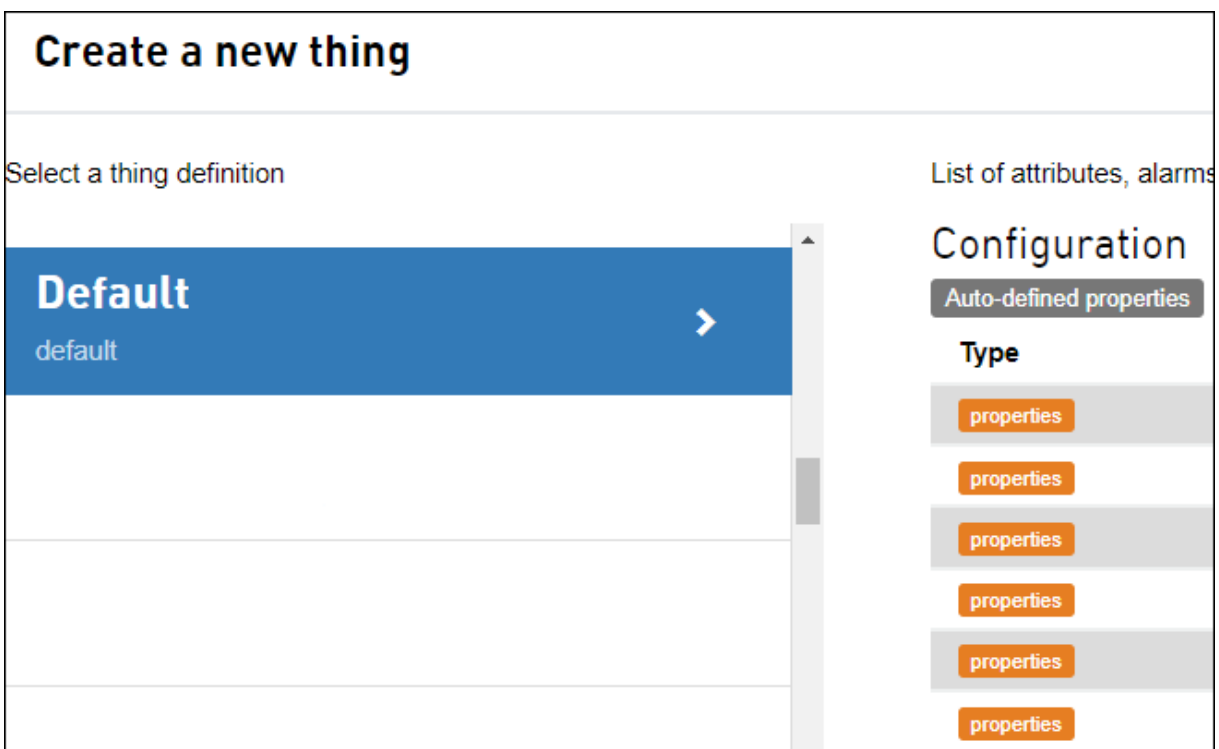


Figure 274

In the following screen, provide the Telit ID as “Identifier” Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the

screenshot below

Figure 275

Click **“Add”** to complete the new thing creation procedure.

If the Thing already exists, its device profile can be changed by following the steps shown in the picture below

Figure 276

3.6.29.5 Application execution example

```
Starting lwm2m demo. This is v1.1.10 built on Oct 22 2021 10:49:52.  
On OneEdge portal, be sure that the demo Device Profile has been uploaded on the IoT Portal  
Initializing resources...  
  
LWM2M enable result OK  
[DEBUG] 38.20 lwm2m_demo:998 - lwm2mIndicationCB{pubTspt_0}$ Monitoring enabled.  
  
Waiting LWM2M Registering (120 seconds timeout)...  
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING  
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED  
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING  
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED  
GET STATUS.  
IF Status: M2MB_LWM2M_IF_STATE_ACTIVE  
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED
```

Figure 277

```
=====
READ-ONLY RESOURCES
=====

Setting integer resource {32010/0/2} value to 50 on LWM2M client.
Reading integer resource {32010/0/2} value on LWM2M client.
Resource /32010/0/2/0 changed!
Reading integer resource {32010/0/2/0} value on LWM2M client.
---Integer value is now 50
Integer data in {32010/0/2/0} resource was updated to new value: 50
-----

Setting integer resource {32010/0/22/0} value to 10 on LWM2M client.
Resource /32010/0/22/0 changed!
Reading integer resource {32010/0/22/0} value on LWM2M client.
Integer data in {32010/0/22/0} resource was updated to new value: 10
Setting integer resource {32010/0/22/1} value to 11 on LWM2M client.
Resource /32010/0/22/1 changed!
Reading integer resource {32010/0/22/1} value on LWM2M client.
Integer data in {32010/0/22/1} resource was updated to new value: 11
-----

Setting double resource {32010/0/3} value to 20.500000 on LWM2M client.
Reading double resource {32010/0/3} value on LWM2M client.
Resource /32010/0/3/0 changed!
Reading double resource {32010/0/3/0} value on LWM2M client.
---Double value is now 20.500000
Float data in {32010/0/3/0} resource was updated to new value: 20.500000
-----

Setting boolean resource {32010/0/4} value to 1 on LWM2M client.
Reading boolean resource {32010/0/4} value on LWM2M client.
Resource /32010/0/4/0 changed!
Reading boolean resource {32010/0/4/0} value on LWM2M client.
---Boolean value is now true
Boolean data in {32010/0/4/0} resource was updated to new value: true
-----
```

```

Current time in seconds from the epoch: 1634900084
Setting timestamp resource {32010/0/6} value to 1634900084 on LWM2M client.
Reading timestamp resource {32010/0/6} value on LWM2M client.
Resource /32010/0/6/0 changed!
Reading timestamp resource {32010/0/6/0} value on LWM2M client.
---Timestamp value is now 1634900084
Time data in {32010/0/6/0} resource was updated to new value: 1634900084 (2021/10/22T10:54:44+00:00)
-----

Setting opaque resource {32010/0/5} on LWM2M client.
Resource /32010/0/5/0 changed!
Reading opaque resource {32010/0/5/0} on LWM2M client.
Opaque data in {32010/0/5/0} resource was updated to new content (10 bytes):
00 01 02 03 04 05 06 07 08 09
Reading opaque resource {32010/0/5} on LWM2M client.
---Opaque content is now (10 bytes): 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09
-----

Setting string resource {32010/0/1} value to Hello World! on LWM2M client.
Reading string resource {32010/0/1} value on LWM2M client.
Resource /32010/0/1/0 changed!
Reading integer resource {32010/0/1/0} value on LWM2M client.
---String content is now: <Hello World!>
String data in {32010/0/1/0} resource was updated to new content: <Hello World!>
-----

```



```
=====
READ-WRITE RESOURCES
=====

Writing integer resource {32010/0/12} value to 50 on LWM2M client.
Reading integer resource {32010/0/12} value on LWM2M client.
Resource /32010/0/12/0 changed!
Reading integer resource {32010/0/12/0} value on LWM2M client.
---Integer value is now 50
Integer data in {32010/0/12/0} resource was updated to new value: 50
-----

Writing double resource {32010/0/13} value to 20.500000 on LWM2M client.
Reading double resource {32010/0/13} value on LWM2M client.
Resource /32010/0/13/0 changed!
Reading double resource {32010/0/13/0} value on LWM2M client.
---Double value is now 20.500000
Float data in {32010/0/13/0} resource was updated to new value: 20.500000
-----

Writing boolean resource {32010/0/14} value to 1 on LWM2M client.
Reading boolean resource {32010/0/14} value on LWM2M client.
Resource /32010/0/14/0 changed!
Reading boolean resource {32010/0/14/0} value on LWM2M client.
---Boolean value is now true
Boolean data in {32010/0/14/0} resource was updated to new value: true
-----
```

```

Current time in seconds from the epoch: 1634900125
Writing timestamp resource {32010/0/16} value to 1634900125 on LWM2M client.
Reading timestamp resource {32010/0/16} value on LWM2M client.
Resource /32010/0/16/0 changed!
Reading timestamp resource {32010/0/16/0} value on LWM2M client.
---Timestamp value is now 1634900125
Time data in {32010/0/16/0} resource was updated to new value: 1634900125 (2021/10/22T10:55:25+00:00)
-----
Writing opaque resource {32010/0/15} on LWM2M client.
Resource /32010/0/15/0 changed!
Reading opaque resource {32010/0/15/0} on LWM2M client.
Opaque data in {32010/0/15/0} resource was updated to new content (10 bytes):
00 01 02 03 04 05 06 07 08 09
Reading opaque resource {32010/0/15} on LWM2M client.
---Opaque content is now (10 bytes): 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09
-----
Writing string resource {32010/0/11} value to <Hello World!> on LWM2M client.
Reading string resource {32010/0/11} value on LWM2M client.
Resource /32010/0/11/0 changed!
Reading integer resource {32010/0/11/0} value on LWM2M client.
---String content is now: <Hello World!>

```

```
=====
WRITE-ONLY RESOURCES
=====

Writing integer resource {32010/0/42} value to 50 on LWM2M client.
String data in {32010/0/11/0} resource was updated to new content: <Hello World!>
Getting integer resource {32010/0/42} value on LWM2M client.
Resource /32010/0/42/0 changed!
Getting integer resource {32010/0/42/0} value on LWM2M client.
---Integer value is now 50
Integer data in {32010/0/42/0} resource was updated to new value: 50
-----

Writing double resource {32010/0/43} value to 20.500000 on LWM2M client.
Getting double resource {32010/0/43} value on LWM2M client.
Resource /32010/0/43/0 changed!
Getting double resource {32010/0/43/0} value on LWM2M client.
---Double value is now 20.500000
Float data in {32010/0/43/0} resource was updated to new value: 20.500000
-----

Writing boolean resource {32010/0/44} value to 1 on LWM2M client.
Getting boolean resource {32010/0/44} value on LWM2M client.
Resource /32010/0/44/0 changed!
Getting boolean resource {32010/0/44/0} value on LWM2M client.
---Boolean value is now true
Boolean data in {32010/0/44/0} resource was updated to new value: true
-----
```

```

Current time in seconds from the epoch: 1634900163
Writing timestamp resource {32010/0/46} value to 1634900163 on LWM2M client.
Getting timestamp resource {32010/0/46} value on LWM2M client.
Resource /32010/0/46/0 changed!
Getting timestamp resource {32010/0/46/0} value on LWM2M client.
---Timestamp value is now 1634900163
Time data in {32010/0/46/0} resource was updated to new value: 1634900163 (2021/10/22T10:56:03+00:00)
-----
Writing opaque resource {32010/0/45} on LWM2M client.
Resource /32010/0/45/0 changed!
Getting opaque resource {32010/0/45/0} on LWM2M client.
Opaque data in {32010/0/45/0} resource was updated to new content (10 bytes):
00 01 02 03 04 05 06 07 08 09
Getting opaque resource {32010/0/45} on LWM2M client.
---Opaque content is now (10 bytes): 0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09
-----
Writing string resource {32010/0/41} value to <Hello World!> on LWM2M client.
Getting string resource {32010/0/41} value on LWM2M client.
Resource /32010/0/41/0 changed!
Getting integer resource {32010/0/41/0} value on LWM2M client.
---String content is now: <Hello World!>
String data in {32010/0/41/0} resource was updated to new content: <Hello World!>
-----
Resources operations examples done.
-----
-----
Will perform a SET with notify ACK enabled.
-----
Enable notify ack
Setting integer resource {32010/0/2} value to 60 on LWM2M client.
Resource /32010/0/2/0 changed!
Reading integer resource {32010/0/2/0} value on LWM2M client.
Integer data in {32010/0/2/0} resource was updated to new value: 60
ACK received from server!
Reading integer resource {32010/0/2} value on LWM2M client.
---Integer value is now 60
Disable notify ack
Done.
Waiting for events from the OneEdge portal. Please write on monitored resources or call an 'exec' one.

```


For example, executing the two Exec Resources at the bottom of the list, the application will react accordingly:

```
Info Exec Ind: 32010/0/101/0
Asked to execute resource 101

Resource /32010/0/11/0 changed!
```

Figure 280

Writing a string resource (id /32010/0/11), the application will notify the change

```
Reading integer resource {32010/0/11/0} value on LWM2M client.

String data in {32010/0/11/0} resource was updated to new content: <Hello from the IoT Portal!>
```

Figure 281

3.6.30 LWM2M FOTA ACK management

Sample application showcasing LWM2M client FOTA events and ACKs management via APIs. Debug prints on **MAIN UART**

Features

- Configure LWM2M client and enable it (if not already running)
- Enable monitoring on object 5 (Firmware Update)
- Configure lwm2m FOTA parameters to enable ACK requirement for each operation (Download, Update)
- Wait for an externally generated FOTA request to handle the steps by sending the acknowledgements when needed

Please refer to “80654NT11889A OneEdge Firmware Management Application Note” for further details about FOTA workflow. Get in touch with TS-OneEdge support to receive the document.

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6","<user apn>"
```

Depending on the Mobile Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.6.30.1 Device Profile upload

Minimal FOTA profile (short lifetime) device profile must be imported and selected to improve the responsiveness of the FOTA operations

To do so, import the file `json/lwm2m_fota_profile_short.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

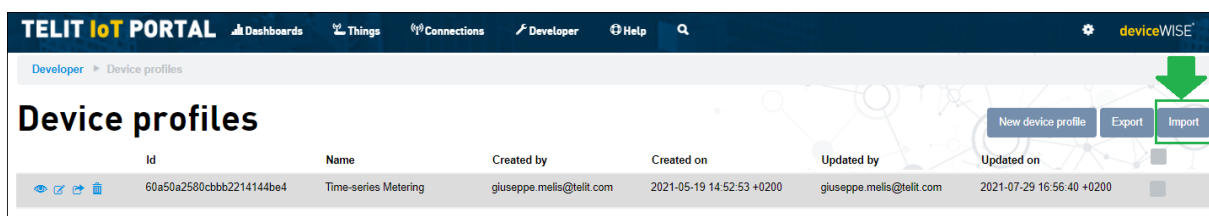


Figure 282

3.6.30.2 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on **“Things”** section, click **“New Thing”** button in the top right corner.

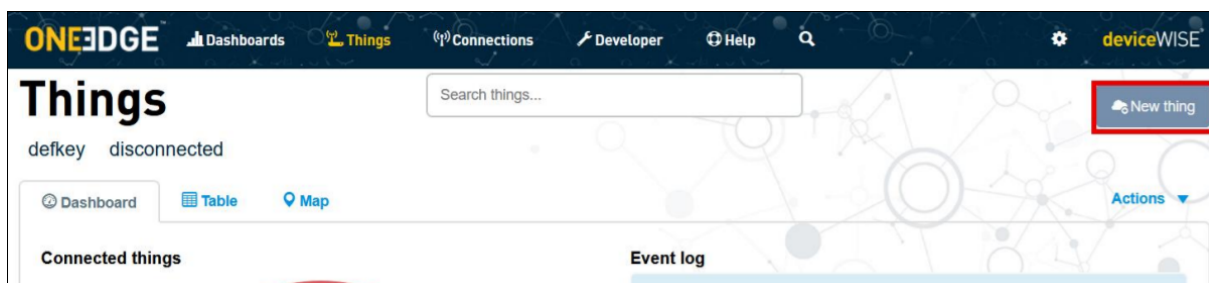


Figure 283

In the Create a new thing dialog, select “Telit Module”

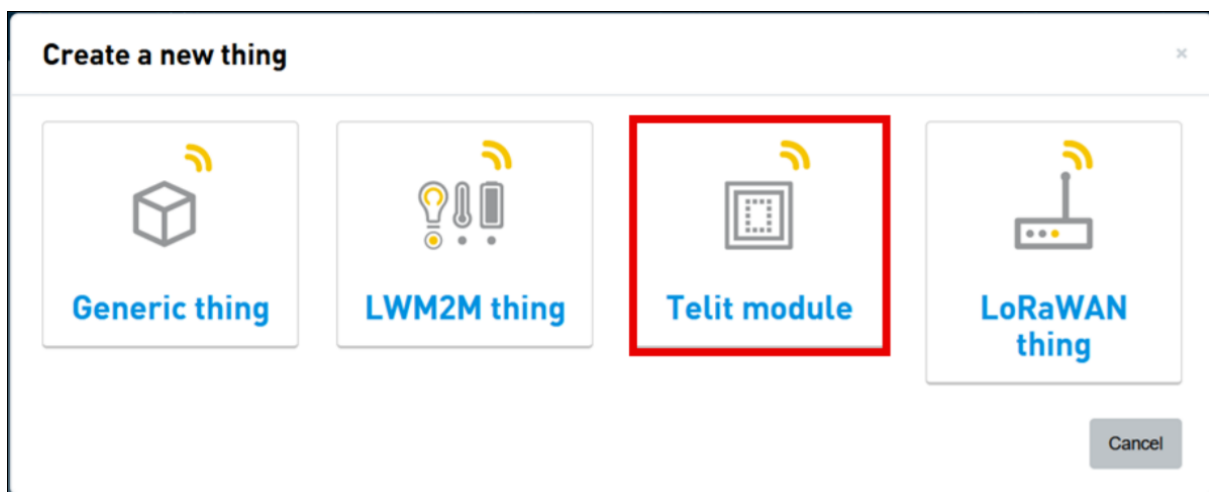


Figure 284

A dialog appears: select “Default” thing definition

Create a new thing

Select a thing definition

Default
default

List of attributes, alarms

Configuration

Auto-defined properties

Type

properties

properties

properties

properties

properties

properties

Figure 285

In the following screen, provide the Telit ID as “Identifier” Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the screenshot below

Create a new thing

Setup

LWM2M

Identification

VAR

Other

Back

LWM2M

Device profile*

Minimal FOTA profile (short lifetime)

Add Cancel

Figure 286

Click **“Add”** to complete the new thing creation procedure.

3.6.30.3 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask

- Initialize LWM2M client,
- Enable unsolicited messages from client
- Enable FOTA ACK configuration
- Create a task (lwm2m_taskCB is its callback function)to manage events from Portal
- Check if Lwm2M client is running, if not, enable it
- Wait for client to register to Portal
- Wait for FOTA events to arrive, and when the Update request arrives, notify the user that everything must stop before sending the last ACK (as the module will start updating and then reboot)

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)
- Run lwm2m_taskCB when a monitored resource changes, to manage the action to be done

3.6.30.4 Application execution example

```
Starting lwm2m FOTA management demo. This is v1.1.14 built on Jan 21 2022 14:39:47.
Initializing resources...
[DEBUG] 22.14 lwm2m_demo:283 - at_cmd_async_init{LWM2M_TASK}$ m2mb_ati_init() on instance 1

IF Status: M2MB_LWM2M_IF_STATE_DISABLED
Client Status: M2MB_LWM2M_CL_STATE_DISABLED
Enable lwm2m client
LWM2M enable result OK
Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

Waiting for events from the OneEdge portal. Please start a FOTA process.
```

Figure 287

```
Resource /5/0/1/0 changed!
[DEBUG] 108.42 lwm2m_demo:544 - lwm2m_taskCB{LWM2M_TASK}$ Firmware update event!
FOTA process is waiting for Download ACK, sending it!
Waiting for Update ACK!
Before sending the ACK to apply FOTA, stop any ongoing operation (filesystem, sockets)

Asked to perform UPDATE step, stopping all operations now.
Application complete.

----- Info, DEVICE REBOOT FOR FOTA -----
?????????
#OTAEV: Module Upgraded To New Fw
```

FOTA request arrives from server, ACK is sent for Download and Update. Device restarts with the new firmware version

```
Starting lwm2m FOTA management demo. This is v1.1.14 built on Jan 21 2022 14:39:47.
Initializing resources...
[DEBUG] 26.61 lwm2m_demo:283 - at_cmd_async_init{LWM2M_TASK}$ m2mb_ati_init() on instance 1
[WARN ] 34.98 lwm2m_demo:738 - lwm2mIndicationCB{pubTspt_0}$ write res 1

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
Enable already done, skip. State: M2MB_LWM2M_IF_STATE_ACTIVE
Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

Waiting for events from the OneEdge portal. Please start a FOTA process.
```

Application restarts and is ready for a new FOTA operation

3.6.31 LWM2M FOTA ACK management (AT URCs)

Sample application showcasing LWM2M client FOTA events and ACKs management via AT URCs. Debug prints on **MAIN UART**

Features

- Configure LWM2M client and enable it (if not already running)
- Enable monitoring on object 5 (Firmware Update)
- Configure lwm2m FOTA parameters to enable ACK requirement for each operation (Download, Update)
- Wait for an externally generated FOTA request to handle the steps by sending the acknowledgements when needed

Please refer to “80654NT11889A OneEdge Firmware Management Application Note” for further details about FOTA workflow. Get in touch with TS-OneEdge support to receive the document.

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6","<user apn>"
```

Depending on the Mobile Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.6.31.1 Device Profile upload

Minimal FOTA profile (short lifetime) device profile must be imported and selected to improve the responsiveness of the FOTA operations

To do so, import the file `json/lwm2m_fota_profile_short.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

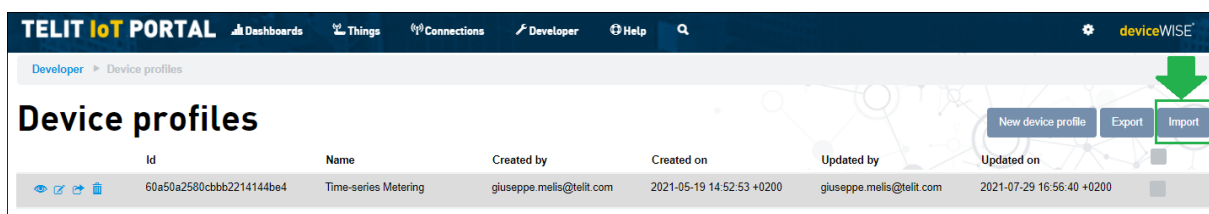


Figure 288

3.6.31.2 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **XXXXXXXXXXXXXXXXXXXXXXXXXXXX**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on **“Things”** section, click **“New Thing”** button in the top right corner.

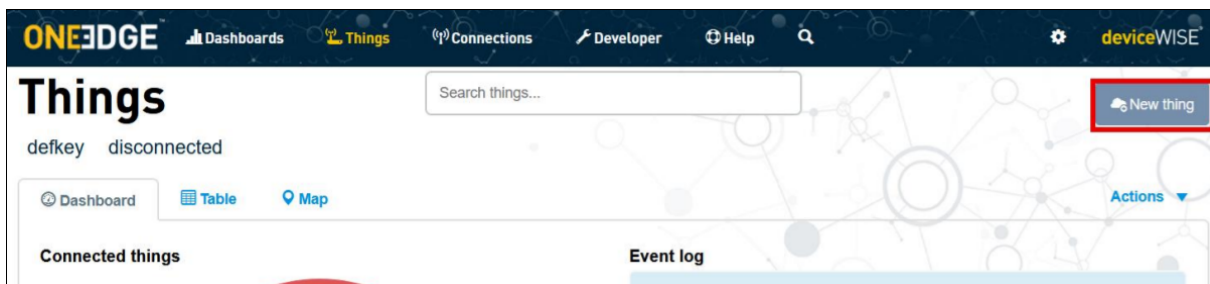


Figure 289

In the Create a new thing dialog, select “Telit Module”

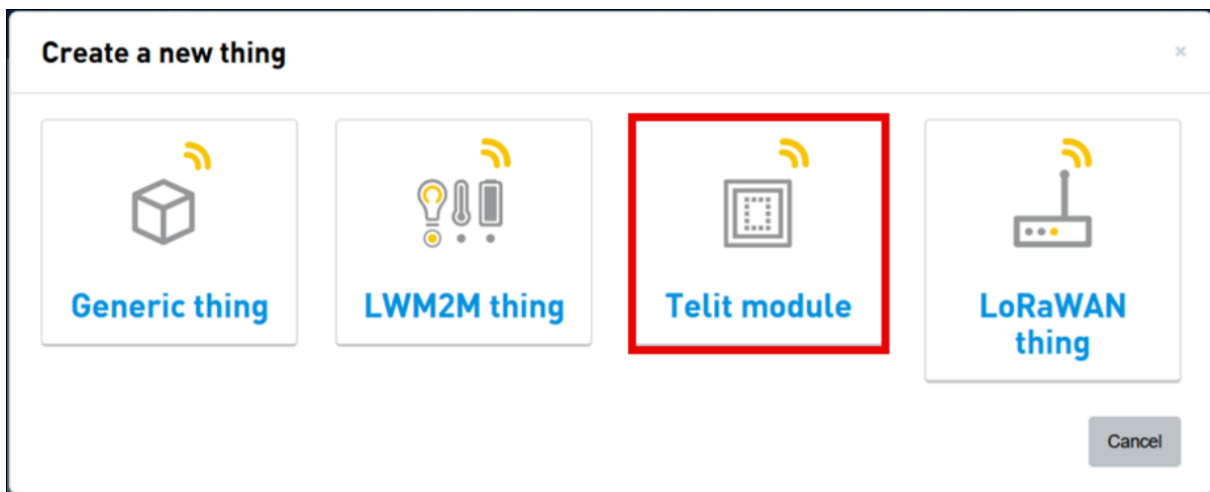


Figure 290

A dialog appears: select “Default” thing definition

Create a new thing

Select a thing definition

Default
default

List of attributes, alarms

Configuration

Auto-defined properties

Type

properties

properties

properties

properties

properties

properties

Figure 291

In the following screen, provide the Telit ID as “Identifier” Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the screenshot below

Create a new thing

Setup

LWM2M

Identification

VAR

Other

Back

LWM2M

Device profile*

Minimal FOTA profile (short lifetime)

Add Cancel

Figure 292

Click **“Add”** to complete the new thing creation procedure.

3.6.31.3 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask

- Initialize LWM2M client,
- Enable unsolicited messages from client
- Enable FOTA ACK configuration
- Create a task (lwm2m_taskCB is its callback function)to manage events from Portal
- Check if Lwm2M client is running, if not, enable it
- Wait for client to register to Portal
- Wait for FOTA events to arrive, and when the Update request arrives, notify the user that everything must stop before sending the last ACK (as the module will start updating and then reboot)

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)
- Run lwm2m_taskCB when a monitored resource changes, to manage the action to be done

3.6.31.4 Application execution example

```
Starting lwm2m FOTA management demo. This is v1.1.14 built on Jan 21 2022 14:39:47.
Initializing resources...
[DEBUG] 22.14 lwm2m_demo:283 - at_cmd_async_init{LWM2M_TASK}$ m2mb_ati_init() on instance 1

IF Status: M2MB_LWM2M_IF_STATE_DISABLED
Client Status: M2MB_LWM2M_CL_STATE_DISABLED
Enable lwm2m client
LWM2M enable result OK
Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

Waiting for events from the OneEdge portal. Please start a FOTA process.
```

Figure 293

```
Resource /5/0/1/0 changed!
[DEBUG] 108.42 lwm2m_demo:544 - lwm2m_taskCB{LWM2M_TASK}$ Firmware update event!
FOTA process is waiting for Download ACK, sending it!
Waiting for Update ACK!
Before sending the ACK to apply FOTA, stop any ongoing operation (filesystem, sockets)

Asked to perform UPDATE step, stopping all operations now.
Application complete.

----- Info, DEVICE REBOOT FOR FOTA -----
?????????
#OTAEV: Module Upgraded To New Fw
```

FOTA request arrives from server, ACK is sent for Download and Update. Device restarts with the new firmware version

```
Starting lwm2m FOTA management demo. This is v1.1.14 built on Jan 21 2022 14:39:47.
Initializing resources...
[DEBUG] 26.61 lwm2m_demo:283 - at_cmd_async_init{LWM2M_TASK}$ m2mb_ati_init() on instance 1
[WARN ] 34.98 lwm2m_demo:738 - lwm2mIndicationCB{pubTspt_0}$ write res 1

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
Enable already done, skip. State: M2MB_LWM2M_IF_STATE_ACTIVE
Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED

IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

Waiting for events from the OneEdge portal. Please start a FOTA process.
```

Application restarts and is ready for a new FOTA operation

3.6.32 LWM2M OBJ_GET AND OBJ_SET

Sample application showcasing LWM2M client m2mb_lwm2m_objget and m2mb_lwm2m_objset M2MB APIs usage. Debug prints on **MAIN UART**

Features

- Configure LWM2M client and enable it
- Create an instance of a custom object
- Create a Json string
- Set string, integer, float, boolean, timestamp and opaque values with m2mb_lwm2m_objset
- Get all resources values with m2mb_lwm2m_objget
- Manage write, read and monitoring resources changed from the portal

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6", "<user apn>"
```

Depending on the Mobiler Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.6.32.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask - Check registration status

- Initialize LWM2M client,
- Check for XML file fo custom object
- Enable unsolicited messages from client
- Create a task (lwm2m_taskCB is its callback function)to manage events from Portal
- Enable Lwm2M client

- Create a new instance for the custom object
- Wait for client to register to Portal
- Performs obj_set and obj_get operations on the related resources
- Wait for events from server

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)
- Run lwm2m_taskCB when a monitored resource changes, to manage the action to be done

3.6.32.2 Device Profile upload

LWM2M resources demo device profile must be imported to have a real-time update of resources values on the LWM2M browser.

To do so, import the file `json/lwm2m_resources_demo.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

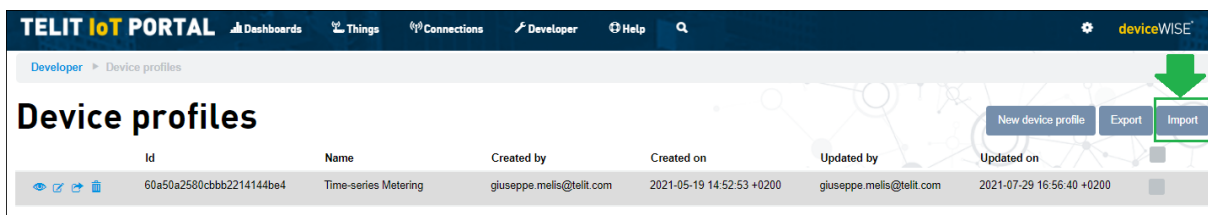


Figure 294

3.6.32.3 Custom Object configuration

The XML file content must be loaded on the Telit IoT Portal for the demo application to be fully executed.

First, enter Developer section from the top menu



Figure 295

Choose Object Registry

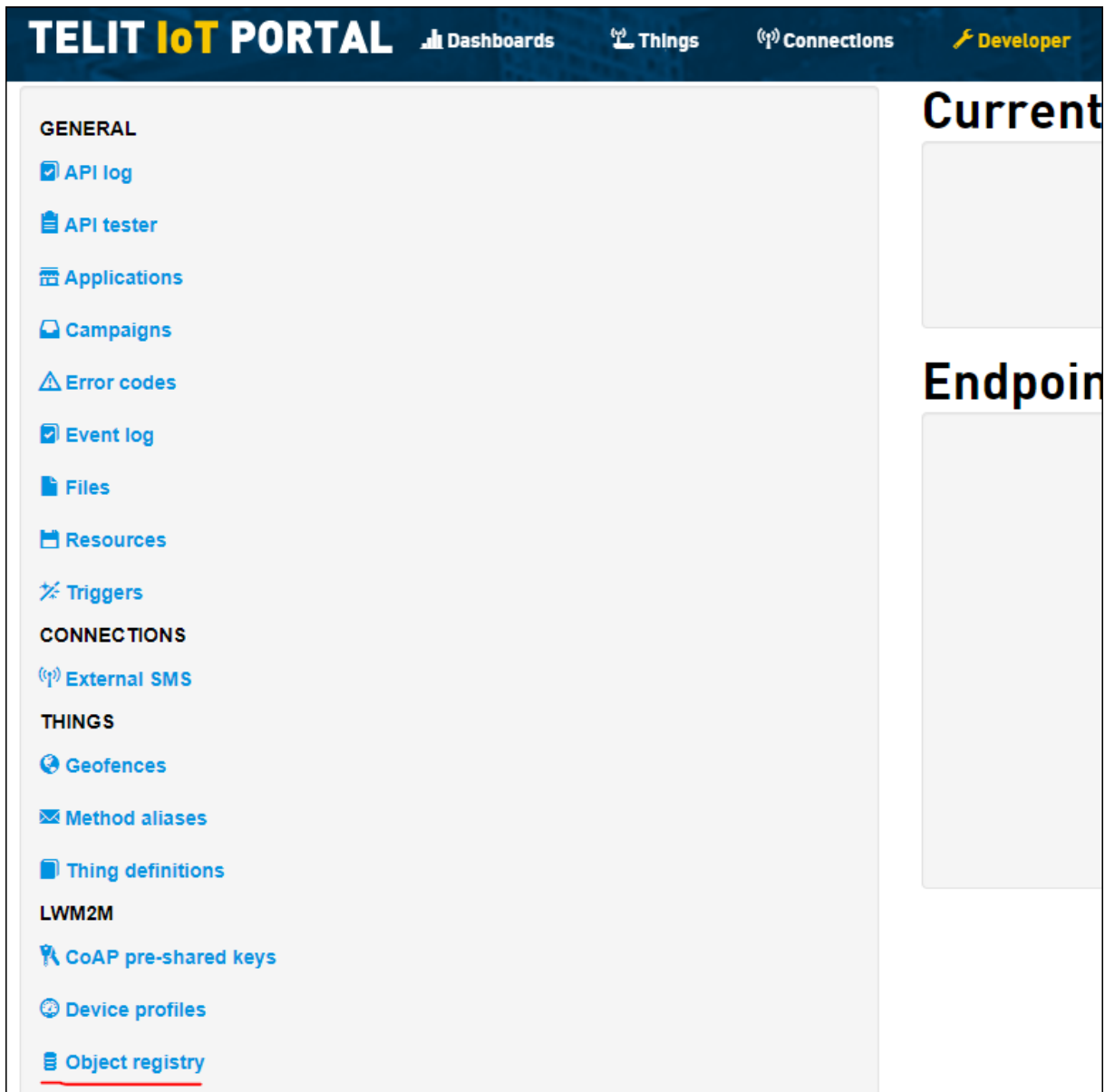


Figure 296

Create a New Object

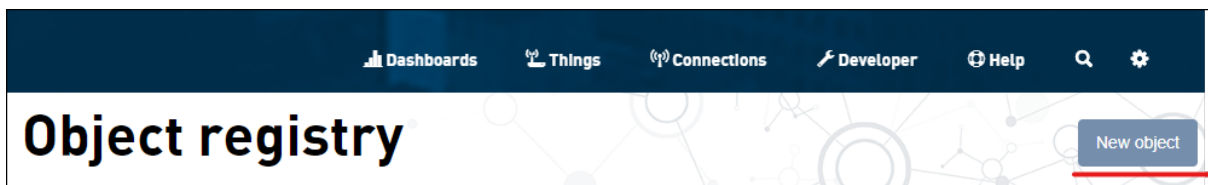
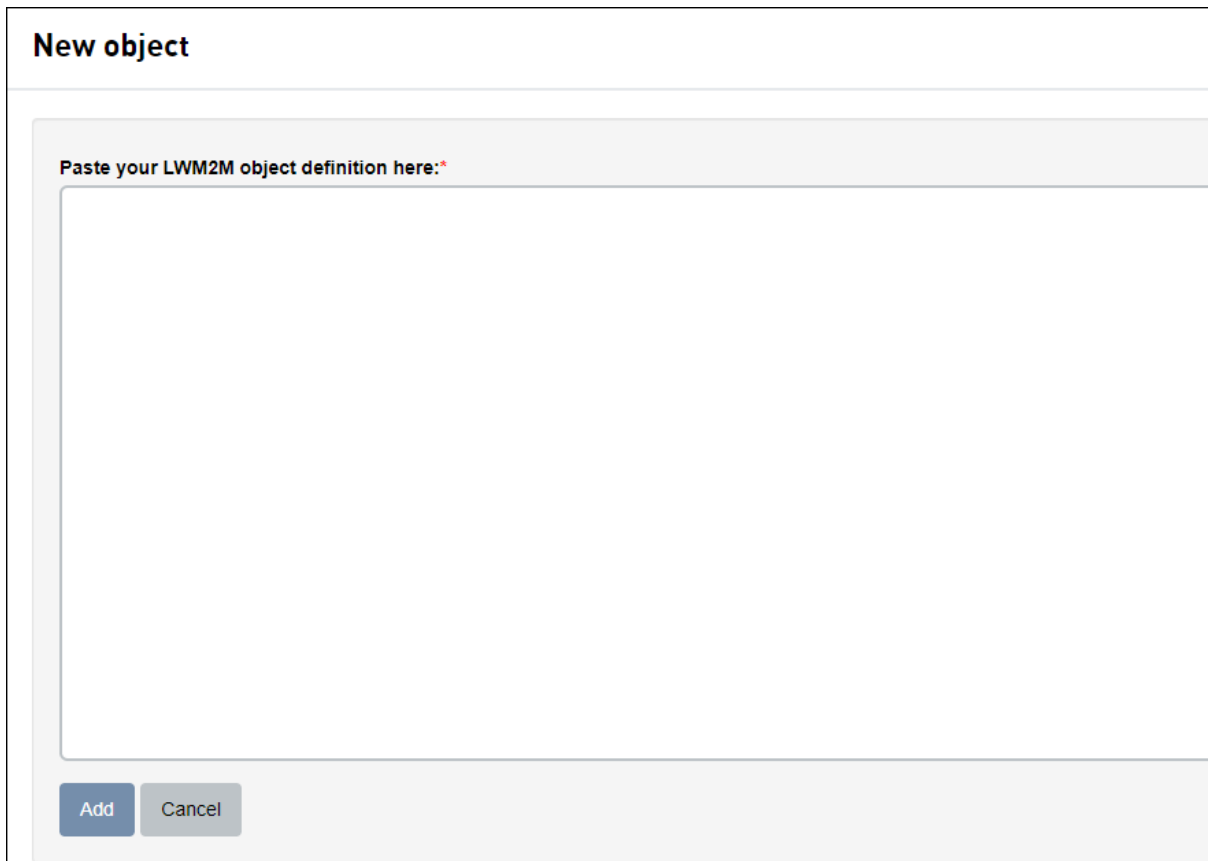


Figure 297

Copy the xml file content and paste it in the new Object form



New object

Paste your LWM2M object definition here:*

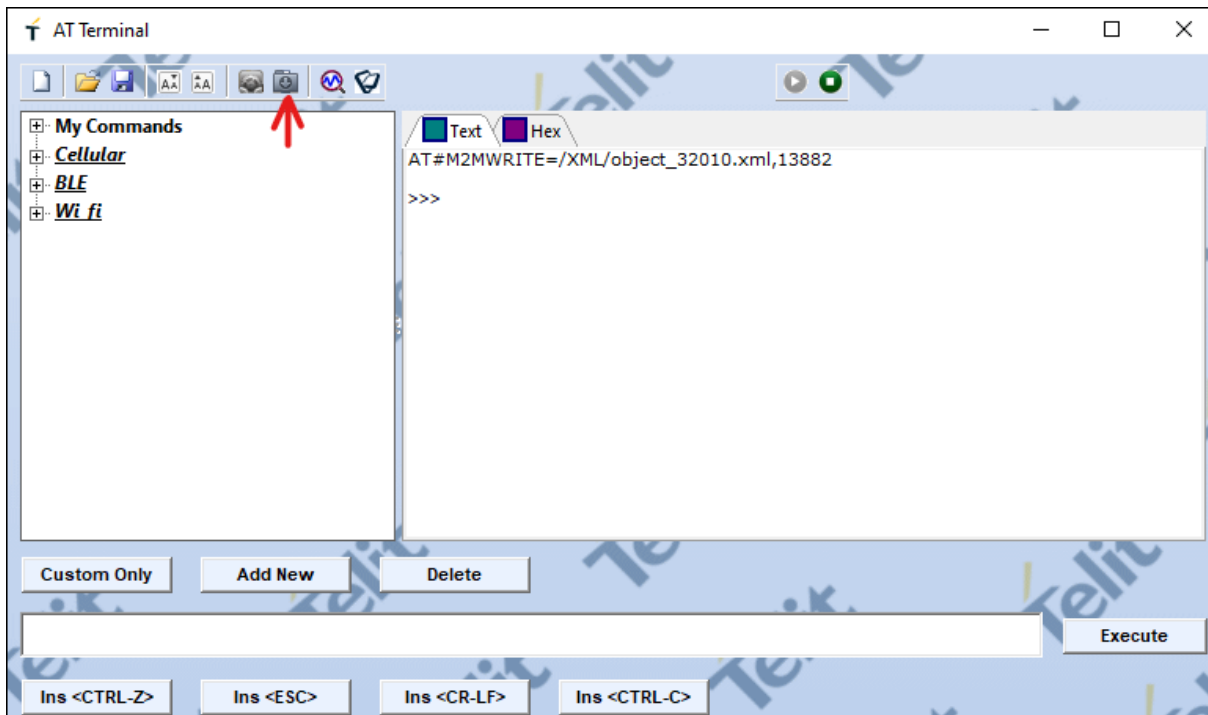
Add Cancel

Figure 298

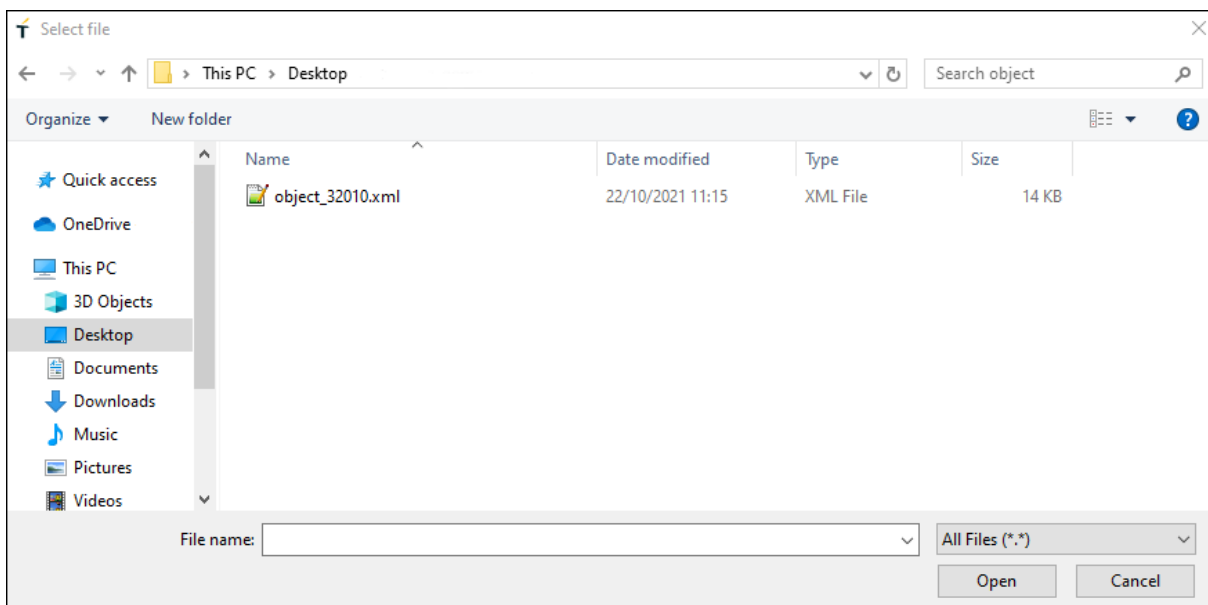
Also, the application requires the XML file `xml/object_32011.xml` (provided with the sample files) to be stored in module's `/XML/` folder. It can be done with

```
AT#M2MWRITE=/XML/object_32011.xml,<size_in_bytes>
```

To load the XML file in the module, Telit AT Controller (TATC) can be used. Once the command above is issued, press the load content button:

**Figure 299**

Select the file from your computer

**Figure 300**

The file is successfully loaded on the module

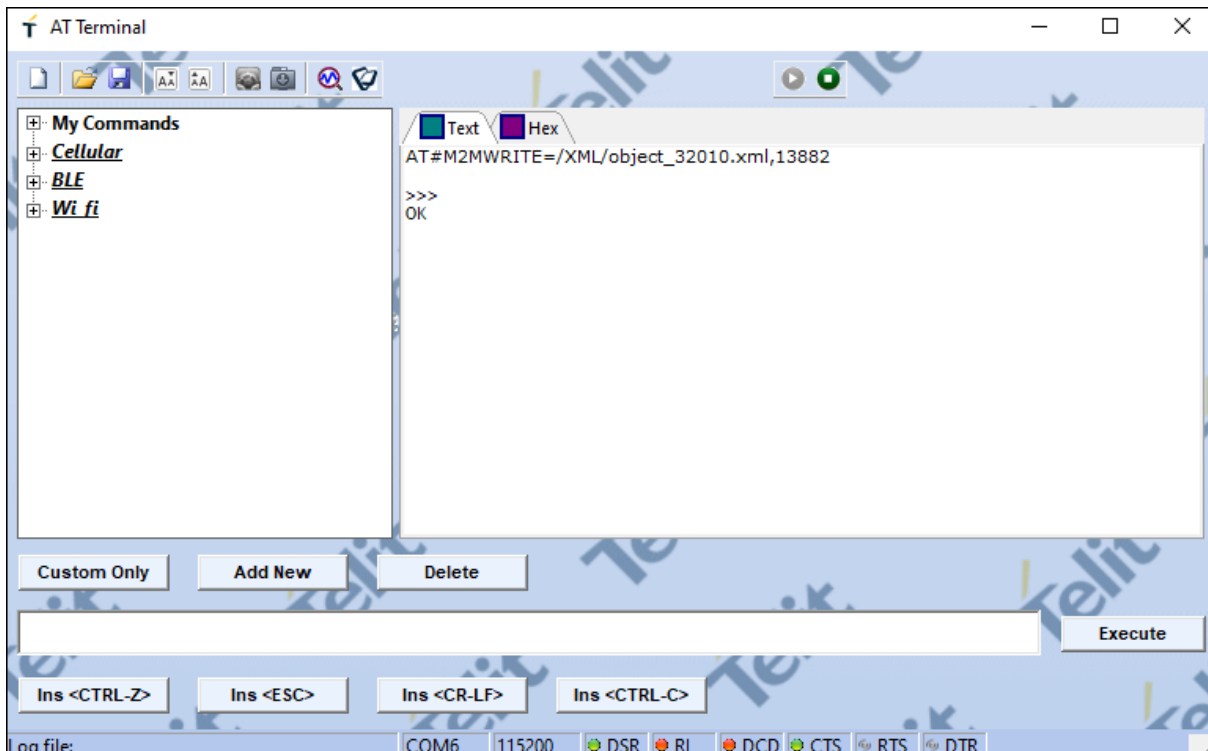


Figure 301

3.6.32.4 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on “Things” section, click “New Thing” button in the top right corner.

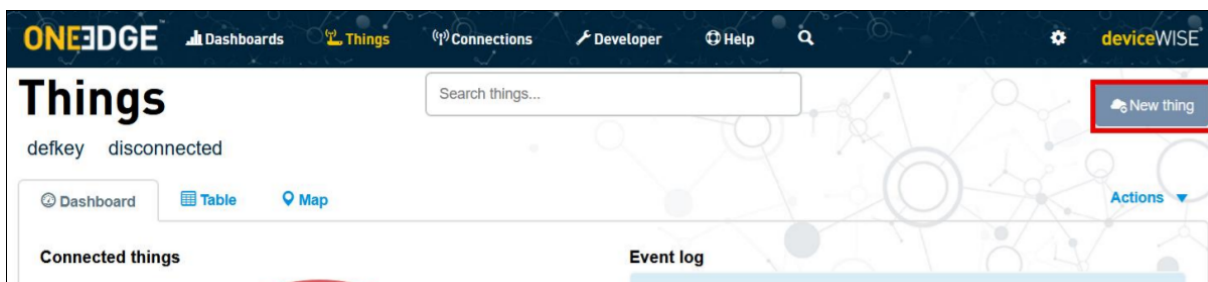


Figure 302

In the Create a new thing dialog, select “Telit Module”

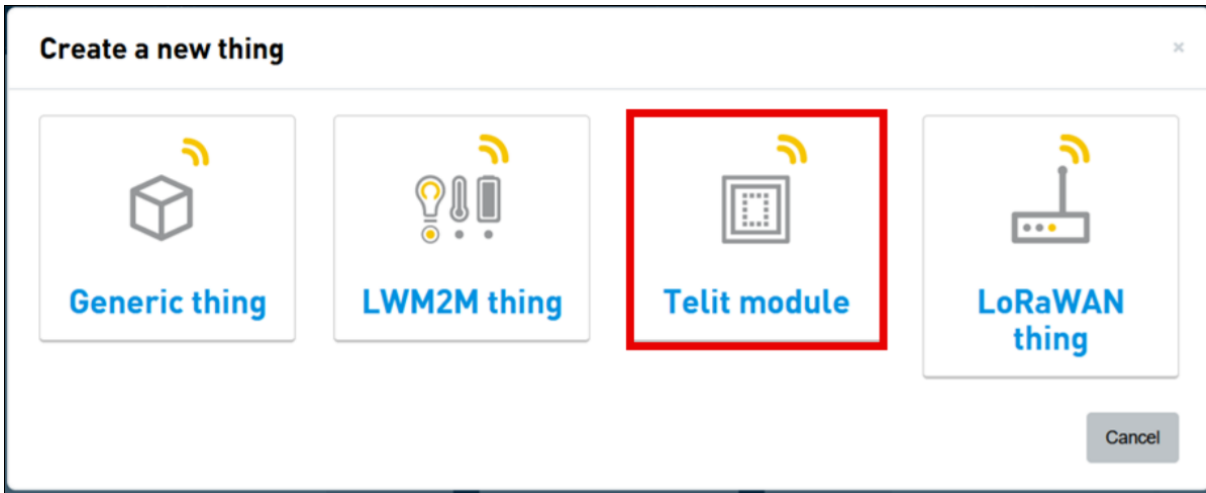


Figure 303

A dialog appears: select “Default” thing definition

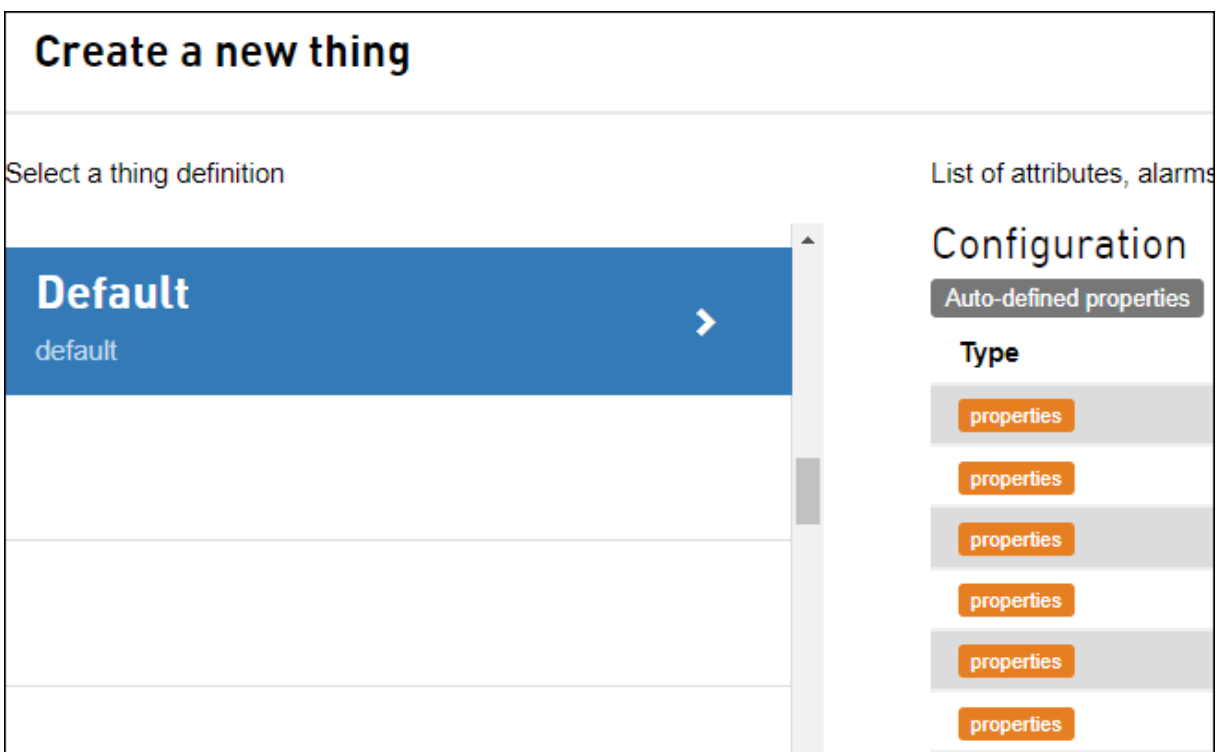


Figure 304

In the following screen, provide the Telit ID as “Identifier” Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the

screenshot below

Figure 305

Click **“Add”** to complete the new thing creation procedure.

If the Thing already exists, its device profile can be changed by following the steps shown in the picture below

Figure 306

3.6.32.5 Application execution example


```
Starting lwm2m demo. This is v1.1.10 built on Oct 22 2021 10:49:52.
On OneEdge portal, be sure that the demo Device Profile has been uploaded on the IoT Portal
Initializing resources...

LWM2M enable result OK
[DEBUG] 38.20 lwm2m_demo:998 - lwm2mIndicationCB{pubTspt_0}$ Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED
GET STATUS.
IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED
```

Figure 307

```
Starting lwm2m demo. This is v1.1.11 built on Nov 24 2021 13:32:52.
On OneEdge portal, be sure that the 'lwm2m resources objset - objget demo' Device Profile has been uploaded on the IoT Portal

Initializing resources...
LWM2M enable result OK
[DEBUG] 29.42 lwm2m_demo:593 - lwm2mIndicationCB{pubTspt_0}$ Monitoring enabled.

Waiting LWM2M Registering (120 seconds timeout)...
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPING
resp->info == M2MB_LWM2M_CL_STATE_BOOTSTRAPPED
resp->info == M2MB_LWM2M_CL_STATE_REGISTERING
resp->info == M2MB_LWM2M_CL_STATE_REGISTERED
GET STATUS.
IF Status: M2MB_LWM2M_IF_STATE_ACTIVE
Client Status: M2MB_LWM2M_CL_STATE_REGISTERED

m2mb_lwm2m_objset SUCCESS

Resource /32011/0/11/0 changed!

Reading string resource {32011/0/11/0} value on LWM2M client.

Resource /32011/0/12/0 changed!
Resource /32011/0/13/0 changed!
Resource /32011/0/15/0 changed!
Resource /32011/0/16/0 changed!
Resource /32011/0/31/0 changed!
Resource /32011/0/31/1 changed!
Resource /32011/0/31/2 changed!
Resource /32011/0/34/0 changed!
Resource /32011/0/34/1 changed!
Resource /32011/0/34/2 changed!
Resource /32011/0/34/3 changed!
```

```
String data in {32011/0/11/0} resource was updated to new content: <hello world>
Reading integer resource {32011/0/12/0} value on LWM2M client.
Integer data in {32011/0/12/0} resource was updated to new value: 70
Reading double resource {32011/0/13/0} value on LWM2M client.
Float data in {32011/0/13/0} resource was updated to new value: -12.500000
Reading opaque resource {32011/0/15/0} on LWM2M client.
Opaque data in {32011/0/15/0} resource was updated to new content (11 bytes):
68 65 6C 6C 6F 20 77 6F 72 6C 64
Reading timestamp resource {32011/0/16/0} value on LWM2M client.
Time data in {32011/0/16/0} resource was updated to new value: 1620828702 (2021/05/12T14:11:42+00:00)
Reading string resource {32011/0/31/0} value on LWM2M client.
String data in {32011/0/31/0} resource was updated to new content: <Greetings>
Reading string resource {32011/0/31/1} value on LWM2M client.
String data in {32011/0/31/1} resource was updated to new content: <from>
Reading string resource {32011/0/31/2} value on LWM2M client.
String data in {32011/0/31/2} resource was updated to new content: <AppZone!>
Reading boolean resource {32011/0/34/0} value on LWM2M client.
Boolean data in {32011/0/34/0} resource was updated to new value: true
Reading boolean resource {32011/0/34/1} value on LWM2M client.
Boolean data in {32011/0/34/1} resource was updated to new value: false
Reading boolean resource {32011/0/34/2} value on LWM2M client.
Boolean data in {32011/0/34/2} resource was updated to new value: true
Reading boolean resource {32011/0/34/3} value on LWM2M client.
Boolean data in {32011/0/34/3} resource was updated to new value: false
```

Waiting 5 seconds before the m2mb_lwm2m_objget

```
Waiting: 5
Waiting: 4
Waiting: 3
Waiting: 2
Waiting: 1
```

m2mb_lwm2m_objget SUCCESS

```
Content:
<{"bn":"/32011/0/",
  "e":[
    {"n":"11","sv":"hello world"},
    {"n":"12","v":70},
    {"n":"13","v":-12.500000},
    {"n":"14","bv":false},
    {"n":"15","sv":"aGVsbG8gd29ybGQ="},
    {"n":"16","v":1620828702},
    {"n":"17","ov":"0:0"},
    {"n":"31/0","sv":"Greetings"},
    {"n":"31/1","sv":"from"},
    {"n":"31/2","sv":"AppZone!"},
    {"n":"31/3","sv":""},
    {"n":"31/4","sv":""},
    {"n":"31/5","sv":""},
    {"n":"31/6","sv":""},
    {"n":"31/7","sv":""},
    {"n":"31/8","sv":""},
    {"n":"31/9","sv":""},
    {"n":"34/0","bv":true},
    {"n":"34/1","bv":false},
    {"n":"34/2","bv":true},
    {"n":"34/3","bv":false},
    {"n":"34/4","bv":false},
    {"n":"34/5","bv":false},
    {"n":"34/6","bv":false},
    {"n":"34/7","bv":false},
    {"n":"34/8","bv":false},
    {"n":"34/9","bv":false}]
}>
```

After the Demo completes the initialization, it is possible to access the object re-

sources from the section Things: select your device, then “LwM2M” tab of OneEdge IoT portal:

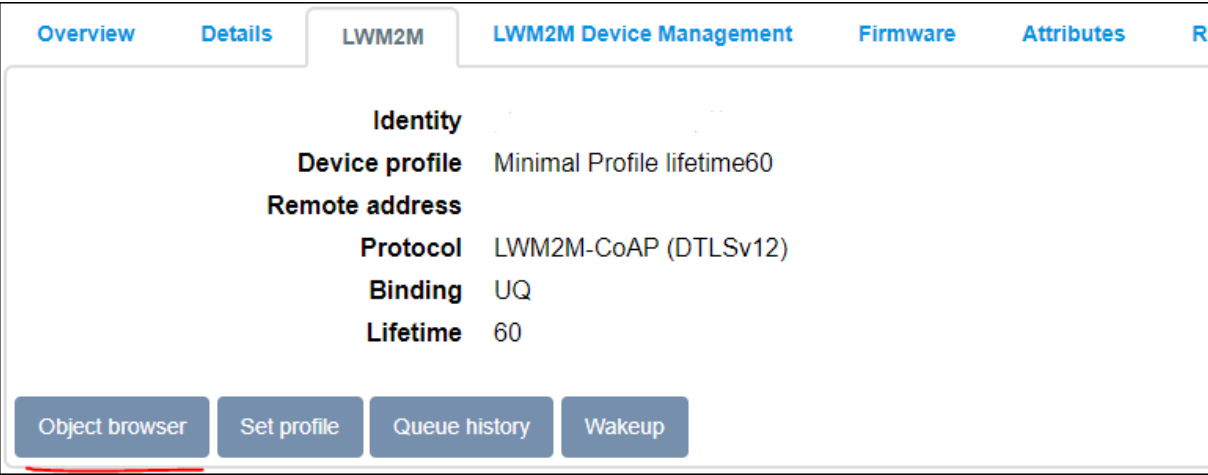


Figure 308

An instance of the object will be present and the resources can be modified.

m2mb lwm2m objset-get demo		/32011					
Instance 0	/32011/0	Observe	Read	Write	Attributes	Delete	
Read Write String	/32011/0/11	Observe	Read	Write	Attributes		hello world
Read Write Integer	/32011/0/12	Observe	Read	Write	Attributes		70
Read Write Float	/32011/0/13	Observe	Read	Write	Attributes		-12.5
Read Write Boolean	/32011/0/14	Observe	Read	Write	Attributes		false
Read Write Opaque	/32011/0/15	Observe	Read	Write	Attributes		view binary data
Read Write Time	/32011/0/16	Observe	Read	Write	Attributes		2021-05-12T14:11:42Z
Read Write Objlink	/32011/0/17	Observe	Read	Write	Attributes		instid=0, objId=0
Read Write Multiinstance String	/32011/0/31	Observe	Read	Write	Attributes		0=Greetings, 1=from, ... Read more
Read Write Multiinstance Boolean	/32011/0/34	Observe	Read	Write	Attributes		0=true, 1=false, 2=tr... Read more

Figure 309

3.6.33 LWM2M REG

Sample application showcasing LWM2M client registration management using M2MB API. Debug prints on **MAIN UART**

Features

- Configure LWM2M client and enable it
- Get LWM2M server information using REG apis
- Deregister from LWM2M server using REG apis
- Register to the LWM2M server using REG apis
- Update registration on LWM2M server using REG apis
- Deregister from LWM2M server using REG apis

Requirements

This application expects the user to configure the PDP context ID 1 with the proper APN. it can be done with the following AT command:

```
AT+CGDCONT=1,"IPV4V6","<user apn>"
```

Depending on the Mobile Network Operator and Access Technology, the APN might be automatically set by the network itself. In this case, nothing must be done by the user.

3.6.33.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the LWM2M client and start it

lwm2m_demo.c

msgLWM2MTask - Check registration status

- Initialize LWM2M client,
- Check for XML file for custom object
- Enable LWM2M client
- Wait for client to register to Portal
- Get the LWM2M server information
- Performs client portal deregistration
- Performs client portal registration

- Performs client portal registration Update
- Performs client portal deregistration

lwm2mIndicationCB

- Manage events arriving from client (operations completion status and unsolicited events)

3.6.33.2 Application execution example

```
Starting LWM2M REG demo. This is v1.0.00 built on Sep 11 2023 15:11:48.
On OneEdge portal, be sure that the 'lwm2m resources demo' Device Profile has been uploaded on the IoT Portal

Initializing resources...
LWM2M enable result OK
Waiting LWM2M Registering (120 seconds timeout)...
sshid[99] status[2] lastReg[1694446138] nextReg[1694532238] lastActivity[1694446138]
Deregistration action using LWM2M REG api!
Waiting deregistration event!
Deregistration event received!
Registration action using LWM2M REG api!
Waiting registration event!
Registration event received!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Update registration action using LWM2M REG api!
Deregistration action using LWM2M REG api!
Waiting deregistration event!
Deregistration event received!
Application complete.
```

Figure 310

3.6.33.3 Device Profile upload

LWM2M resources demo device profile must be imported to have a real-time update of resources values on the LWM2M browser.

To do so, import the file `json/lwm2m_resources_demo.json` (provided with the sample files) on section Developer > Device profiles of OneEdge IoT portal:

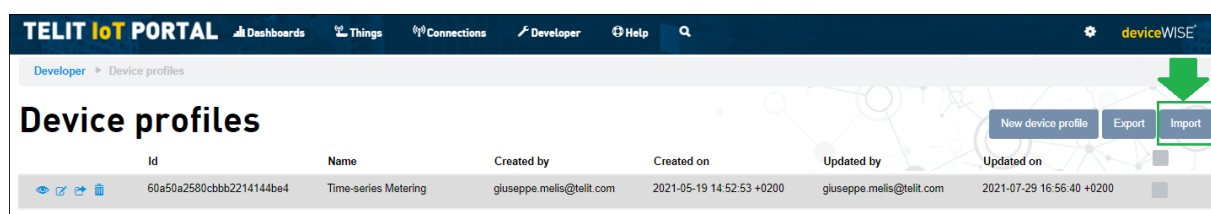


Figure 311

3.6.33.4 Custom Object configuration

The XML file content must be loaded on the Telit IoT Portal for the demo application to be fully executed.

First, enter Developer section from the top menu



Figure 312

Choose Object Registry

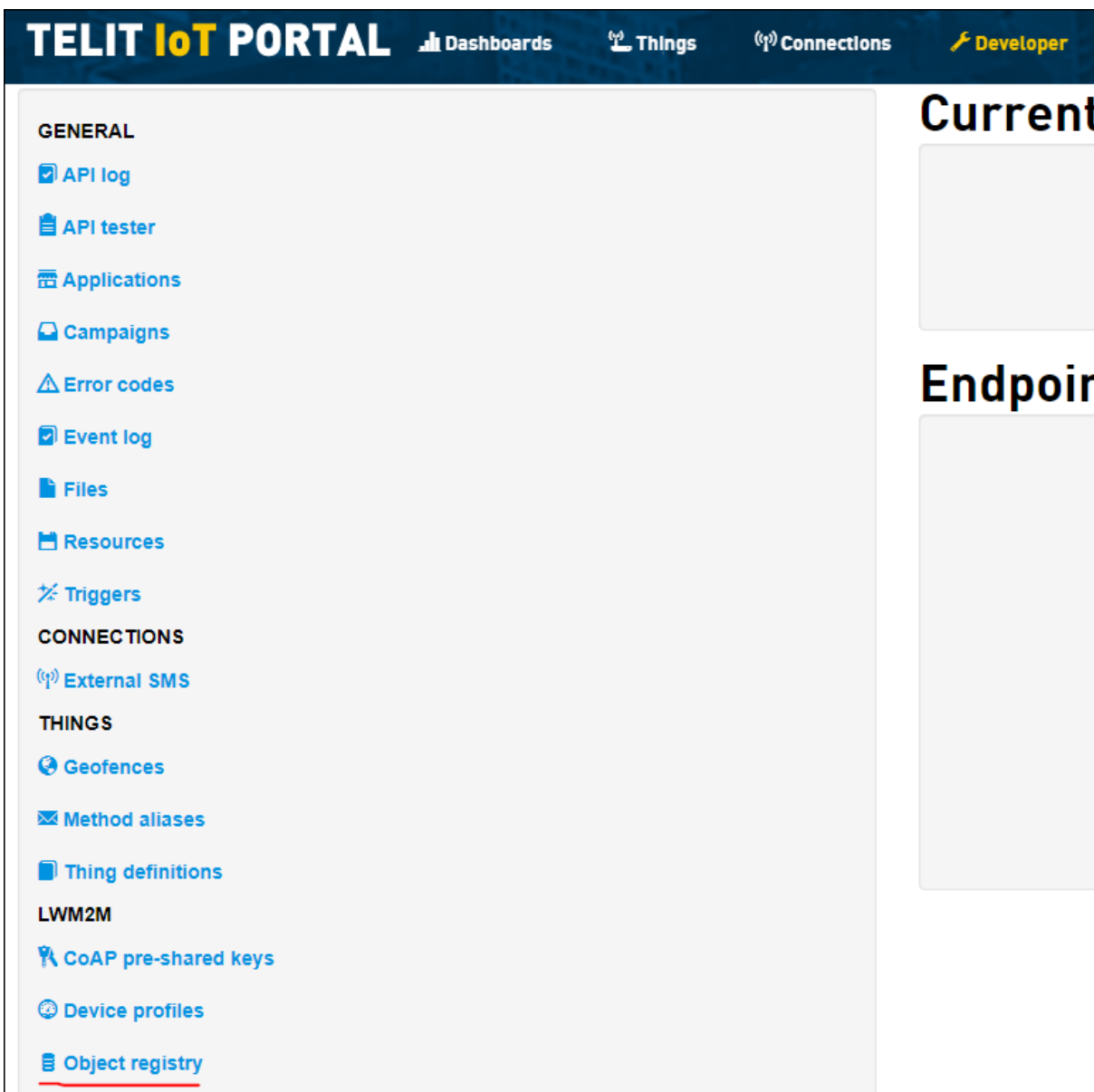


Figure 313

Create a New Object

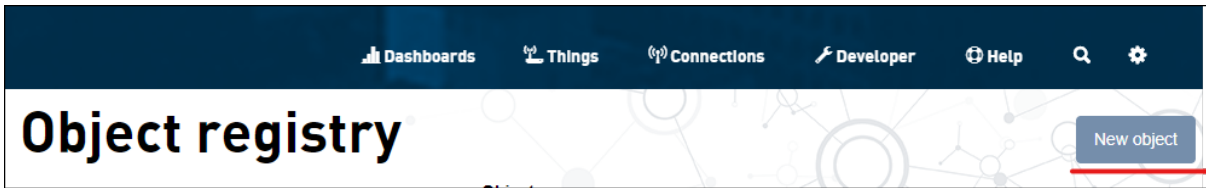


Figure 314

Copy the xml file content and paste it in the new Object form

Figure 315

Also, the application requires the XML file `xml/object_32010.xml` (provided with the sample files) to be stored in module's `/XML/` folder. It can be done with

```
AT#M2MWRITE=/XML/object_32010.xml,<size_in_bytes>
```

To load the XML file in the module, Telit AT Controller (TATC) can be used. Once the command above is issued, press the load content button:

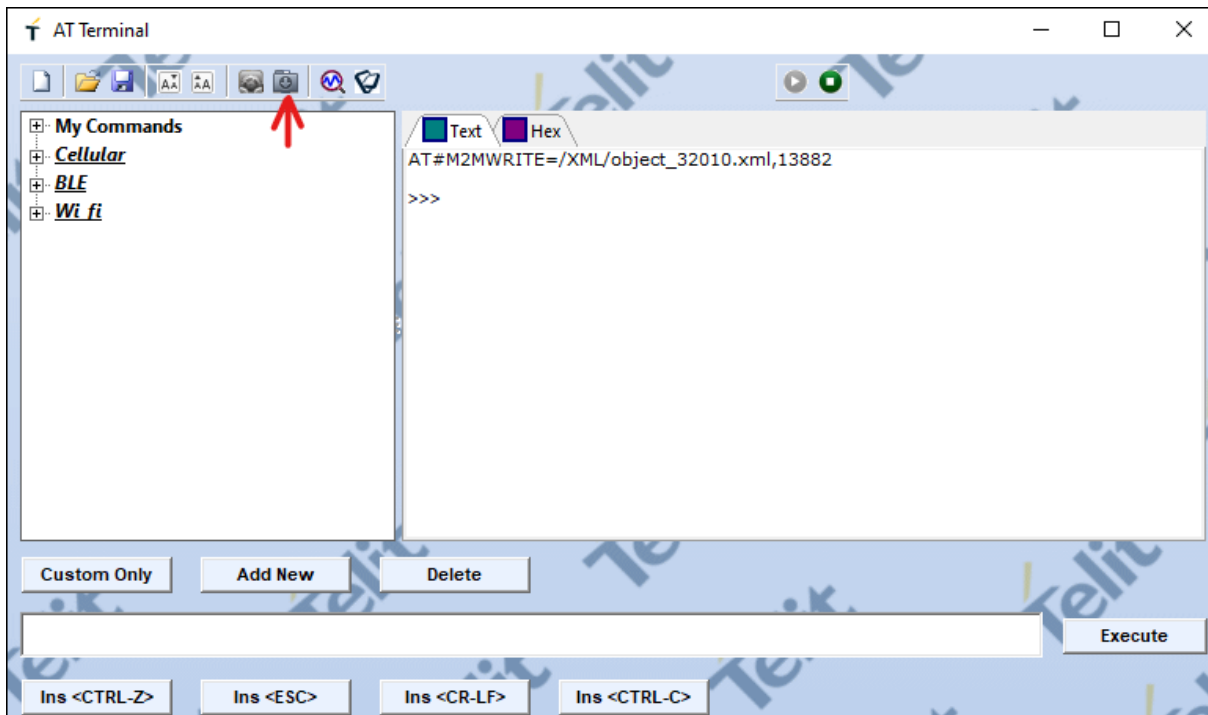


Figure 316

Select the file from your computer

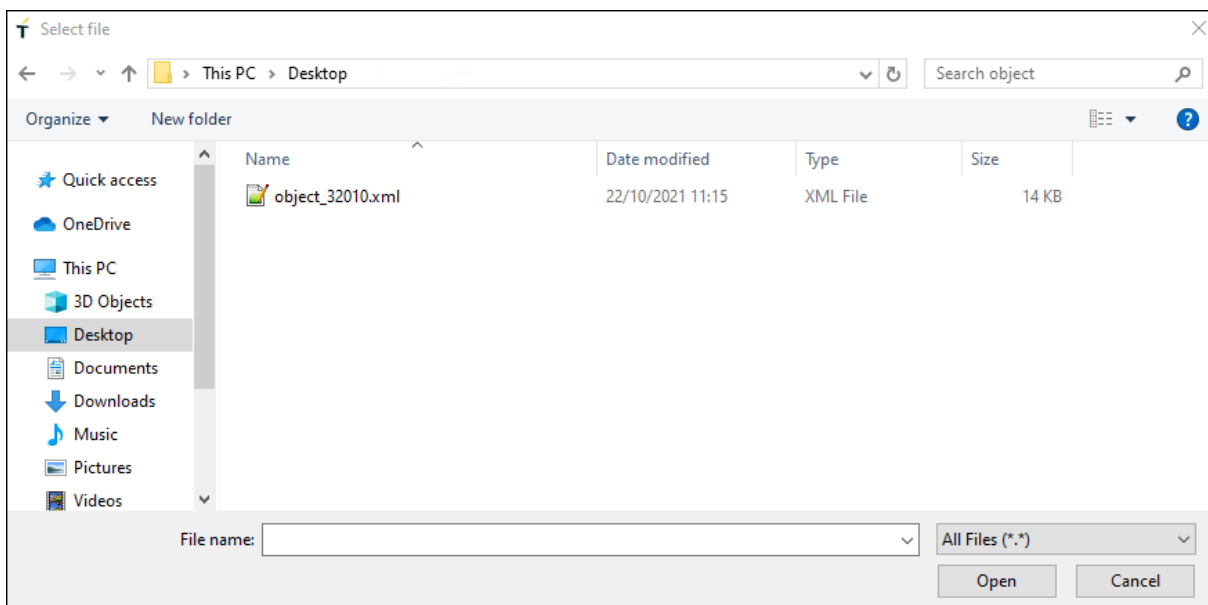


Figure 317

The file is successfully loaded on the module

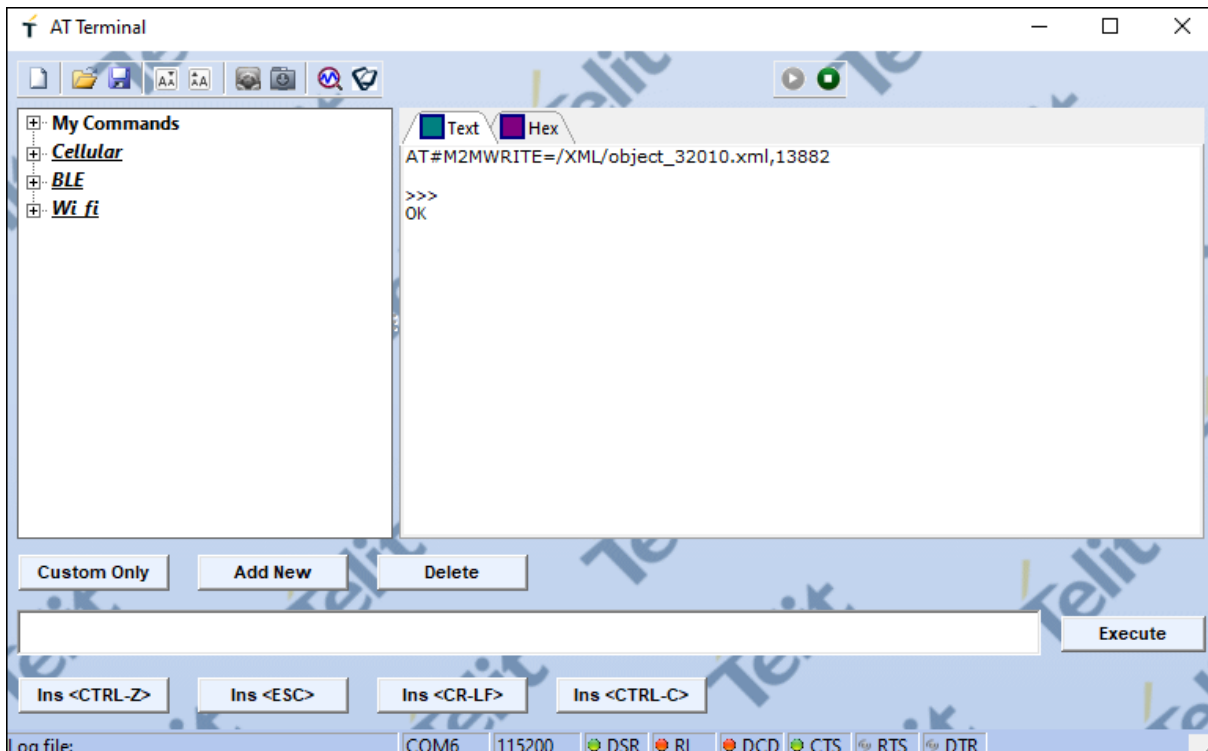


Figure 318

3.6.33.5 Onboard the device

Get the Telit ID

To retrieve the Telit ID data, issue AT#TID to get the Telit ID. The command response will be similar to

#TID: **xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx**,1 OK

Take note of the Telit ID highlighted in **bold** above (or copy it on a text editor): this ID it will be needed for the onboarding process.

Create a new Thing

From the OneEdge portal, on “**Things**” section, click “**New Thing**” button in the top right corner.

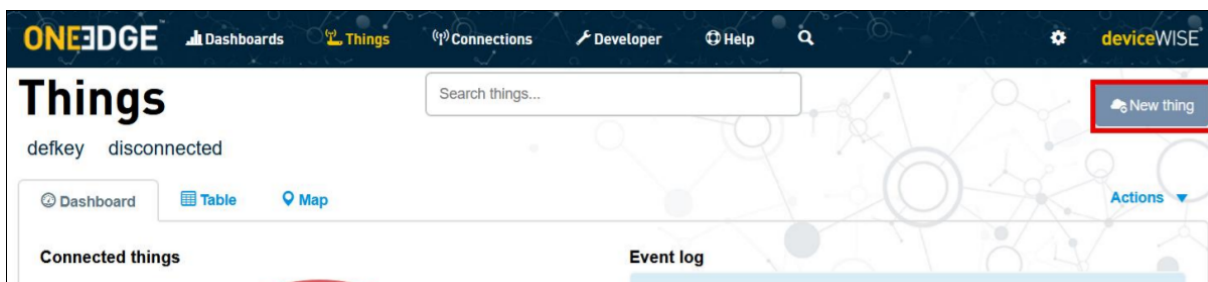


Figure 319

In the Create a new thing dialog, select “Telit Module”

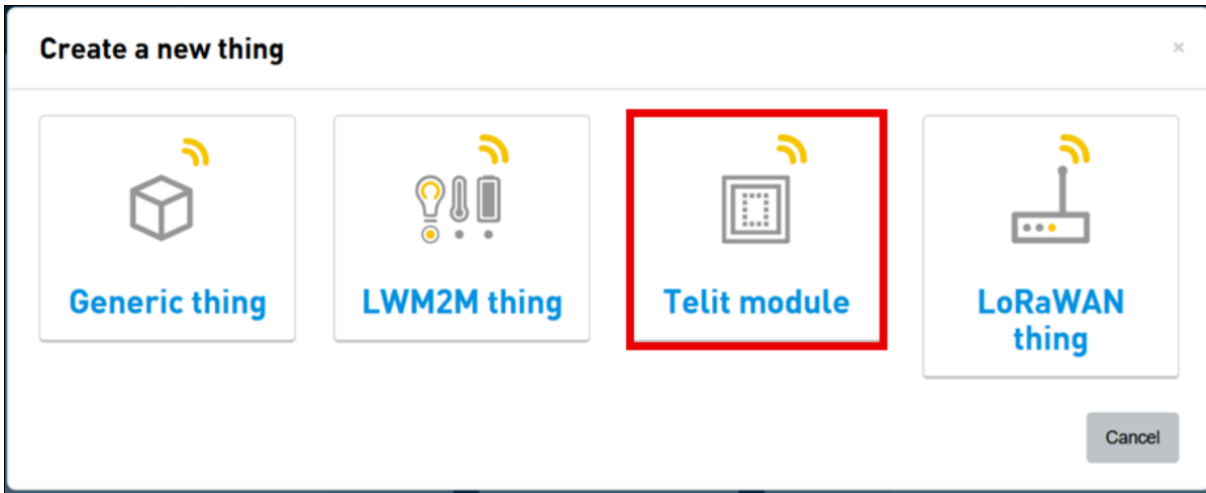


Figure 320

A dialog appears: select “Default” thing definition

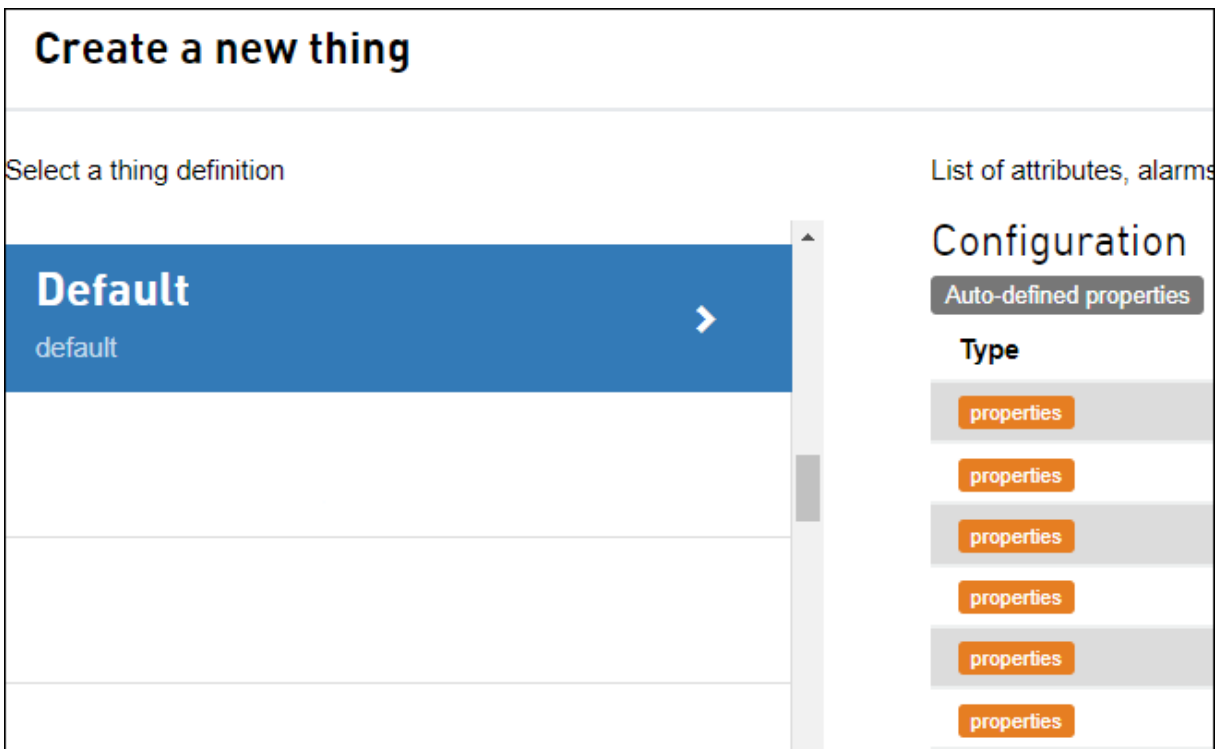


Figure 321

In the following screen, provide the Telit ID as “Identifier”. Click on “Find” and make sure that model, firmware and the other details are properly populated.

Click on lwm2m tab and set the device profile previously imported as shown in the screenshot below

Create a new thing

Setup
LWM2M
 Identification
 VAR
 Other
 Back

LWM2M
 Device profile*
 Lwm2m resources demo

Add Cancel

Figure 322

Click **“Add”** to complete the new thing creation procedure.

If the Thing already exists, its device profile can be changed by following the steps shown in the picture below

TELIT IoT PORTAL

Things > ep_353080091125356

ep_353080091125356

Overview Details **LWM2M** LWM2M Device Management Firmware Attributes Remote AT Events Files Tunnels API usage Actions

Identity NDWfxhWyZGatw114
 Device profile Device Management Profile - Troubleshooting
 Remote address 141.226.117.241:3702
 Protocol LWM2M-CoAP (DTLSv12)
 Binding UQ
 Lifetime 630

Object browser **Set profile** Queue history Wakeup

Set LWM2M profile

Change profile*
 Lwm2m resources demo

Set Cancel

Figure 323

3.6.34 Logging Demo

Sample application showing how to print on one of the available output interfaces.
Debug prints on **MAIN UART**

Features

- How to open a logging channel
- How to set a logging level
- How to use different logging macros

3.6.34.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Print a message with every log level

```
Starting Logging demo app. This is v1.0.7 built on Mar 26 2020 13:57:06.  
[WARN ] 20.17 M2MB_main.c:74 - M2MB_main{M2M_DamsStart}$ This is a WARNING MESSAGE  
[ERROR] 20.18 M2MB_main.c:76 - M2MB_main{M2M_DamsStart}$ THIS IS AN ERROR MESSAGE  
[CRITICAL] 20.19 M2MB_main.c:78 - M2MB_main{M2M_DamsStart}$ THIS IS AN CRITICAL MESSAGE  
[DEBUG] 20.19 M2MB_main.c:80 - M2MB_main{M2M_DamsStart}$ This is a DEBUG message  
[TRACE] 20.20 M2MB_main.c:82 - M2MB_main{M2M_DamsStart}$ This is a TRACE message  
END.
```

Figure 324

3.6.35 MD5 example

Sample application showing how to compute MD5 hashes using m2mb crypto. Debug prints on **MAIN UART**

Features

- Compute MD5 hash of a file
- Compute MD5 hash of a string

3.6.35.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a temporary file with the expected content
- Compute MD5 hash of the provided text file
- Compare the hash with the expected one
- Compute MD5 hash of a string
- Compare the hash with the expected one
- Delete test file

```
Starting MD5 demo app. This is v1.0.7 built on Apr 7 2020 10:19:54.  
Buffer written successfully into file. 45 bytes were written.  
  
Computing hash from file...  
Computed hash: bb0fa6eff92c305f166803b6938dd33a  
Expected hash: bb0fa6eff92c305f166803b6938dd33a  
Hashes are the same!  
  
Computing hash from string...  
Computed hash: bb0fa6eff92c305f166803b6938dd33a  
Expected hash: bb0fa6eff92c305f166803b6938dd33a  
Hashes are the same!
```

Figure 325

3.6.36 MQTT Client

Sample application showcasing MQTT client functionalities (with SSL). Debug prints on **MAIN UART**

Features

- How to check module registration and enable PDP context
- How to configure MQTT client parameters
- How to connect to a broker with SSL and exchange data over a subscribed topic

3.6.36.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage MQTT client and start it

mqtt_demo.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Init MQTT client
- Configure it with all parameters (Client ID, username, password, PDP context ID, keepalive timeout...)
- Connect MQTT client to broker
- Subscribe to two topics
- Publish 10 messages with increasing counter. Even messages are sent to topic 1, odd messages on topic 2.
- Print received message in mqtt_topc_cb function
- Disconnect MQTT client and deinit it
- Disable PDP context

```

Starting MQTT demo app. This is v1.0.7 built on Apr 7 2020 10:34:08.
[DEBUG] 16.18 mqtt_demo.c:192 - MQTT_Task(MQTT_TASK)$ INIT
[DEBUG] 16.18 mqtt_demo.c:206 - MQTT_Task(MQTT_TASK)$ m2mb_os_ev_init success
[DEBUG] 16.19 mqtt_demo.c:214 - MQTT_Task(MQTT_TASK)$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 16.19 mqtt_demo.c:221 - MQTT_Task(MQTT_TASK)$ Waiting for registration...
[DEBUG] 16.20 mqtt_demo.c:131 - NetCallback{pubTspt_0}$ Module is registered
[DEBUG] 16.21 mqtt_demo.c:232 - MQTT_Task(MQTT_TASK)$ Pdp context activation
[DEBUG] 18.26 mqtt_demo.c:246 - MQTT_Task(MQTT_TASK)$ Activate PDP with APN web.omnitel.it on CID 3....
[DEBUG] 18.95 mqtt_demo.c:155 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 18.96 mqtt_demo.c:159 - PdpCallback{pubTspt_0}$ IP address: 37.118.201.56
[DEBUG] 18.96 mqtt_demo.c:268 - MQTT_Task(MQTT_TASK)$ Init MQTT
[DEBUG] 18.97 mqtt_demo.c:278 - MQTT_Task(MQTT_TASK)$ m2mb_mqtt_init succeeded

Connecting to broker <api-dev.devicewise.com>:1883...
Done.
Subscribing to test_topic and test_topic2..
[DEBUG] 20.35 mqtt_demo.c:367 - MQTT_Task(MQTT_TASK)$ Done.

[DEBUG] 20.36 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 2> to topic test_topic
[DEBUG] 20.37 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 20.71 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 27
[DEBUG] 20.72 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 2>
[DEBUG] 23.37 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 3> to topic test_topic2
[DEBUG] 23.38 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 23.92 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 27
[DEBUG] 23.93 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 3>
[DEBUG] 26.40 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 4> to topic test_topic
[DEBUG] 26.41 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 26.93 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 27
[DEBUG] 26.93 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 4>
[DEBUG] 29.42 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 5> to topic test_topic2
[DEBUG] 29.43 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 29.99 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 27
[DEBUG] 30.00 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 5>
[DEBUG] 32.46 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 6> to topic test_topic
[DEBUG] 32.48 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 33.00 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 27
[DEBUG] 33.01 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 6>
[DEBUG] 35.47 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 7> to topic test_topic2
[DEBUG] 35.48 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 36.01 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 27
[DEBUG] 36.02 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 7>
[DEBUG] 38.50 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 8> to topic test_topic
[DEBUG] 38.51 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 39.15 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 27
[DEBUG] 39.16 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 8>
[DEBUG] 41.52 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 9> to topic test_topic2
[DEBUG] 41.53 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 42.10 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 27
[DEBUG] 42.12 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 9>
[DEBUG] 44.56 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 10> to topic test_topic
[DEBUG] 44.57 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 45.09 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic; data len: 28
[DEBUG] 45.11 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 10>
[DEBUG] 47.58 mqtt_demo.c:392 - MQTT_Task(MQTT_TASK)$ PUBLISHING <Hello from M2MB MQTT! ID: 11> to topic test_topic2
[DEBUG] 47.59 mqtt_demo.c:397 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 48.12 mqtt_demo.c:103 - mqtt_topic_cb(MQTT_Async)$ MQTT Message on Topic test_topic2; data len: 28
[DEBUG] 48.13 mqtt_demo.c:107 - mqtt_topic_cb(MQTT_Async)$ Message: <Hello from M2MB MQTT! ID: 11>

Disconnecting from MQTT broker..
[DEBUG] 50.60 mqtt_demo.c:414 - MQTT_Task(MQTT_TASK)$ Done.
[DEBUG] 50.61 mqtt_demo.c:443 - MQTT_Task(MQTT_TASK)$ application exit
[DEBUG] 50.62 mqtt_demo.c:453 - MQTT_Task(MQTT_TASK)$ m2mb_pdp_deactivate returned success
[DEBUG] 50.63 mqtt_demo.c:457 - MQTT_Task(MQTT_TASK)$ Application complete.
[DEBUG] 51.23 mqtt_demo.c:164 - PdpCallback{pubTspt_0}$ Context deactivated!

```

Figure 326

3.6.37 MultiTask

Sample application showcasing multi tasking functionalities with M2MB API. Debug prints on **MAIN UART**

Features

- How to create tasks using azx utilities
- How to use send messages to tasks
- How to use a semaphore to synchronize two tasks

3.6.37.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create three tasks with the provided utility (this calls public m2mb APIs)
- Send a message to the task1, its callback function azx_msgTask1 will be called

azx_msgTask1

- Print received parameters from main
- Send modified parameters to task2 (its callback function azx_msgTask2 will be called)
- wait for an InterProcess Communication semaphore to be available (released by task3)
- Once the semaphore is available, print a message and return

azx_msgTask2

- Print received parameters from caller
- If first parameter is bigger than a certain value, Send modified parameters to task3
- Else, use the second parameter as a task handle and print the corresponding name plus the value of the first parameter

azx_msgTask3

- Print received parameters from task 2
- release IPC semaphore
- send message to task 2 with first parameter below the threshold and second parameter with task3 handle


```
Starting MultiTask demo app. This is v1.0.12-C1 built on Jun 23 2020 15:36:31.
Inside "myTask1" user callback function. Received parameters from MAIN: 3 4 5
Task1 - Sending a message to task 2 with modified parameters...
Task1 - Waiting for semaphore to be released by task 3 now...

Inside "myTask2" user callback function. Received parameters: 5 7 10
Task2 - Sending a message to task 3 with modified parameters...
Task2 - Done.

Inside "myTask3" user callback function. Received parameters from Task 2: 15 14 9
Task3 - Releasing IPC semaphore...

Task1 - After semaphore! return...

Task3 - IPC semaphore released.
Task3 - Sending a message to task 2 with specific 'type' parameter value of 0 and task 3 handle as param1...

Inside "myTask2" user callback function. Received parameters: 0 1073951320 9
Task3 - Done.
Task2 - Received type 0 from task "myTask3"
Task2 - Done.
```

Figure 327

3.6.38 MutEx

Sample application showing mutex usage, with ownership and prioritization usage.
Debug prints on **MAIN UART**

Features

- How to create a mutex
- How to use the mutex with tasks having different priorities
- how to reorder the pending tasks queue for the mutex

3.6.38.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create four tasks with the provided utility (this calls public m2mb APIs). The first task is a “producer”, putting data on a shared buffer. The second is a “consumer” of said data, the other two are used for prioritization demo
- run producer and consumer tasks at the same pace. the shared buffer will stay empty, because the resource is consumed right after creation
- run producer twice as fast as consumer. The buffer is slowly filled
- run consumer twice as fast as publisher. The buffer is always empty.
- reserve the mutex in the main task and run producer, support and support2 tasks (in this order). Then release the mutex and check the execution order. It should be by arrival.
- reserve the mutex in the main task and run the same three task, but before releasing the mutex, call the prioritization API. the task with highest priority (producer) is put as first in the queue.

```

Starting Mutex app. This is v1.0.12-C1 built on Jul 1 2020 08:37:15.
[DEBUG] 14.50 M2MB_main:90 - mutex_init{M2M_DamsStart}$ [MUTEX] Mutex initialized

[CASE 1 ] Producer and consumer have same idle time

[DEBUG] 14.51 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 14.52 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 14.53 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 14.53 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 14.54 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 14.54 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 14.55 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 14.56 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 15.56 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 15.56 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 15.57 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 15.58 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 15.58 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 15.59 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 15.60 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 15.60 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 16.61 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 16.61 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 16.62 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 16.63 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 16.63 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 16.64 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 16.64 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 16.65 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released

```

Figure 328

```

[CASE 2 ] Producer has double idle time

[DEBUG] 17.56 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 17.56 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 17.57 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 17.58 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 17.58 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 17.59 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 17.59 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 17.60 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 18.63 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 18.64 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 0 items
[DEBUG] 18.64 M2MB_main:268 - msgConsumer{CONSUMER}$ Can't consume anything, buffer size is 0
[DEBUG] 18.65 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 19.62 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 19.62 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 19.63 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 19.64 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 19.68 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 19.69 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 19.69 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 19.70 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 20.73 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 20.74 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 0 items
[DEBUG] 20.75 M2MB_main:268 - msgConsumer{CONSUMER}$ Can't consume anything, buffer size is 0
[DEBUG] 20.75 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 21.67 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 21.68 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 21.68 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 21.69 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 21.77 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 21.79 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 21.80 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 21.80 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released

```

Figure 329

```
[CASE 3 ] Producer has half idle time

[DEBUG] 22.62 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 22.63 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 22.64 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 22.64 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 22.65 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 22.65 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 22.66 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 22.67 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 23.67 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 23.68 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 23.68 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 23.69 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 24.71 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 24.72 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 1 items
[DEBUG] 24.72 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 0
[DEBUG] 24.73 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 24.74 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 24.74 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 0 items
[DEBUG] 24.75 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 0
[DEBUG] 24.76 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 25.79 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 25.79 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 1 items
[DEBUG] 25.80 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 1
[DEBUG] 25.81 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
[DEBUG] 26.78 M2MB_main:250 - msgConsumer{CONSUMER}$ Mutex acquired
[DEBUG] 26.78 M2MB_main:251 - msgConsumer{CONSUMER}$ Now there are 2 items
[DEBUG] 26.79 M2MB_main:261 - msgConsumer{CONSUMER}$ [CONSUMER]I consumed 99 from index 1
[DEBUG] 26.79 M2MB_main:308 - msgConsumer{CONSUMER}$ Mutex released
[DEBUG] 26.84 M2MB_main:119 - msgProducer{PRODUCER}$ Mutex acquired
[DEBUG] 26.84 M2MB_main:120 - msgProducer{PRODUCER}$ Now there are 1 items
[DEBUG] 26.85 M2MB_main:125 - msgProducer{PRODUCER}$ Produced item 99 at index 1
[DEBUG] 26.86 M2MB_main:176 - msgProducer{PRODUCER}$ Mutex released
```

Figure 330

```
[CASE 4 ] NO HTPF

Reserve MUTEX so all tasks are enqueued
[DEBUG] 30.77 M2MB_main:387 - msgSupport{HPTF_SUPPORT}$ freepos = 0 | evaluate[freepos]= 3
[DEBUG] 30.78 M2MB_main:416 - msgSupport2{HPTF_SUPPORT2}$ freepos = 1 | evaluate[freepos]= 4
[DEBUG] 30.79 M2MB_main:223 - msgProducer{PRODUCER}$ producer: freepos = 2 | evaluate[freepos]= 1
[DEBUG] 35.85 M2MB_main:586 - M2MB_main{M2M_DamsStart}$ EVALUATE SEQUENCE IS 3 4 1. Expected: 3 4 1
NO HTPF OK

[CASE 4.1 ] HTPF USED

Reserve MUTEX so all tasks are enqueued
m2mb_os_mtx_hptf OK
[DEBUG] 41.98 M2MB_main:223 - msgProducer{PRODUCER}$ producer: freepos = 0 | evaluate[freepos]= 1
[DEBUG] 41.99 M2MB_main:387 - msgSupport{HPTF_SUPPORT}$ freepos = 1 | evaluate[freepos]= 3
[DEBUG] 42.00 M2MB_main:416 - msgSupport2{HPTF_SUPPORT2}$ freepos = 2 | evaluate[freepos]= 4
[DEBUG] 44.03 M2MB_main:650 - M2MB_main{M2M_DamsStart}$ EVALUATE SEQUENCE IS 1 3 4, expected 1 3 4
HTPF DEMO OK
The application has ended...
```

Figure 331

3.6.39 NTP example

The application connects to an NTP server, gets current date and time and updates module's internal clock. Debug prints on **MAIN UART**

Features

- How to get current date and time from an NTP server
- How to set current date and time on module

3.6.39.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Send message to ntpTask

ntp_task.c

NTP_task() - Waits module registration - When module is registered, initializes ntp setting CID, server url and timeout - When PDP context is correctly opened, a query to NTP server is done to get current date and time - On SET_MODULE_RTC message type reception, module RTC is set with date time value got from NTP server.

m2mb_ntp_ind_callback() - As soon as M2MB_NTP_VALID_TIME event is received, current date and time is printed and a message (with SET_MODULE_RTC type) is sent to NTP_task

```
Start NTP demo application. This is v1.0 built on Apr 16 2021 09:36:12.
Waiting for registration...
Module is registered!

Activate PDP context with APN ibox.tim.it on CID 3
Context activated, IP address: 2.195.170.123
Get current time from server 0.pool.ntp.org, PORT: 123

Current time is: Friday 2021-04-16, 07:37:33

Current time correctly set on module
Module system time is: 2021-04-16, 07:37:33
```

Figure 332

3.6.40 ON OFF button management example

Sample application on how to handle ON/OFF button. Debug prints on **MAIN UART**

Features

- How to enable ON/OFF button handling
- How to turn off module after ON/OFF button press/release

Application workflow

M2MB_main.c

- Open gpio_ON_OFF
- Configure the interrupt and the callback
- Init power functionality
- Wait for on/off button to be pressed
- If button has been pressed for more than 3 seconds (and then released) then module is turned off.

```
Starting ON OFF button handling demo app. This is v1.1.8 built on Feb  9 2023 11:56:05.  
Waiting for ON/OFF button to be pressed...  
onoff pressed!  
onoff released!  
ON/OFF pressed form more than 3 sec, turn off module!
```

Figure 333

3.6.41 RTC example

Sample application that shows RTC apis functionalities: how to get/set module system time and timestamp. Debug prints on **MAIN UART**

Features

- How to read module timestamp
- How to read module system time
- How to set new system time

3.6.41.1 Application workflow

M2MB_main.c

- Init log azx and print a welcome message
- Init net functionality and wait for module registration
- Init RTC functionality and get module time in timestamp format (seconds from the epoch)
- Get module system time in date/time format
- Add 1 hour to timestamp, convert it to system time and set it to module

```
Start RTC demo application. This is v1.0 built on Oct  1 2021 15:01:40.
Waiting for registration...
Module is registered!

Current time in seconds from the epoch: 1633101266
Module system time is: 2021-10-01, 15:14:26

Get current time and add an hour

Current time in seconds from the epoch: 1633101266
New time to be set : 2021-10-01, 16:14:26, tz:4, dst:0

Set new time and check the setting
NEW module system time is: 2021-10-01, 16:14:26
```

Figure 334

3.6.42 SIM event handler example

Sim Event Demo application. Debug prints on **MAIN UART**, using AZX log example functions

Features

- How to use ATI function for asynchronous management
- How to catch URC from an AppZone application
- How to catch SIM related events and handle them

Application workflow

M2MB_main.c

- Print welcome message
- Initialize AT interface
- Initialize AT URC manager task
- Initialize SIM event manager task
- Send "AT#SIMPR=1" to activate SIM URCs
- Insert SIM in SIM slot 1 and receive SIM inserted message
- Remove SIM from SIM slot 1 and receive SIM removed message

```
Starting SIM Presence Demo app. This is v1.1.17 built on Dec 16 2022 15:14:28.  
Please ensure SIM is not inserted before starting this sample application  
Sending command AT#SIMPR=1 to enable SIM presence URC messages...  
Command set.  
SIM 0 state changed to 1!  
  
SIM CARD HAS BEEN INSERTED!  
  
SIM 0 state changed to 0!  
  
SIM CARD HAS BEEN REMOVED!
```

Figure 335

3.6.43 SMS PDU

Sample application showcasing how to create and decode PDUs to be used with m2mb_sms_* API set. A SIM card and antenna must be present. Debug prints on **MAIN UART**

Features

- How to enable SMS functionality
- How to use encode an SMS PDU to be sent with m2mb_api
- How to decode a received SMS response from PDU to ASCII mode.

3.6.43.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Init sms functionality
- Create PDU from text message
- Send message to destination number
- Wait for response
- When SMS PDU response is received, decode it and print information about it, plus the message content

```
m2mb_sms_init() succeeded

Sending message <How are you?>...
m2mb_sms_send() - succeeded
M2MB_SMS_SEND_RESP Callback
Send resp msg ID 10
SMS received!
SMS correctly received!

Reading SMS from memory...
m2mb_sms_read() request succeeded

--- SMS read ---
SMS tag M2MB_SMS_TAG_MT_NOT_READ
SMS format M2MB_SMS_FORMAT_3GPP
Code type: 0
Sender type: 145
Msg len: 12
Msg bytes: 11
Msg date 19/7/17 16:7:58 (timezone: 2)
Received SMS, content: <<Fine thanks >>
Sender: +[REDACTED]
```

Figure 336

3.6.44 SMS_atCmd example

Sample application showcasing how to receive an SMS containing an AT command, process the AT command and send its answer to sender (configurable in sms_config.txt). A SIM card and antenna must be present. Debug prints on **MAIN UART**

Features

- How to receive an SMS with an AT command as text inside
- How to send AT command to parser and read the answer
- How to send the AT command answer back to sender via SMS

Optional configuration file to be put in /mod folder, copy sms_config.txt file into your module running the following AT command:

```
AT#M2MWRITE="/mod/sms_config.txt",138
>>> here receive the prompt; then type or send the file, sized 138 bytes
```

3.6.44.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Init SMS functionality
- Read configuration file sms_config.txt (send SMS with AT command answer back, delete SMS received)
- Init AT command parser
- Create a task to handle SMS parsing and AT command sending
- Wait for an incoming SMS

callbacks.c

msgSMSparse()

- When SMS has been received, content is decoded and printed. If there is an AT command inside, command is executed and answer printed and sent back to sender as an SMS (depending on sms_config.txt setting)

```

yStarting SMS with AT command demo app. This is v1.0.13-C1 built on Mar 18 2021 12:42:22.
[DEBUG] 16.61 M2MB_main:135 - M2MB_main{M2M_DamsStart}$ m2mb_os_ev_init success
m2mb_sms_init() succeeded
[DEBUG] 16.62 M2MB_main:168 - M2MB_main{M2M_DamsStart}$ M2MB_SMS_INCOMING_IND indication enabled
[DEBUG] 16.63 M2MB_main:179 - M2MB_main{M2M_DamsStart}$ M2MB_SMS_INCOMING_IND MEMORY FULL indication enabled
[DEBUG] 16.64 M2MB_main:196 - M2MB_main{M2M_DamsStart}$ Storage set to M2MB_SMS_STORAGE_SM
[DEBUG] 16.65 callbacks:114 - readConfigFromFile{M2M_DamsStart}$ Reading parameters from file
[DEBUG] 16.66 callbacks:116 - readConfigFromFile{M2M_DamsStart}$ Opening /mod/sms_config.txt in read mode..
Default: SMS with answer sending DISABLED, delete sms DISABLED
[DEBUG] 16.67 at_async:115 - at_cmd_async_init{M2M_DamsStart}$ m2mb_ati_init() on instance 0
Please send an SMS with a configuration as ("ATCMD: <atcmd>")...

```

Figure 337

3.6.45 SMTP Client

Sample application showing SMTP echo demo with M2MB API. Debug prints on **MAIN UART**

Features

- How to check module registration and activate PDP context
- How to open a SMTP client
- How to send a mail

Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage SMTP client and start it

M2MB_main.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Initialize SMTP client and connect to SMTP server
- Prepare email and send it
- Close SMTP client
- Disable PDP context

```
Starting SMTP demo app. This is v1.1.17 built on Jan  9 2023 09:57:51.
Reading parameters from file...
Set APN to: <<nxt17.net>>
Set APN USER to: <<>>
Set APN PASS to: <<>>
Set PDP_CTX to: 1
Set MAIL SERVER to: <<          >>
Set MAIL PORT to: <<          >>
Set MAIL USER to: <<          >>
Set MAIL PASS to: <<          >>
Set MAIL FROM to: <<          >>
Set MAIL FROM NAME to: << >>
Set MAIL SUBJECT to: <<Hello from AppZone>>
Set MAIL BODY to: <<AppZone Test>>
Set MAIL TO to: <<          >>
Set MAIL TO NAME to: << >>
Waiting for registration...
Pdp context activation
Connecting to SMTP Server...
SMTP Client connected!
Preparing example's mail...
Sending example's mail...
Example's mail sent!
SMTP client closed
Application complete.
```

Figure 338

3.6.46 SPI Echo

Sample application showing how to communicate over SPI with m2mb API. Debug prints on **MAIN UART**

Features

- How to open an SPI bus. MOSI and MISO will be shorted, to have an echo.
- How to communicate over SPI bus

3.6.46.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Open SPI bus, set parameters
- Send data on MOSI and read the same in MISO

Notes:

For LE910Cx (both Linux and ThreadX based devices), AT#SPIEN=1 command must be sent once before running the app

```
Starting SPI demo app. This is v1.0.7 built on Apr  1 2020 13:48:05.  
Transfer successful. Received: hello from spi echo
```

Figure 339

3.6.47 SPI sensors

Sample application showing SPI usage, configuring two ST devices: a magnetometer (ST LIS3MDL) and a gyroscope (ST L3G4200D). The application will read values from both devices using GPIO4 and 3 (respectively) as magnetometer CS and gyro CS. Debug prints on **MAIN UART**

Features

- How to open an SPI bus with a slave device
- How to communicate with the device over the SPI bus

Setup

- Connect sensor VDD to 3v8 supply (e.g. Vbatt on the module)
- Connect sensor GND to a GND pin of the module
- Connect sensors MOSI to module SPI_MOSI
- Connect sensors MISO to module SPI_MISO
- Connect sensors CLK to module SPI_CLK
- Connect magnetometer CS to module GPIO 2
- Connect gyroscope CS to module GPIO 3

3.6.47.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Open SPI bus, set parameters
- Configure GPIO 2 and GPIO 3 as output, set them high (idle)
- Set registers to configure magnetometer
- Read in a loop (10 iterations) the registers carrying the 3 axes values and show the gauss value for each of them. A metal object is put close to the sensor to change the read values.
- Set registers to configure gyroscope
- Read in a loop (10 iterations) the registers carrying the 3 axes values and show the degrees per second value for each of them. The board is rotated to change the read values.

Notes:

For LE910Cx (both Linux and ThreadX based devices), AT#SPIEN=1 command must be sent once before running the app

```
Starting SPI demo app. This is v1.0.7 built on Apr 1 2020 13:58:25.
SPI start

Magnetometer SPI Demo start
Reading Magnetometer WHOAMI. Expected: 0x3D
Expected response received!
Setting continuous conversion mode...
Continuous conversion mode successfully set.
Setting 10 Hz Output Data Rate, Medium performance mode X Y axis...
Magnetometer Enabled. 10Hz ODR, Medium Perf. Mode (X,Y).
Setting Medium performance for Z axis, little endian...
Medium Perf. Mode (Z), little endian.
Setting complete, starting reading loop...

X: 0.204 gauss
Y: -0.321 gauss
Z: 0.305 gauss

X: 0.290 gauss
Y: -0.103 gauss
Z: 0.043 gauss

X: -2.513 gauss
Y: -0.353 gauss
Z: -4.000 gauss

X: 1.980 gauss
Y: 0.174 gauss
Z: -1.945 gauss

X: 4.000 gauss
Y: -0.090 gauss
Z: -4.000 gauss

X: -0.605 gauss
Y: -0.154 gauss
Z: 0.210 gauss

X: -0.580 gauss
Y: 2.004 gauss
Z: -0.047 gauss

X: 0.177 gauss
Y: -0.359 gauss
Z: 0.295 gauss

X: 0.173 gauss
Y: -0.356 gauss
Z: 0.301 gauss

X: 0.174 gauss
Y: -0.356 gauss
Z: 0.298 gauss
Reading complete.
```

Figure 340

3.6.48 SW Timer (Software Timer)

The sample application shows how to use SW Timers M2MB API. Debug prints on **MAIN UART**

Features

- How to open configure a SW timer
- How to use the timer to manage recurring events

3.6.48.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create sw timer structure
- Configure it with 4 seconds timeout, periodic timer (auto fires when expires)
- Init the timer with the parameters
- Start the timer
- Wait 10 seconds
- Stop the timer

timerCb

- Print a message with inside the callback

```
Starting SW Timers demo app. This is v1.0.7 built on Apr 7 2020 09:51:25.  
timer expired!  
[DEBUG] 21.41 M2MB_main.c:59 - timerCb{pubTspt_0}$ timer handle: 0x4002b004  
timer expired!  
[DEBUG] 25.47 M2MB_main.c:59 - timerCb{pubTspt_0}$ timer handle: 0x4002b004  
stopping the timer  
Stop a running timer: success  
Application end
```

Figure 341

3.6.49 Secure MicroService

Sample application showcasing how to manage secure microservice functionalities.
Debug prints on **MAIN UART**

Features

- Write data in Secure Data Area (SDA), non protected
- Read the written data and compare with the original buffer
- Write a cripty key in Secure Data Area (SDA), non protected
- Perform a rotate of the written key data
- Perform MD5 sum of written data from TZ file
- Compare computed digest with expected one
- Write data in trust zone as a trusted object (it will not be possible to read it again but only use its content for crypto operations)
- Try to read the trusted object and verify it fails
- Rotate trusted item and verify retrieving the content fails
- compute MD5 sum of trusted item and compare with the expected one
- Try to pass data from a trusted item to a non trusted item using untrusted TZ buffers, and verify it fails

3.6.49.1 Application workflow

M2MB_main.c

- Write a buffer in a SDA item using `m2mb_secure_ms_write`
- Read the same item using `m2mb_secure_ms_read`
- Write a buffer containing some cripty key in a SDA item using `m2mb_secure_ms_write`
- Rotate the content of the key item
- Read it with `m2mb_secure_ms_read`
- Load the key content using `m2mb_secure_ms_crypto_alloc` and `m2mb_secure_crypto_add_it` in a SECURE_MS buffer
- Compute MD digest with `m2mb_secure_ms_crypto_md`
- Write a buffer containing some cripty key in a SDA item using `m2mb_secure_ms_write` but with **TRUSTED** option in `m2mb_secure_ms_open`
- Verify that `m2mb_secure_ms_read` on the trusted item fails
- Verify that `m2mb_secure_ms_crypto_rotate` fails for the trusted item
- Verify the MD5 digest
- Try to copy the trusted item data in a SECURE_MS buffer with `m2mb_secure_ms_crypto_alloc` and `m2mb_secure_crypto_add_item`, then load it in an untrusted object with `m2mb_secure_ms_crypto_write`, and verify it fails.

```

Starting secure ms demo app. This is v1.0.13-C1 built on Jul 30 2020 12:19:02.
Writing data in normal item
Stored input data in Secure Data Area
Reading data from normal item
Data length in SDA: 11 bytes
Securely loaded the data from the SDA
Read 11 bytes: <hello world>
original and retrieved strings are the same

Writing key in normal item
Stored input data in Secure Data Area

Rotate data in normal item
Original key: AA_THIS_IS_MY_SECRET_KEY_BB
Rotated key:

Compute MD5 of data in normal item
Data length in SDA: 27 bytes
MD5: 8EDAD26E26E1C74C7C02386C1C7F541D
hash is the expected one!

Writing data in trusted item
Stored input data in Secure Data Area
Reading data from trusted item (should fail!)
Data length in SDA: 27 bytes
m2mb_secure_ms_read() failed for trusted item, as expected!

Rotate data in trusted item
[ERROR] 17.01 M2MB_main:329 - read_rotate{M2M_DamsStart}$ Cannot read data from SECURE_MS_BUFFER to user buffer
Original key: AA_THIS_IS_MY_SECRET_KEY_BB
Rotated key:

Compute MD5 of data in trusted item
Data length in SDA: 27 bytes
MD5: 8EDAD26E26E1C74C7C02386C1C7F541D
Hash is the expected one!

Try to pass data from trusted to untrusted through TZ buffers
Cannot store data from SECURE_MS_BUFFER to SDA 'non-trusted', as expected

```

Figure 342

3.6.50 TCP IP

Sample application showcasing TCP echo demo with M2MB API. Debug prints on **MAIN UART**

Features

- How to check module registration and activate PDP context
- How to open a TCP client socket
- How to communicate over the socket

3.6.50.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage socket and start it

m2m_tcp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and link it to the PDP context id
- Connect to the server
- Send data and receive response
- Close socket
- Disable PDP context

```
Starting TCP-IP demo app. This is v1.0.7 built on Mar 26 2020 16:20:30.
[DEBUG] 21.23 m2m_tcp_test.c:201 - M2M_msgTCPTask{TCP_TASK}$ INIT
[DEBUG] 21.25 m2m_tcp_test.c:217 - M2M_msgTCPTask{TCP_TASK}$ m2mb_os_ev_init success
[DEBUG] 21.26 m2m_tcp_test.c:223 - M2M_msgTCPTask{TCP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 21.26 m2m_tcp_test.c:231 - M2M_msgTCPTask{TCP_TASK}$ Waiting for registration...
[DEBUG] 21.28 m2m_tcp_test.c:128 - NetCallback{pubTspt_0}$ Module is registered to cell 0x816B!
[DEBUG] 21.29 m2m_tcp_test.c:244 - M2M_msgTCPTask{TCP_TASK}$ Pdp context activation
[DEBUG] 21.30 m2m_tcp_test.c:248 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.34 m2m_tcp_test.c:263 - M2M_msgTCPTask{TCP_TASK}$ Activate PDP with APN web.omnitel.it....
[DEBUG] 24.52 m2m_tcp_test.c:155 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 24.52 m2m_tcp_test.c:158 - PdpCallback{pubTspt_0}$ IP address: 83.225.44.56
[DEBUG] 24.54 m2m_tcp_test.c:273 - M2M_msgTCPTask{TCP_TASK}$ Creating Socket...
[DEBUG] 24.54 m2m_tcp_test.c:284 - M2M_msgTCPTask{TCP_TASK}$ Socket created
[DEBUG] 24.55 m2m_tcp_test.c:294 - M2M_msgTCPTask{TCP_TASK}$ Socket ctx set to 3
[DEBUG] 24.95 m2m_tcp_test.c:307 - M2M_msgTCPTask{TCP_TASK}$ Retrieved IP: 185.86.42.218
[DEBUG] 25.17 m2m_tcp_test.c:322 - M2M_msgTCPTask{TCP_TASK}$ Socket Connected!
[DEBUG] 25.18 m2m_tcp_test.c:329 - M2M_msgTCPTask{TCP_TASK}$ Sending data over socket..
[DEBUG] 25.19 m2m_tcp_test.c:342 - M2M_msgTCPTask{TCP_TASK}$ Data send successfully (16 bytes)
[DEBUG] 27.20 m2m_tcp_test.c:356 - M2M_msgTCPTask{TCP_TASK}$ trying to receive 16 bytes..
[DEBUG] 27.21 m2m_tcp_test.c:364 - M2M_msgTCPTask{TCP_TASK}$ Data received (16): <hello from m2mb!>
[DEBUG] 27.21 m2m_tcp_test.c:373 - M2M_msgTCPTask{TCP_TASK}$ application exit
[DEBUG] 27.22 m2m_tcp_test.c:385 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 27.24 m2m_tcp_test.c:388 - M2M_msgTCPTask{TCP_TASK}$ Application complete.
[DEBUG] 29.43 m2m_tcp_test.c:164 - PdpCallback{pubTspt_0}$ Context successfully deactivated!
```

Figure 343

3.6.51 TCP non blocking example

Sample application that shows how to configure and connect a TCP-IP non blocking socket. Debug prints on **MAIN UART**

Features

- How to check module registration and activate PDP context
- How to open a TCP client non Blocking socket
- How to communicate over the socket

Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage socket and start it

m2m_tcp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and link it to the PDP context id
- Set the socket as non Blocking and connect to server. Uses `m2mb_socket_bsd_select`, `m2mb_socket_bsd_fd_isset_func` to check when socket is connected.
- Send data and receive response
- Close socket
- Disable PDP context

```

Starting TCP-IP non blocking demo app. This is v1.1.9 built on Mar 22 2022 15:35:56.
[DEBUG] 19.24 m2mb_tcp_te:215 - M2M_msgTCPTask{TCP_TASK}$ INIT
[DEBUG] 19.25 m2mb_tcp_te:231 - M2M_msgTCPTask{TCP_TASK}$ m2mb_os_ev_init success
[DEBUG] 19.25 m2mb_tcp_te:237 - M2M_msgTCPTask{TCP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 19.26 m2mb_tcp_te:245 - M2M_msgTCPTask{TCP_TASK}$ Waiting for registration...
[DEBUG] 19.27 m2mb_tcp_te:142 - NetCallback{pubTspt_0}$ Module is registered to cell 0x20!
[DEBUG] 19.28 m2mb_tcp_te:258 - M2M_msgTCPTask{TCP_TASK}$ Pdp context activation
[DEBUG] 19.29 m2mb_tcp_te:262 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 21.34 m2mb_tcp_te:277 - M2M_msgTCPTask{TCP_TASK}$ Activate PDP with APN ibox.tim.it....
[DEBUG] 21.89 m2mb_tcp_te:169 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 21.90 m2mb_tcp_te:172 - PdpCallback{pubTspt_0}$ IP address: 2.195.160.174
[DEBUG] 21.90 m2mb_tcp_te:287 - M2M_msgTCPTask{TCP_TASK}$ Creating Socket...
[DEBUG] 21.91 m2mb_tcp_te:299 - M2M_msgTCPTask{TCP_TASK}$ Socket created
[DEBUG] 21.92 m2mb_tcp_te:309 - M2M_msgTCPTask{TCP_TASK}$ Socket ctx set to 1
[DEBUG] 22.10 m2mb_tcp_te:322 - M2M_msgTCPTask{TCP_TASK}$ Retrieved IP: 185.86.42.218
[DEBUG] 22.11 m2mb_tcp_te:332 - M2M_msgTCPTask{TCP_TASK}$ Socket set to nonBlocking
[DEBUG] 22.11 m2mb_tcp_te:351 - M2M_msgTCPTask{TCP_TASK}$ Connection in progress...
[DEBUG] 22.29 m2mb_tcp_te:377 - M2M_msgTCPTask{TCP_TASK}$ Socket <0x400343a0> is connected!
[DEBUG] 22.30 m2mb_tcp_te:398 - M2M_msgTCPTask{TCP_TASK}$ Socket Connected!
[DEBUG] 22.32 m2mb_tcp_te:411 - M2M_msgTCPTask{TCP_TASK}$ Sending data over socket..
[DEBUG] 22.32 m2mb_tcp_te:424 - M2M_msgTCPTask{TCP_TASK}$ Data send successfully (16 bytes)
[DEBUG] 24.37 m2mb_tcp_te:438 - M2M_msgTCPTask{TCP_TASK}$ trying to receive 16 bytes..
[DEBUG] 24.38 m2mb_tcp_te:446 - M2M_msgTCPTask{TCP_TASK}$ Data received (16): <hello from m2mb!>
[DEBUG] 24.39 m2mb_tcp_te:456 - M2M_msgTCPTask{TCP_TASK}$ application exit
[DEBUG] 24.43 m2mb_tcp_te:468 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 24.45 m2mb_tcp_te:471 - M2M_msgTCPTask{TCP_TASK}$ Application complete.
[DEBUG] 24.79 m2mb_tcp_te:178 - PdpCallback{pubTspt_0}$ Context successfully deactivated!

```

Figure 344

3.6.52 TCP Socket status

Sample application showcasing how to check a TCP connected socket current status.
Debug prints on **MAIN UART**

Features

- How to check module registration and activate PDP context
- How to open a TCP client socket
- How to check if the TCP socket is still valid

3.6.52.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage socket and start it

m2m_tcp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and link it to the PDP context id
- Connect to the server
- Check in a loop the current socket status using the adv_select function with a 2 seconds timeout
- Close socket when the remote host closes it
- Disable PDP context


```

Starting TCP socket status check demo app. This is v1.0.14-C1 built on Sep  8 2020 14:59:25.
[DEBUG] 21.33 m2m_tcp_tes:324 - M2M_msgTCPTask{TCP_TASK}$ INIT
[DEBUG] 21.34 m2m_tcp_tes:338 - M2M_msgTCPTask{TCP_TASK}$ m2mb_os_ev_init success
[DEBUG] 21.34 m2m_tcp_tes:344 - M2M_msgTCPTask{TCP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 21.35 m2m_tcp_tes:352 - M2M_msgTCPTask{TCP_TASK}$ Waiting for registration...
[DEBUG] 21.36 m2m_tcp_tes:365 - M2M_msgTCPTask{TCP_TASK}$ Pdp context activation
[DEBUG] 21.37 m2m_tcp_tes:369 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 23.41 m2m_tcp_tes:384 - M2M_msgTCPTask{TCP_TASK}$ Activate PDP with APN NXT17.NET....
[DEBUG] 24.09 m2m_tcp_tes:281 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 24.10 m2m_tcp_tes:284 - PdpCallback{pubTspt_0}$ IP address: 100.77.5.223
[DEBUG] 24.10 m2m_tcp_tes:394 - M2M_msgTCPTask{TCP_TASK}$ Creating Socket...
[DEBUG] 24.11 m2m_tcp_tes:405 - M2M_msgTCPTask{TCP_TASK}$ Socket created
[DEBUG] 24.11 m2m_tcp_tes:415 - M2M_msgTCPTask{TCP_TASK}$ Socket ctx set to 3
[DEBUG] 24.60 m2m_tcp_tes:428 - M2M_msgTCPTask{TCP_TASK}$ Retrieved IP: 185.86.42.218
[DEBUG] 24.93 m2m_tcp_tes:443 - M2M_msgTCPTask{TCP_TASK}$ Socket Connected!
[DEBUG] 26.98 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 29.03 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
...
[DEBUG] 82.18 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 84.23 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 86.28 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 88.31 m2m_tcp_tes:461 - M2M_msgTCPTask{TCP_TASK}$ Socket does not have any event, try again...
[DEBUG] 88.90 m2m_tcp_tes:154 - adv_select{TCP_TASK}$ Data is available on socket <0x40032b3c>
[DEBUG] 88.92 m2m_tcp_tes:160 - adv_select{TCP_TASK}$ There are <0> pending bytes on socket
Socket was closed by remote!
[DEBUG] 88.92 m2m_tcp_tes:494 - M2M_msgTCPTask{TCP_TASK}$ application exit
[DEBUG] 88.94 m2m_tcp_tes:506 - M2M_msgTCPTask{TCP_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 88.94 m2m_tcp_tes:509 - M2M_msgTCPTask{TCP_TASK}$ Application complete.
[DEBUG] 89.31 m2m_tcp_tes:290 - PdpCallback{pubTspt_0}$ Context successfully deactivated!

```

Figure 345

3.6.53 TCP Server

Sample application showcasing TCP listening socket demo with M2MB API. Debug prints on **MAIN UART**

Features

- How to check module registration and activate PDP context
- How to open a TCP listening socket
- How to manage external hosts connection and exchange data

3.6.53.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task to manage socket and start it

m2m_tcp_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and set it in non-blocking mode
- Bind the socket to the listening port
- Start listening for incoming connection
- Check if a connection is incoming using m2mb_socket_bsd_select function
- If a client connects, perform accept on the child socket
- Send a "START" message to the client
- Send some data
- Wait for data from client and print it
- Close the child socket
- Start listening again, up to 3 times
- Close listening socket
- Disable PDP context

Debug Log

```

Starting TCP Server demo app. This is v1.0.7 built on Apr 7 2020 13:28:24.
[DEBUG] 14.55 m2m_tcp_test.c:220 - M2M_msgTCPTask(TCP_TASK)$ INIT
[DEBUG] 14.55 m2m_tcp_test.c:236 - M2M_msgTCPTask(TCP_TASK)$ m2mb_os_ev_init success
[DEBUG] 14.57 m2m_tcp_test.c:242 - M2M_msgTCPTask(TCP_TASK)$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 14.57 m2m_tcp_test.c:250 - M2M_msgTCPTask(TCP_TASK)$ Waiting for registration...
[DEBUG] 14.58 m2m_tcp_test.c:138 - NetCallback(pubTspt_0)$ Module is registered to cell 0x5222!
[DEBUG] 14.59 m2m_tcp_test.c:263 - M2M_msgTCPTask(TCP_TASK)$ Pdp context activation
[DEBUG] 14.60 m2m_tcp_test.c:267 - M2M_msgTCPTask(TCP_TASK)$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 16.57 m2m_tcp_test.c:282 - M2M_msgTCPTask(TCP_TASK)$ Activate PDP with APN ibox.tim.it...
[DEBUG] 17.16 m2m_tcp_test.c:165 - PdpCallback(pubTspt_0)$ Context activated!
[DEBUG] 17.17 m2m_tcp_test.c:168 - PdpCallback(pubTspt_0)$ IP address: 2.195.165.137

-----
| Start TCP server |
|-----|

[DEBUG] 19.15 m2m_tcp_test.c:301 - M2M_msgTCPTask(TCP_TASK)$ Creating Socket...
[DEBUG] 19.15 m2m_tcp_test.c:312 - M2M_msgTCPTask(TCP_TASK)$ Socket created
[DEBUG] 19.16 m2m_tcp_test.c:313 - M2M_msgTCPTask(TCP_TASK)$ m2mb_socket_bsd_socket(): valid socket ID [0x4002E79C] - PASS
[DEBUG] 20.16 m2m_tcp_test.c:319 - M2M_msgTCPTask(TCP_TASK)$ issuing m2m_socket_bsd_ioctl() to set non-blocking mode ...
[DEBUG] 20.17 m2m_tcp_test.c:331 - M2M_msgTCPTask(TCP_TASK)$ Binding Socket...
[DEBUG] 22.12 m2m_tcp_test.c:343 - M2M_msgTCPTask(TCP_TASK)$ Socket Bind Pass

Start TCP listening on port 6500...
[DEBUG] 24.13 m2m_tcp_test.c:368 - M2M_msgTCPTask(TCP_TASK)$ select...
Select result: 0
[DEBUG] 28.13 m2m_tcp_test.c:368 - M2M_msgTCPTask(TCP_TASK)$ select...
Select result: 1

TCP Server Coming Connection
--> Accept
[DEBUG] 30.52 m2m_tcp_test.c:397 - M2M_msgTCPTask(TCP_TASK)$ Socket Accept Pass

Connected! (socket dial n.1)

[DEBUG] 30.53 m2m_tcp_test.c:403 - M2M_msgTCPTask(TCP_TASK)$ Client Source Address: 185.86.42.254
[DEBUG] 30.54 m2m_tcp_test.c:404 - M2M_msgTCPTask(TCP_TASK)$ Client Port: 58658
[DEBUG] 30.54 m2m_tcp_test.c:405 - M2M_msgTCPTask(TCP_TASK)$ Client Family: 2
[DEBUG] 31.56 m2m_tcp_test.c:410 - M2M_msgTCPTask(TCP_TASK)$

-----
[DEBUG] 31.57 m2m_tcp_test.c:411 - M2M_msgTCPTask(TCP_TASK)$ | Send/receive data test |
[DEBUG] 31.57 m2m_tcp_test.c:412 - M2M_msgTCPTask(TCP_TASK)$ -----

[DEBUG] 32.58 m2m_tcp_test.c:416 - M2M_msgTCPTask(TCP_TASK)$
--> issuing m2mb_socket_bsd_send(): transmit "START" packet...
[DEBUG] 32.59 m2m_tcp_test.c:423 - M2M_msgTCPTask(TCP_TASK)$ --> done (11 have been transmitted)
[DEBUG] 32.60 m2m_tcp_test.c:425 - M2M_msgTCPTask(TCP_TASK)$ ALL data transmitted - PASS
[DEBUG] 32.61 m2m_tcp_test.c:430 - M2M_msgTCPTask(TCP_TASK)$
--> issuing m2mb_socket_bsd_send(): transmit 58 bytes...
[DEBUG] 32.62 m2m_tcp_test.c:437 - M2M_msgTCPTask(TCP_TASK)$ --> done (58 have been transmitted)
[DEBUG] 32.63 m2m_tcp_test.c:440 - M2M_msgTCPTask(TCP_TASK)$ ALL data transmitted - PASS
[DEBUG] 32.64 m2m_tcp_test.c:448 - M2M_msgTCPTask(TCP_TASK)$
Waiting for data...

[DEBUG] 39.64 m2m_tcp_test.c:457 - M2M_msgTCPTask(TCP_TASK)$ test
[DEBUG] 99.61 m2m_tcp_test.c:465 - M2M_msgTCPTask(TCP_TASK)$
m2mb_socket_bsd_recv() has received 6 bytes

[DEBUG] 102.60 m2m_tcp_test.c:469 - M2M_msgTCPTask(TCP_TASK)$
Server TCP is closing the current connection ...

```

Figure 346

Data on a PuTTY terminal

```
START  
aaaaaaaaa-bbbbbbbbbb-ccccccccc-ddddddddd-eeeeeeeeee  
test  
█
```

Figure 347

3.6.54 TLS SSL Client

Sample application showcasing TLS/SSL with client certificates usage with M2MB API. Debug prints on **MAIN UART**

Features

- How to check module registration and enable PDP context
- How to open a SSL client socket
- How to communicate over SSL socket

3.6.54.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Create a task to manage the connection and start it

ssl_test.c

- Initialize Network structure and check registration
- Initialize PDP structure and start PDP context
- Create socket and link it to the PDP context id
- Connect to the server over TCP socket
- Initialize the TLS parameters (TLS1.2) andh auth mode (server+client auth in the example)
- Create SSL context
- Read certificates files and store them
- Create secure socket and connect to the server using SSL
- Send data and receive response
- Close secure socket
- Close socket
- Delete SSL context
- Disable PDP context

The application requires the certificates to be stored in /mod/ssl_certs/ folder. It can be created with

AT#M2MMKDIR=/mod/ssl_certs

Certificates can then be loaded with

AT#M2MWRITE="/mod/ssl_certs/modulesCA.crt",1740

and providing the file content in RAW mode (for example using the "Transfer Data" button in Telit AT Controller)

For client certificates (if required), the commands will be

AT#M2MWRITE="/mod/ssl_certs/modulesClient.crt",1651

AT#M2MWRITE="/mod/ssl_certs/modulesClient_pkcs1.key",1679

PLEASE NOTE: always verify the file sizes to be used in the commands above as they might change

```
Starting TLS-SSL demo app. This is v1.1.2 built on Mar  3 2021 10:15:00.
[DEBUG] 10.85 ssl_test:252 - msgHTTPSTask{TLS_TASK}$ INIT
[DEBUG] 10.85 ssl_test:266 - msgHTTPSTask{TLS_TASK}$ m2mb_os_ev_init success
[DEBUG] 10.85 ssl_test:270 - msgHTTPSTask{TLS_TASK}$ Init SSL session test app
[DEBUG] 10.85 ssl_test:285 - msgHTTPSTask{TLS_TASK}$ m2mb_ssl_create_config sslConfigHnd1 = 0x40037958, sslRes= 0
[DEBUG] 10.85 ssl_test:294 - msgHTTPSTask{TLS_TASK}$ m2mb_ssl_create_config PASSED
[DEBUG] 10.85 ssl_test:306 - msgHTTPSTask{TLS_TASK}$ m2mb_ssl_create_ctxt PASSED
[DEBUG] 10.85 ssl_test:311 - msgHTTPSTask{TLS_TASK}$ loading CA CERT from file /mod/ssl_certs/modulesCA.crt
[DEBUG] 10.85 ssl_test:315 - msgHTTPSTask{TLS_TASK}$ file size: 1740
[DEBUG] 10.85 ssl_test:328 - msgHTTPSTask{TLS_TASK}$ Reading content from file. Size: 1740
Buffer successfully received from file. 1740 bytes were loaded.
Closing file.
[DEBUG] 10.85 ssl_test:361 - msgHTTPSTask{TLS_TASK}$ loading client CERT from file /mod/ssl_certs/modulesClient.crt
[DEBUG] 10.85 ssl_test:365 - msgHTTPSTask{TLS_TASK}$ file size: 1651
[DEBUG] 10.85 ssl_test:378 - msgHTTPSTask{TLS_TASK}$ Reading content from file. Size: 1651
Buffer successfully received from file. 1651 bytes were loaded.
Closing file.
[DEBUG] 10.85 ssl_test:401 - msgHTTPSTask{TLS_TASK}$ loading client KEY from file /mod/ssl_certs/modulesClient_pkcs1.key
[DEBUG] 10.85 ssl_test:405 - msgHTTPSTask{TLS_TASK}$ file size: 1679
[DEBUG] 10.85 ssl_test:418 - msgHTTPSTask{TLS_TASK}$ Reading content from file. Size: 1679
Buffer successfully received from file. 1679 bytes were loaded.
Closing file.
[DEBUG] 10.85 ssl_test:448 - msgHTTPSTask{TLS_TASK}$ certificates successfully stored!
[DEBUG] 10.85 ssl_test:457 - msgHTTPSTask{TLS_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 10.85 ssl_test:465 - msgHTTPSTask{TLS_TASK}$ Waiting for registration...
[DEBUG] 10.86 ssl_test:171 - NetCallback{pubTspt_0}$ Module is registered to cell 0x468E!
[DEBUG] 10.86 ssl_test:477 - msgHTTPSTask{TLS_TASK}$ Pdp context activation
[DEBUG] 10.86 ssl_test:481 - msgHTTPSTask{TLS_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 12.87 ssl_test:496 - msgHTTPSTask{TLS_TASK}$ Activate PDP with APN web.omnitel.it...
[DEBUG] 13.71 ssl_test:197 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 13.71 ssl_test:200 - PdpCallback{pubTspt_0}$ IP address: 2.41.76.63
[DEBUG] 13.71 ssl_test:514 - msgHTTPSTask{TLS_TASK}$ Creating Socket...
[DEBUG] 13.71 ssl_test:525 - msgHTTPSTask{TLS_TASK}$ Socket created
[DEBUG] 13.71 ssl_test:535 - msgHTTPSTask{TLS_TASK}$ Socket ctx set to 3
[DEBUG] 13.92 ssl_test:548 - msgHTTPSTask{TLS_TASK}$ Retrieved IP: 185.86.42.218
[DEBUG] 14.05 ssl_test:562 - msgHTTPSTask{TLS_TASK}$ Socket Connected!
[DEBUG] 15.97 ssl_test:587 - msgHTTPSTask{TLS_TASK}$ m2mb_ssl_connect ret 0
[DEBUG] 17.99 ssl_test:593 - msgHTTPSTask{TLS_TASK}$ Sending bytes..
```

```
[DEBUG] 17.99 ssl_test:593 - msgHTTPSTask{TLS_TASK}$ Sending bytes..
[DEBUG] 17.99 ssl_test:596 - msgHTTPSTask{TLS_TASK}$ SSL write result = 44
[DEBUG] 22.03 ssl_test:608 - msgHTTPSTask{TLS_TASK}$ pending bytes: 1087
[DEBUG] 22.03 ssl_test:612 - msgHTTPSTask{TLS_TASK}$ trying to receive 1087 bytes..
[DEBUG] 22.03 ssl_test:618 - msgHTTPSTask{TLS_TASK}$ Server response: (269)<HTTP/1.1 200 OK
Date: Wed, 03 Mar 2021 09:18:22 GMT
Server: Apache/2.2.15 (CentOS)
Last-Modified: Mon, 22 Jan 2018 10:57:39 GMT
ETag: "1fffc-27f-5635b4c6f12b3"
Accept-Ranges: bytes
Content-Length: 639
Connection: close
Content-Type: text/html; charset=UTF-8
>
[DEBUG] 22.03 ssl_test:634 - msgHTTPSTask{TLS_TASK}$ pending bytes: 762
[DEBUG] 22.03 ssl_test:638 - msgHTTPSTask{TLS_TASK}$ trying to receive remaining 762 bytes..
[DEBUG] 22.03 ssl_test:644 - msgHTTPSTask{TLS_TASK}$ Server response: (639)<<html>
<head>
<title>module.telit.com</title>
<meta content="text/html; charset=utf-8" />
</head>
<body>
<table border=0 align=center>
<tr>
<td height="100" align=center><h2>modules.telit.com - Test HTML page</h2></td>
</tr>
<tr>
<td align=center><img src=Telit.jpg alt="Telit logo" height="126" width="410"></img></td>
</tr>
<tr>
<td height="200" align=center> This is a simple HTML page, </br>
made with simple HTML code,</br>
just for test!
</td>
</tr>
<tr>
<td height="100" align=center><font size="3">Telit &copy; 2015 - 2017 All rights reserved</font></td>
</tr>
</table>
</body>
</html>
>
[DEBUG] 22.03 ssl_test:662 - msgHTTPSTask{TLS_TASK}$ application exit
[DEBUG] 22.03 ssl_test:680 - msgHTTPSTask{TLS_TASK}$ m2mb_pdp_deactivate returned success
[DEBUG] 22.03 ssl_test:683 - msgHTTPSTask{TLS_TASK}$ Application complete.
[DEBUG] 22.77 ssl_test:206 - PdpCallback{pubTspt_0}$ Context deactivated!
```

3.6.55 Uart To Server

Sample application showcasing how to send data from main UART to a connected TCP server. Debug messages are printed on AUX UART port.

Features

- How to open main UART to receive data
- How to connect to a server
- How to transmit received data from the UART to the server and viceversa

3.6.55.1 Application workflow

M2MB_main.c

- Open UART for data and USB1 for debug
- Init socket, activate PDP context and connect to server
- Init UART, set its callback function, create tasks to handle input from UART and response from server (optional)
- Send a confirmation on UART
- Wait for data, when it is received, send it to the server
- When a response is received, print it on UART.

Main UART:

```
Ready to receive data and send to socket.
<<<test message
<<<test 2
```

Figure 348

Debug log on USB1:

```
Starting. This is build: Jul 17 2019 16:39:24. MASK: 000F
Waiting for registration...
Activate PDP with APN internet.wind.biz....
Context activated!
Socket created
Server IP address: 185.86.42.218
Socket Connected and ready to receive data!
Uart opened, setting callback for data..
Waiting for data from uart.
UART IN: <test message>. Sending to socket...
Data sent to socket!
Response from server (12 bytes): <test message>
UART IN: <test 2>. Sending to socket...
Data sent to socket!
Response from server (6 bytes): <test 2>
```

Figure 349

3.6.56 UDP client

Sample application showcasing UDP echo demo with M2MB API. Debug prints on **MAIN UART**

Features

- How to check module registration and activate PDP context
- How to open a UDP client socket
- How to communicate over the socket

3.6.56.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Print welcome message
- Create a task and start it

m2m_udp_test.c - Initialize Network structure and check registration - Initialize PDP structure and start PDP context - Create socket and link it to the PDP context id - Send data and receive response - Close socket - Disable PDP context

```
Starting UDP client demo app. This is v1.0.7 built on Apr 1 2020 14:57:13.
INIT
[DEBUG] 21.23 m2m_udp_test.c:223 - M2M_msgUDPTask{UDP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
Waiting for registration...
[DEBUG] 21.25 m2m_udp_test.c:131 - NetCallback{pubTspt_0}$ Module is registered to cell 0xC4CF!
[DEBUG] 21.26 m2m_udp_test.c:241 - M2M_msgUDPTask{UDP_TASK}$ Pdp context initialization
[DEBUG] 21.26 m2m_udp_test.c:245 - M2M_msgUDPTask{UDP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
Activate PDP with APN web.omnitel.it...
[DEBUG] 24.11 m2m_udp_test.c:157 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 24.11 m2m_udp_test.c:160 - PdpCallback{pubTspt_0}$ IP address: 109.113.222.12
[DEBUG] 24.12 m2m_udp_test.c:268 - M2M_msgUDPTask{UDP_TASK}$ Creating Socket...
[DEBUG] 24.13 m2m_udp_test.c:280 - M2M_msgUDPTask{UDP_TASK}$ Socket created
Socket ctx set to 3
[DEBUG] 24.41 m2m_udp_test.c:306 - M2M_msgUDPTask{UDP_TASK}$ Retrieved IP: 185.86.42.218
Socket ready.
Data successfully sent (16 bytes)
Socket rcv...
[DEBUG] 26.47 m2m_udp_test.c:352 - M2M_msgUDPTask{UDP_TASK}$ m2mb_socket_bsd_set_sock_opt() M2MB_SOCKET_BSD_SO_RCVTIMEO - success
trying to receive 16 bytes..
Data received (16): <hello from m2mb!>
[DEBUG] 26.48 m2m_udp_test.c:377 - M2M_msgUDPTask{UDP_TASK}$ application exit
Socket Closed
[DEBUG] 26.49 m2m_udp_test.c:399 - M2M_msgUDPTask{UDP_TASK}$ m2mb_pdp_deactivate returned success
Application complete.
[DEBUG] 27.04 m2m_udp_test.c:166 - PdpCallback{pubTspt_0}$ Context successfully deactivated!
```

Figure 350

3.6.57 UDP_Server example

Sample application that shows UDP listening socket demo with m2mb apis. Debug prints on **MAIN UART**

Features

- How to configure an UDP socket into listen mode
- How to receive data using m2mb_socket_bsd_select
- How to read data received and send data to client

Application workflow

M2MB_main.c

- Print welcome message
- Init task apis and create M2M_msgUDPTask to handle UDP socket

m2mb_udp_test.c

M2M_msgUDPTask

- Wait for module registration
- Activate PDP context
- Create UDP listen socket
- Wait for incoming data from client using m2mb_socket_bsd_select
- When there are data on socket, read them and send some data back to client

```
Starting UDP Server demo app. This is v1.1.10 built on Mar 3 2022 15:42:39.

[DEBUG] 19.28 m2mb_udp_te:221 - M2M_msgUDPTask{UDP_TASK}$ INIT
[DEBUG] 19.29 m2mb_udp_te:237 - M2M_msgUDPTask{UDP_TASK}$ m2mb_os_ev_init success
[DEBUG] 19.30 m2mb_udp_te:243 - M2M_msgUDPTask{UDP_TASK}$ m2mb_net_init returned M2MB_RESULT_SUCCESS
[DEBUG] 19.31 m2mb_udp_te:251 - M2M_msgUDPTask{UDP_TASK}$ Waiting for registration...
[DEBUG] 19.32 m2mb_udp_te:140 - NetCallback{pubTspt_0}$ Module is registered to cell 0x5222!
[DEBUG] 19.33 m2mb_udp_te:264 - M2M_msgUDPTask{UDP_TASK}$ Pdp context activation
[DEBUG] 19.33 m2mb_udp_te:268 - M2M_msgUDPTask{UDP_TASK}$ m2mb_pdp_init returned M2MB_RESULT_SUCCESS
[DEBUG] 21.39 m2mb_udp_te:283 - M2M_msgUDPTask{UDP_TASK}$ Activate PDP with APN ibox.tim.it...
[DEBUG] 22.00 m2mb_udp_te:167 - PdpCallback{pubTspt_0}$ Context activated!
[DEBUG] 22.01 m2mb_udp_te:170 - PdpCallback{pubTspt_0}$ IP address: 2.195.163.102

---- Start UDP server Listen IP: 2.195.163.102 PORT: 6500 ----

[DEBUG] 24.07 m2mb_udp_te:299 - M2M_msgUDPTask{UDP_TASK}$ Creating Socket...
[DEBUG] 24.08 m2mb_udp_te:310 - M2M_msgUDPTask{UDP_TASK}$ Socket created
[DEBUG] 24.08 m2mb_udp_te:311 - M2M_msgUDPTask{UDP_TASK}$ m2mb_socket_bsd_socket(): valid socket ID [0x40035490] - PASS
[DEBUG] 25.12 m2mb_udp_te:317 - M2M_msgUDPTask{UDP_TASK}$ issuing m2m_socket_bsd_ioctl() to set non-blocking mode ...
[DEBUG] 25.13 m2mb_udp_te:329 - M2M_msgUDPTask{UDP_TASK}$ Binding Socket...
[DEBUG] 27.18 m2mb_udp_te:341 - M2M_msgUDPTask{UDP_TASK}$ Socket Bind Pass

Waiting for Incoming UDP data, Listen IP: 2.195.163.102 PORT: 6500

[DEBUG] 29.25 m2mb_udp_te:359 - M2M_msgUDPTask{UDP_TASK}$ select...
[DEBUG] 32.32 m2mb_udp_te:362 - M2M_msgUDPTask{UDP_TASK}$ Select result: 0
[DEBUG] 33.34 m2mb_udp_te:359 - M2M_msgUDPTask{UDP_TASK}$ select...
[DEBUG] 36.41 m2mb_udp_te:362 - M2M_msgUDPTask{UDP_TASK}$ Select result: 0
[DEBUG] 37.45 m2mb_udp_te:359 - M2M_msgUDPTask{UDP_TASK}$ select...
[DEBUG] 40.52 m2mb_udp_te:362 - M2M_msgUDPTask{UDP_TASK}$ Select result: 0
[DEBUG] 41.54 m2mb_udp_te:359 - M2M_msgUDPTask{UDP_TASK}$ select...
[DEBUG] 44.61 m2mb_udp_te:362 - M2M_msgUDPTask{UDP_TASK}$ Select result: 0
[DEBUG] 45.64 m2mb_udp_te:359 - M2M_msgUDPTask{UDP_TASK}$ select...
[DEBUG] 45.94 m2mb_udp_te:362 - M2M_msgUDPTask{UDP_TASK}$ Select result: 1
[DEBUG] 45.95 m2mb_udp_te:366 - M2M_msgUDPTask{UDP_TASK}$

Incoming UDP data available
---- Receive/send data test ----

trying to receive bytes..
[DEBUG] 46.99 m2mb_udp_te:399 - M2M_msgUDPTask{UDP_TASK}$ Client Source Address: 151.99.1.17
[DEBUG] 47.00 m2mb_udp_te:400 - M2M_msgUDPTask{UDP_TASK}$ Client Port: 36368
[DEBUG] 47.01 m2mb_udp_te:401 - M2M_msgUDPTask{UDP_TASK}$ Client Family: 2
Data received (4): <test>
Data successfully sent (16 bytes)
```

Figure 351

3.6.58 USB Cable Check

Sample application showing how to check if USB cable is plugged in or not. Debug prints on **MAIN UART**

Features

- How to open an USB channel and configure it with a callback function
- How to manage USB cable events in the callback function

3.6.58.1 Application workflow

M2MB_main.c

- Open UART/UART_AUX for debug
- open usb channel and set the callback
- Print greeting message
- Print current usb status

USB_Cb

- if the event is a connection/disconnection, show the current status

```
Starting USB cable check demo app. This is v1.0.0 built on Aug 19 2020 10:27:40.
m2mb_usb_open succeeded
m2mb_usb_ioctl: set usb callback
m2mb_usb_ioctl: got cable status
USB cable CONNECTED, status: 1

Waiting for USB cable to be plugged/unplugged...
Usb cable check event, USB status: 0
Usb cable check event, USB status: 1
Usb cable check event, USB status: 0
Usb cable check event, USB status: 1
```

Figure 352

3.6.59 Basic USB read/write example

Sample application that shows how to use the basic read/write USB apis. Synchronous or asynchronous mode is available setting SYNC to 1 or 0. Debug prints on **MAIN UART**

Features

- Read and write on USB (synchronous mode)
- Read and write on USB (asynchronous mode)

3.6.59.1 Application workflow

M2MB_main.c

- Open USB port (USB0)
- Set rx and tx timeouts
- **SYNC**
- read until some data are available on USB
- as soon as some data are available on USB read them and write on USB data received
- **ASYNC**
- set the USB callback
- write some data on USB and wait for data to be read
- as soon as some data are available on USB M2MB_USB_RX_EVENT is generated and handled by callback. Data are read and printed on serial com port.

```
Starting USB read write demo app. This is v1.0.0 built on Nov  4 2021 15:38:20.

Open USB port
m2mb_usb_open succeeded
m2mb_usb_ioctl: got cable status
USB cable CONNECTED, status: 1
Synchronous read and write
Read until some bytes are received...
rx timeout expired
```

Figure 353

3.6.60 Watchdog example

Sample application that shows how to set and implement a watchdog. Task has been locked waiting for an event with a timeout longer than wd inactivity timeout. If no wd kick or no actions

Features

- Enable watchdog on Task_1
- Handle watchdog kicks with a timer

3.6.60.1 Application workflow

M2MB_main.c

- Create Task_1 that will be put under watchdog control
- In Task_1 watchdog is enabled on it and a timer created to handle watchdog kicks. Then task enters in a loop where kicks are sent. After 5 kicks a task lock is simulated waiting for an event release (with a timeout longer than task watchdog inactivity timeout)
- As soon as watchdog inactivity timeout expired M2MB_WDOG_TIMEOUT_IND is generated and handled in WDcallback (in this case the event is released, no further actions are done)
- No further actions are performed by Task_1 (no kicks are sent) so after watchdog system timeout expiration app is rebooted.

```
Starting WATCHDOG demo app. This is v1.0.0 built on Nov  4 2021 14:34:06.
Event correctly initialized
Task started
Init WatchDog
m2mb_wDog_init OK
Tick duration: 1s
Adding Task under WD control with inactivity timeout of 60s
m2mb_wDog_addTask OK
Enabling the WatchDog
m2mb_wDog_enable OK
Timer creation OK

Start WD kick every 5s...
WD kick
WD kick done
```

Figure 354

3.6.61 ZLIB example

Sample application showing how to compress/uncompress with ZLIB. Debug prints on **MAIN UART**

Features

- How to compress a file
- How to uncompress a file

In order to execute the entire test, copy test.gz file into your module running the following AT command:

```
AT#M2MWRITE="/mod/test.gz",138
```

>>> here receive the prompt; then type or send the file, sized 138 bytes

3.6.61.1 Application workflow

M2MB_main.c

- Open USB/UART/UART_AUX
- Test the compression and decompression of a data string
- Test the decompression of a .gz file (test.gz), expected to be in /mod folder, into its content test.txt. The file must be uploaded by the user (see steps above).

```
Starting Logging demo app. This is v1.0.7 built on Apr  7 2020 09:02:35.
Starting TEST_COMPR_UNCOMPR.
len: 138; comprLen: 57
Compressed message:
W+EHU(,ILIVH*E/ISHÉ PE*I-HMQE/K-R(Ï Êç$VU*Håşê y4RI«¥1.
comprLen: 57; uncomprLen: 138
uncompress():
the quick brown fox jumped over the lazy dog. the quick brown fox jumped over the lazy dog. the quick brown fox jumped over the lazy dog.
Ending TEST_COMPR_UNCOMPR with SUCCESS.
Starting test_uncompress.
Data extracted correctly into the file ./mod/test.txt
test_uncompress finished correctly!
```

Figure 355

4 Installing beta version libraries Plug-in

4.1 New beta plug-in installation

To install a new plug-in for a beta firmware into the IDE, first receive plug-in “.zip” packet, then unzip the file in a local folder and open the SDK IDE.

PLEASE DO NOT USE BETA PLUGINS FOR PRODUCTION DEPLOYMENTS, SOFTWARE IS PROVIDED AS IS AND CUSTOMER ACKNOWLEDGES THAT IT IS POSSIBLE THE DEVICE MAY MISFUNCTION. PLEASE REFER TO Contact Information, Support section

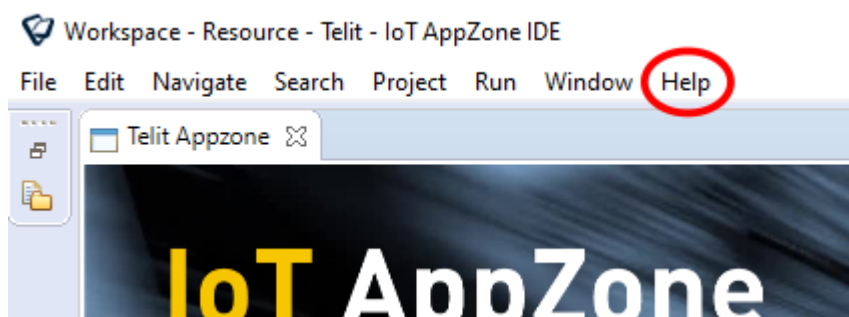
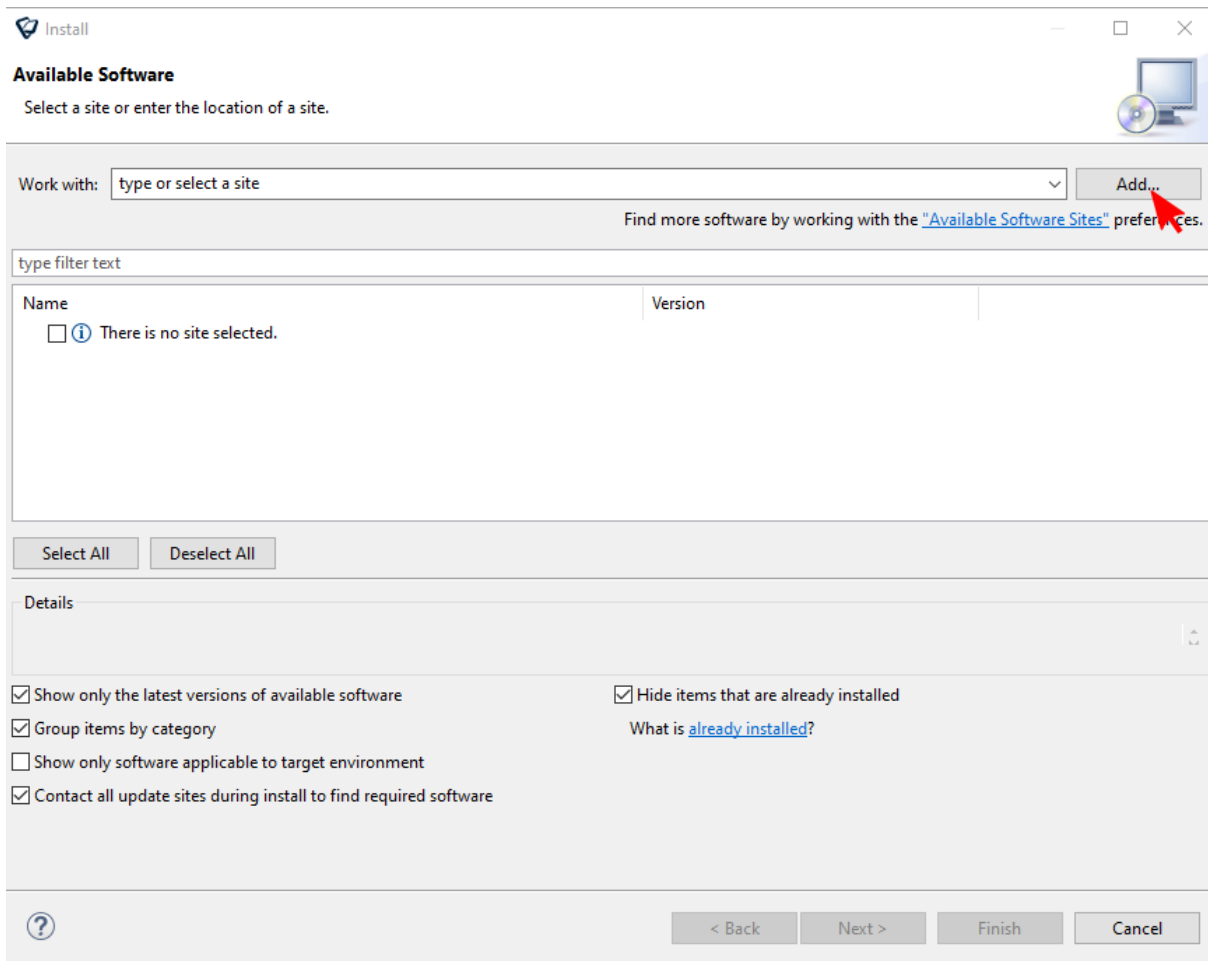
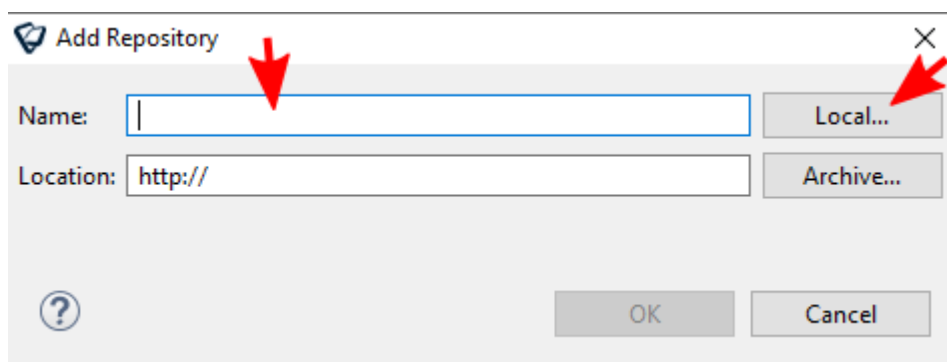


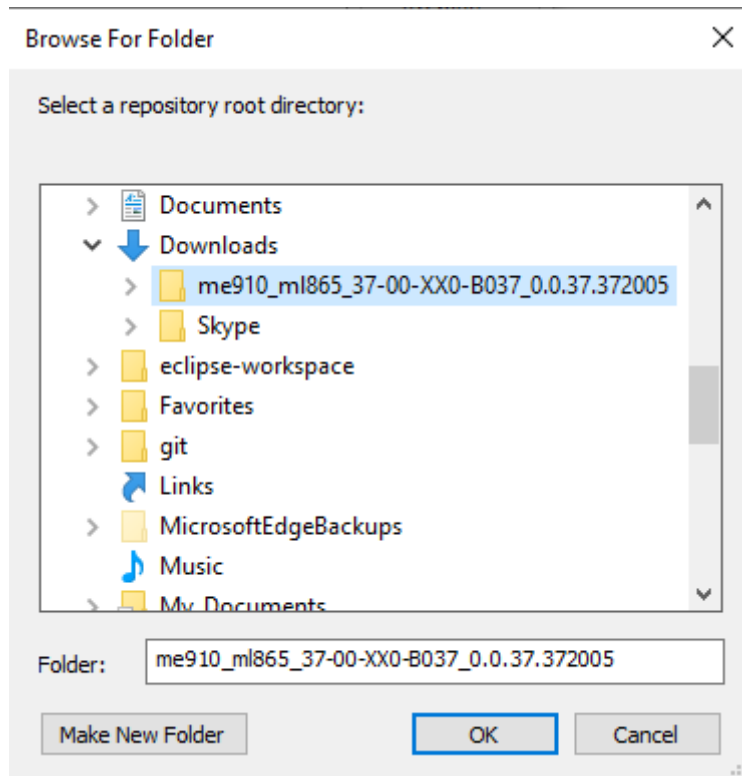
Figure 356

Click on “Help” tag and choose “Install New Software...”. This window will appear:

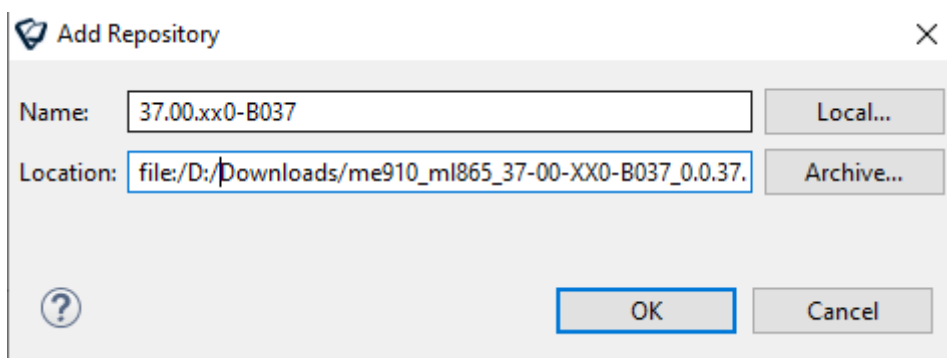
**Figure 357**

Click on "Add..." button and then in the following window click on "Local..." to select the unzipped folder with the plug-in content.

**Figure 358**

**Figure 359**

Once selected the plug-in folder, the “Location:” form will present the selected path. Now in “Name:” write a name for the new libraries (for example 37.00.xx0_B037) and click on “OK” button.

**Figure 360**

The new packet is now ready to be installed: select it and click on “Next >” button until “Review Licenses” window will appear.

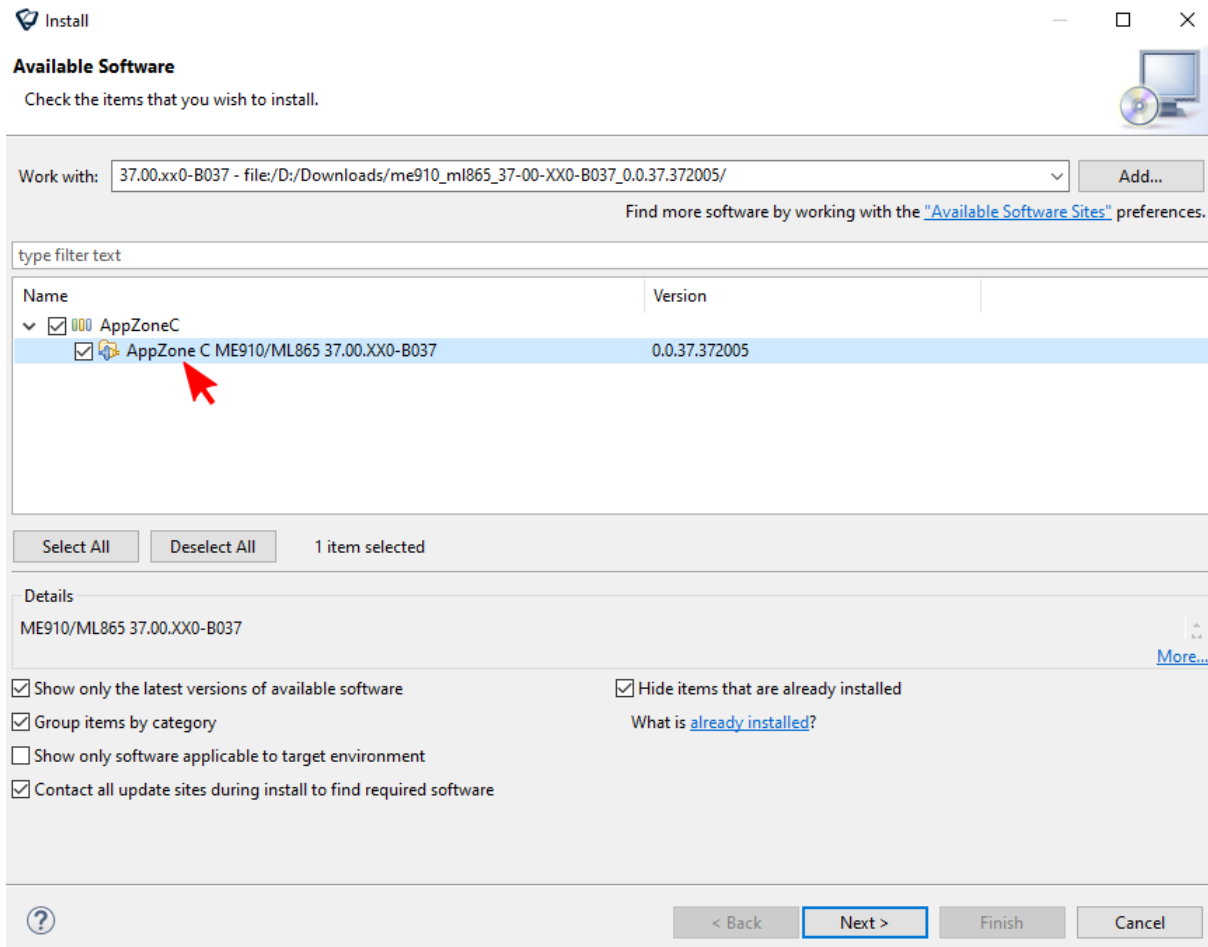


Figure 361

Accept the licenses when required and click on "Finish" button to complete the installation.

4.2 Change existing project libraries

To align an old project to the new libraries, right click on the project and choose "Properties".

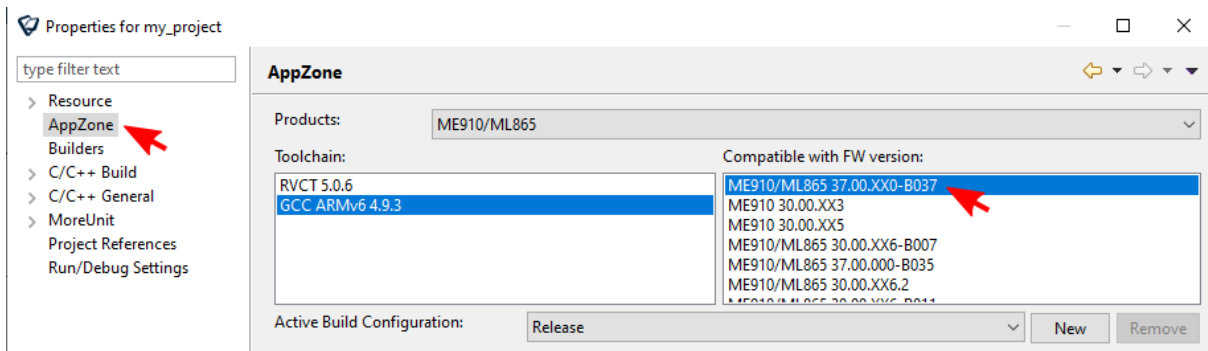


Figure 362

Now select “AppZone” on the left side of the window, and on the right choose the packet with the same name as the firmware version to be used. Then click on “OK” (or “Apply”) button.

4.3 Create a project with the new plug-in

To use the new libraries, create a new project: “File”-> “New” -> “Telit Project”

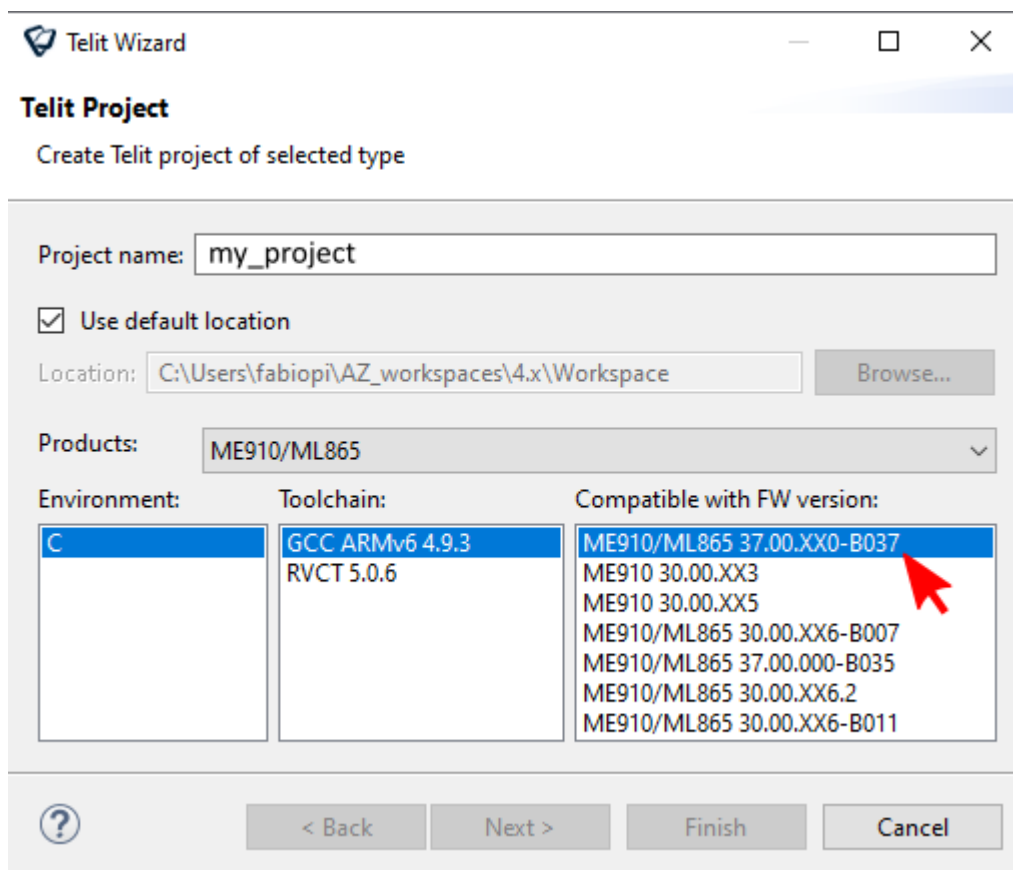


Figure 363

Select the new firmware version (37.00.xx0-B037) and create an empty project.



SUPPORT INQUIRIES

Link to www.telit.com and contact our technical support team for any questions related to technical issues.

www.telit.com



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